Section 7:

Introduction to Geometry, Splines, and Bezier Curves

Computer Graphics and Imaging
UC Berkeley CS184/284A

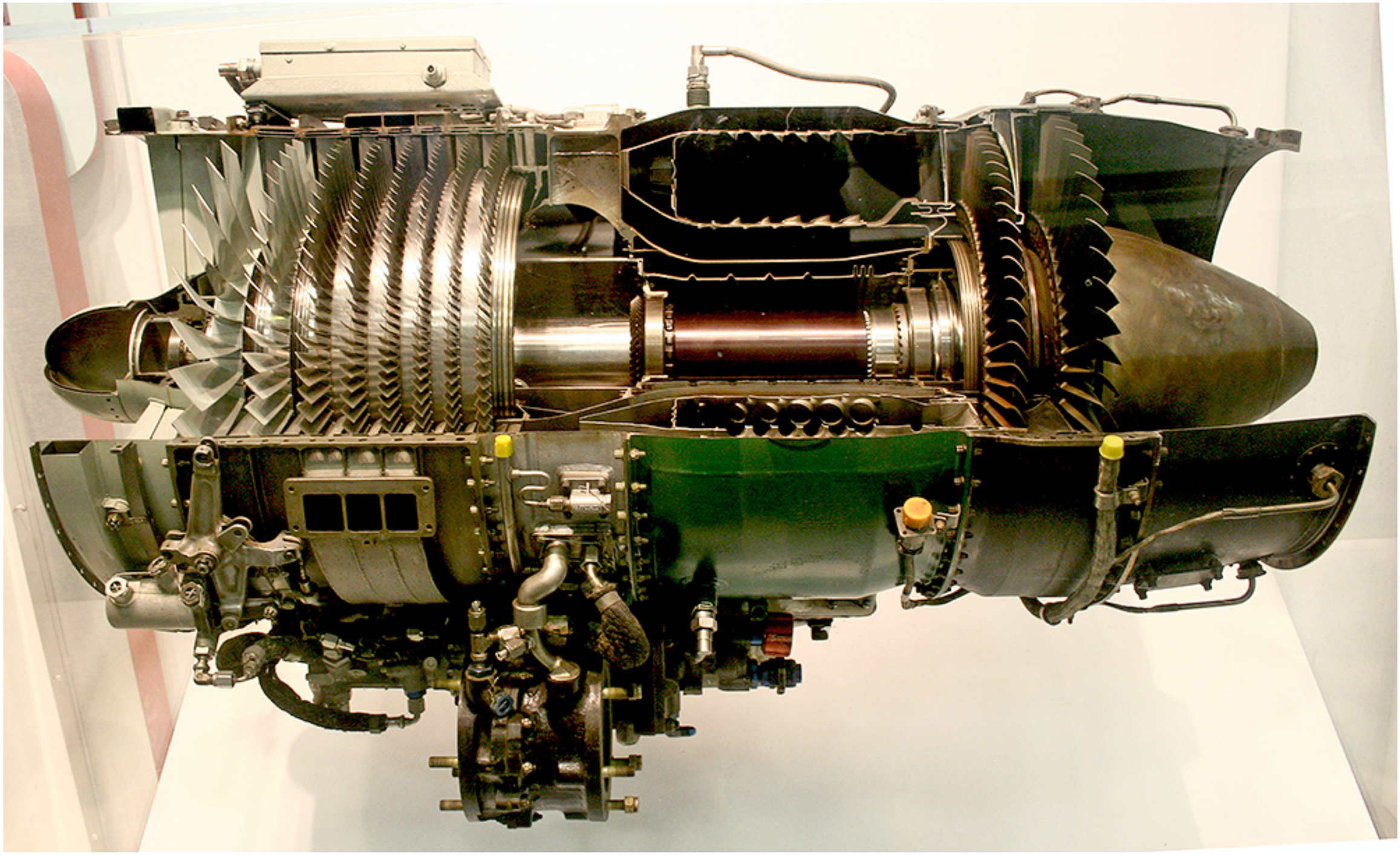
Examples of Geometry



Examples of Geometry



Examples of Geometry



Examples of Geometry





Examples of Geometry



Examples of Geometry



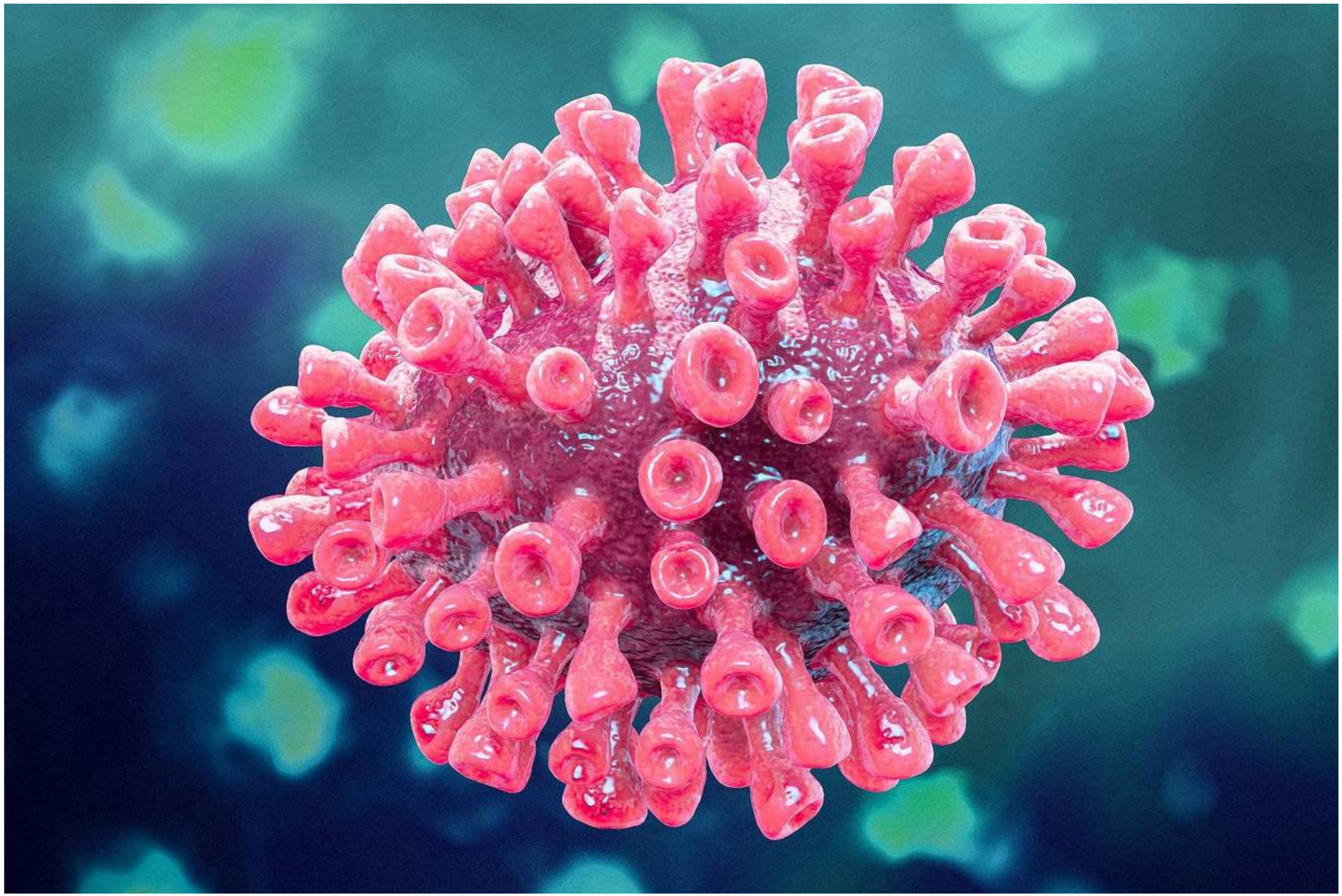
Examples of Geometry



Examples of Geometry

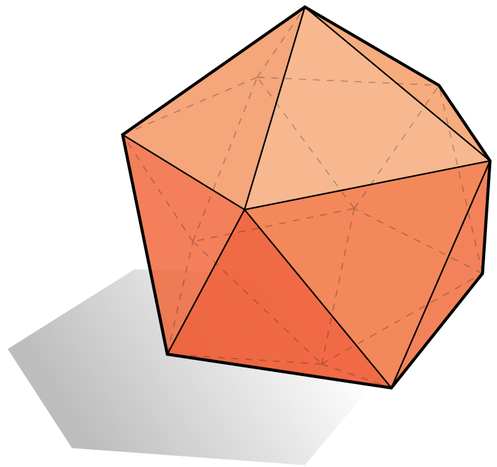


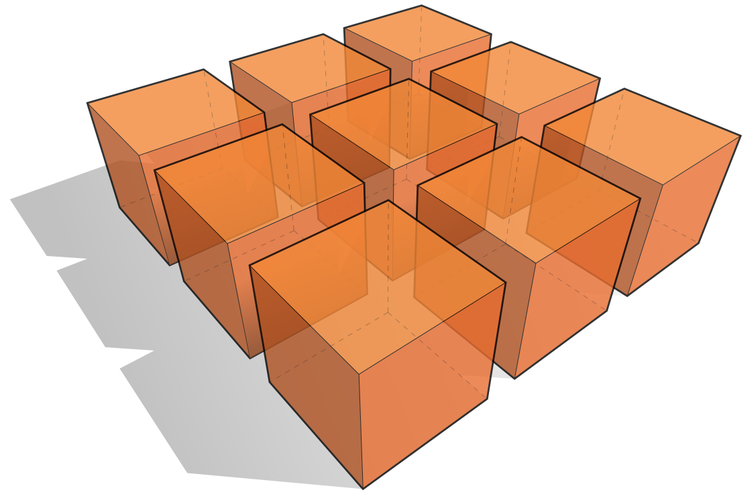
Examples of Geometry



The Lancet

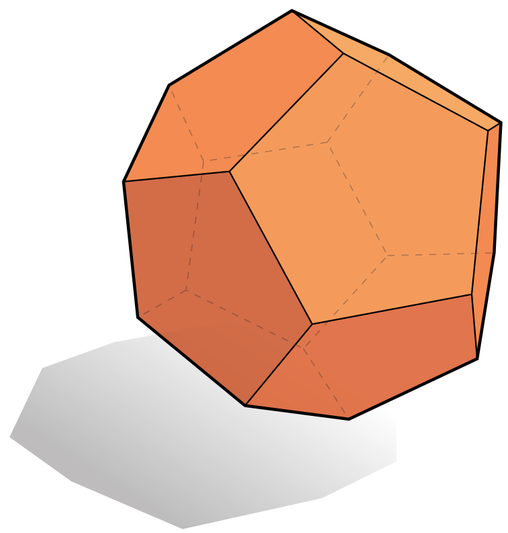
Many Ways to Represent Geometry

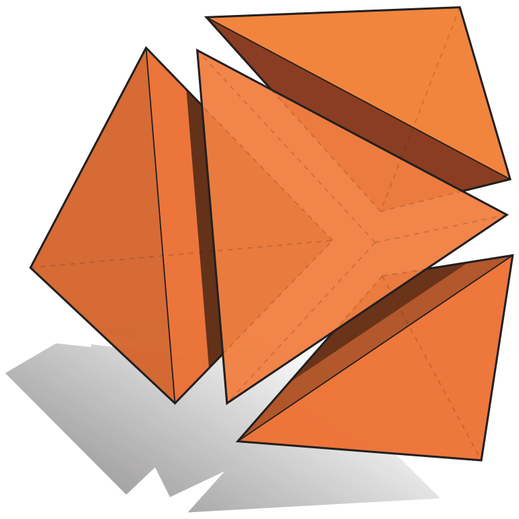




Explicit

- point cloud
- polygon mesh
- subdivision, NURBS
- …





Implicit

- level sets
- algebraic surface
- distance functions
- …

Each choice best suited to a different <u>task</u> or <u>type of geometry</u>

Smooth Curves

Smooth Curves and Surfaces

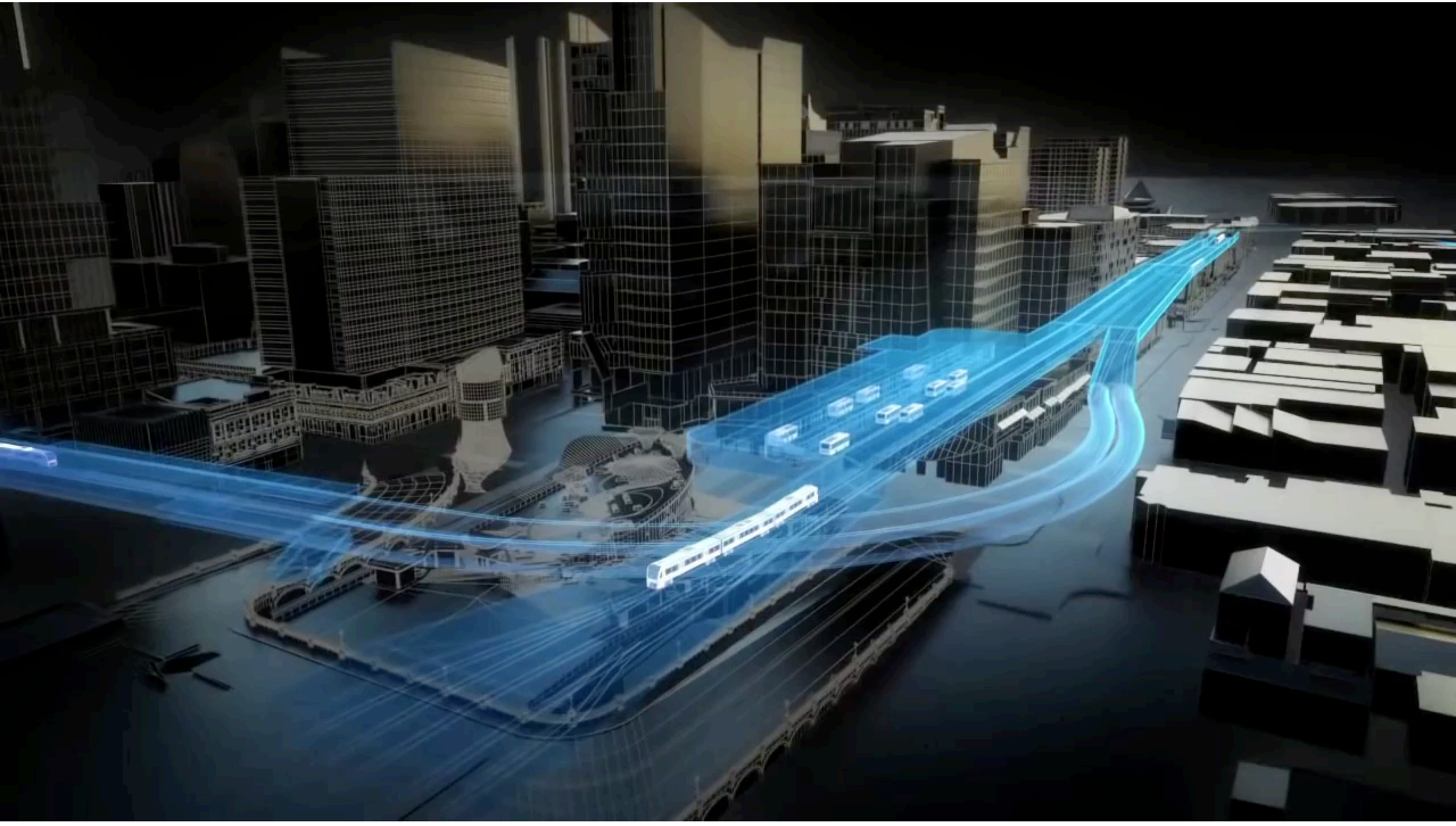
So far we can make:

- Things with corners (lines, triangles, squares, …)
- Specialty shapes (circles, ellipses, …)

Many applications require designed, smooth shapes

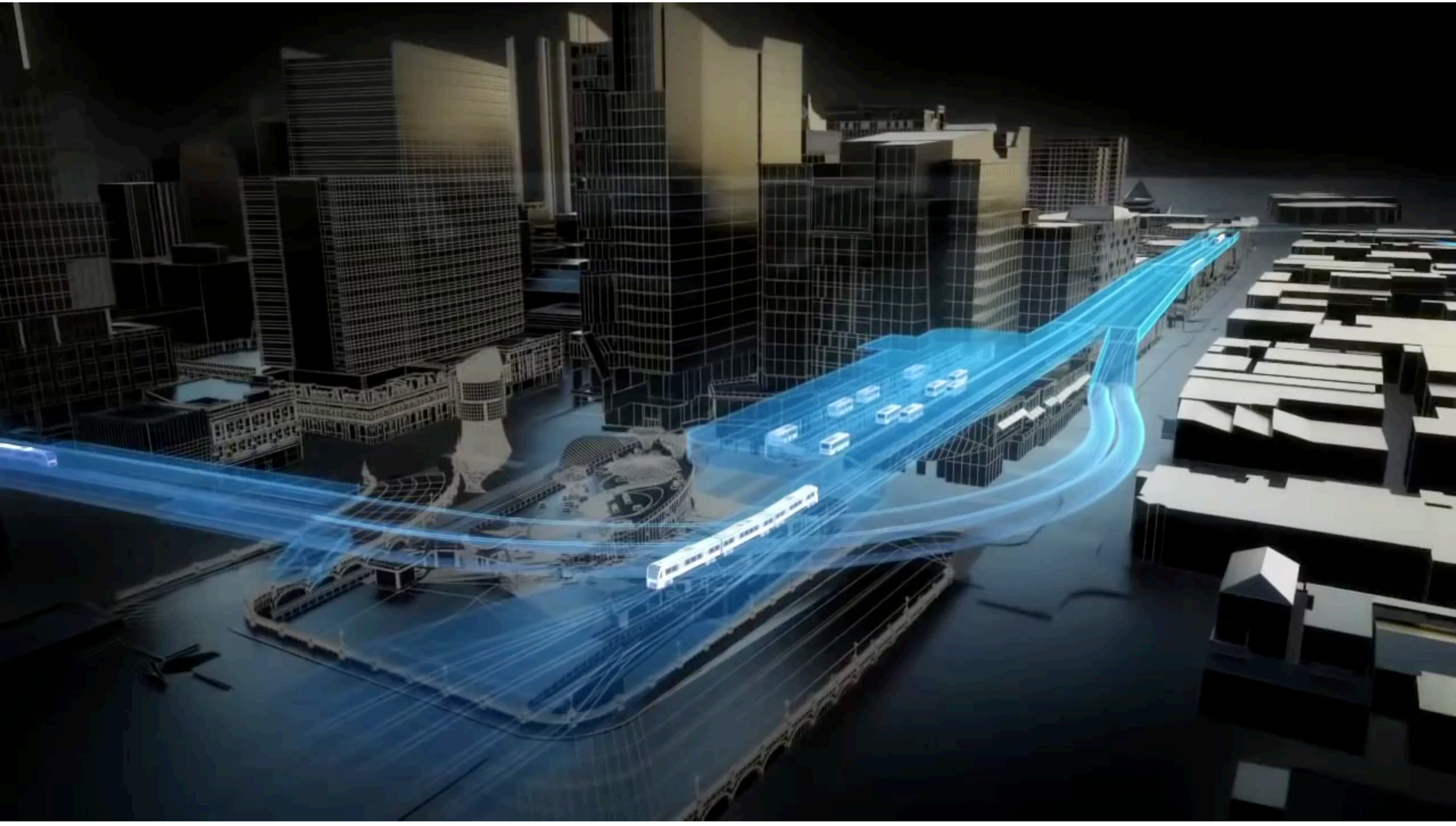
- Camera paths, vector fonts, …
- Resampling filter functions
- CAD design, object modeling, …

Camera Paths



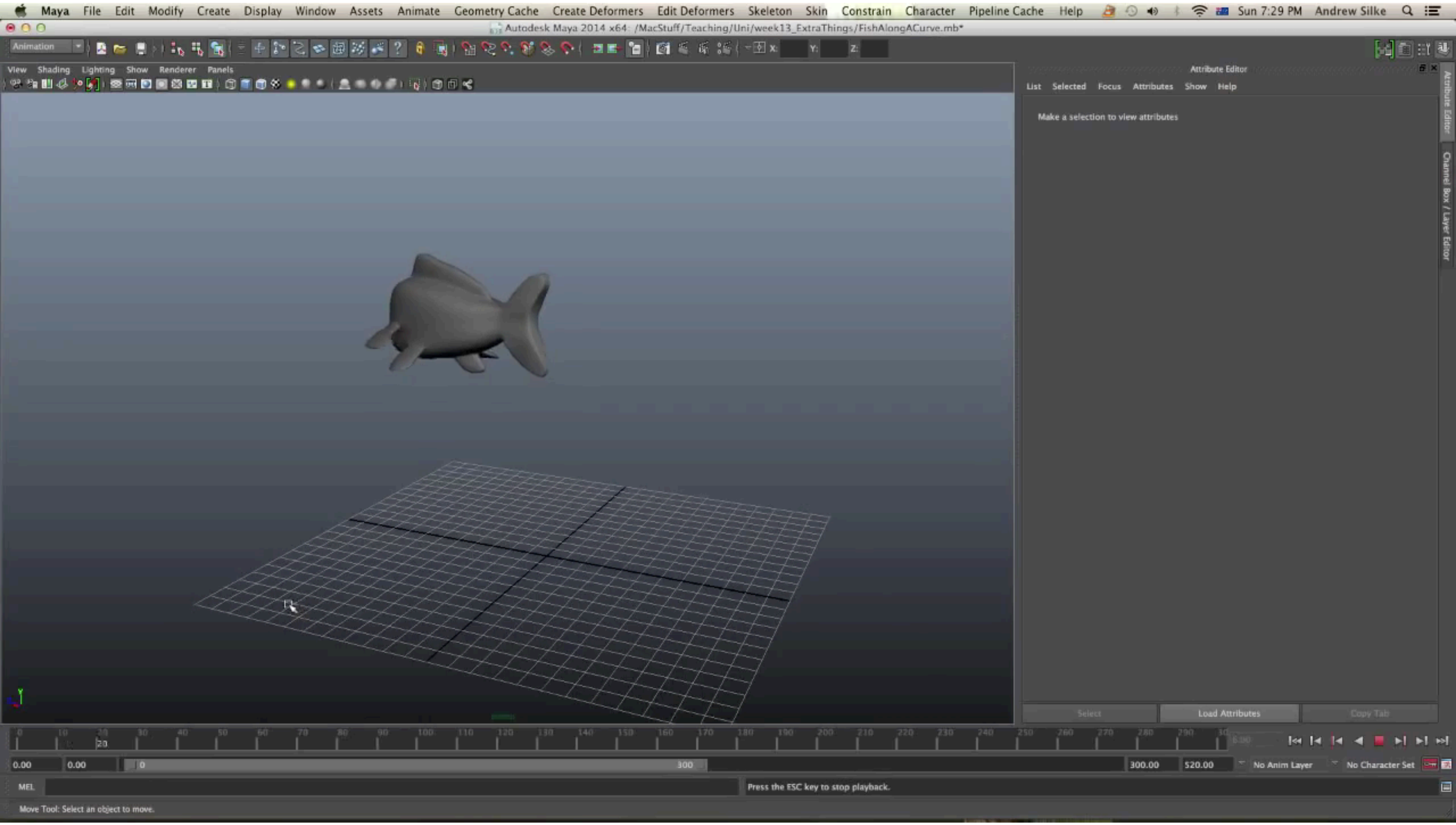
Flythrough of proposed Perth Citylink subway, https://youtu.be/rIJMuQPwr3E

Camera Paths



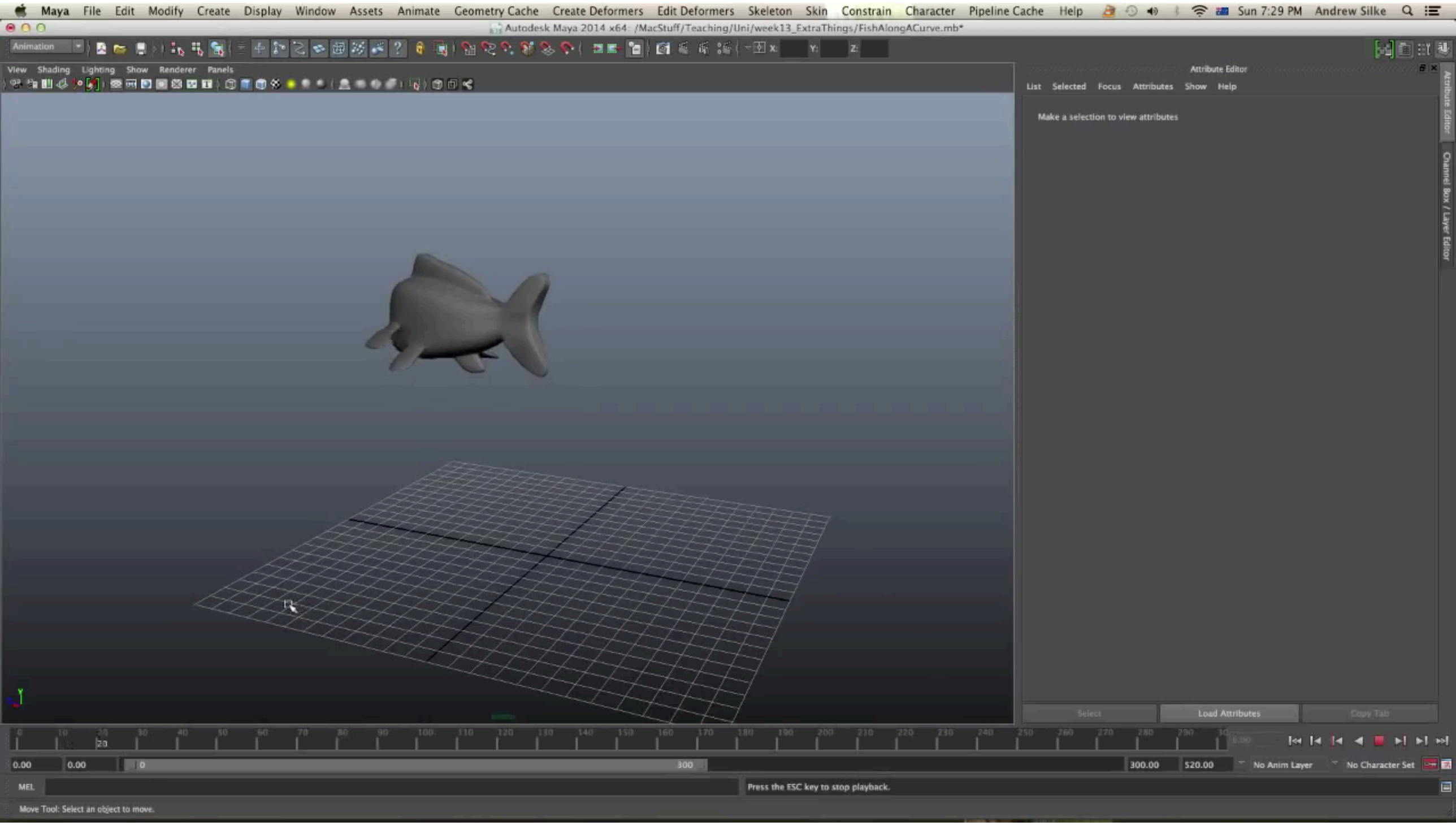
Flythrough of proposed Perth Citylink subway, https://youtu.be/rIJMuQPwr3E

Animation Curves



Maya Animation Tutorial: https://youtu.be/b-o5wtZlJPc

Animation Curves

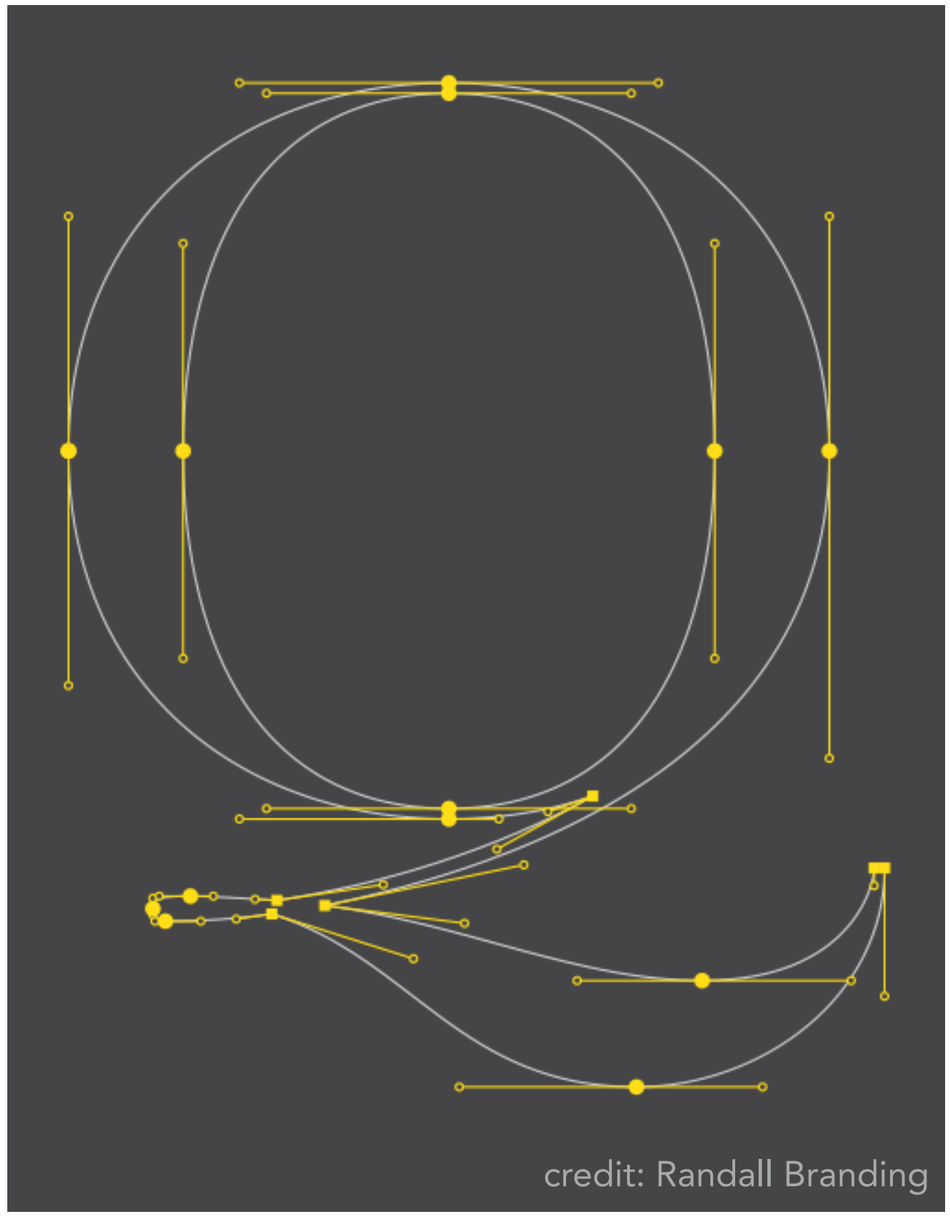


Maya Animation Tutorial: https://youtu.be/b-o5wtZlJPc

Vector Fonts

The Quick Brown Fox Jumps Over The Lazy Dog

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz 0123456789



Baskerville font - represented as cubic Bézier splines

CAD Design



3D Car Modeling with Rhinoceros

CAD Design



3D Car Modeling with Rhinoceros

Parametric Representations

Curves: $\boldsymbol{x} = \boldsymbol{x}(u)$ $\quad \boldsymbol{x} \in \Re^n \quad u \in \Re$

Surfaces: $\boldsymbol{x} = \boldsymbol{x}(u, v)$ $\quad \boldsymbol{x} \in \Re^n \quad u, v \in \Re$

$\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{u})$ $\quad \boldsymbol{u} \in \Re^2$

Volumes: $\boldsymbol{x} = \boldsymbol{x}(u, v, w)$ $\quad \boldsymbol{x} \in \Re^n \quad u, v, w \in \Re$

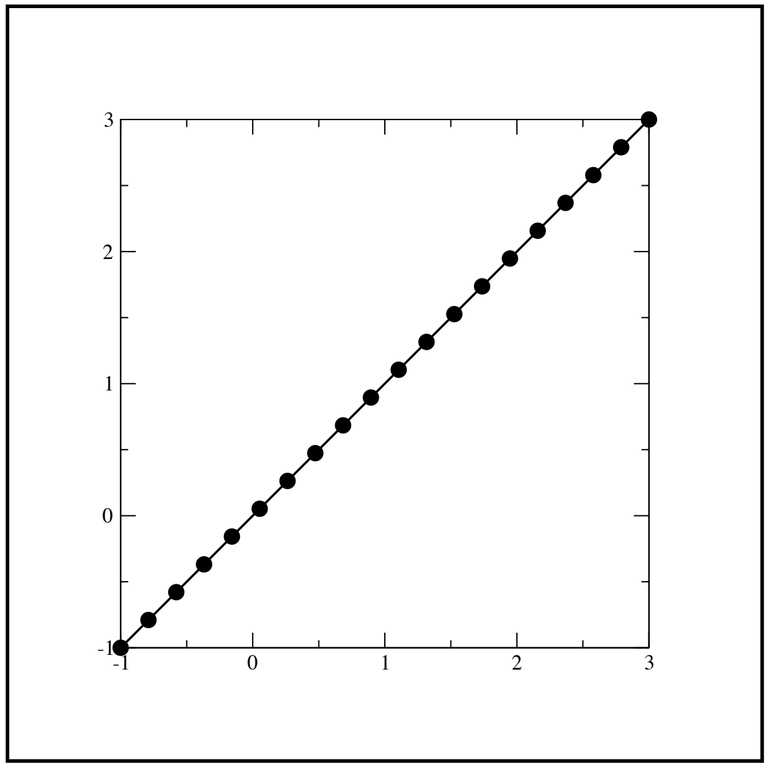
$\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{u})$ $\quad \boldsymbol{u} \in \Re^3$

and so on...

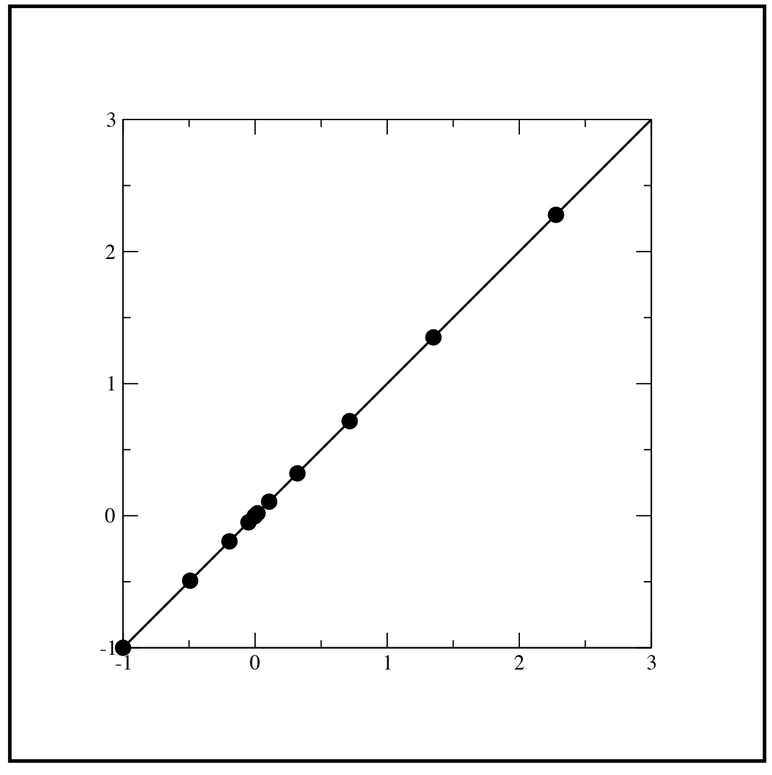
Note: a vector function is equivalently $\boldsymbol{n}$ scalar functions

Parametric Rep. Non-unique

Same curve/surface may have multiple formulae



$$\boldsymbol{x}(u) = [u, u]$$

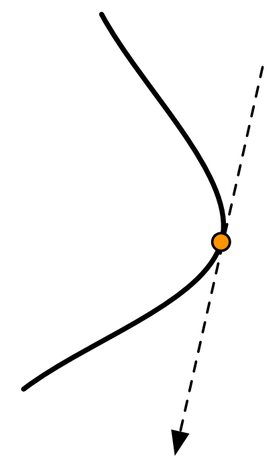


$$\boldsymbol{x}(u) = [u^3, u^3]$$

Some Differential Geometry

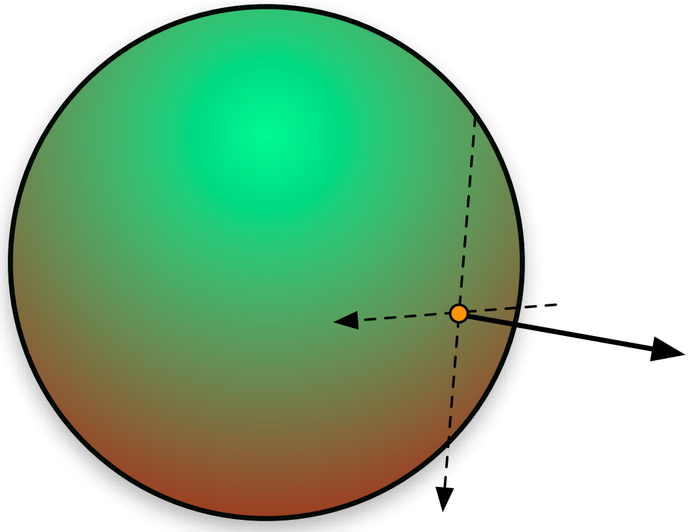
Tangent to curve

$$\boldsymbol{t}(u) = \left.\frac{\partial \boldsymbol{x}}{\partial u}\right|_u$$



Tangents to surface

$$\boldsymbol{t}_u(u,v) = \left.\frac{\partial \boldsymbol{x}}{\partial u}\right|_{u,v} \qquad \boldsymbol{t}_v(u,v) = \left.\frac{\partial \boldsymbol{x}}{\partial v}\right|_{u,v}$$



Normal of surface

$$\hat{\boldsymbol{n}} = \frac{\boldsymbol{t}_u \times \boldsymbol{t}_v}{||\boldsymbol{t}_u \times \boldsymbol{t}_v||}$$

Degeneracies: $\partial \boldsymbol{x}/\partial u = 0$ **or** $\boldsymbol{t}_u \times \boldsymbol{t}_v = 0$

Tangent Space

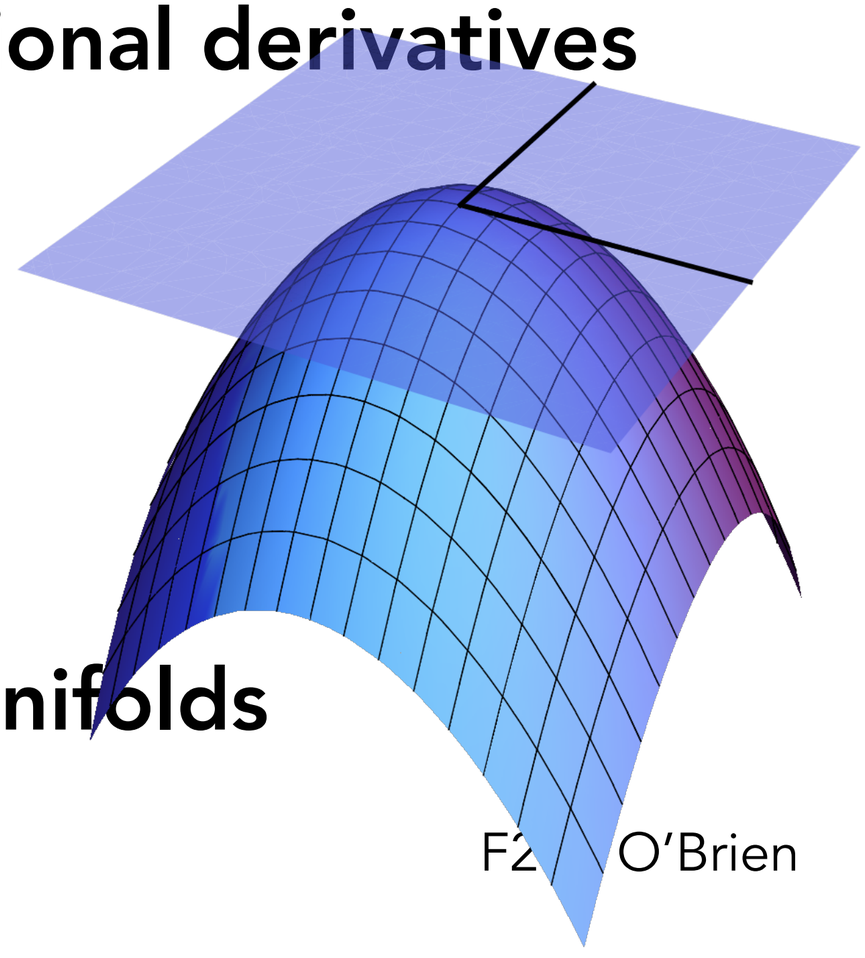
The *tangent space* at a point on a surface is the vector space spanned by

$$\frac{\partial \mathbf{x}(\mathbf{u})}{\partial u} \qquad \frac{\partial \mathbf{x}(\mathbf{u})}{\partial v}$$

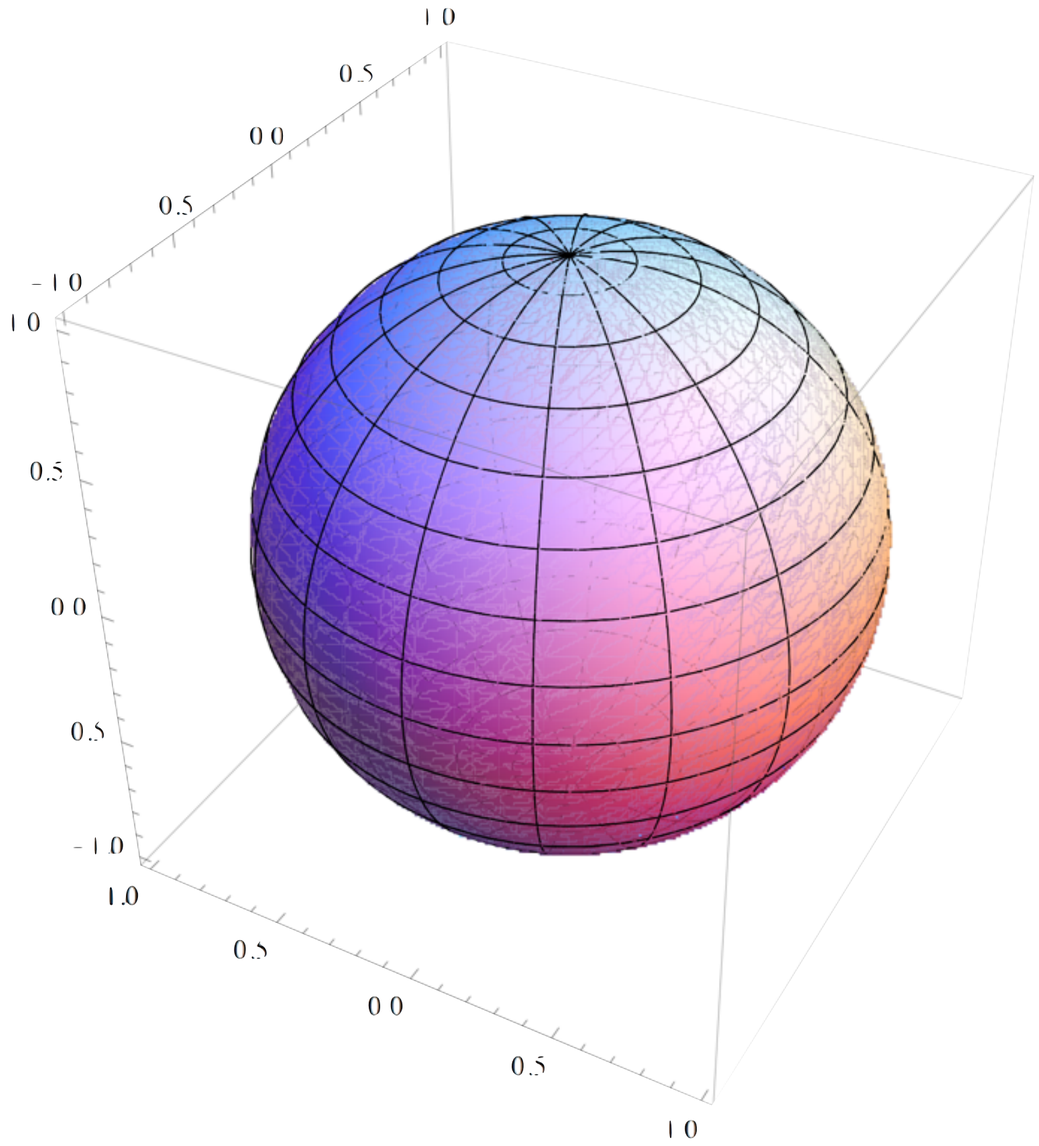
- Definition assumes that these directional derivatives are linearly independent.
- Tangent space of surface may exist even if the parameterization is bad

For surface the space is a plane

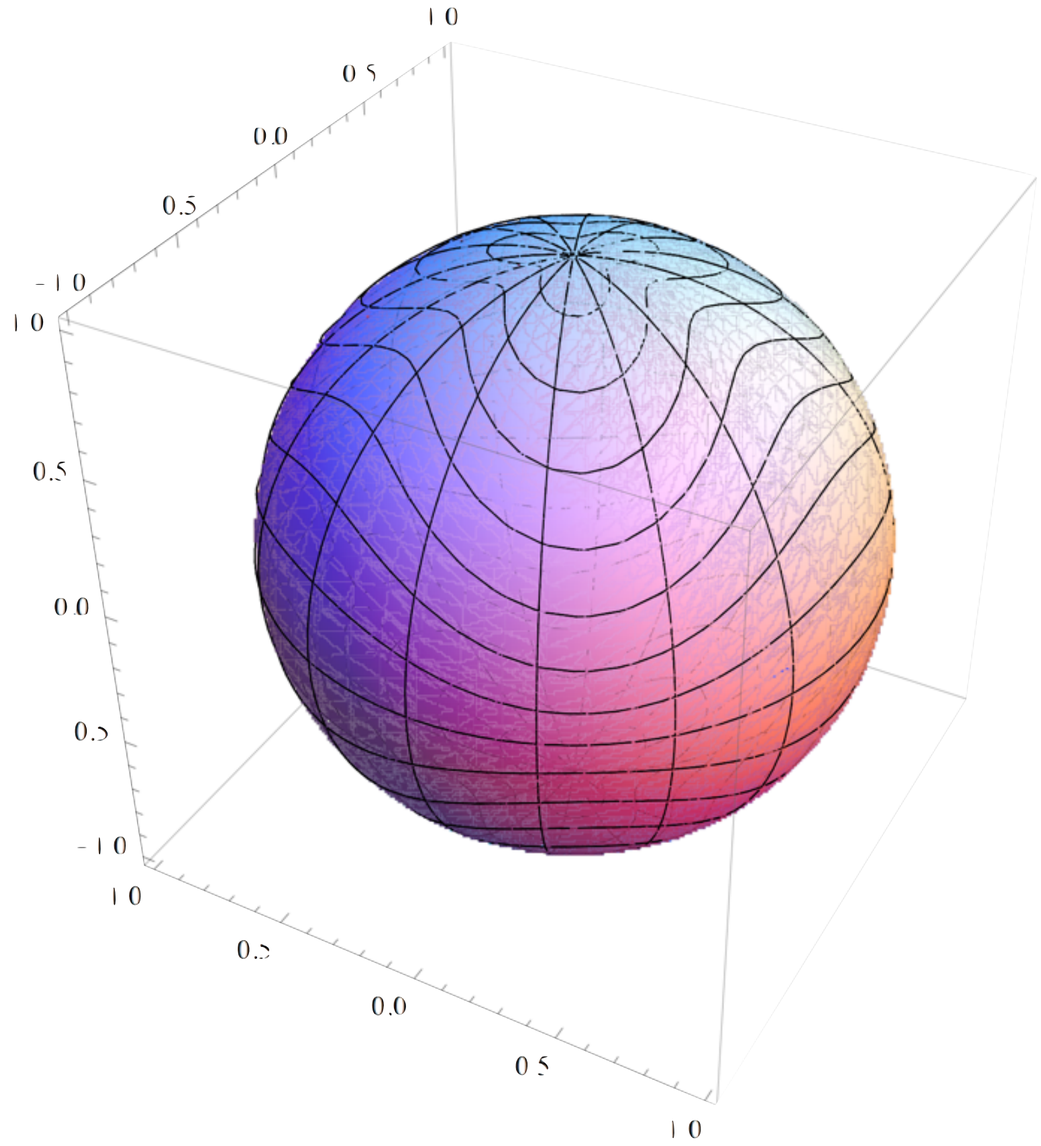
- Generalized to higher dimension manifolds



Non Orthogonal Tangents



$$\begin{bmatrix} \cos(\theta 2\pi)\cos(\phi pi/2) \\ \sin(\theta 2\pi)\cos(\phi\pi/2) \\ \sin(\phi\pi/2) \end{bmatrix}$$

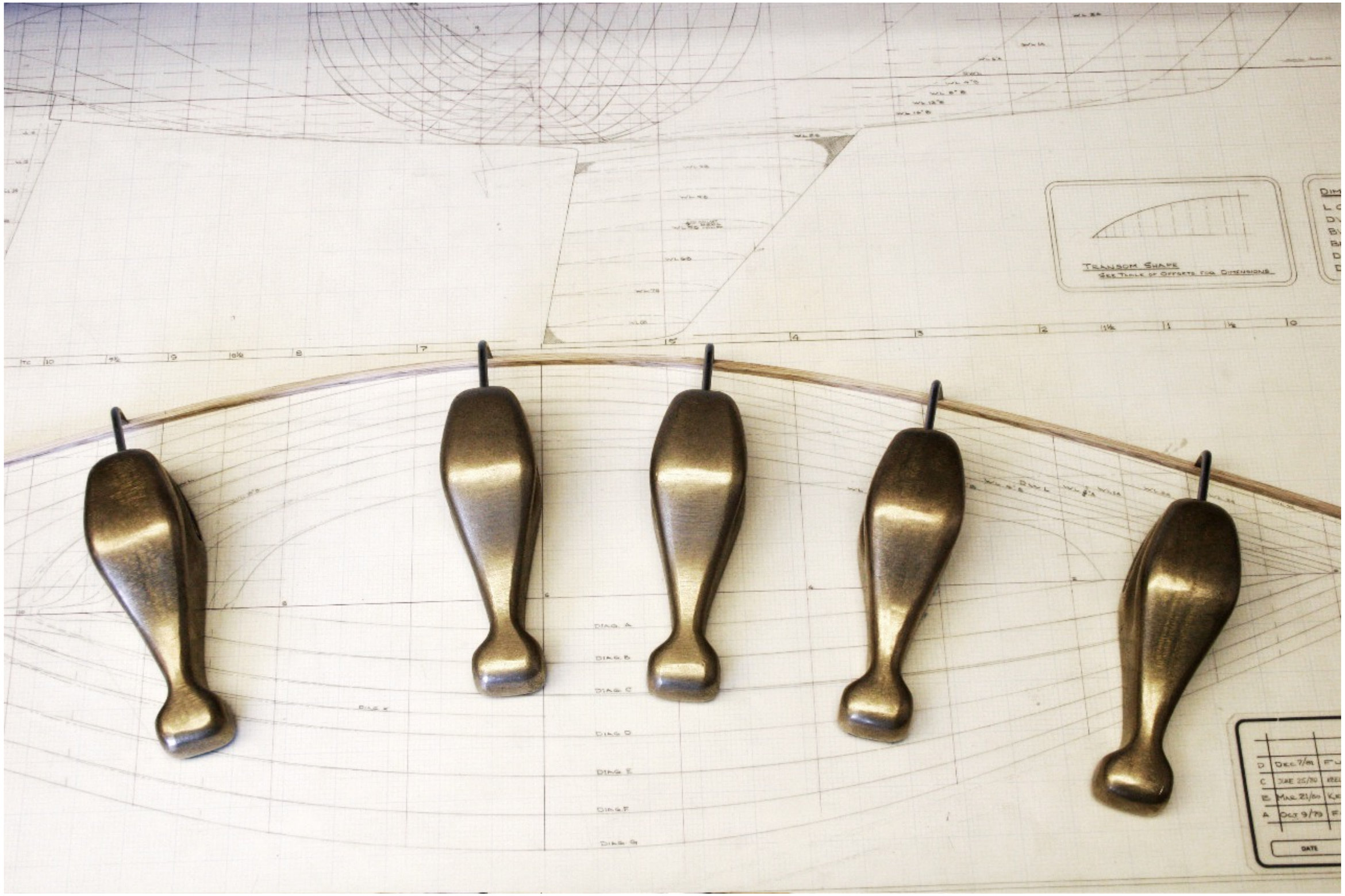


$$\begin{bmatrix} \cos(2\pi\theta)\cos\left(\frac{1}{2}\pi\left(\frac{1}{2}(1-|\phi|)\cos(6\pi\theta)\phi+\phi\right)\right) \\ \cos\left(\frac{1}{2}\pi\left(\frac{1}{2}(1-|\phi|)\cos(6\pi\theta)\phi+\phi\right)\right)\sin(2\pi\theta) \\ \sin\left(\frac{1}{2}\pi\left(\frac{1}{2}(1-|\phi|)\cos(6\pi\theta)\phi+\phi\right)\right) \end{bmatrix}$$

$\theta \in [0..1] \quad \phi \in [-1..1]$

Splines

A Real Draftsperson's Spline



http://www.alatown.com/spline-history-architecture/

Spline Topics

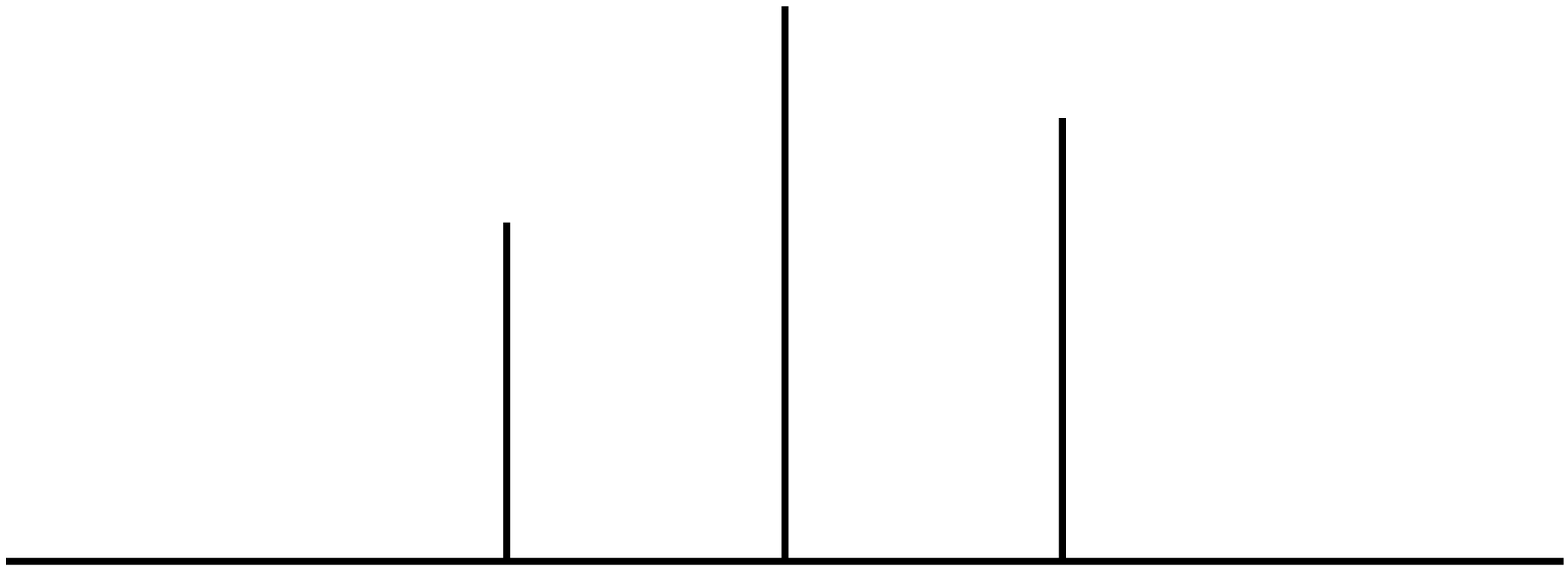
Interpolation

- Cubic Hermite interpolation
- Catmull-Rom interpolation

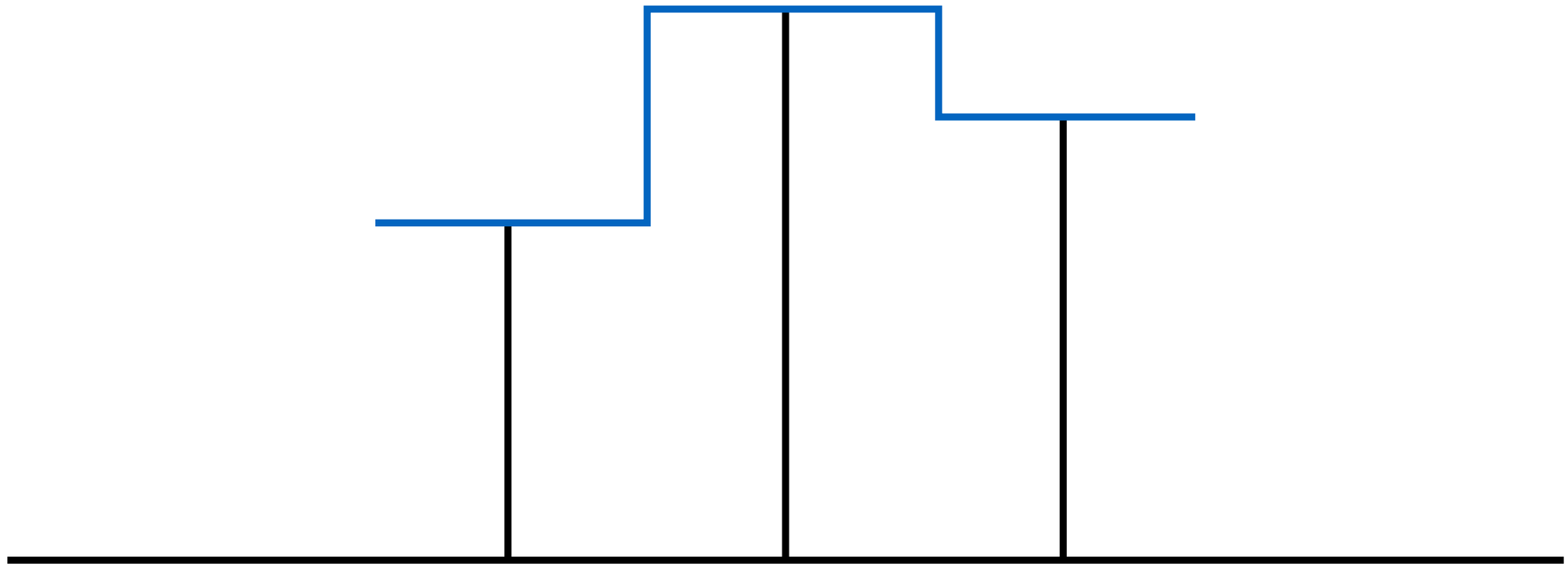
Bezier curves

Bezier surfaces

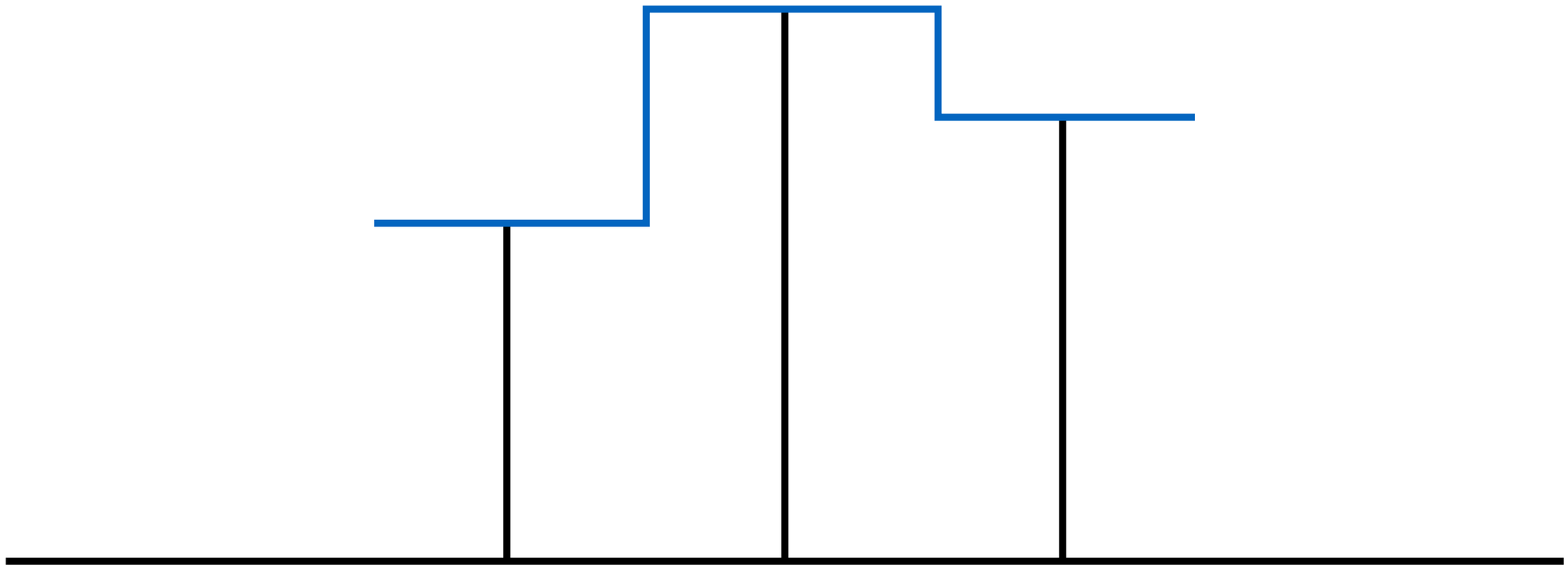
Goal: Interpolate Values



Nearest Neighbor Interpolation

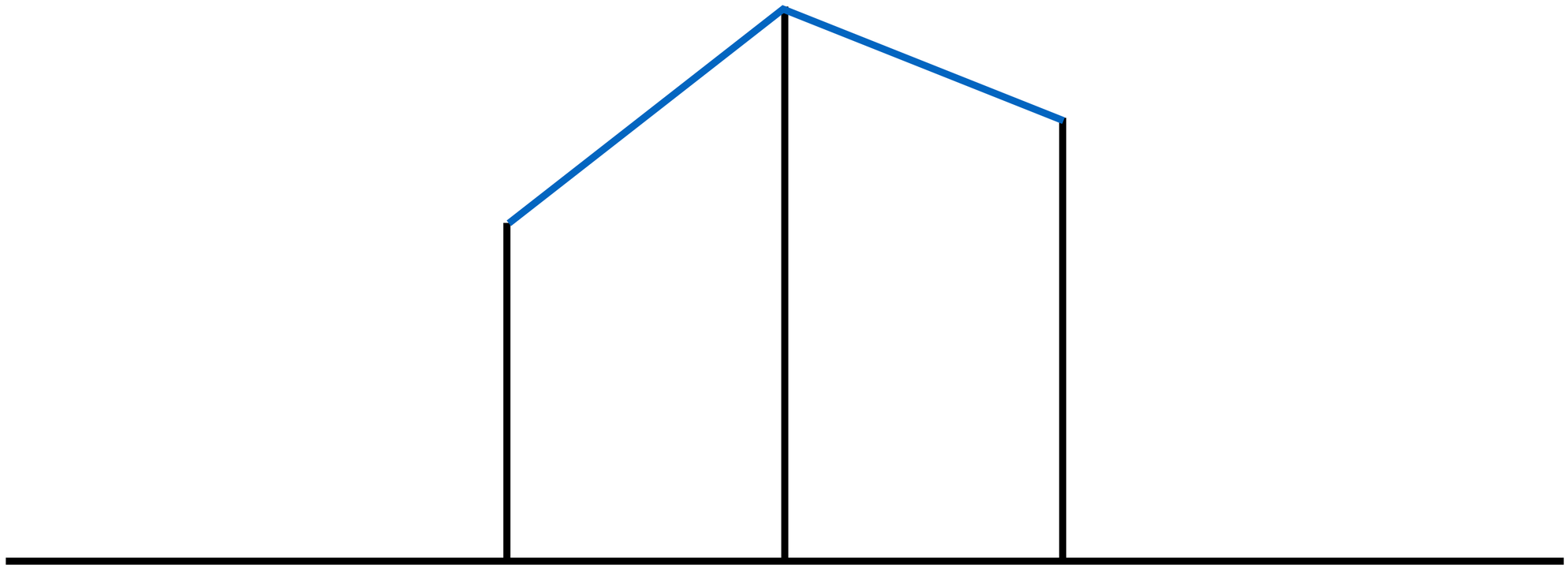


Nearest Neighbor Interpolation

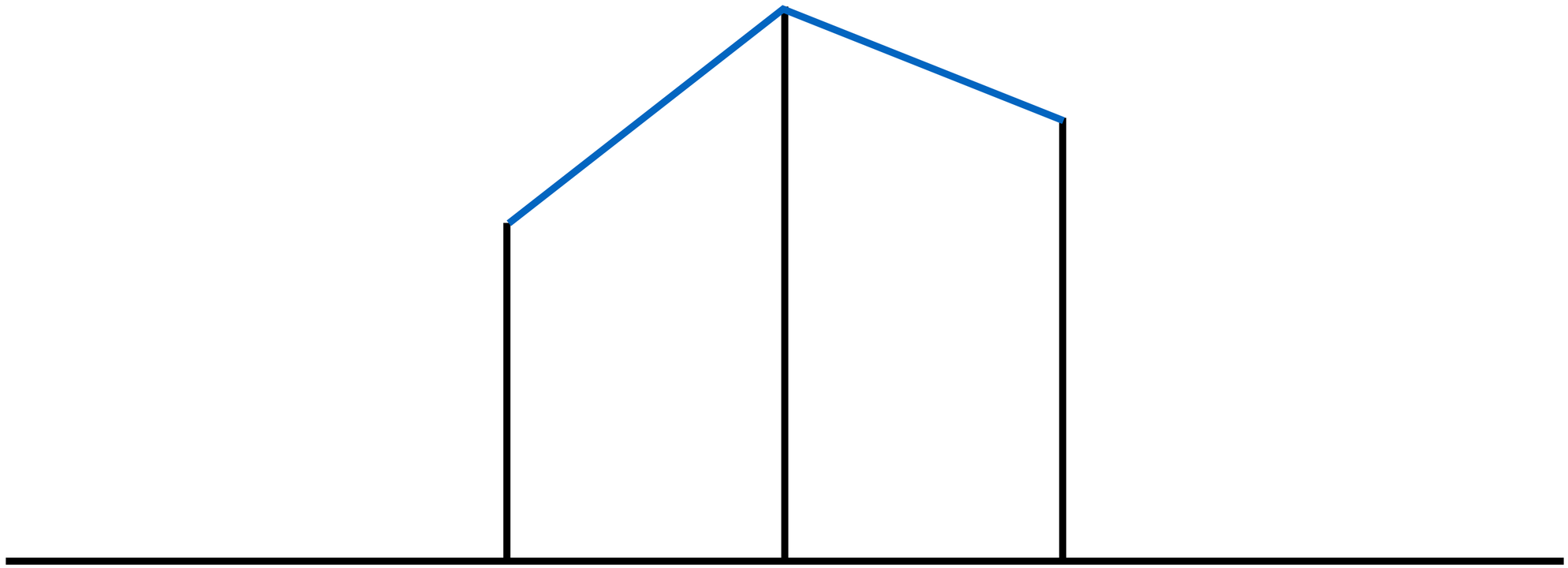


Problem: values not continuous

Linear Interpolation

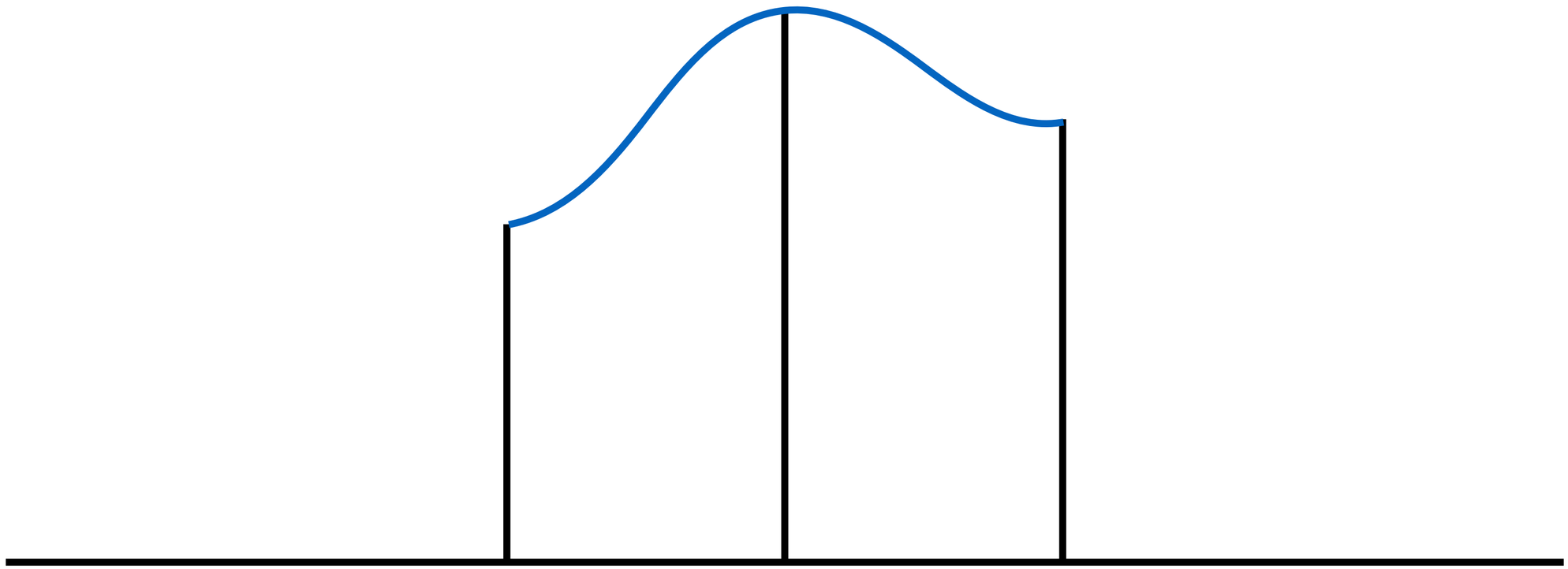


Linear Interpolation



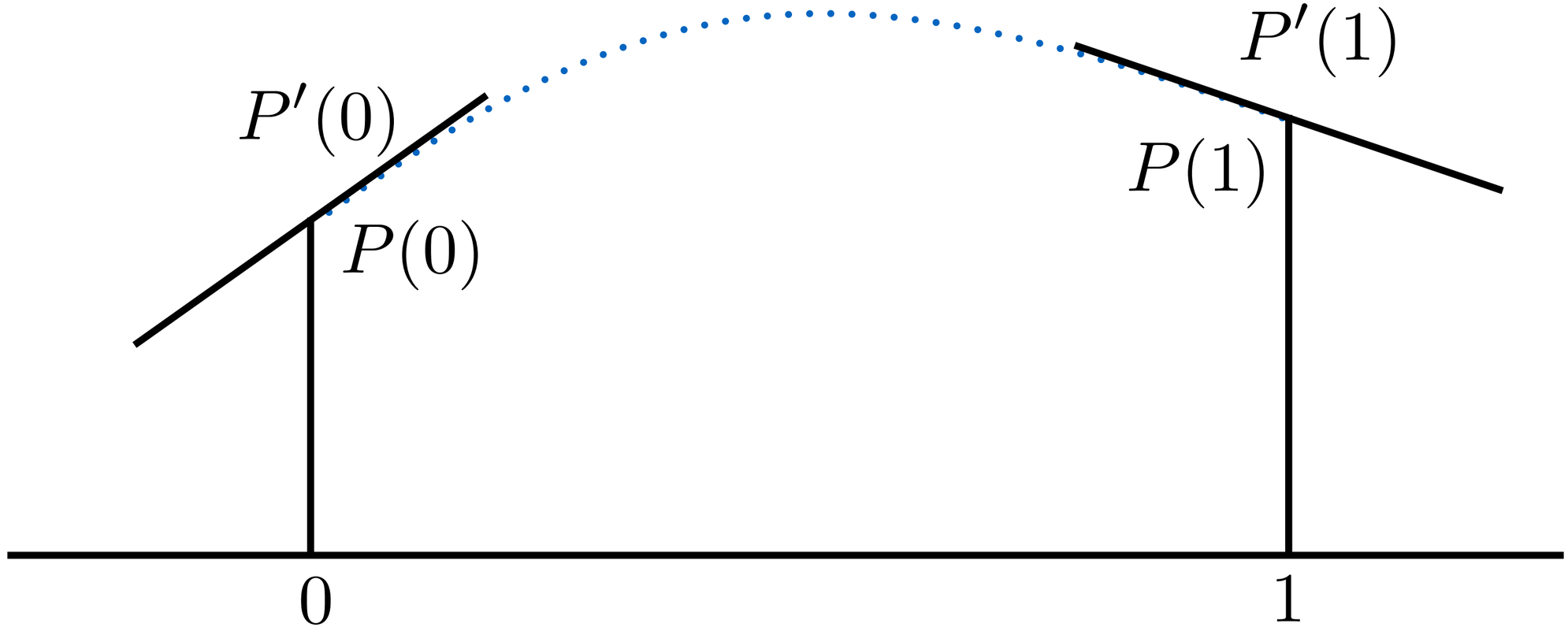
Problem: derivatives not continuous

Smooth Interpolation?



Cubic Hermite Interpolation

Cubic Hermite Interpolation



Inputs: values and derivatives at endpoints

Polynomial Basis Functions

Power Basis

$$x(u) = \sum_{i=0}^{d} c_i \, u^i$$

e.g. a cubic polynomial

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x(u) = \boldsymbol{C} \cdot \boldsymbol{\mathcal{P}}^d$$

$$\boldsymbol{C} = [c_0, c_1, c_2, \ldots, c_d]$$

$$\boldsymbol{\mathcal{P}}^d = [1, u, u^2, \ldots, u^d]$$

The elements of $\boldsymbol{\mathcal{P}}^d$ are ***linearly independent***

i.e. no good approximation $u^k \not\approx \sum_{i \neq k} c_i \, u^i$

Specifying a Curve

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x(u_0) = x_0$$

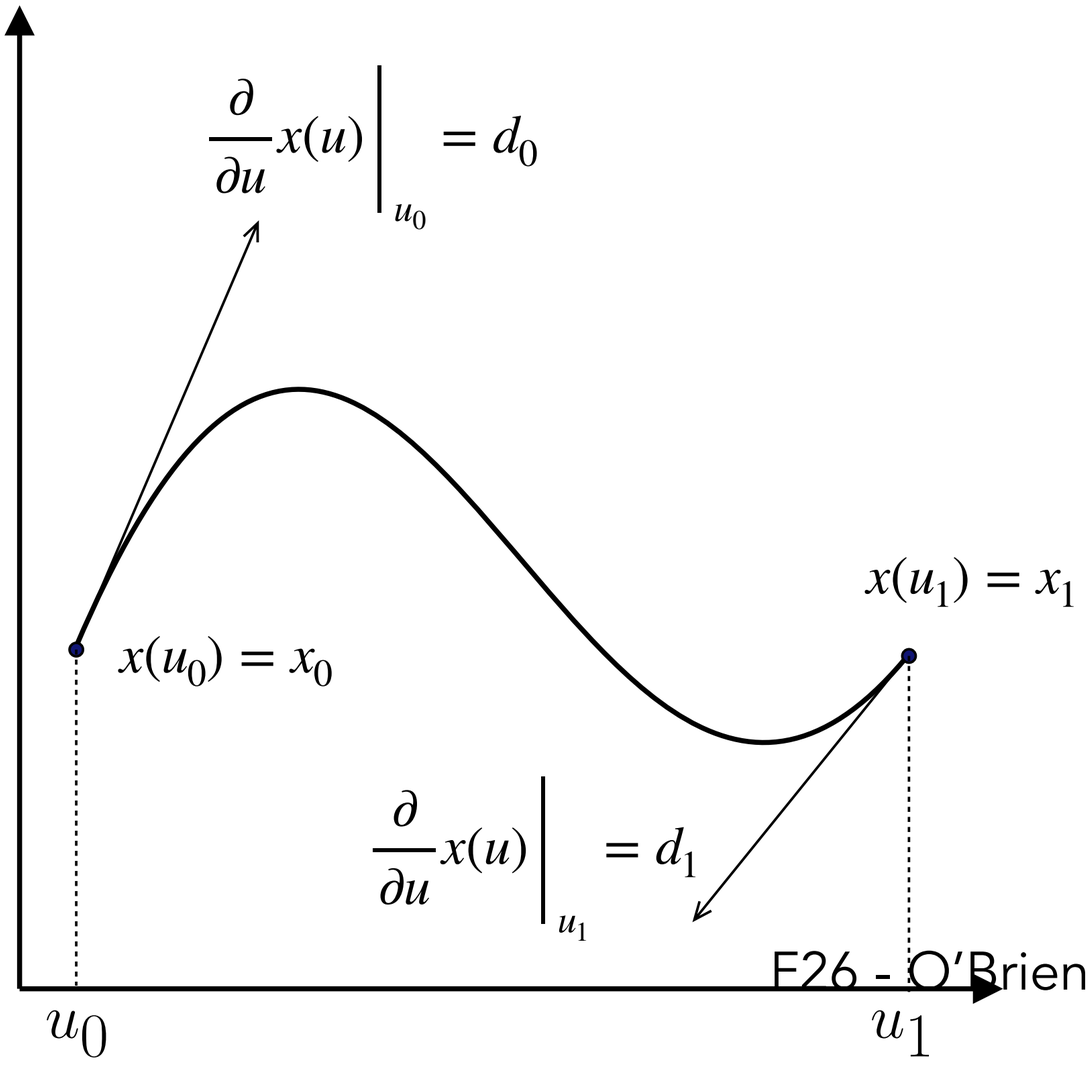
$$x(u_1) = x_1$$

$$\left.\frac{\partial}{\partial u} x(u)\right|_{u_0} = d_0$$

$$\left.\frac{\partial}{\partial u} x(u)\right|_{u_1} = d_1$$

For now, assume

$$u_0 = 0 \quad u_1 = 1$$



Specifying a Curve

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$x(u_0) = x_0$$

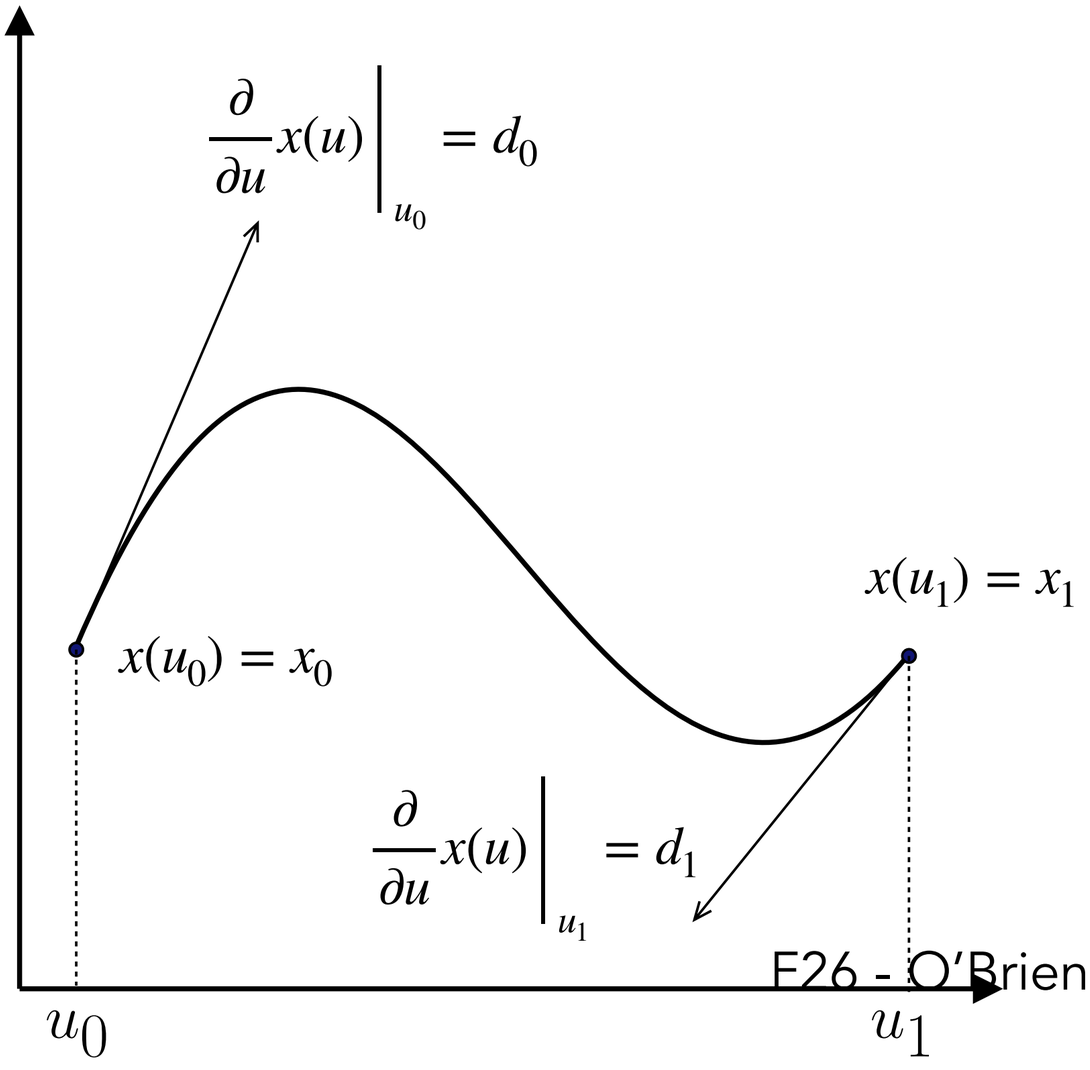
$$x(u_1) = x_1$$

$$x'(u_0) = d_0$$

$$x'(u_1) = d_1$$

For now, assume

$$u_0 = 0 \quad u_1 = 1$$



Specifying a Curve

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$c_0 + c_1 u_0 + c_2 u_0^2 + c_3 u_0^3 = x_0$$

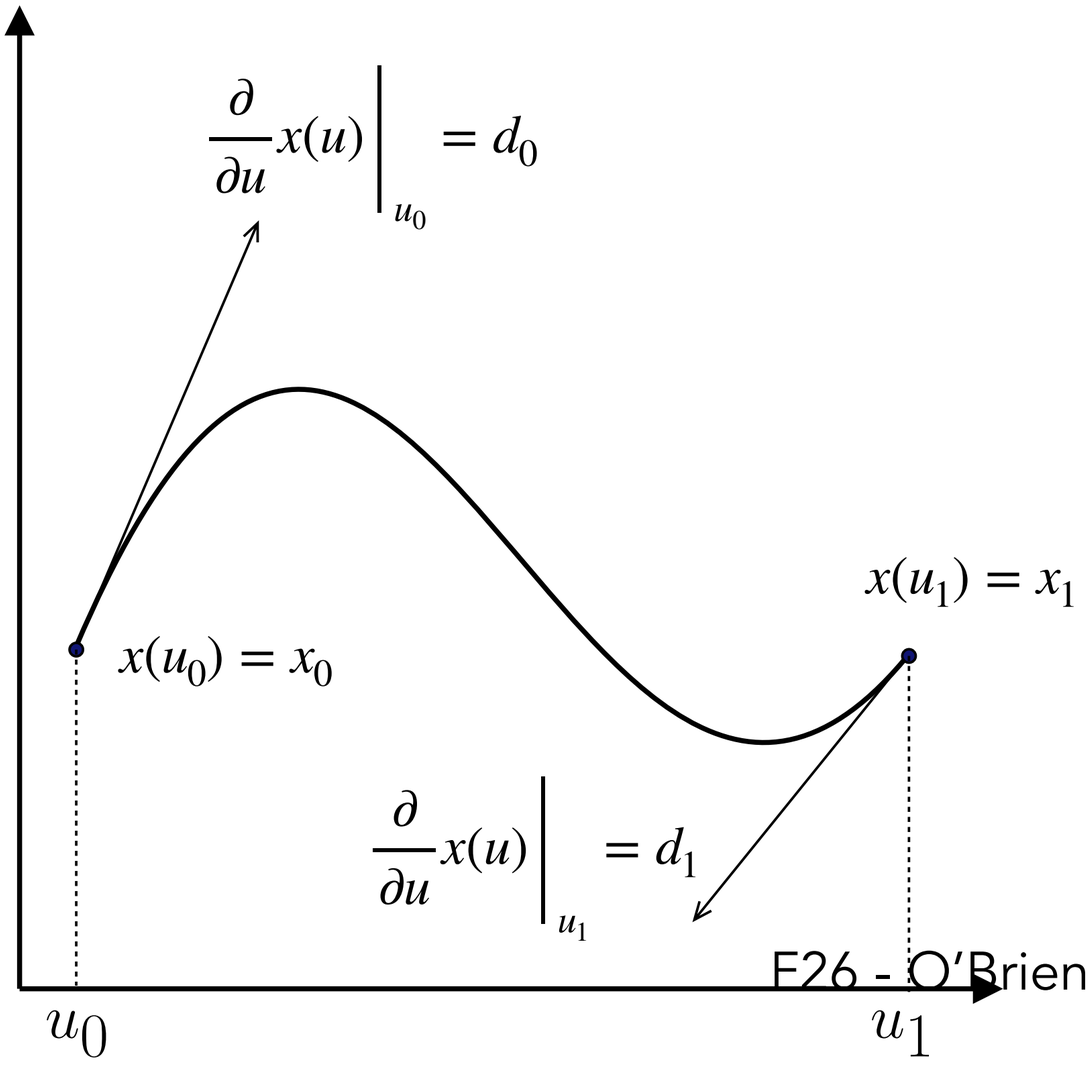
$$c_0 + c_1 u_1 + c_2 u_1^2 + c_3 u_1^3 = x_1$$

$$c_1 + 2c_2 u_0 + 3c_3 u_0^2 = d_0$$

$$c_1 + 2c_2 u_1 + 3c_3 u_1^2 = d_1$$

For now, assume

$$u_0 = 0 \quad u_1 = 1$$



Specifying a Curve

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$c_0 + c_1\, 0 + c_2\, 0 + c_3\, 0 = x_0$$

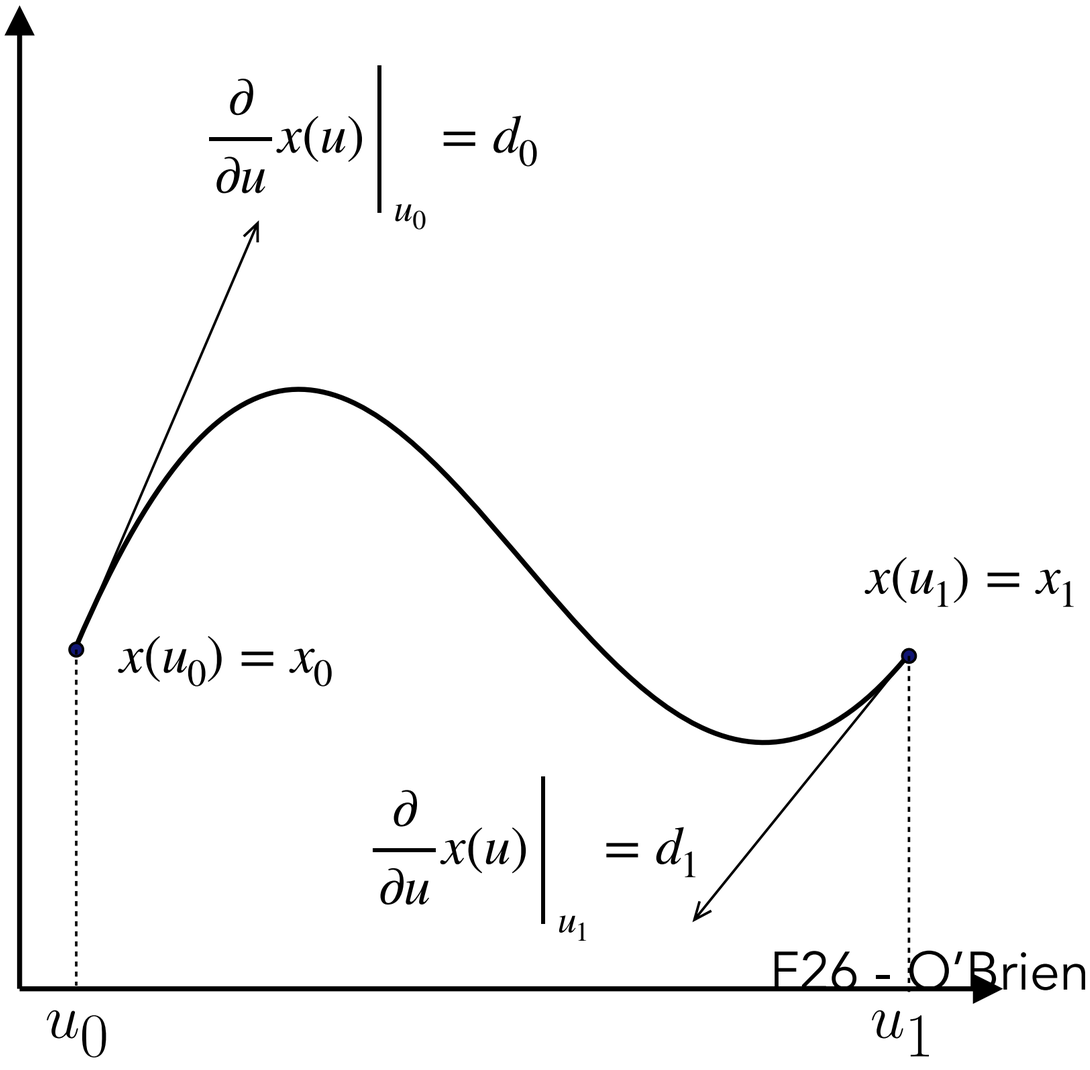
$$c_0 + c_1\, 1 + c_2\, 1 + c_3\, 1 = x_1$$

$$c_1 + 2c_2\, 0 + 3c_3\, 0 = d_0$$

$$c_1 + 2c_2\, 1 + 3c_3\, 1 = d_1$$

For now, assume

$$u_0 = 0 \quad u_1 = 1$$



Specifying a Curve

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

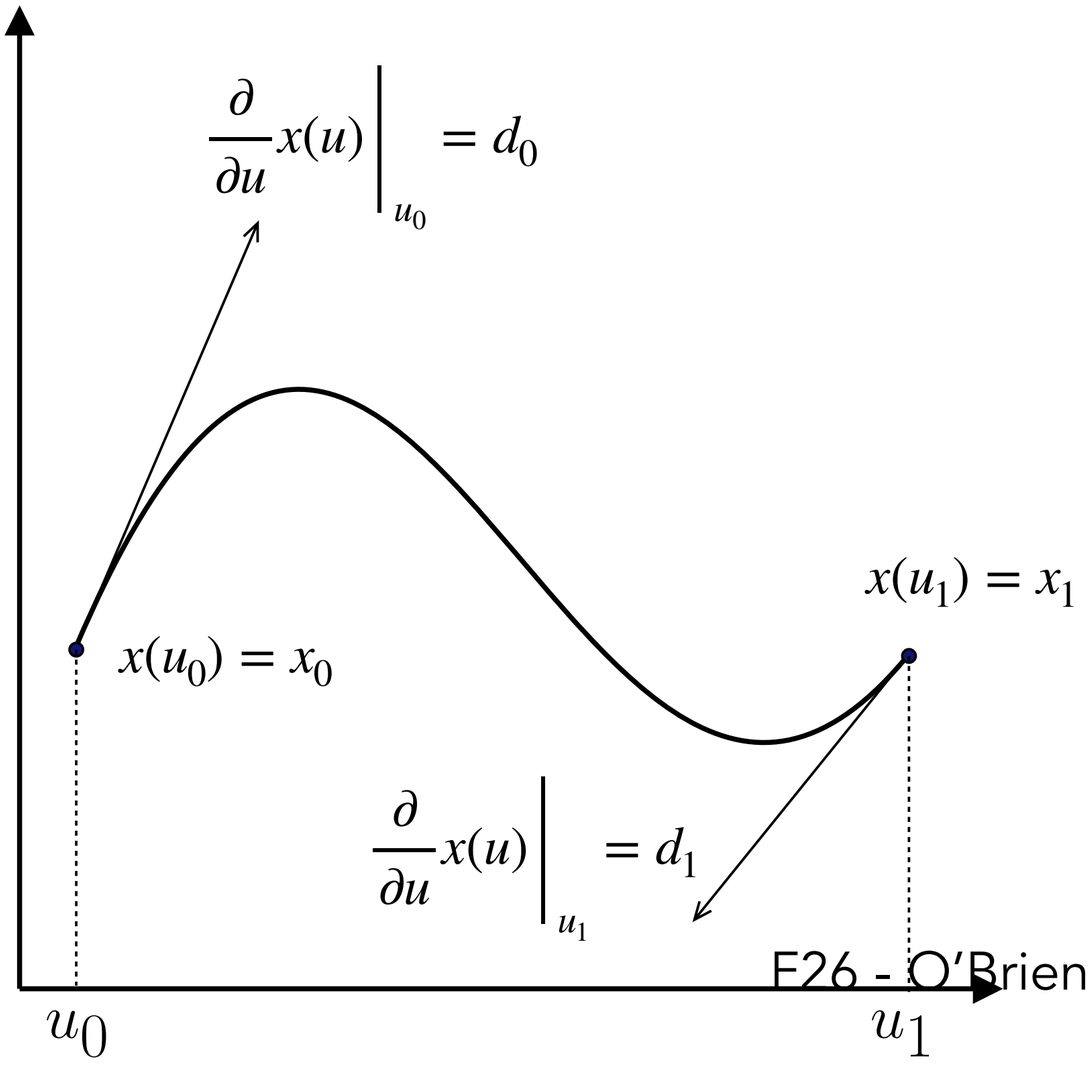
$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$\begin{aligned} c_0 &= x_0 \\ c_0 + c_1 + c_2 + c_3 &= x_1 \\ c_1 &= d_0 \\ c_1 + 2c_2 + 3c_3 &= d_1 \end{aligned}$$

For now, assume

$$u_0 = 0 \quad u_1 = 1$$



Specifying a Curve

(From previous slide for reference.)

$$\begin{aligned} c_0 &= x_0 \\ c_0 + c_1 + c_2 + c_3 &= x_1 \\ c_1 &= d_0 \\ c_1 + 2c_2 + 3c_3 &= d_1 \end{aligned}$$

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$

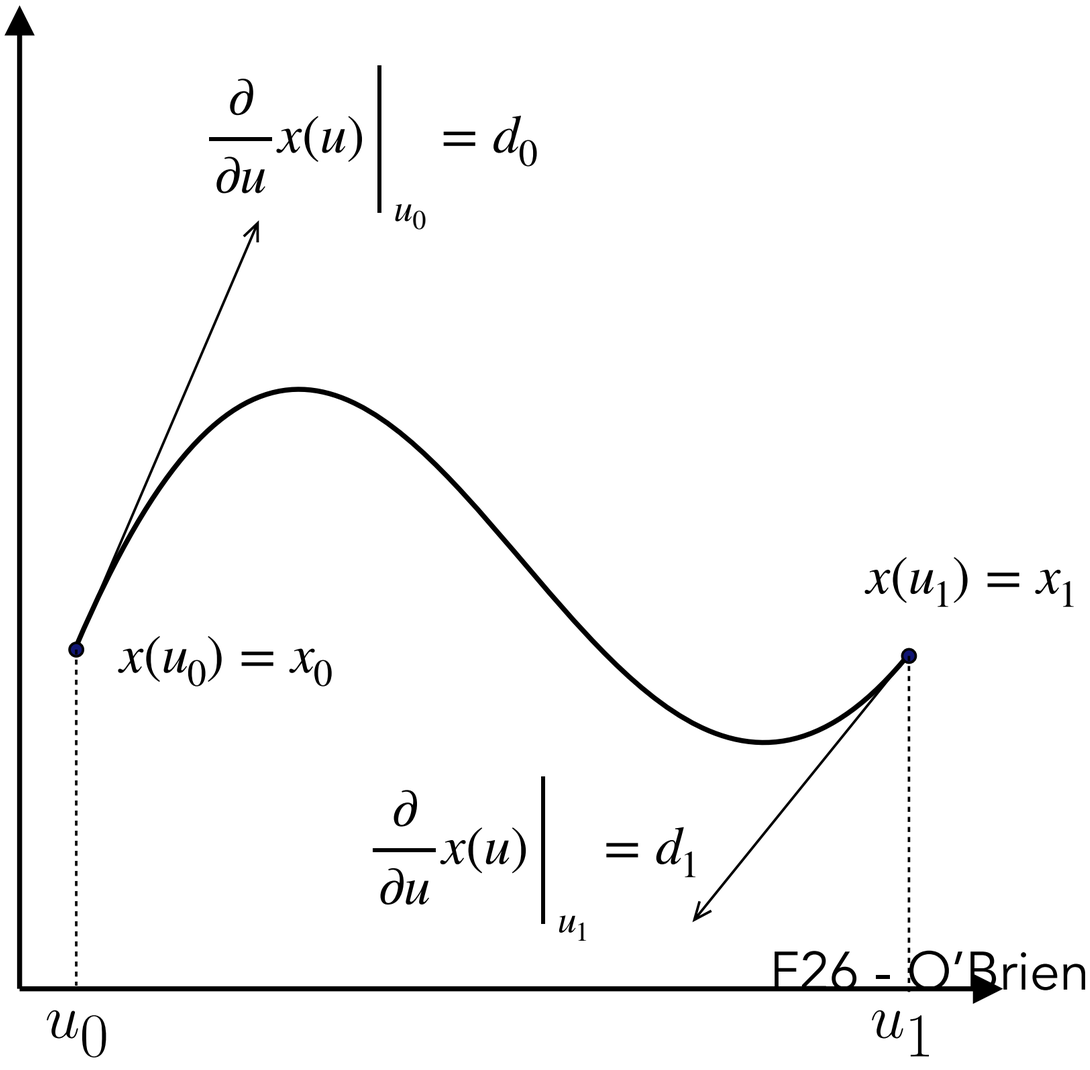
Matrix

Coefficients (unknown)

Constraint Values

For now, assume

$$u_0 = 0 \quad u_1 = 1$$



Specifying a Curve

Given desired values (constraints) how do we determine the coefficients for cubic power basis?

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

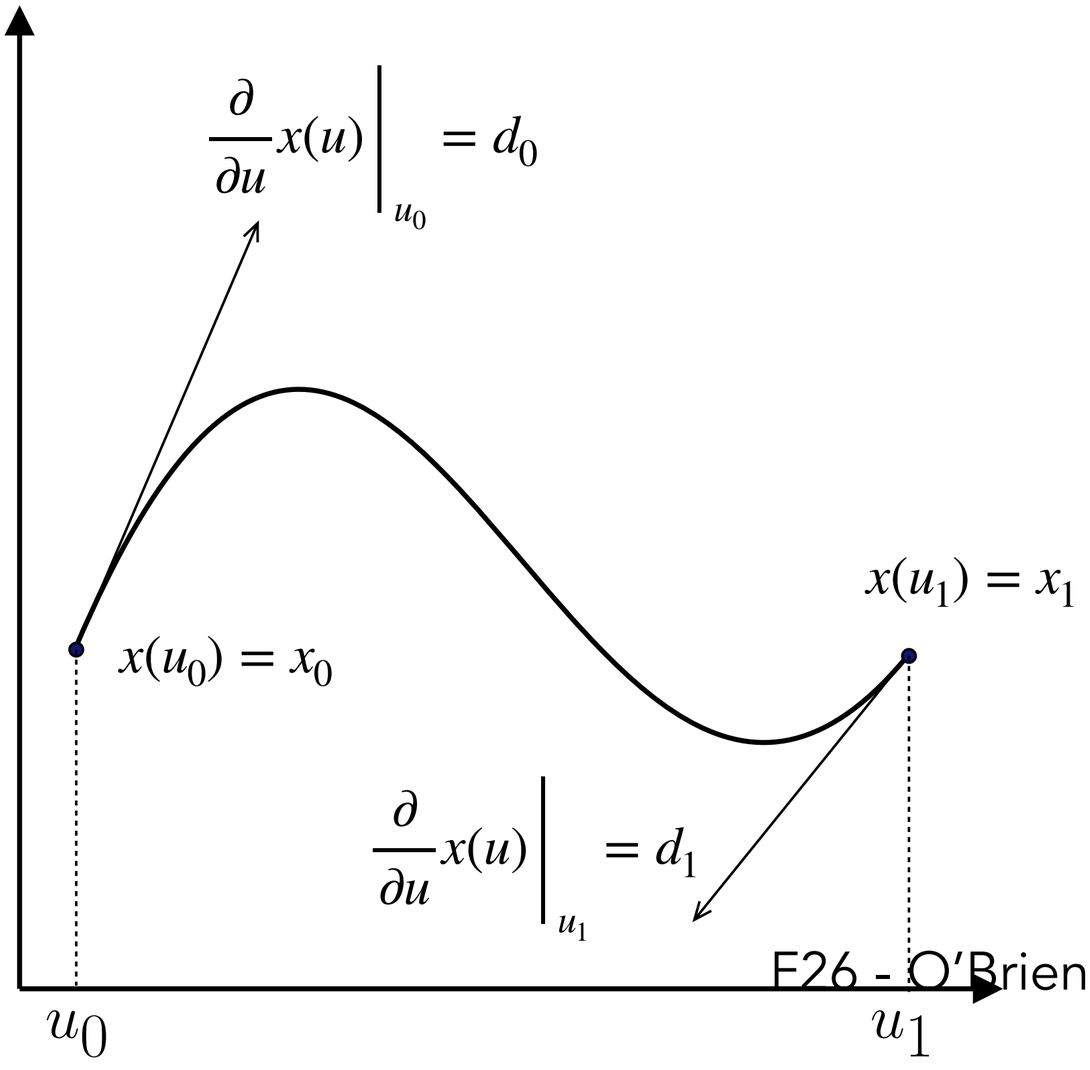
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$

$$\mathbf{B} \cdot \mathbf{c} = \mathbf{h}$$

Often called "control points", but note the *d* in this case are directions, not points.

For now, assume

$$u_0 = 0 \quad u_1 = 1$$

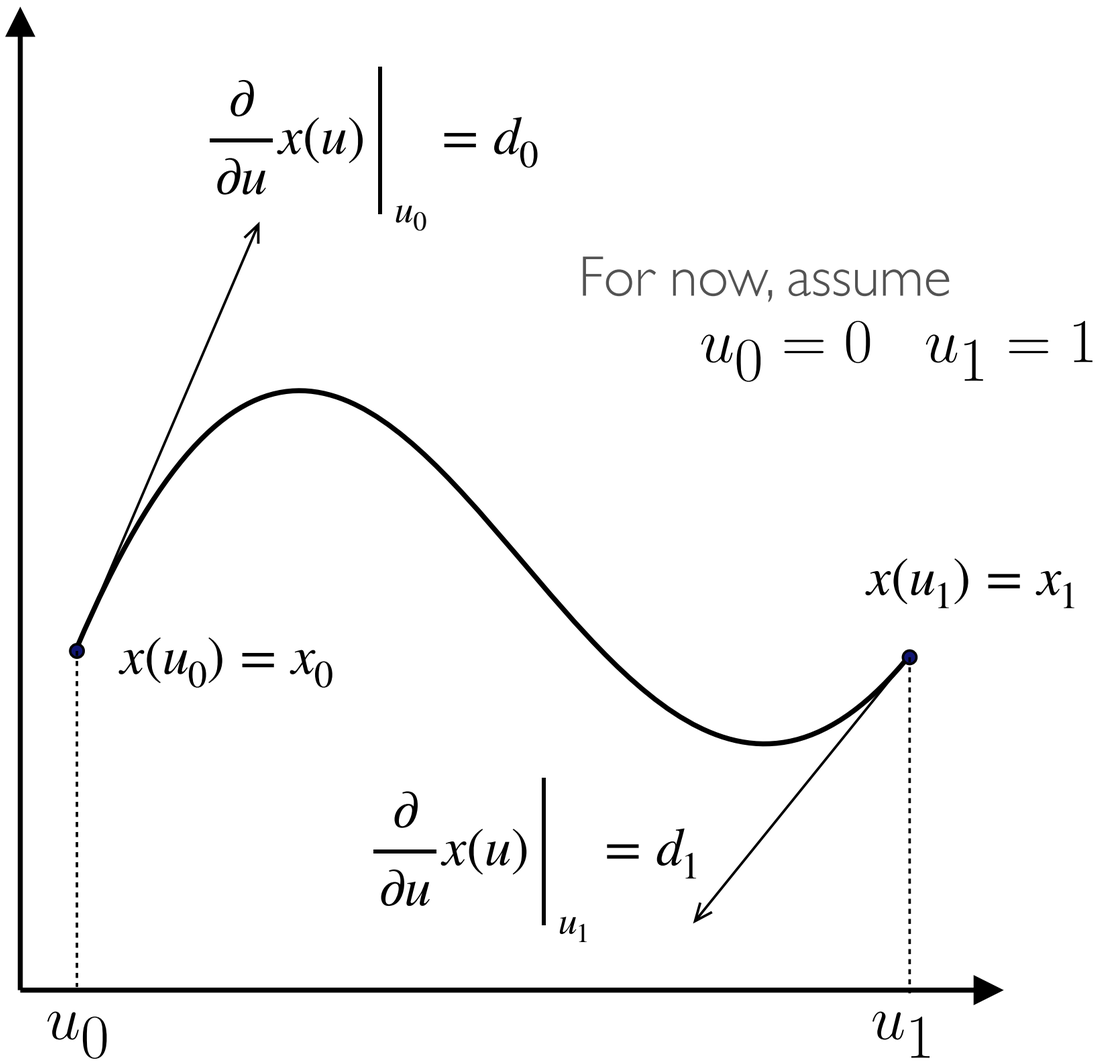


Specifying a Curve

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$



$$\mathbf{B} \cdot \mathbf{c} = \mathbf{h}$$

$$\mathbf{c} = \mathbf{B}^{-1} \cdot \mathbf{h}$$

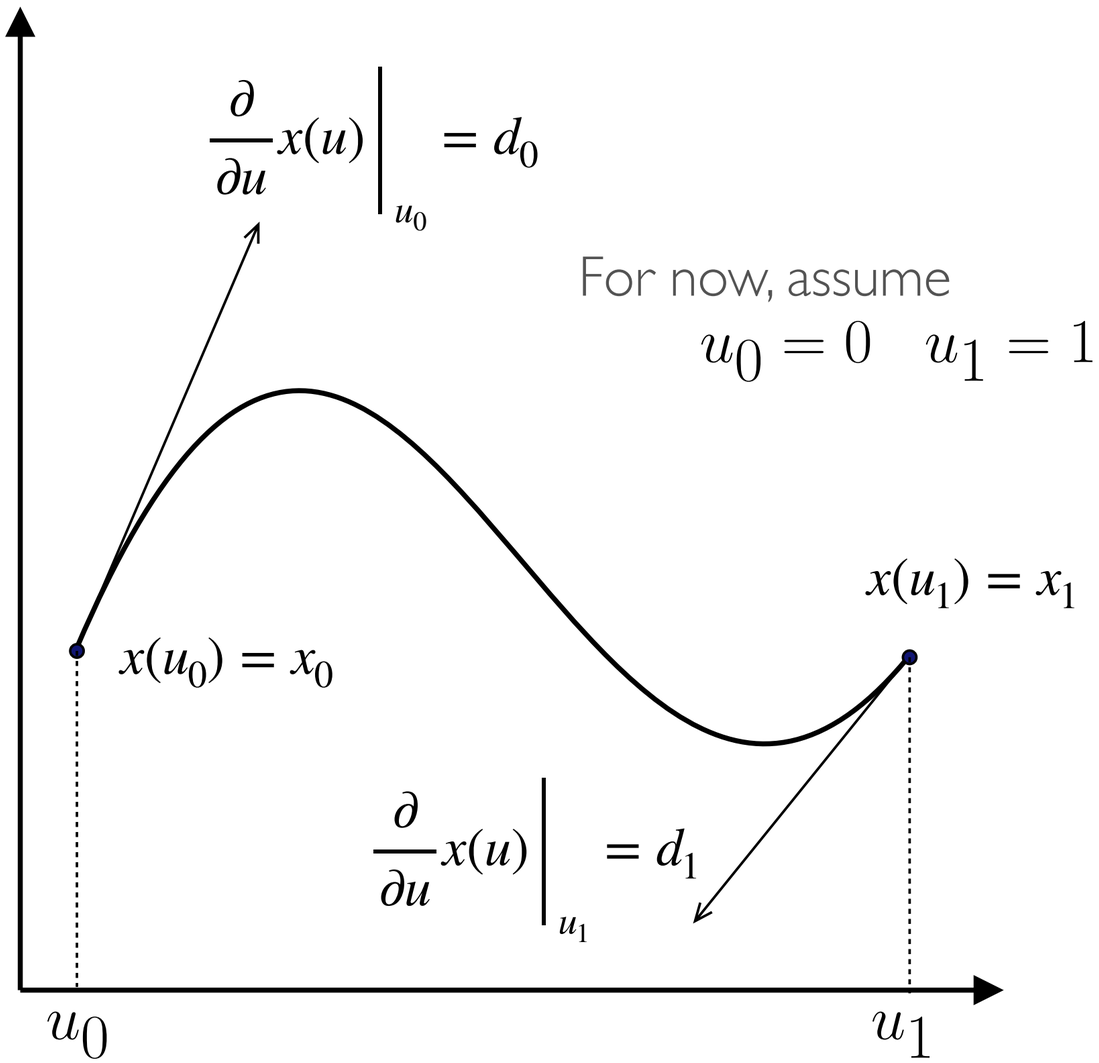
$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 3 & -2 & -1 \\ 2 & -2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$

Specifying a Curve

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x'(u) = c_1 + 2c_2\, u + 3c_3\, u^2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 3 \end{bmatrix} \cdot \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$



$$\mathbf{B} \cdot \mathbf{c} = \mathbf{h}$$

$$\mathbf{c} = \mathbf{B}^{-1} \cdot \mathbf{h}$$

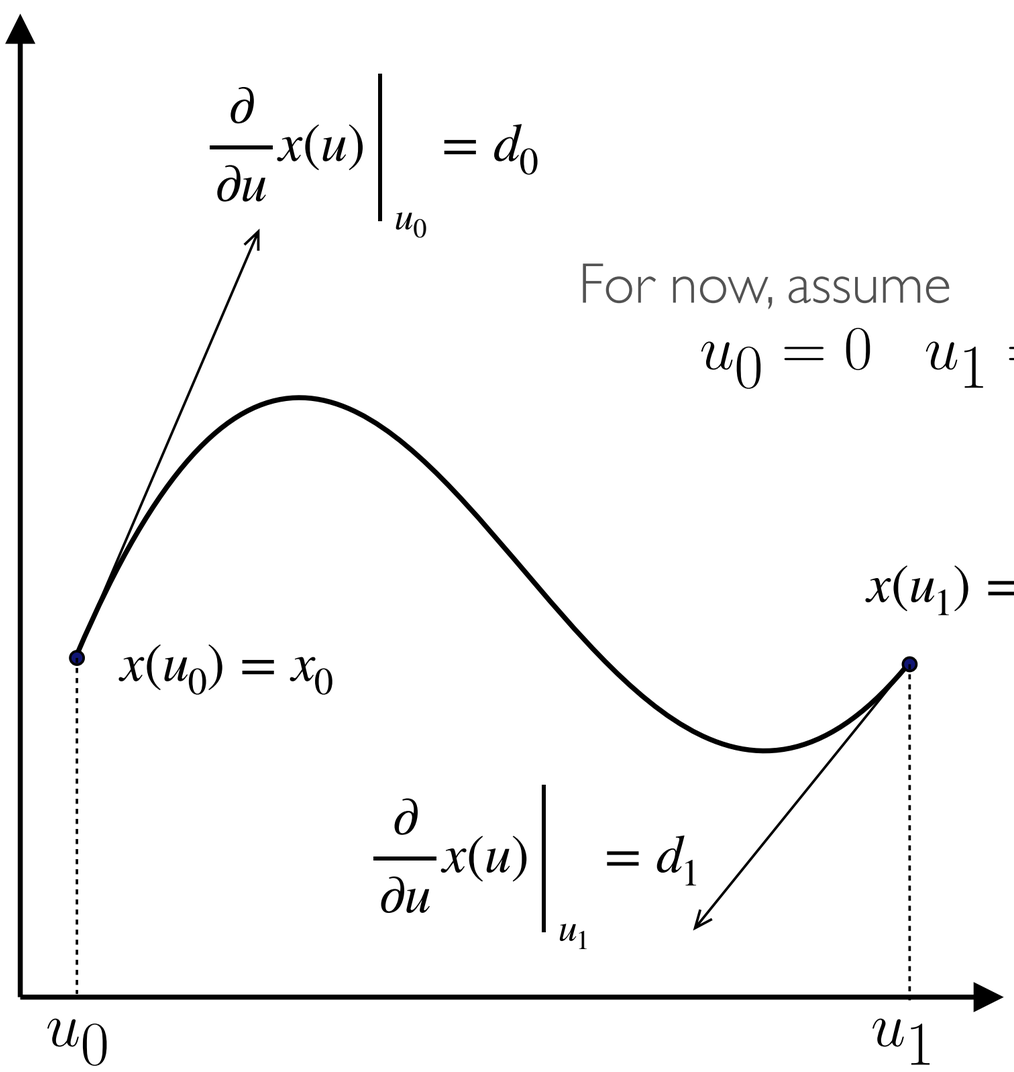
$$\mathbf{c} = \boldsymbol{\beta}_H \cdot \mathbf{h}$$

$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 3 & -2 & -1 \\ 2 & -2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$

Cubic Hermite Basis

$$\mathbf{c} = \boldsymbol{\beta}_H \cdot \mathbf{h}$$

$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 3 & -2 & -1 \\ 2 & -2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$



$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$= \mathscr{P}^3 \cdot \mathbf{c}$$

$$= \mathscr{P}^3 \cdot \boldsymbol{\beta}_H \cdot \mathbf{h}$$

$$x(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{h}$$

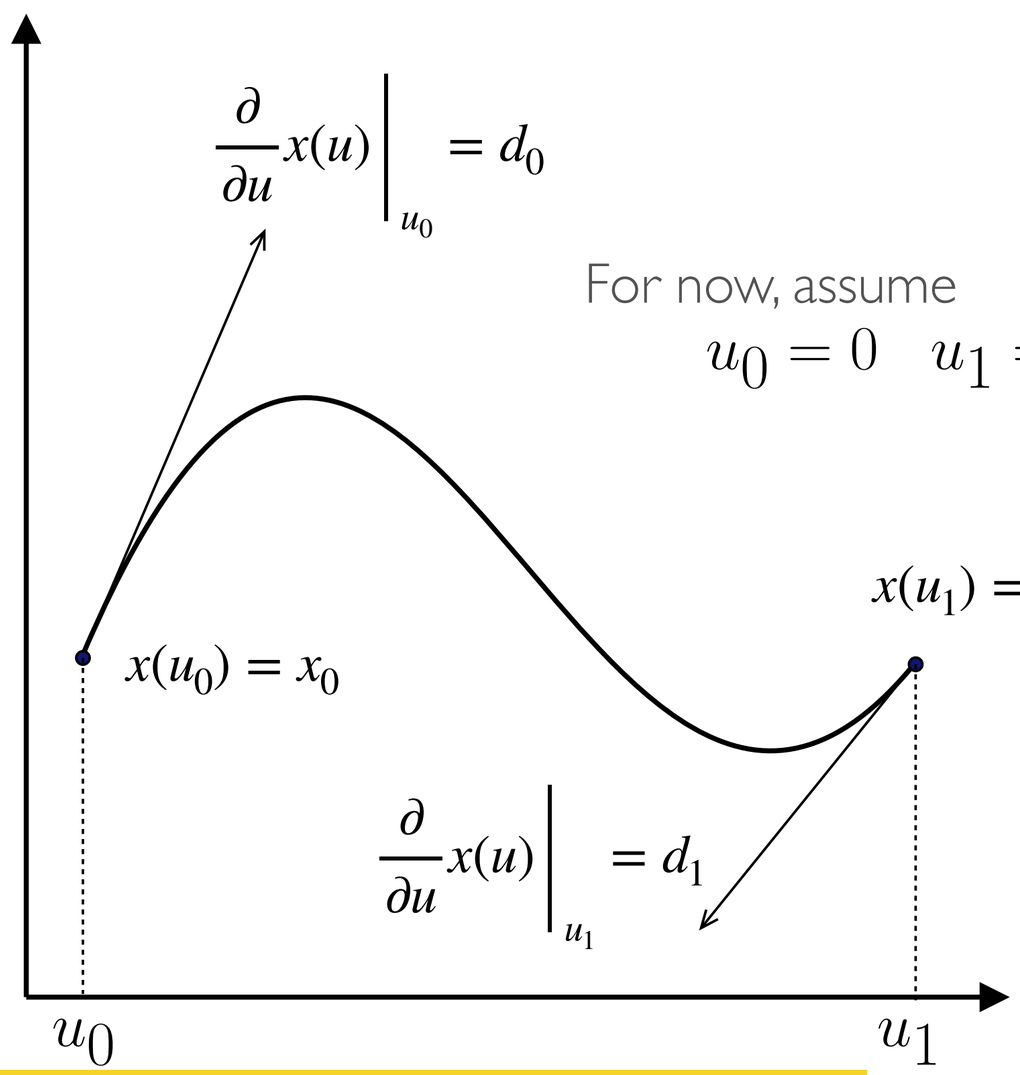
$\boldsymbol{\beta}_H$ ⟵ **Hermite Basis Matrix**

$$\mathscr{P}^3(u) = [1 \; u \; u^2 \; u^3]$$

Cubic Hermite Basis

$$\mathbf{c} = \boldsymbol{\beta}_H \cdot \mathbf{h}$$

$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 3 & -2 & -1 \\ 2 & -2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_0 \\ x_1 \\ d_0 \\ d_1 \end{bmatrix}$$



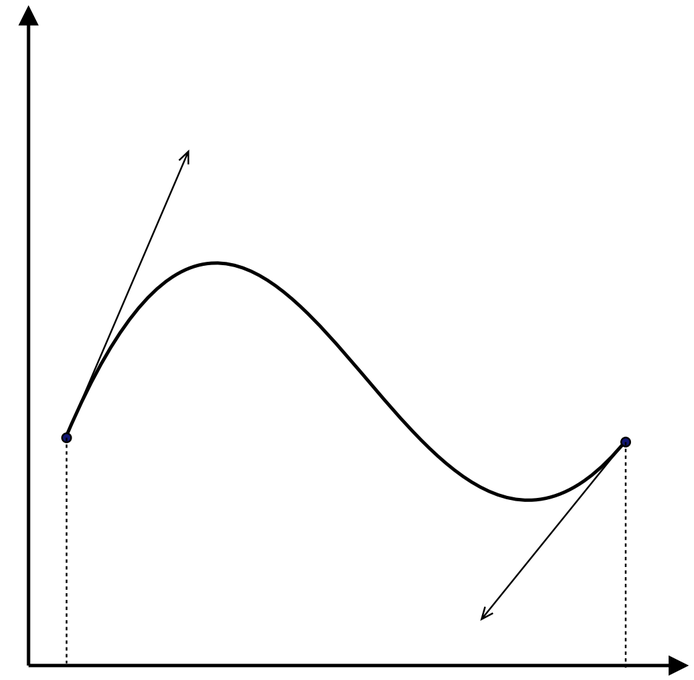
$$x(u) = \boldsymbol{\mathcal{P}}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{h}$$

$$x(u) = \begin{bmatrix} 1 + 0\,u - 3\,u^2 + 2\,u^3 \\ 0 + 0\,u + 3\,u^2 - 2\,u^3 \\ 0 + 1\,u - 2\,u^2 + 1\,u^3 \\ 0 + 0\,u - 1\,u^2 + 1\,u^3 \end{bmatrix}^{\mathsf{T}} \cdot \mathbf{h}$$

Cubic Hermite Basis

$$x(u) = \mathcal{P}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{h}$$

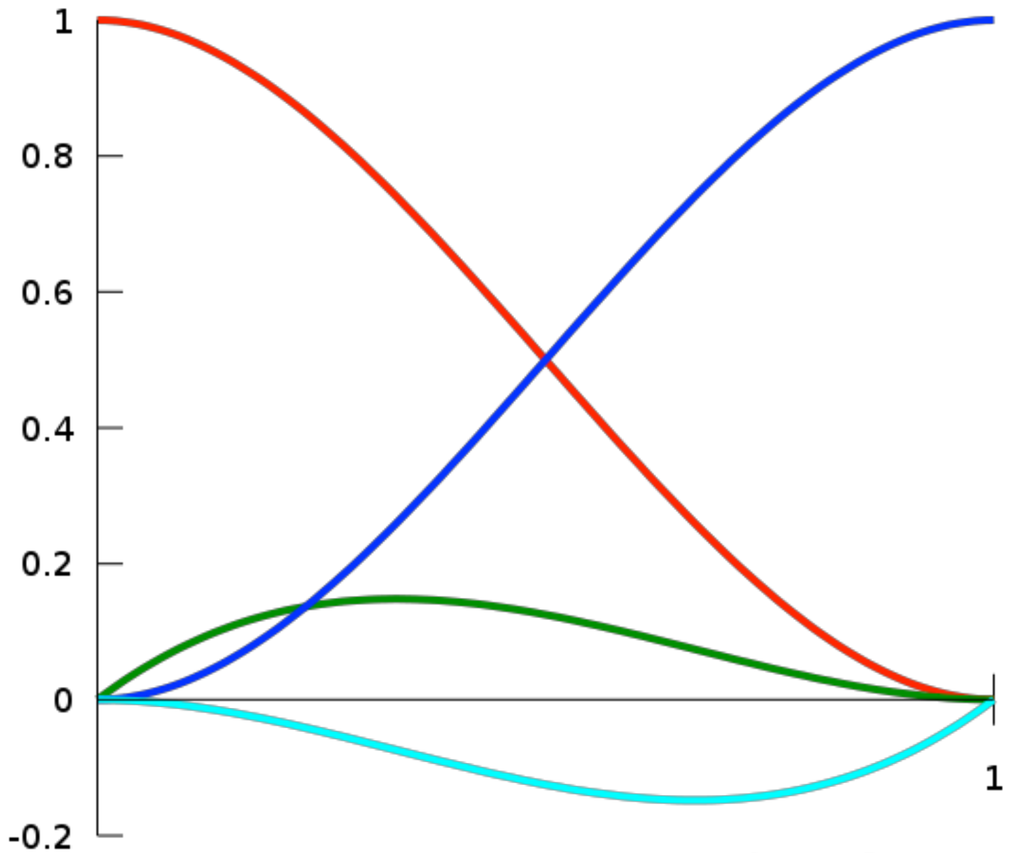
$$= \begin{bmatrix} 1 + 0\,u - 3\,u^2 + 2\,u^3 \\ 0 + 0\,u + 3\,u^2 - 2\,u^3 \\ 0 + 1\,u - 2\,u^2 + 1\,u^3 \\ 0 + 0\,u - 1\,u^2 + 1\,u^3 \end{bmatrix}^T \cdot \mathbf{h}$$



Hermite Basis Functions

Basis matrix depends on u_0 and u_1

We could use this same approach for higher order than cubic.



Cubic Hermite Basis

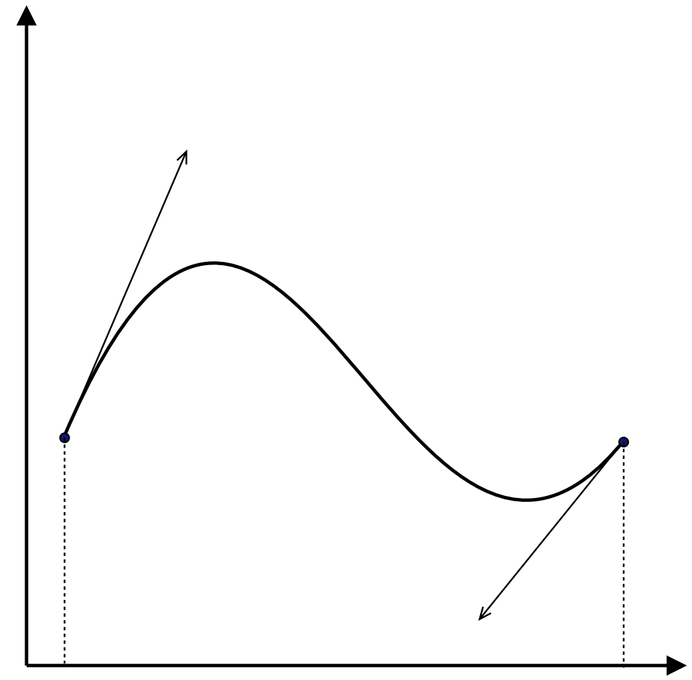
$$x(u) = \mathcal{P}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{h}$$

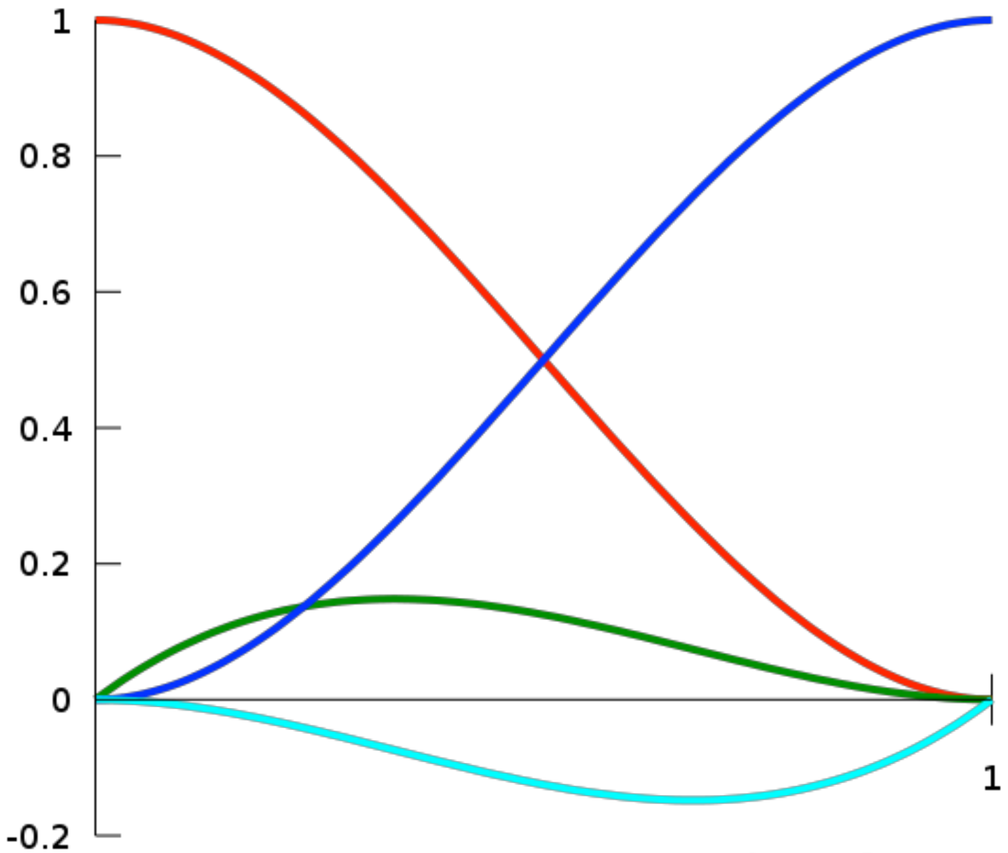
$$= \begin{bmatrix} 1 + 0\,u - 3\,u^2 + 2\,u^3 \\ 0 + 0\,u + 3\,u^2 - 2\,u^3 \\ 0 + 1\,u - 2\,u^2 + 1\,u^3 \\ 0 + 0\,u - 1\,u^2 + 1\,u^3 \end{bmatrix}^T \cdot \mathbf{h}$$

Hermite Basis Functions

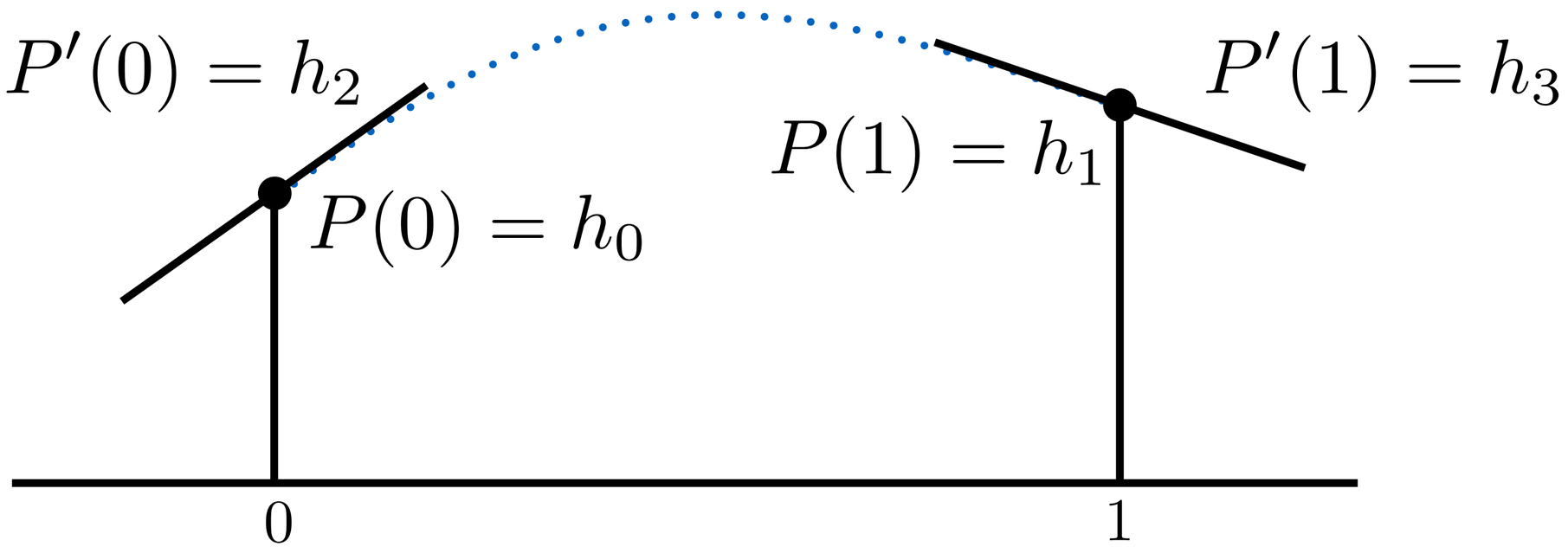
Basis matrix depends on u_0 and u_1

We could use this same approach for higher order than cubic.

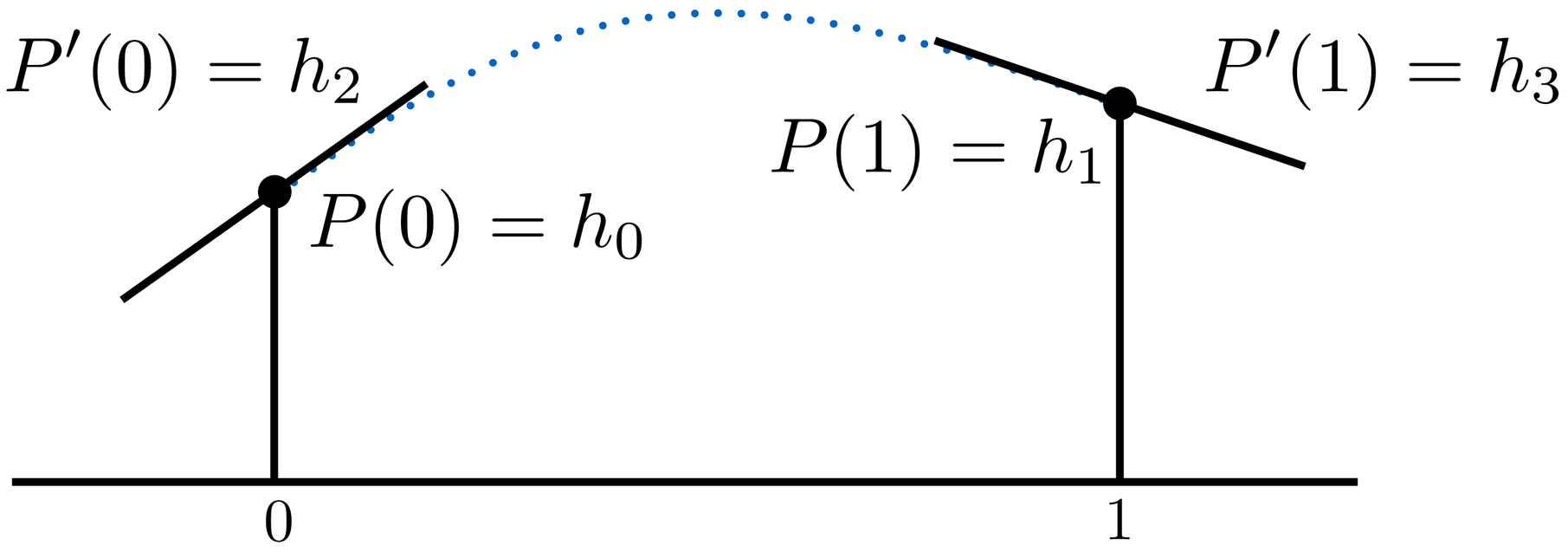




Recap: Cubic Hermite Interpolation

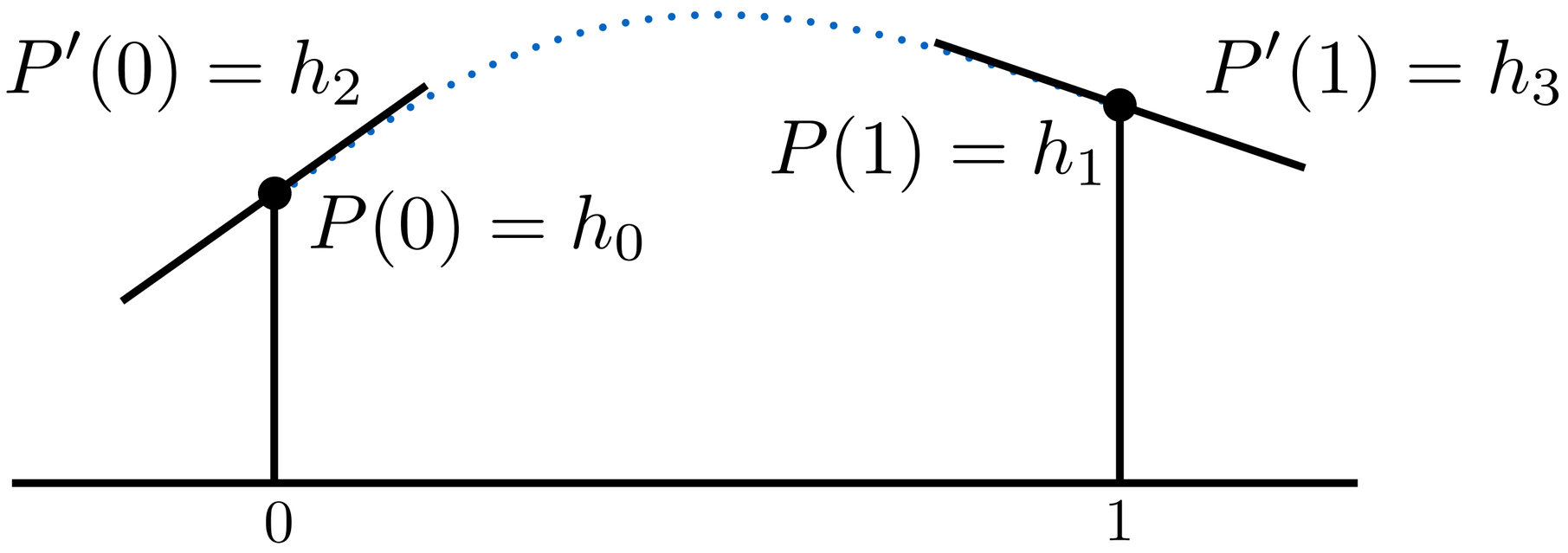


Recap: Cubic Hermite Interpolation



Inputs: values and derivatives at endpoints

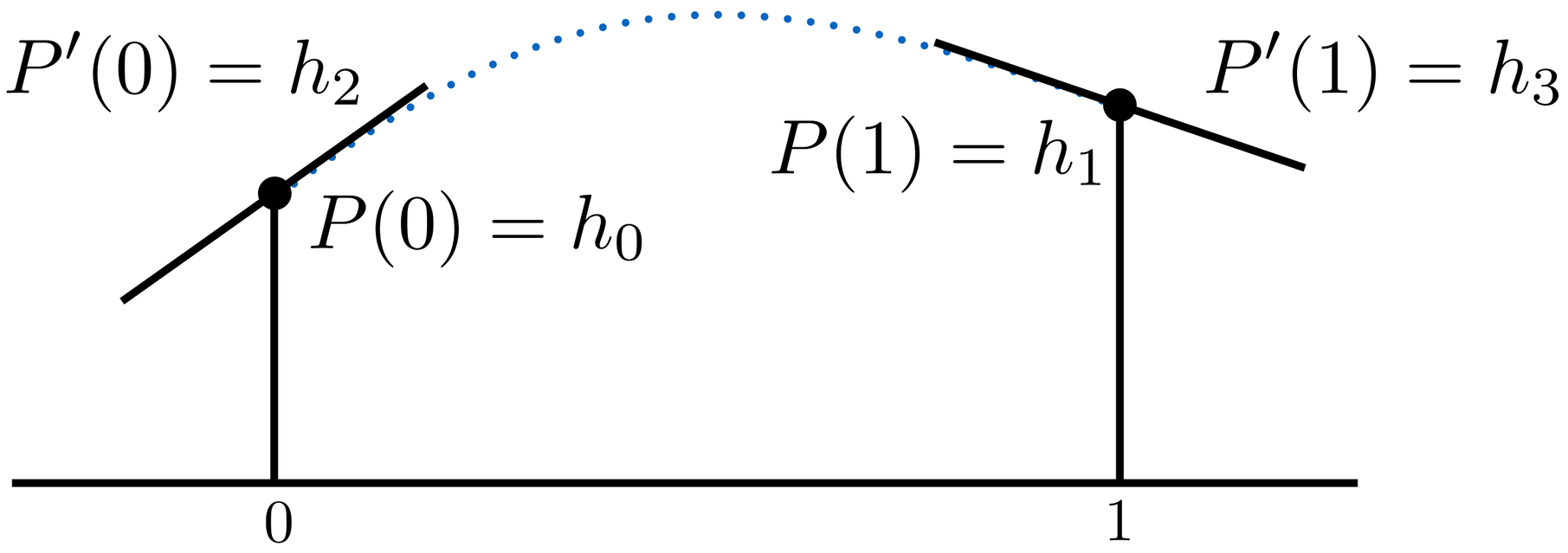
Recap: Cubic Hermite Interpolation



Inputs: values and derivatives at endpoints

Output: cubic polynomial that interpolates

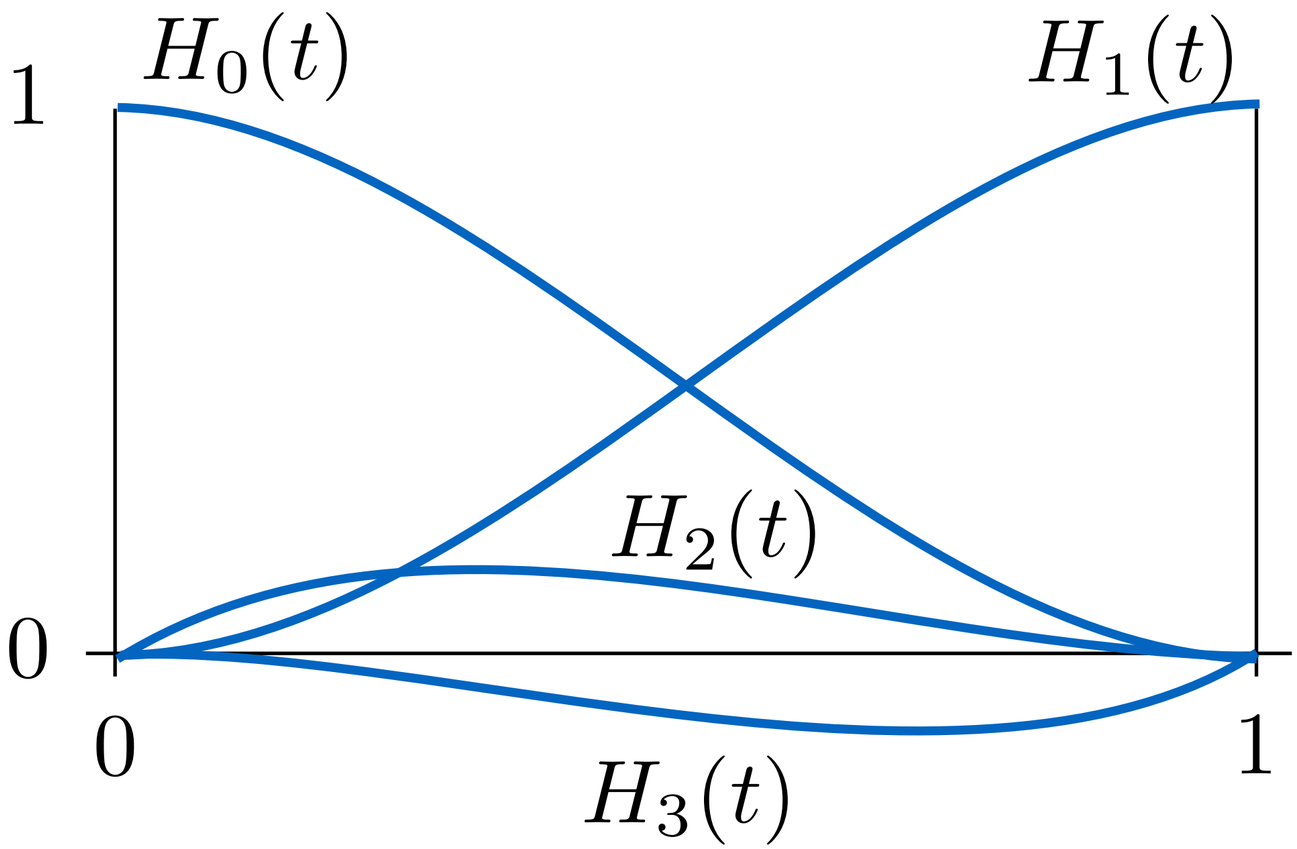
Recap: Cubic Hermite Interpolation



Inputs: values and derivatives at endpoints
Output: cubic polynomial that interpolates
Solution: weighted sum of Hermite basis functions

$$P(t) = h_0\, H_0(t) + h_1\, H_1(t) + h_2\, H_2(t) + h_3\, H_3(t)$$

Recap: Cubic Hermite Interpolation



$$H_0(t) = 2t^3 - 3t^2 + 1$$

$$H_1(t) = -2t^3 + 3t^2$$

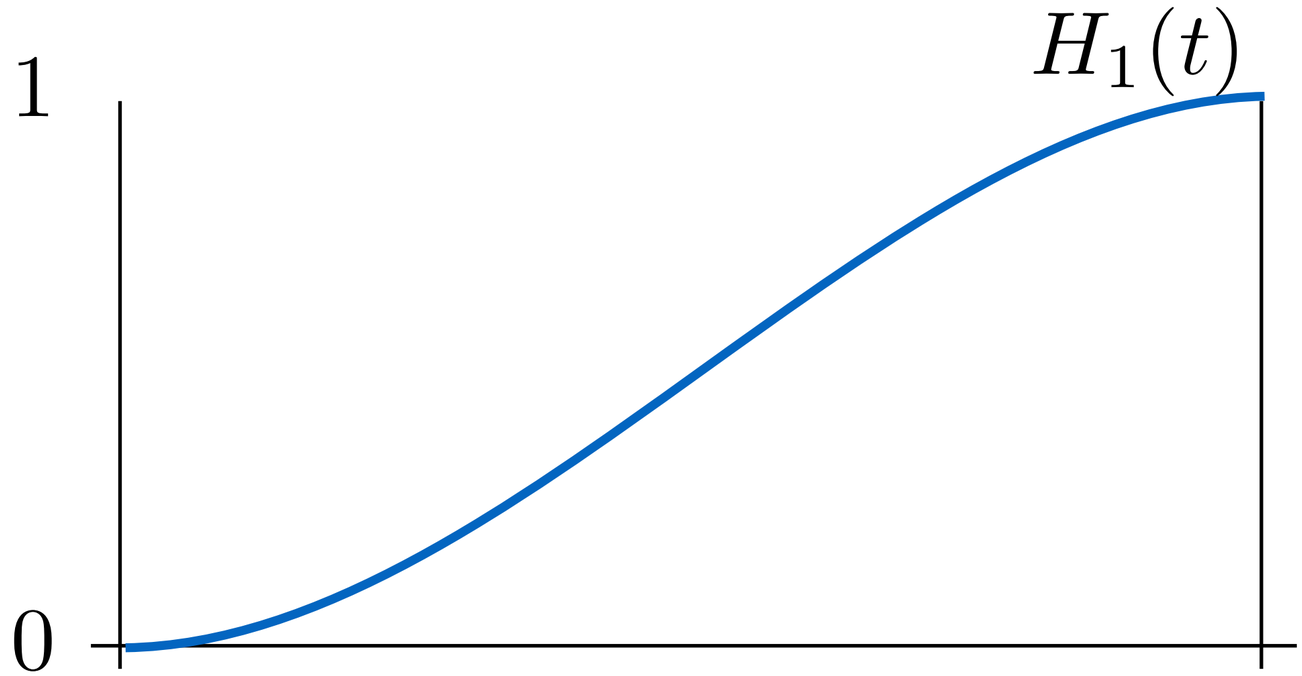
$$H_2(t) = t^3 - 2t^2 + t$$

$$H_3(t) = t^3 - t^2$$

Ease Function

A very useful function

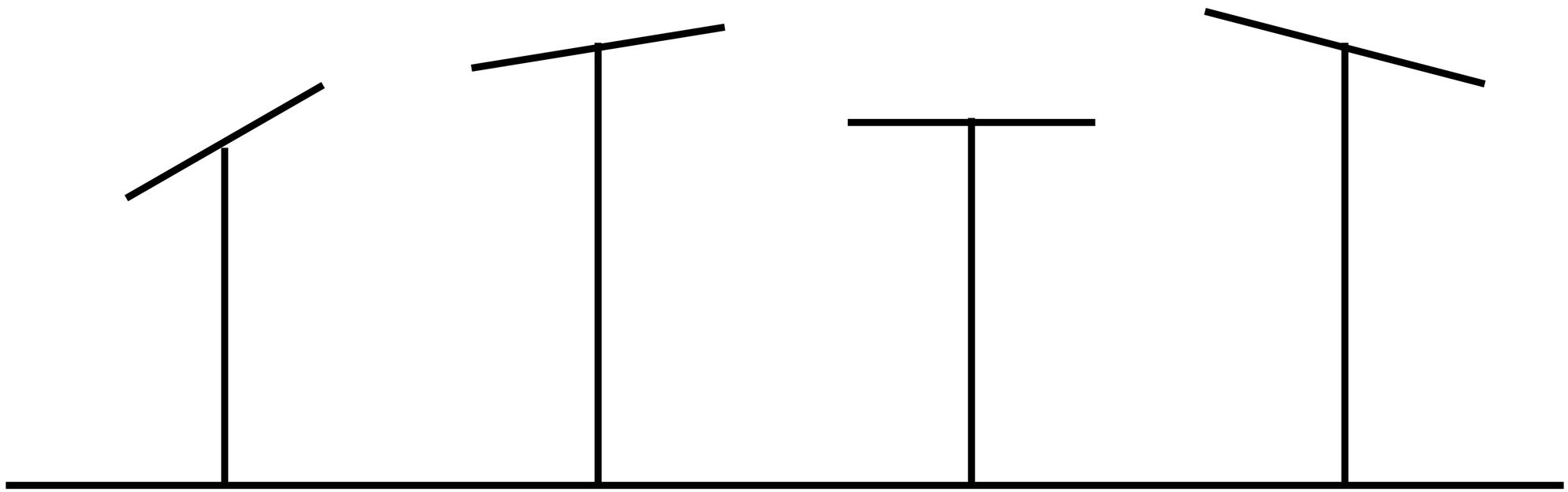
In animation, start and stop gently (zero velocity)



$$H_1(t) = -2t^3 + 3t^2 = t^2(3 - 2t)$$

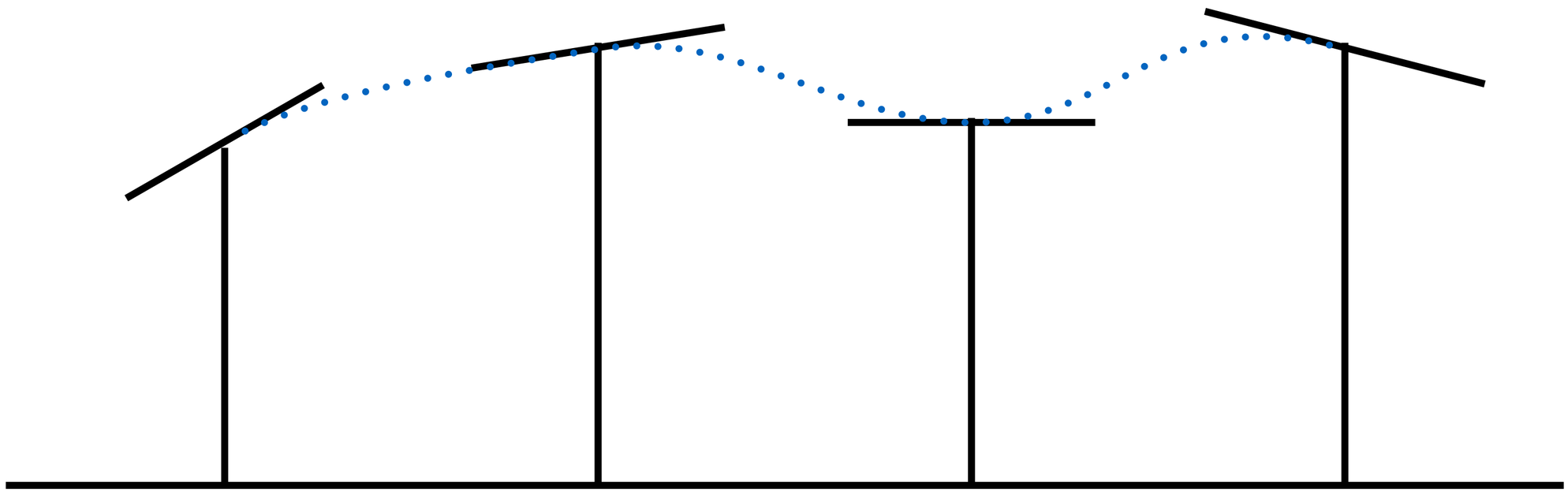
Catmull-Rom Interpolation

Hermite Spline Interpolation



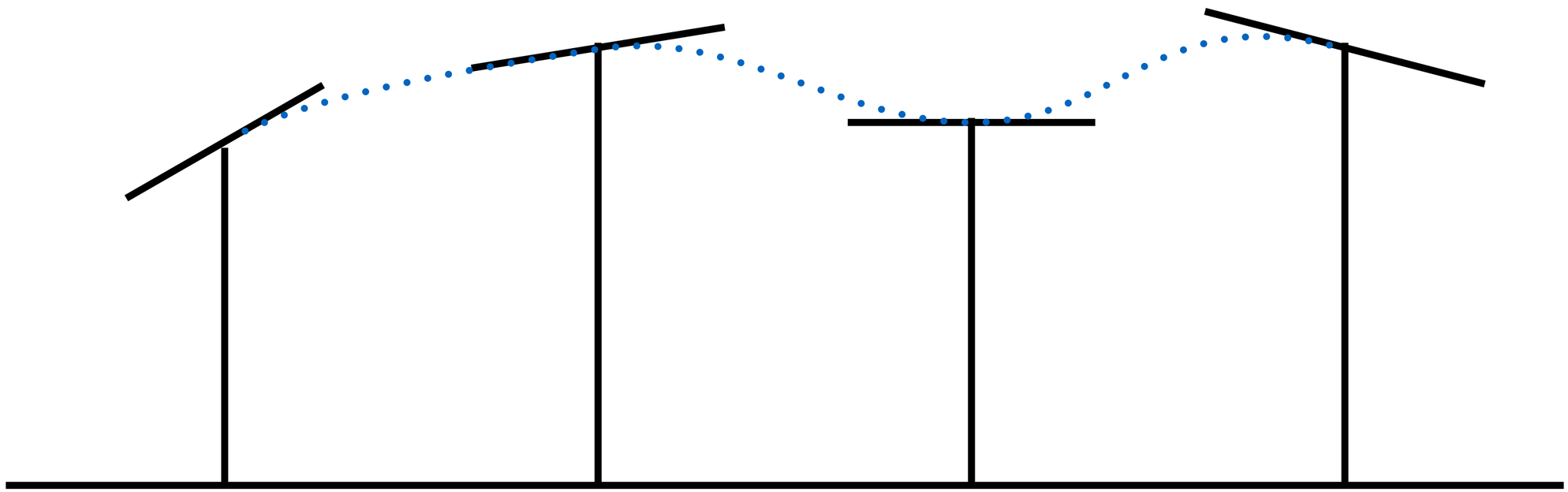
Inputs: sequence of values and derivatives

Hermite Spline Interpolation



Inputs: sequence of values and derivatives

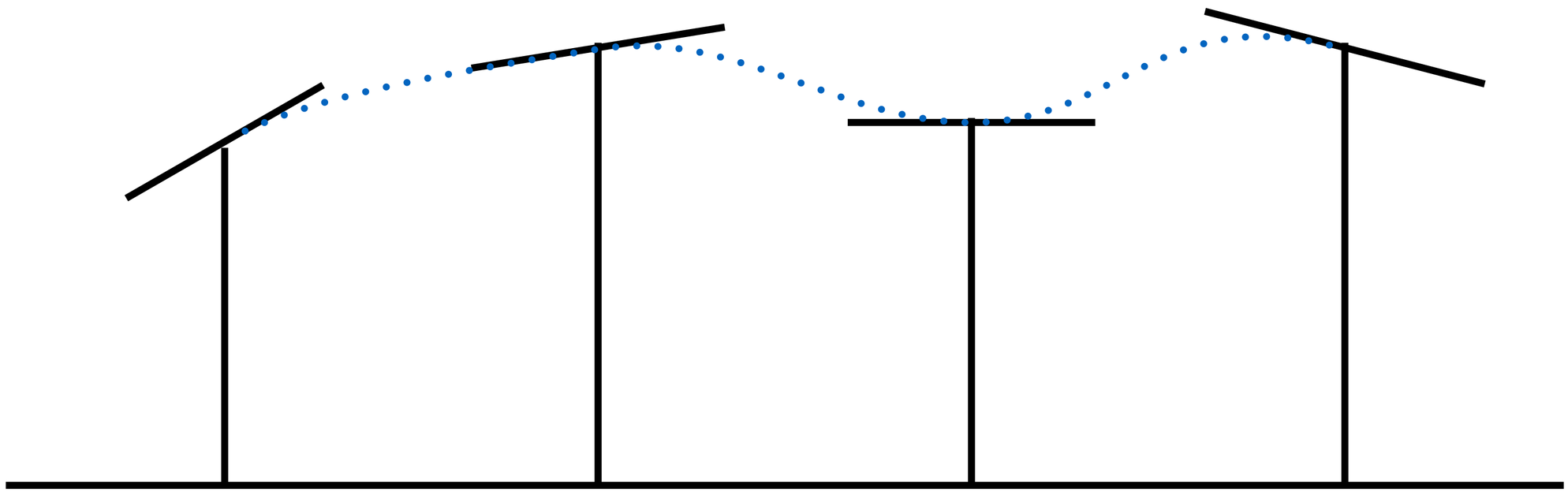
Hermite Spline Interpolation



Inputs: sequence of values and derivatives

Question: Why not use higher-order polynomial?

Hermite Spline Interpolation



Inputs: sequence of values and derivatives

Question: Why not use higher-order polynomial?

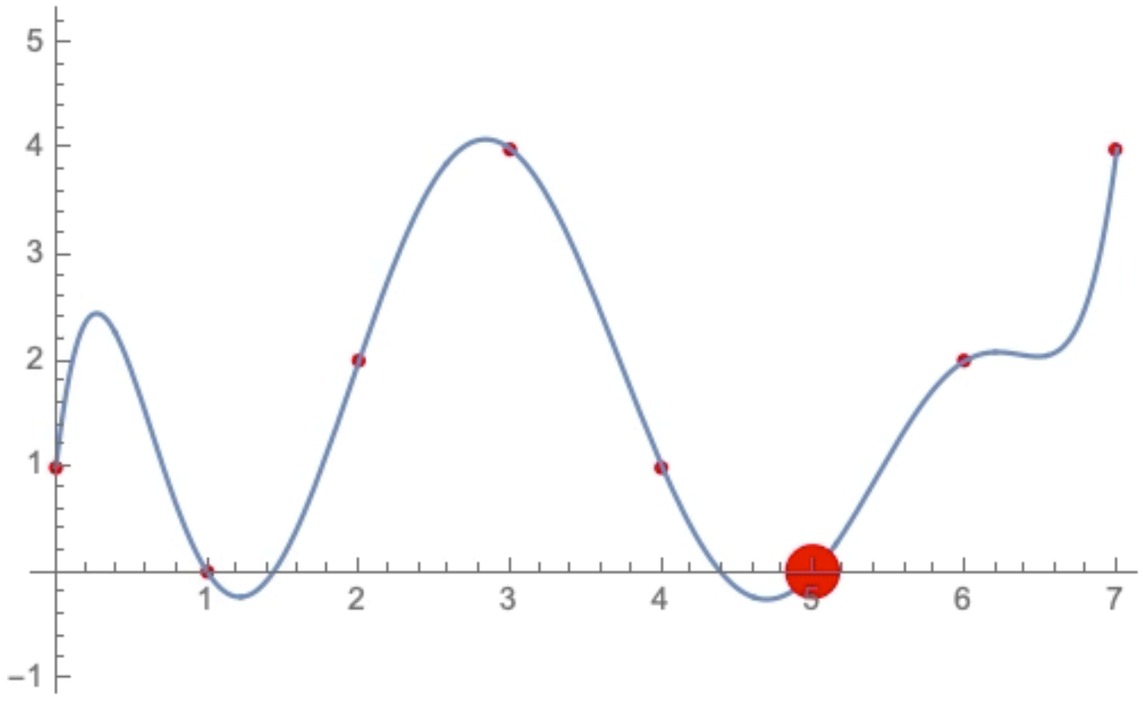
Higher order polynomials tend to wiggle excessively.

Would like to keep effect of edits “local”.

Higher-Order Polynomial?

High-degree polynomials don’t behave well

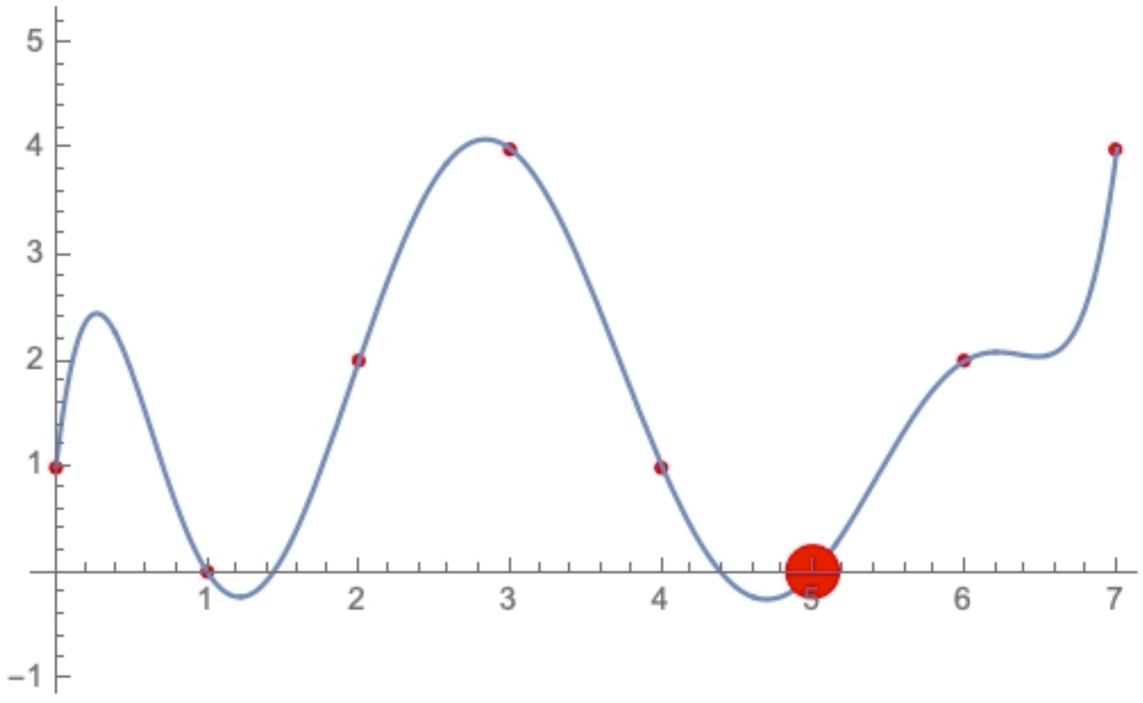
- Unstable
- No local edits
- Expensive



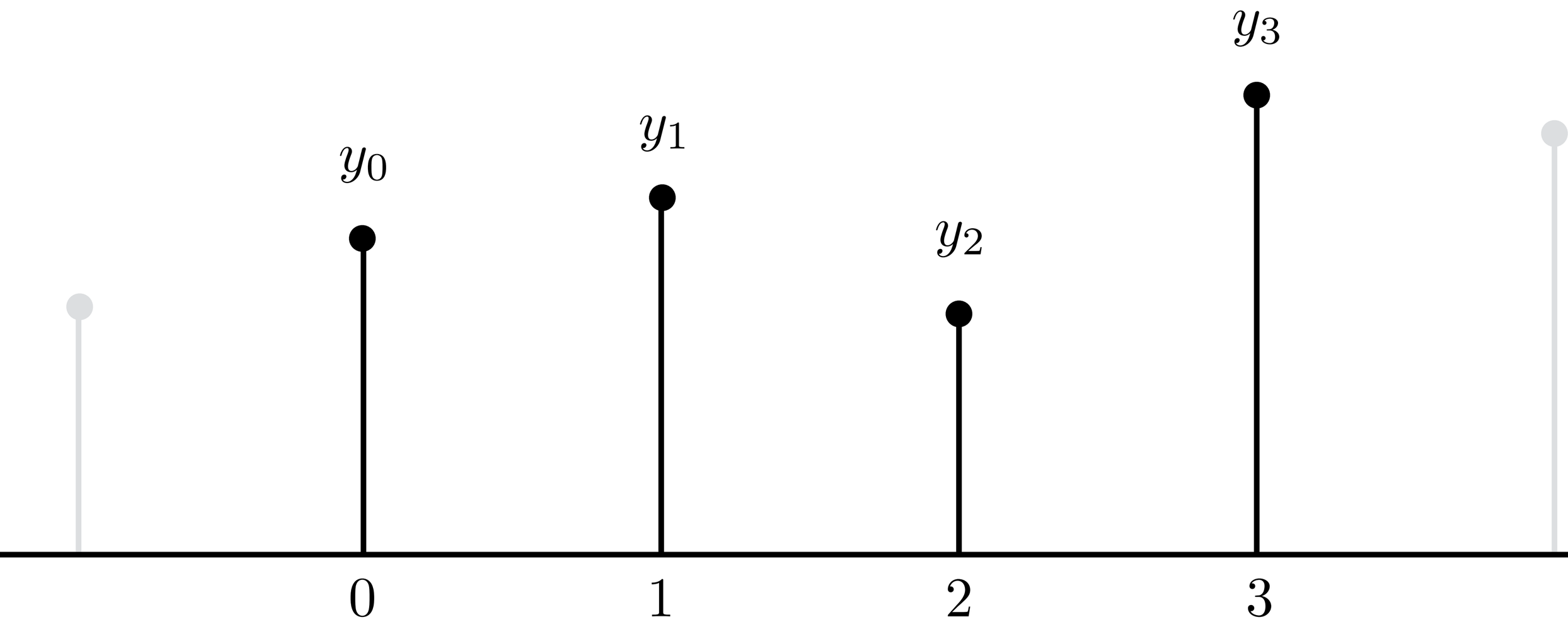
Higher-Order Polynomial?

High-degree polynomials don’t behave well

- Unstable
- No local edits
- Expensive

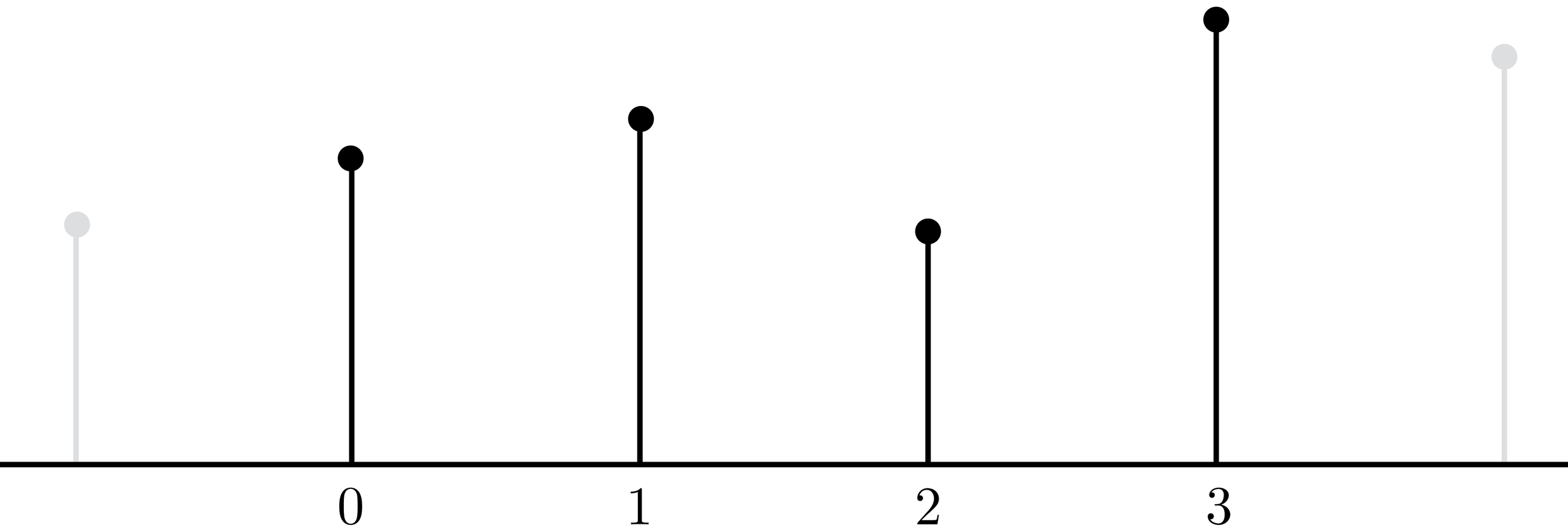


Catmull-Rom Interpolation



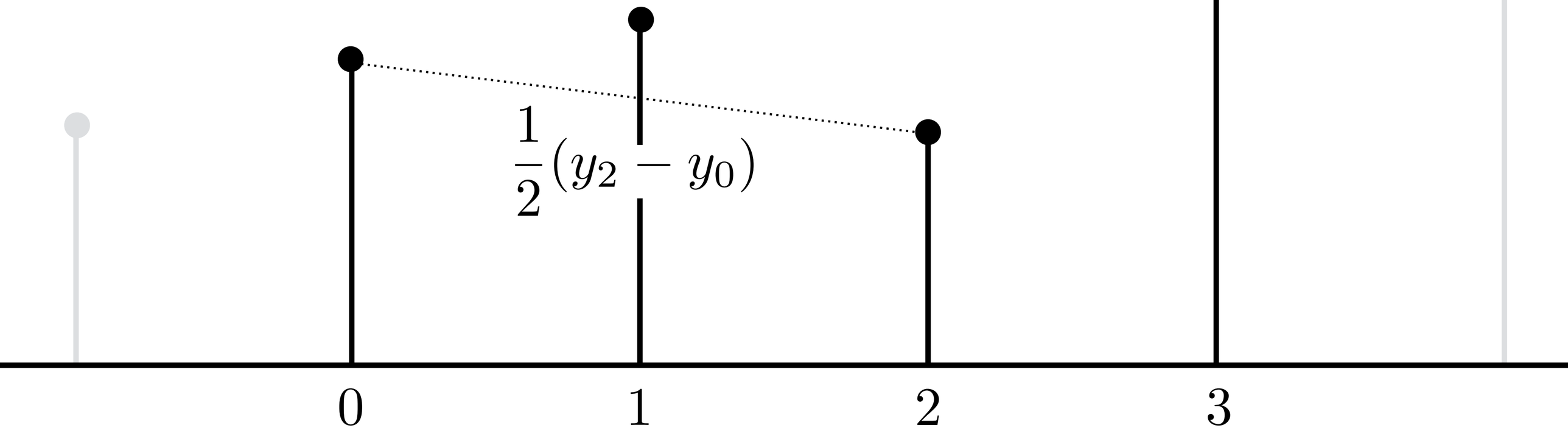
Inputs: sequence of values

Catmull-Rom Interpolation



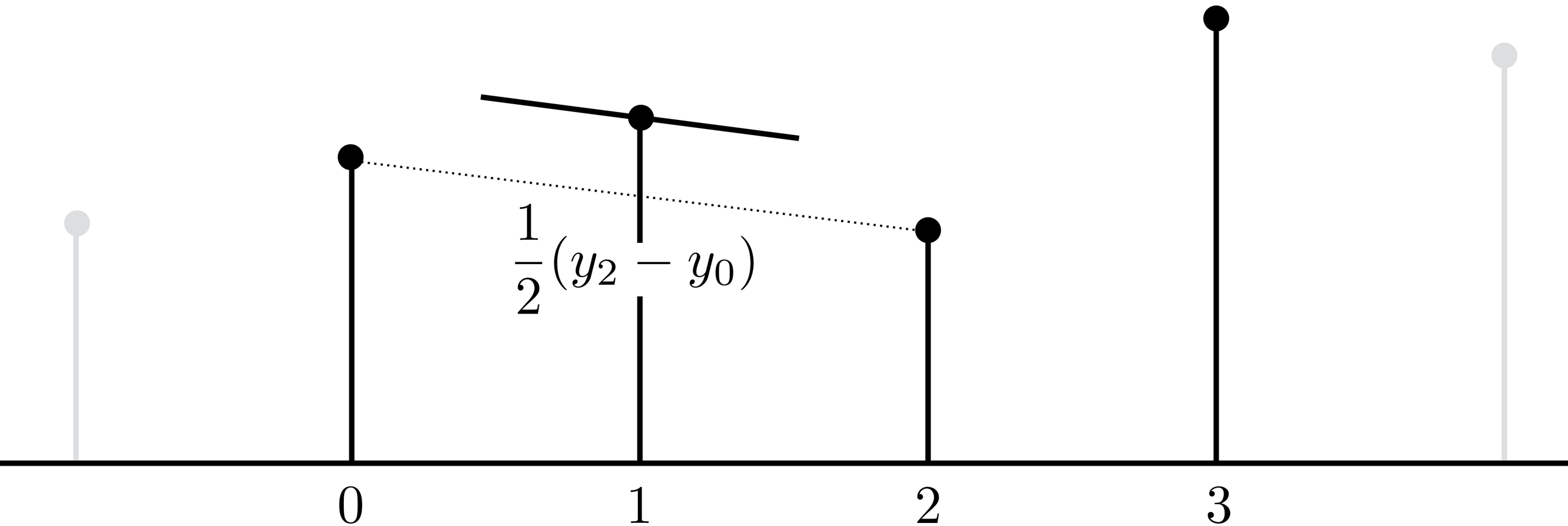
Rule for derivatives:
Match slope between previous and next values

Catmull-Rom Interpolation



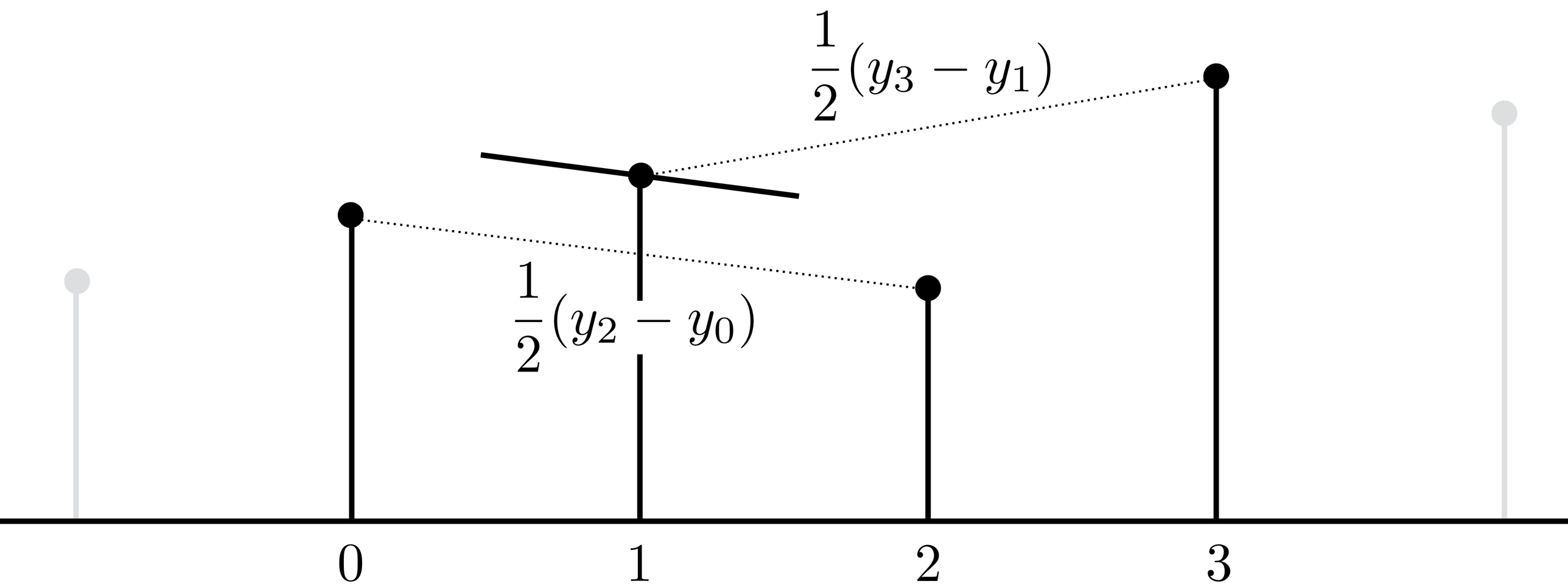
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Catmull-Rom Interpolation



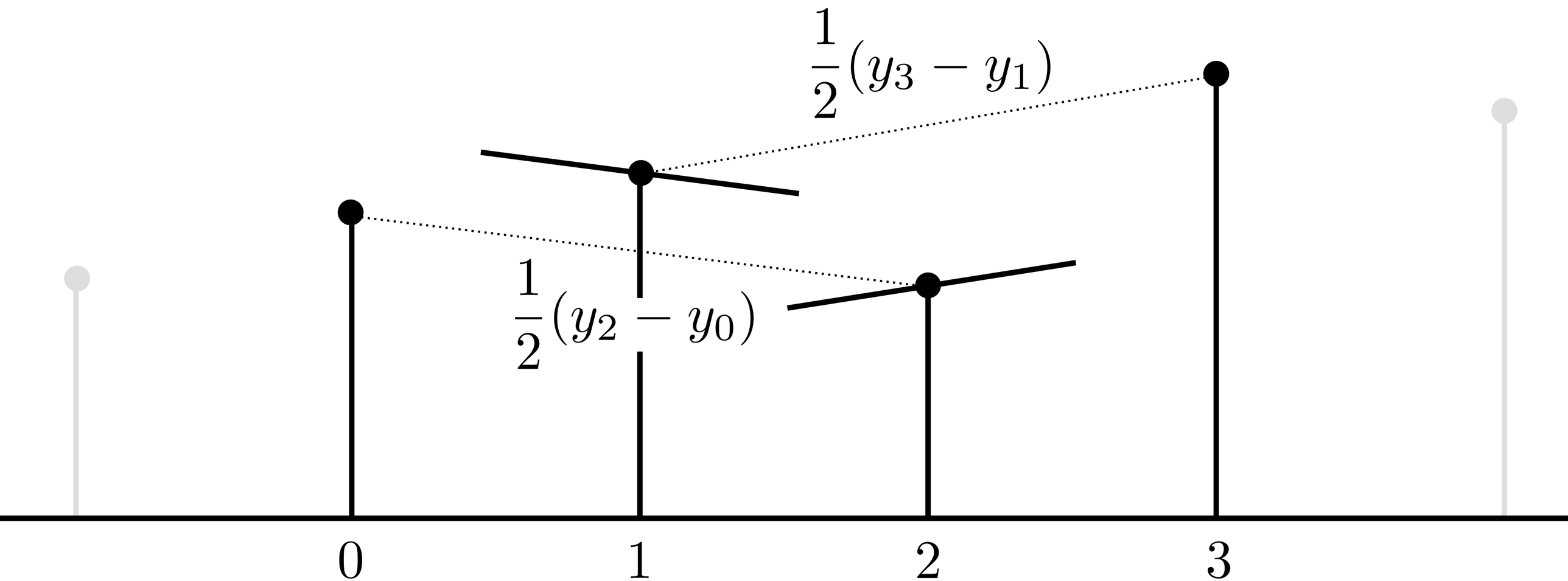
Rule for derivatives:
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Catmull-Rom Interpolation



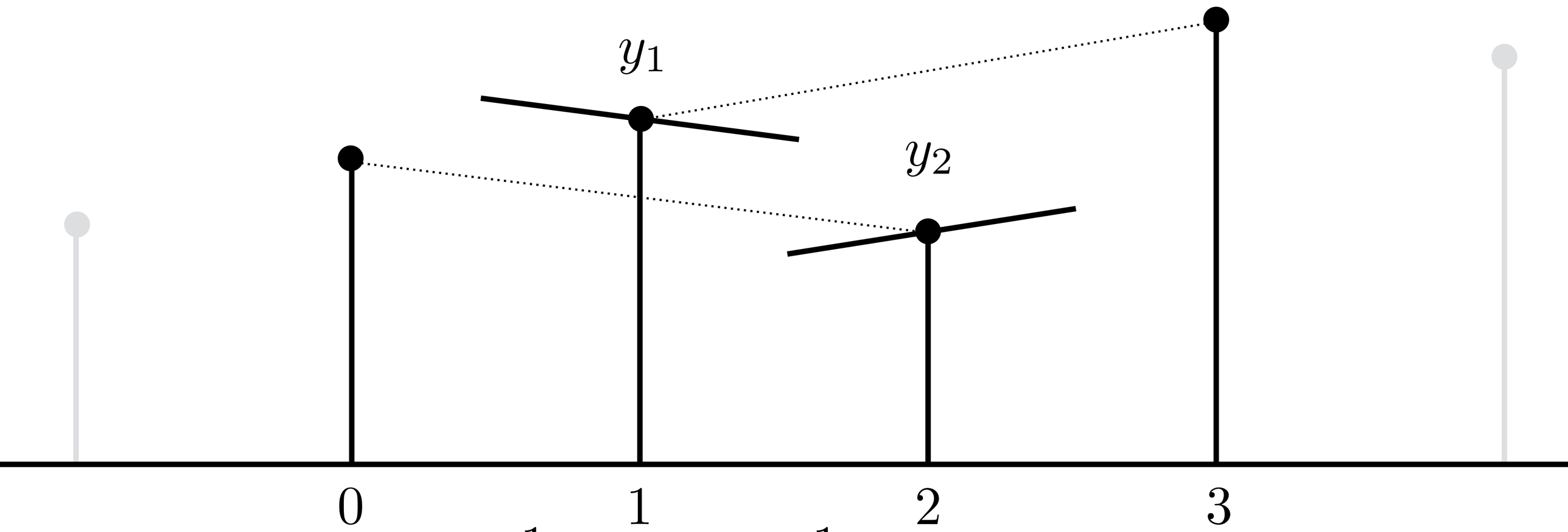
Rule for derivatives:
Match slope between previous and next values

Catmull-Rom Interpolation



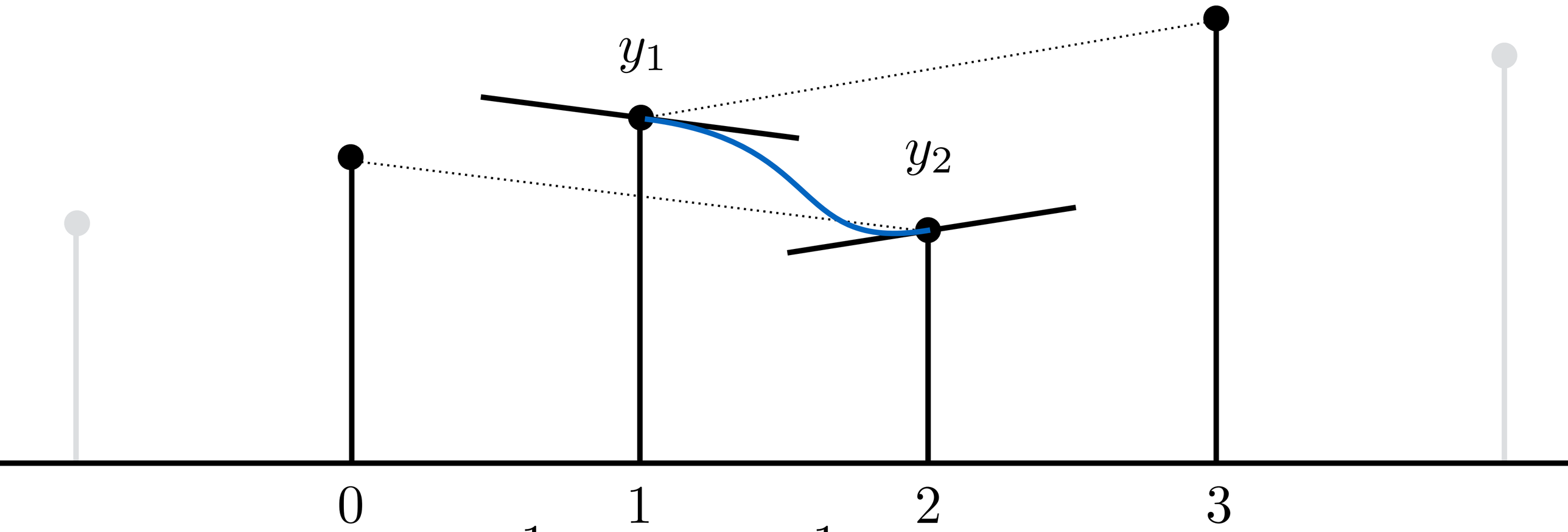
Rule for derivatives:
Match slope between previous and next values

Catmull-Rom Interpolation



Then use Hermite interpolation

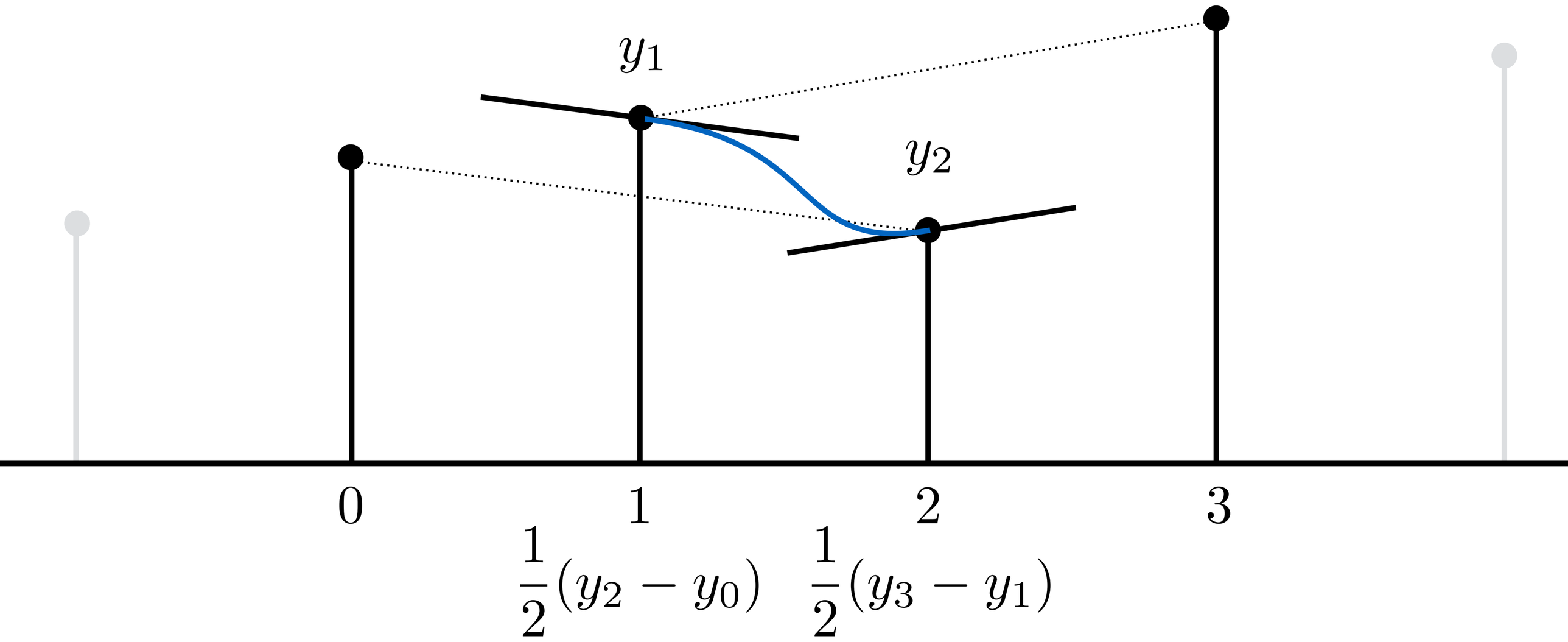
Catmull-Rom Interpolation



Then use Hermite interpolation

Catmull-Rom Interpolation

Note the domain where curve is defined!

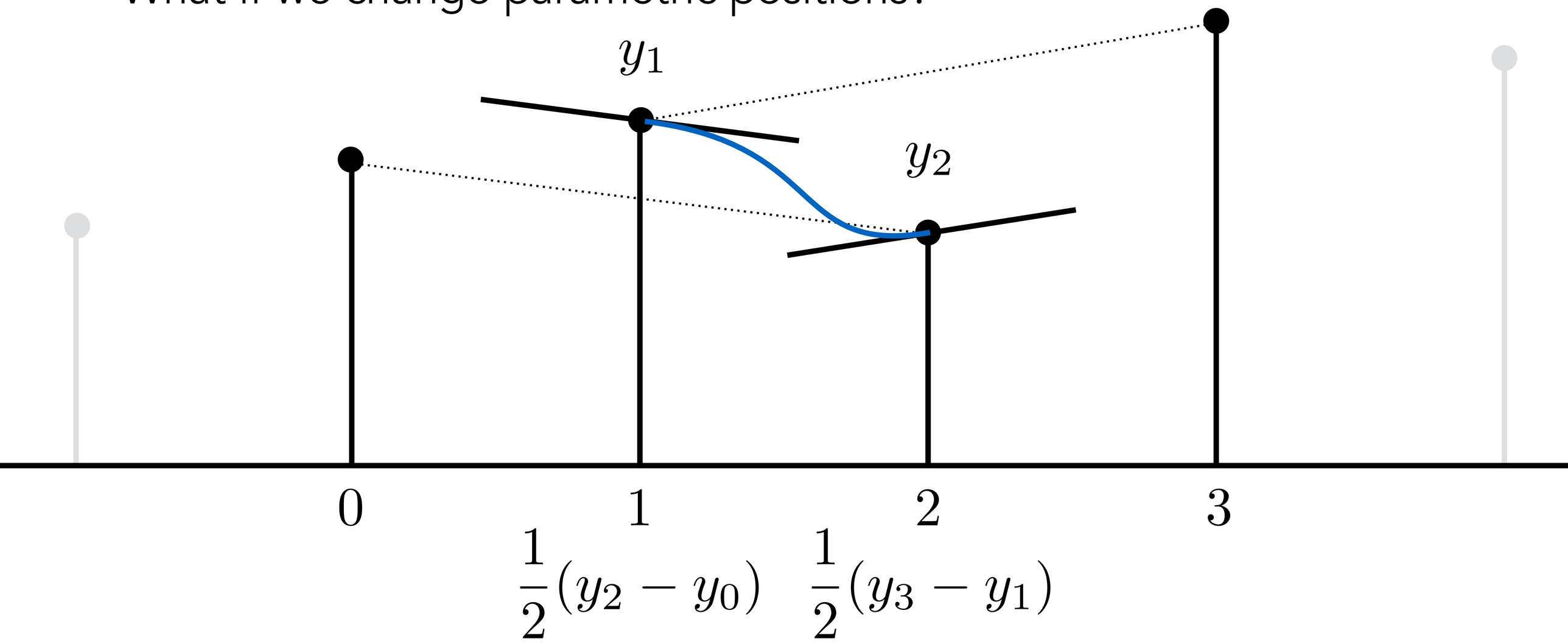


Then use Hermite interpolation

Catmull-Rom Interpolation

Note the domain where curve is defined!

What if we change parametric positions?

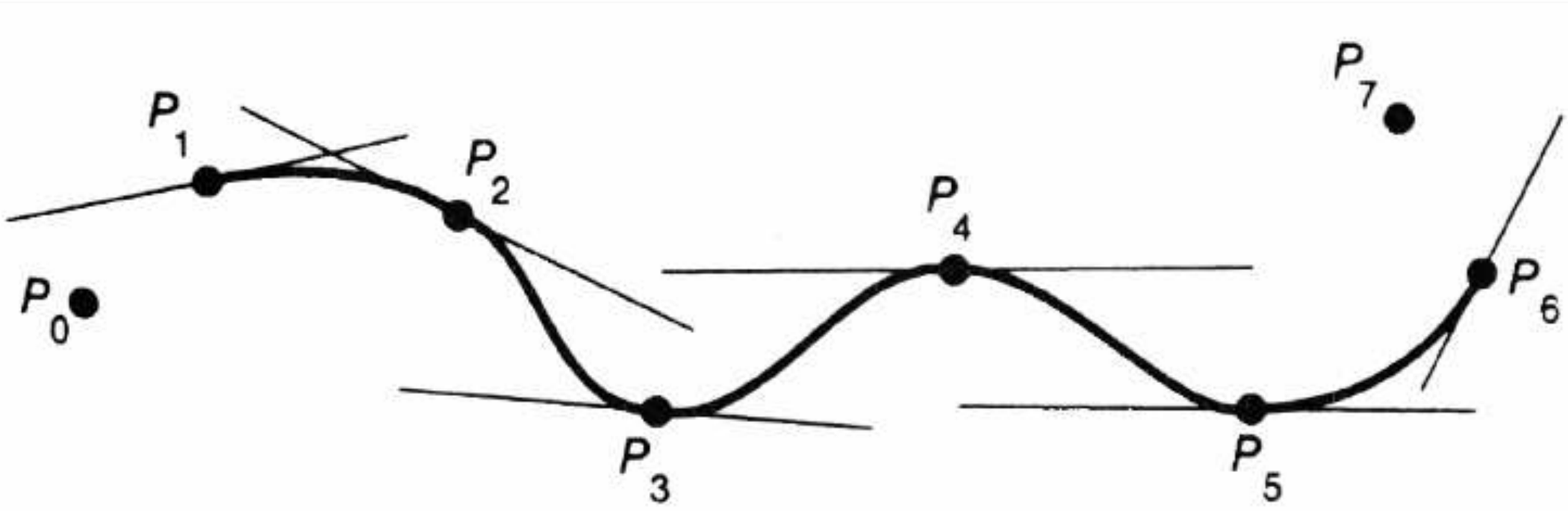


Then use Hermite interpolation

Catmull-Rom Spline

Input: sequence of points

Output: spline that interpolates all points with C1 continuity



- Piecewise Cubic Curve

- C1 continuous!

Matrix Form of Catmull-Rom Space Curve?

Use Hermite matrix form

- Points & tangents given by Catmull-Rom rules

Power Basis

Polynomial Coefficients

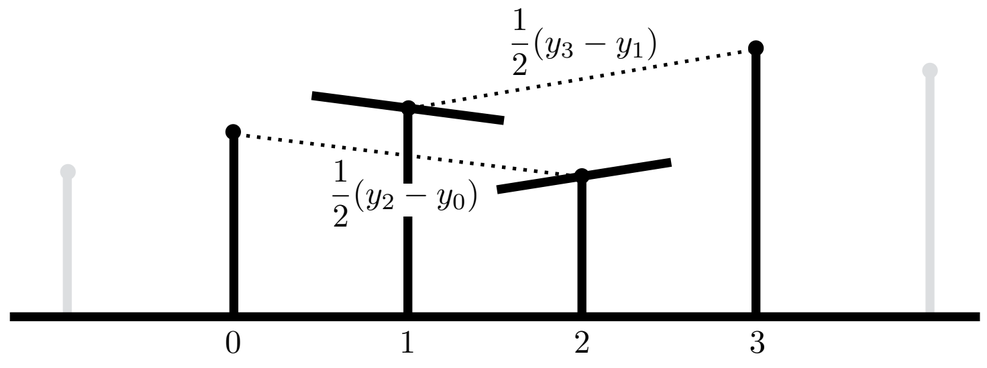
Hermite points

$$\mathbf{h}_0 = \mathbf{p}_1$$
$$\mathbf{h}_1 = \mathbf{p}_2$$

Hermite tangents

$$\mathbf{h}_2 = \tfrac{1}{2}(\mathbf{p}_2 - \mathbf{p}_0)$$
$$\mathbf{h}_3 = \tfrac{1}{2}(\mathbf{p}_3 - \mathbf{p}_1)$$

$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \begin{bmatrix} \mathbf{c}_0 \\ \mathbf{c}_1 \\ \mathbf{c}_2 \\ \mathbf{c}_3 \end{bmatrix}$$



Matrix Form of Catmull-Rom Space Curve?

Use Hermite matrix form

- Points & tangents given by Catmull-Rom rules

Power Basis

Hermite Basis Matrix β_H

Hermite Control Values

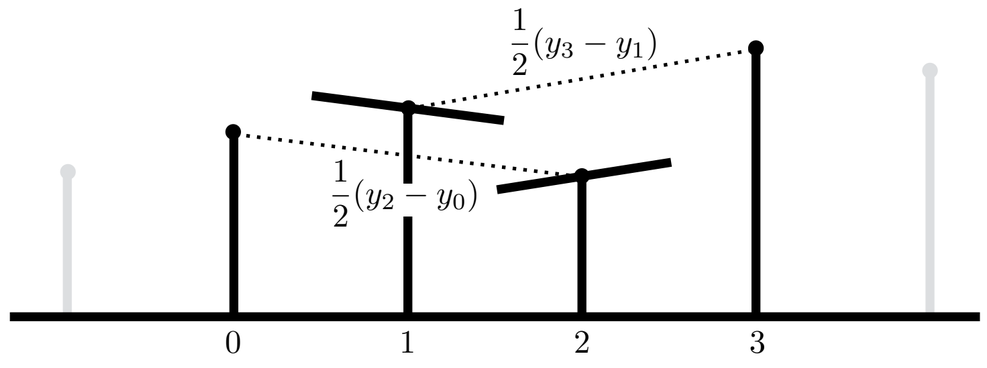
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Hermite tangents

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$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \boldsymbol{\beta}_H \cdot \begin{bmatrix} \mathbf{h}_0 \\ \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix}$$



Matrix Form of Catmull-Rom Space Curve?

Use Hermite matrix form

- Points & tangents given by Catmull-Rom rules

Power Basis

Hermite Basis Matrix β_H

Convert Catmull-Rom to Hermite

Catmull-Rom Control Points

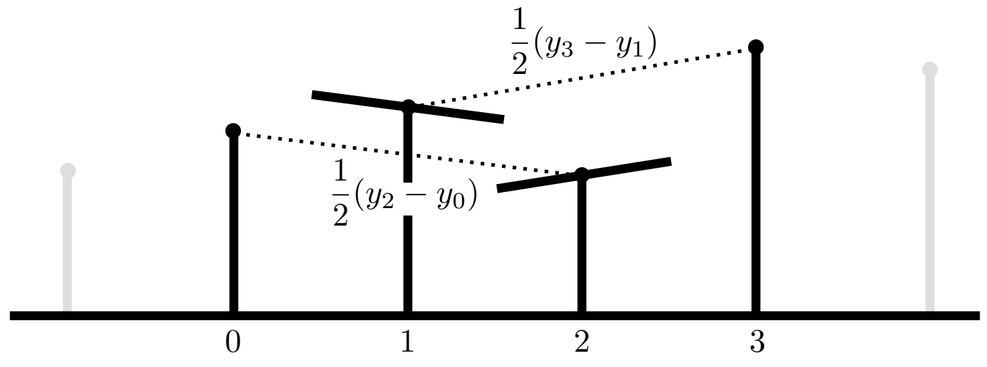
Hermite points

$$\mathbf{h}_0 = \mathbf{p}_1$$
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Hermite tangents

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$$\mathbf{h}_3 = \tfrac{1}{2}(\mathbf{p}_3 - \mathbf{p}_1)$$

$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \boldsymbol{\beta}_H \cdot \mathbf{M}_{CR \to H} \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$



Matrix Form of Catmull-Rom Space Curve?

Use Hermite matrix form

- Points & tangents given by Catmull-Rom rules

Hermite points

$$\mathbf{h}_0 = \mathbf{p}_1$$
$$\mathbf{h}_1 = \mathbf{p}_2$$

Hermite tangents

$$\mathbf{h}_2 = \tfrac{1}{2}(\mathbf{p}_2 - \mathbf{p}_0)$$
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Power Basis

Hermite Basis Matrix β_H

Convert Catmull-Rom to Hermite

Catmull-Rom Control Points

$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -3 & 3 & -2 & -1 \\ 2 & -2 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \frac{-1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & \frac{-1}{2} & 0 & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

Matrix Form of Catmull-Rom Space Curve?

Use Hermite matrix form

- Points & tangents given by Catmull-Rom rules

Hermite points

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Hermite tangents

$$\mathbf{h}_2 = \tfrac{1}{2}(\mathbf{p}_2 - \mathbf{p}_0)$$
$$\mathbf{h}_3 = \tfrac{1}{2}(\mathbf{p}_3 - \mathbf{p}_1)$$

Power Basis

Catmull-Rom Basis Matrix β_{CR}

Catmull-Rom Control Points

$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-1}{2} & 0 & \frac{1}{2} & 0 \\ 1 & -2\frac{1}{2} & 2 & \frac{-1}{2} \\ \frac{-1}{2} & 1\frac{1}{2} & -1\frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

Matrix Form of Catmull-Rom Space Curve?

Use Hermite matrix form

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Hermite tangents

$$\mathbf{h}_2 = \tfrac{1}{2}(\mathbf{p}_2 - \mathbf{p}_0)$$
$$\mathbf{h}_3 = \tfrac{1}{2}(\mathbf{p}_3 - \mathbf{p}_1)$$

Power Basis

Catmull-Rom Basis Matrix β_{CR}

Catmull-Rom Control Points

$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \boldsymbol{\beta}_{CR} \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

$$\mathbf{p}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_{CR} \cdot \mathbf{p}$$

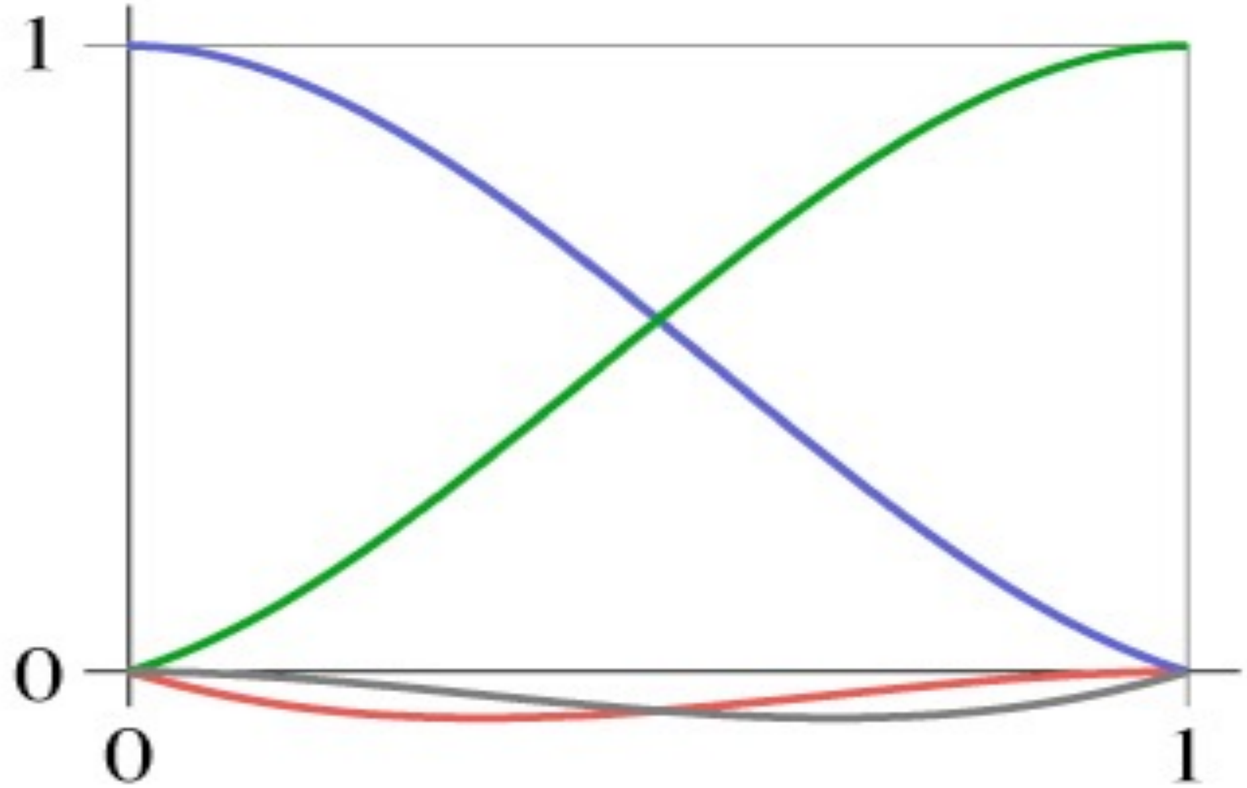
Matrix Form of Catmull-Rom Space Curve

$$\mathbf{p}(u) = \begin{bmatrix} 1 \\ u \\ u^2 \\ u^3 \end{bmatrix}^{\mathsf{T}} \cdot \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{-1}{2} & 0 & \frac{1}{2} & 0 \\ 1 & -2\frac{1}{2} & 2 & \frac{-1}{2} \\ \frac{-1}{2} & 1\frac{1}{2} & -1\frac{1}{2} & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

$$\mathbf{p}(u) = \boldsymbol{\mathcal{P}}^3(u) \cdot \boldsymbol{\beta}_{CR} \cdot \mathbf{p}$$

Matrix columns are Catmull-Rom basis functions

Compare to Hermite



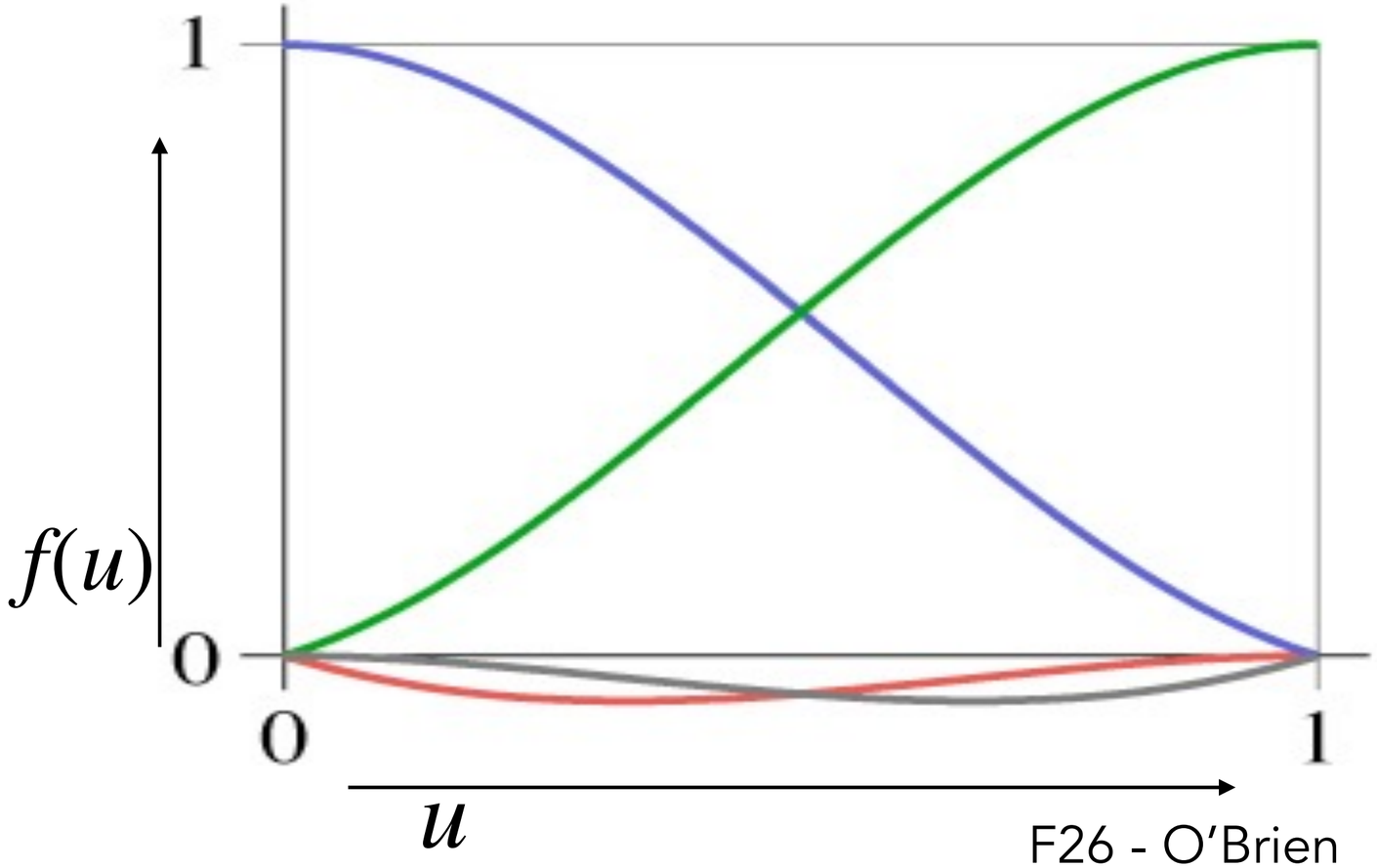
Catmull-Rom Basis Functions

$$\mathbf{p}(u) = \begin{bmatrix} 0 - \frac{1}{2}u + 1u^2 - \frac{1}{2}u^3 \\ 1 + 0u - 2\frac{1}{2}u^2 + 1\frac{1}{2}u^3 \\ 0 + \frac{1}{2}u + 2u^2 - 1\frac{1}{2}u^3 \\ 0 + 0u - \frac{1}{2}u^2 + \frac{1}{2}u^3 \end{bmatrix}^T \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

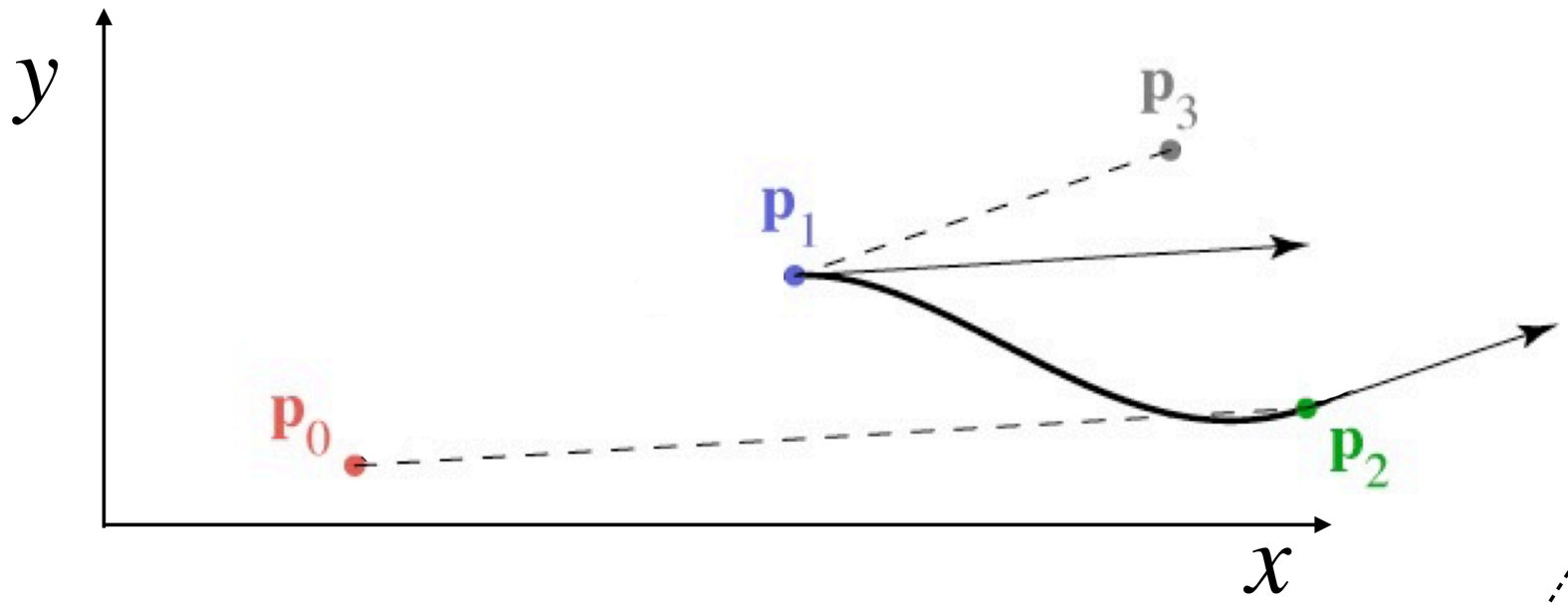
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Compare to Hermite

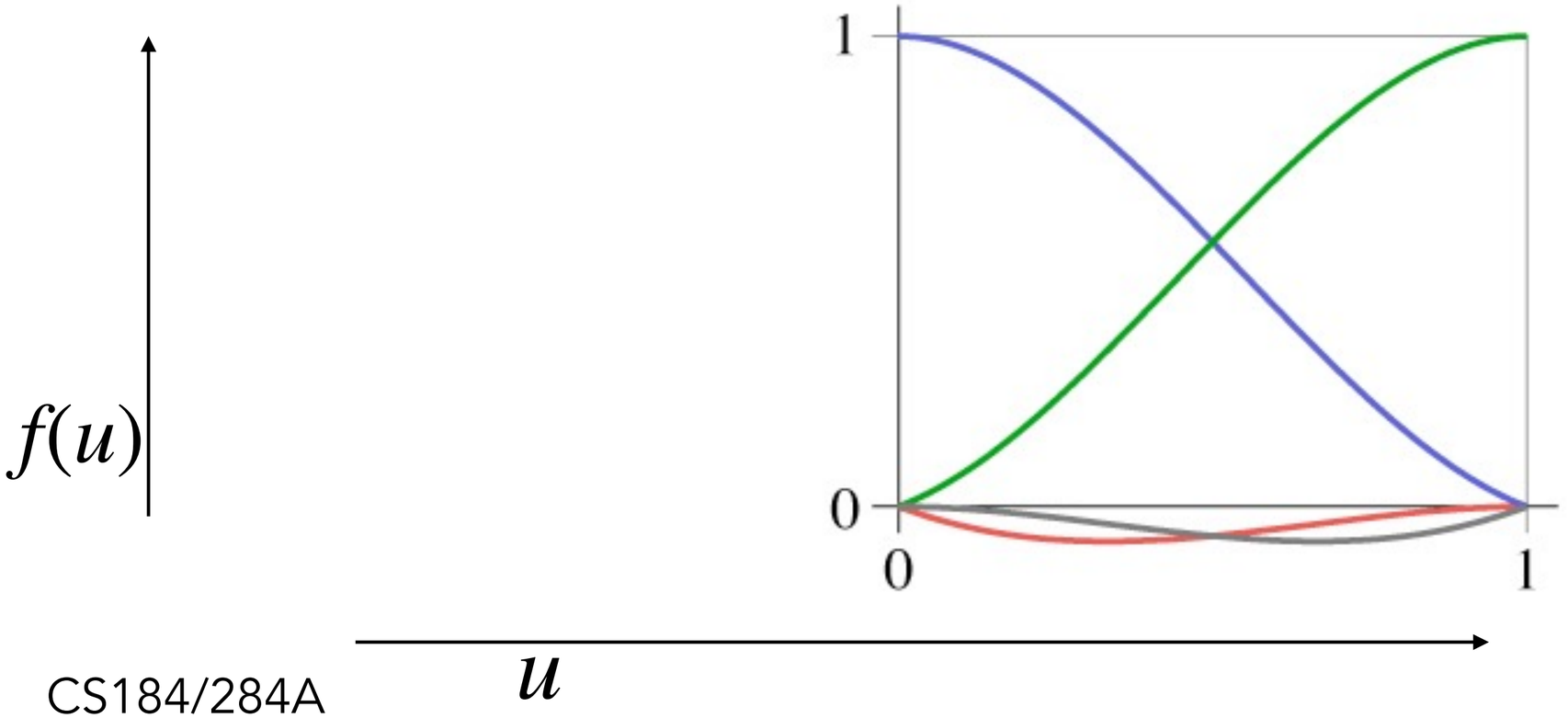


Catmull-Rom Basis in “Hump” form

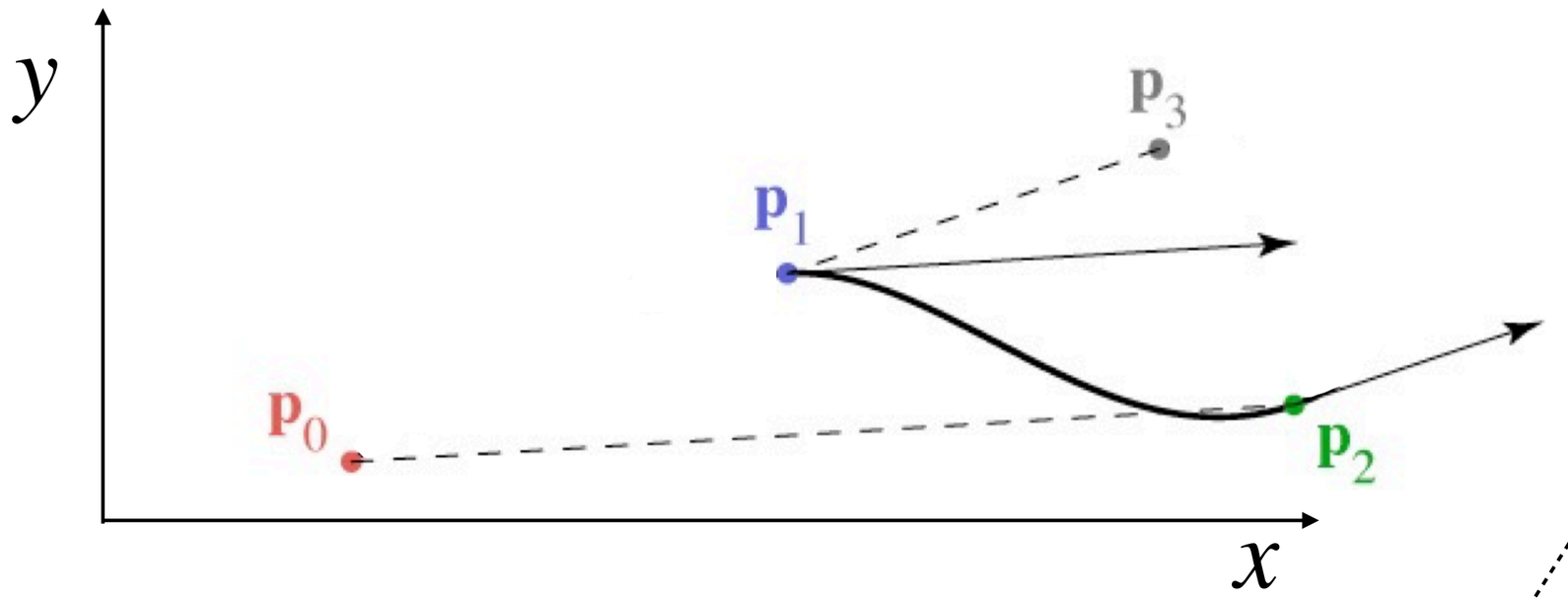


Object space

Parameter space

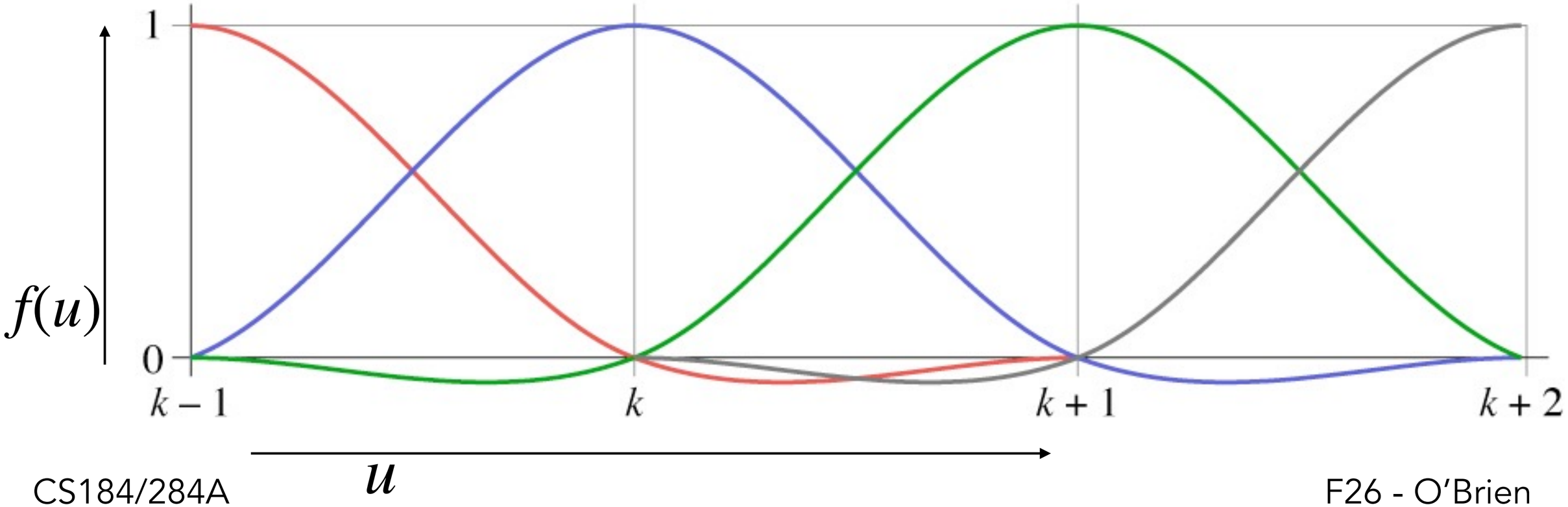


Catmull-Rom Basis in “Hump” form

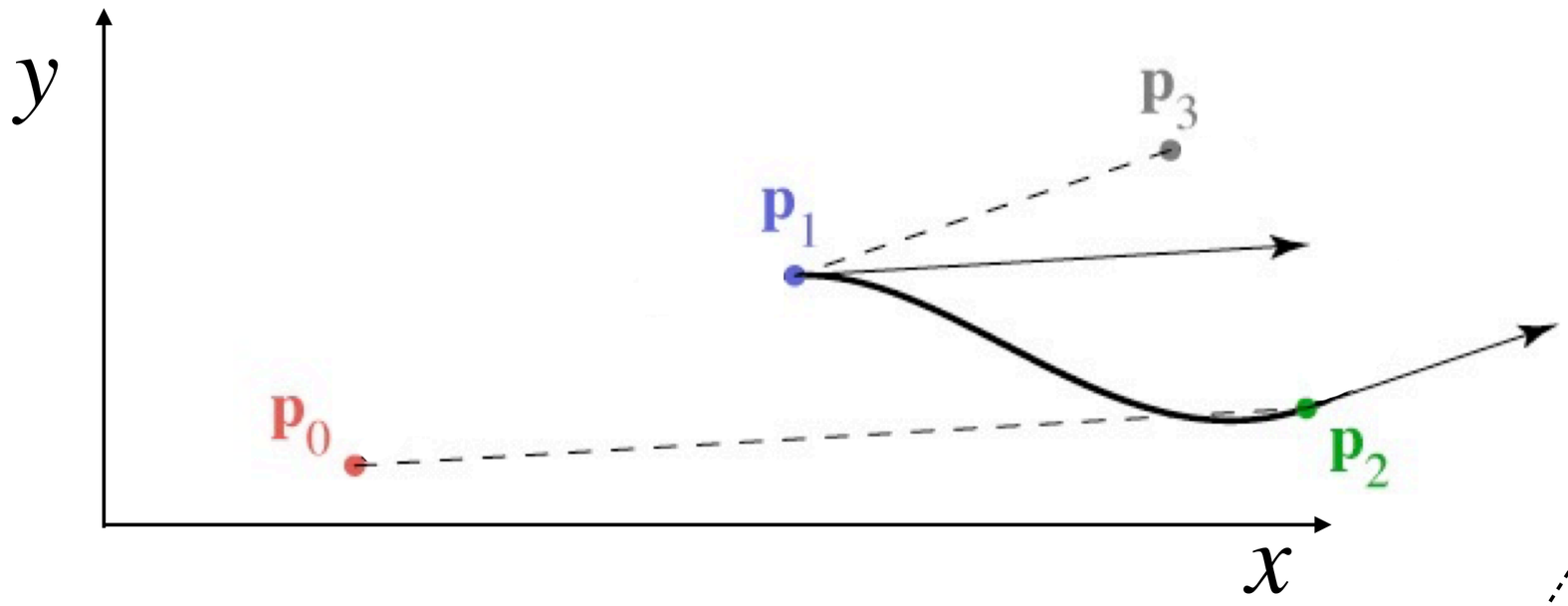


Object space

Parameter space



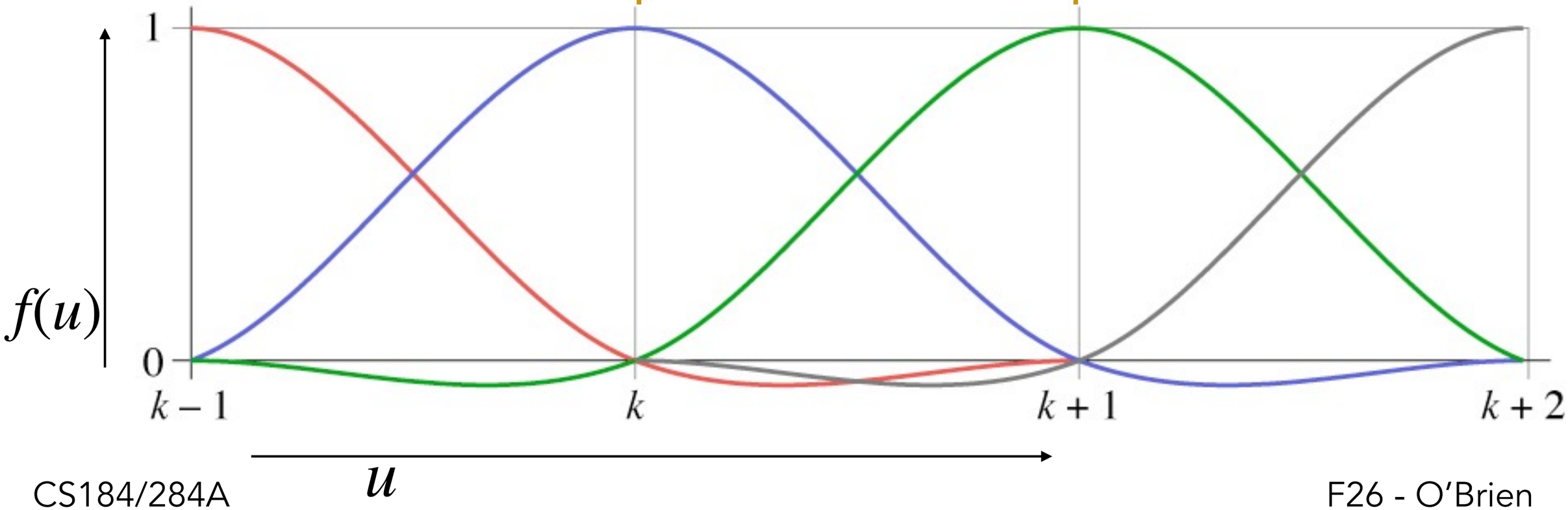
Catmull-Rom Basis in “Hump” form



Object space

Parameter space

Region where fully defined

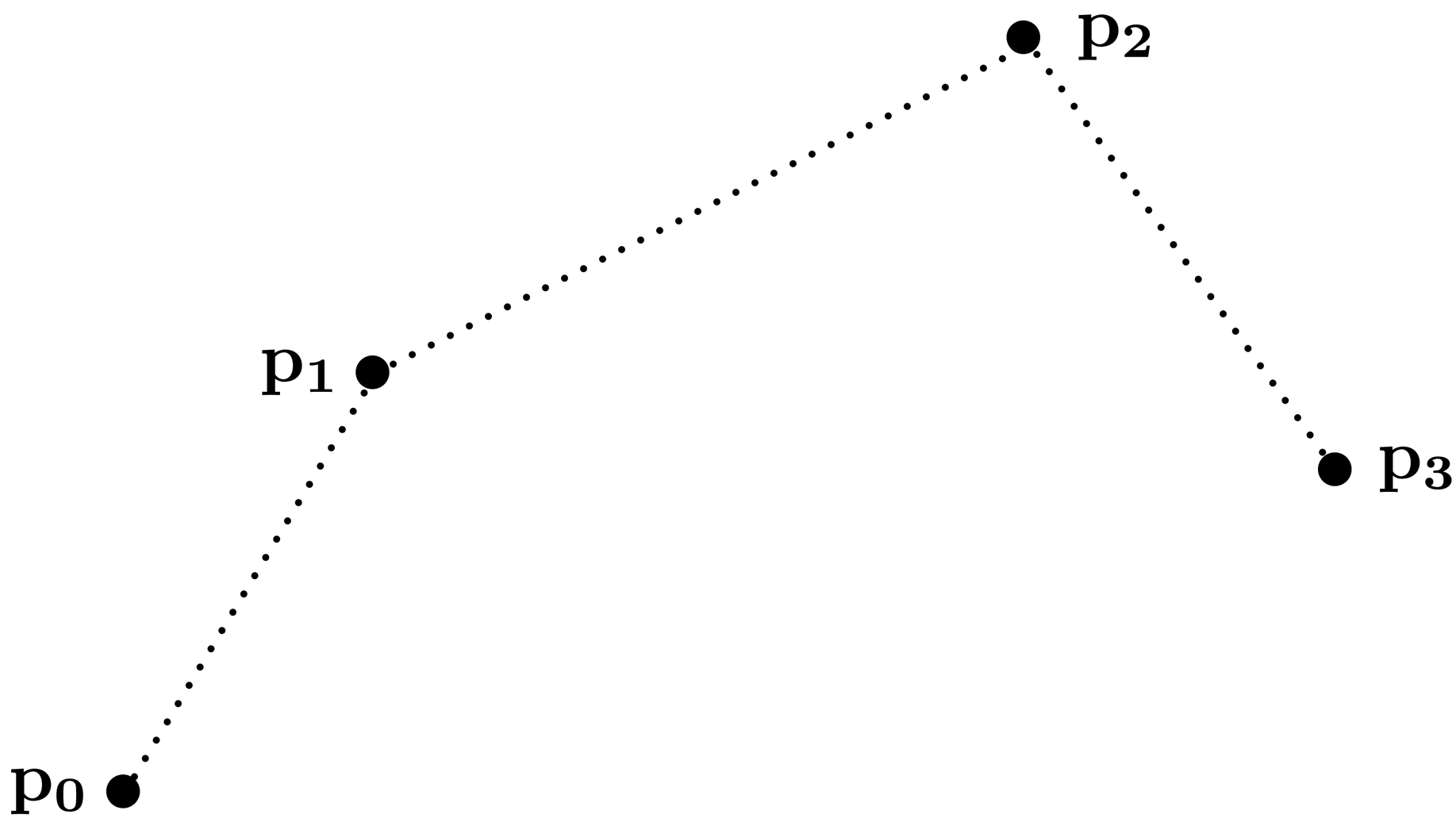


Bézier Curves

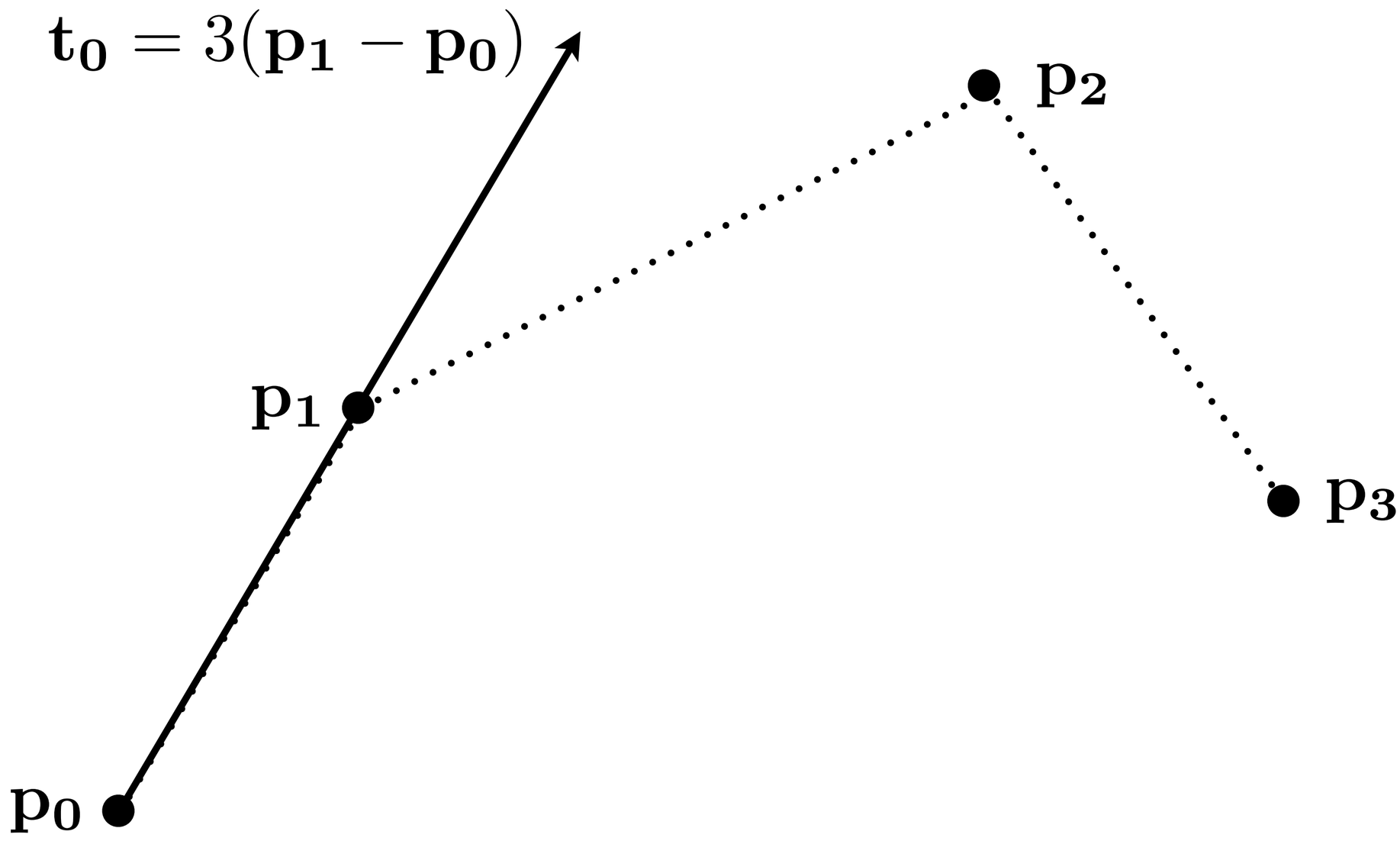
Examples of Geometry



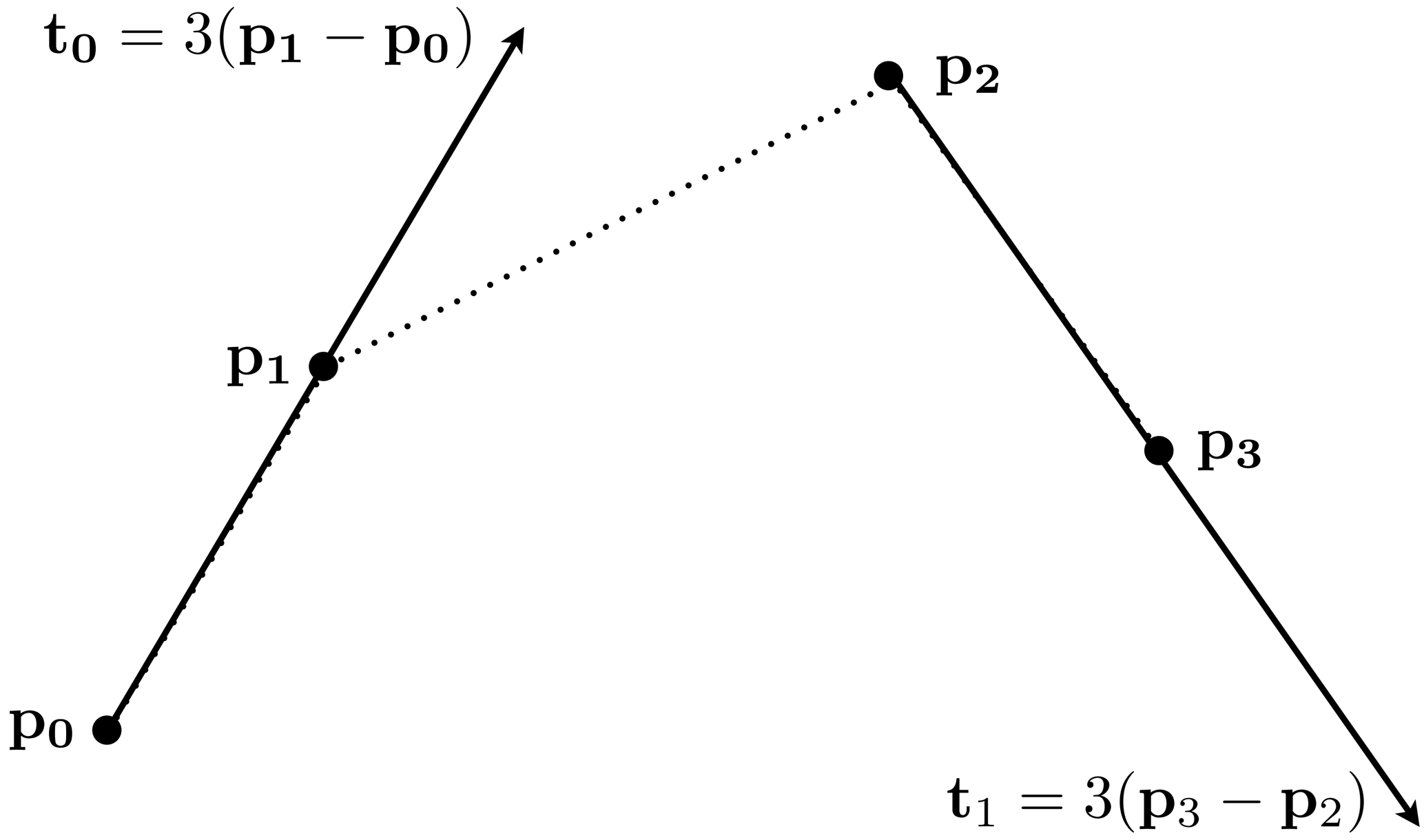
Defining Cubic Bézier Curve With Tangents



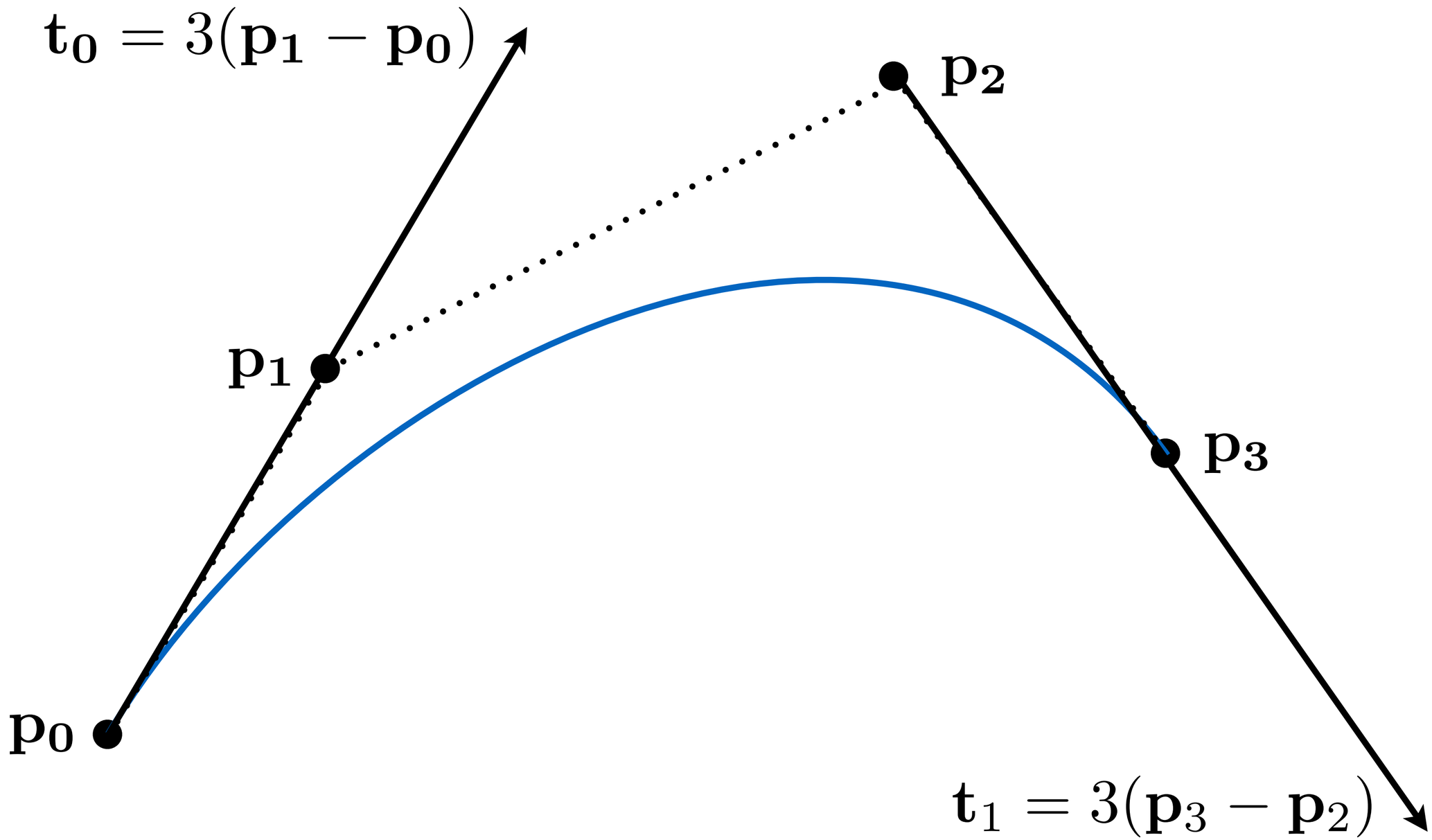
Defining Cubic Bézier Curve With Tangents



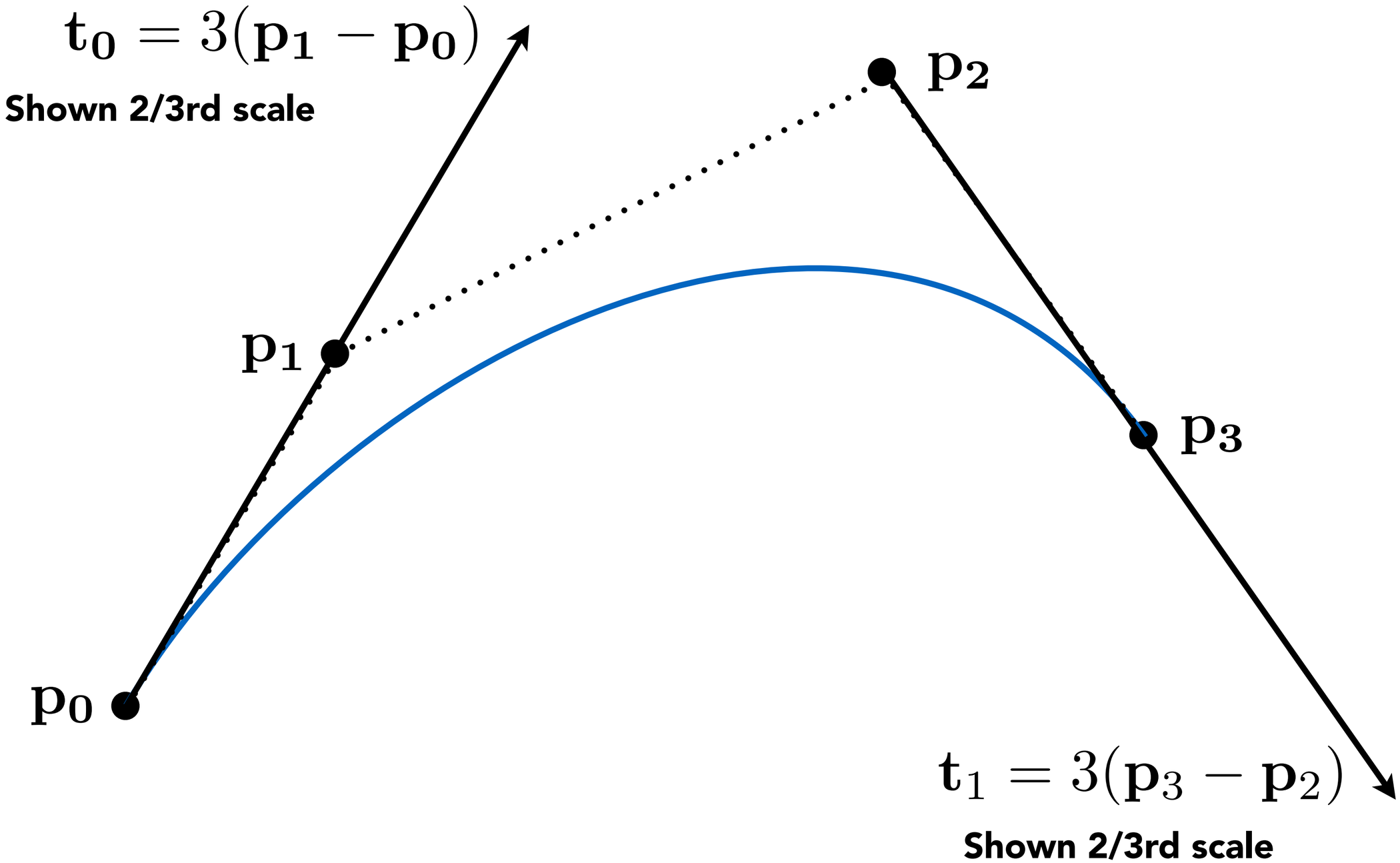
Defining Cubic Bézier Curve With Tangents



Defining Cubic Bézier Curve With Tangents



Defining Cubic Bézier Curve With Tangents



Cubic Bézier

Similar to Hermite, but specify tangents indirectly

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

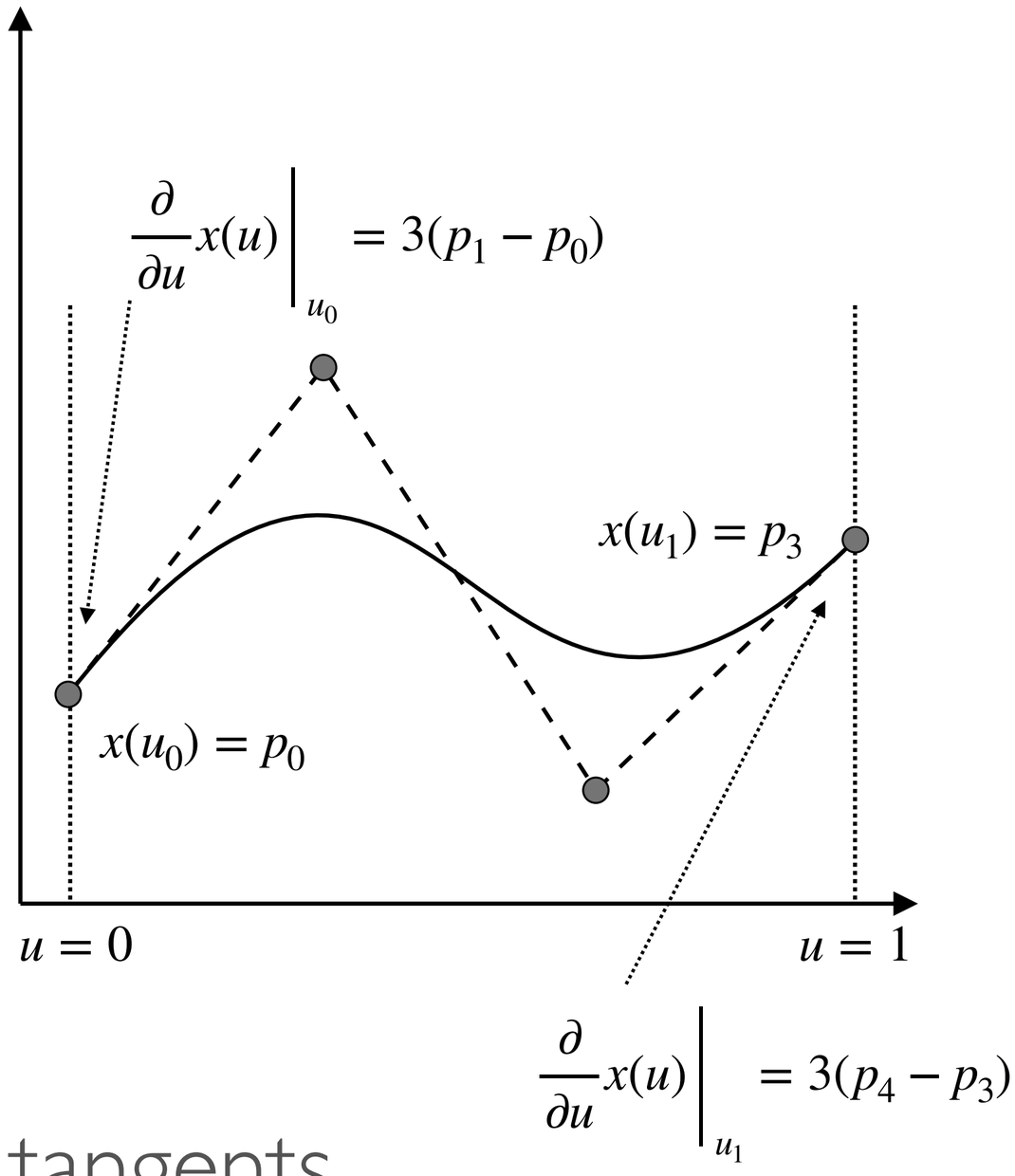
$$x(u_0) = h_0 = p_0$$

$$x(u_1) = h_1 = p_3$$

$$x'(u_0) = h_2 = 3(p_1 - p_2)$$

$$x'(u_1) = h_3 = 3(p_3 - p_2)$$

$$\mathbf{x}(u) = \boldsymbol{\mathcal{P}}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{h}$$



Note: all the control points are actually points in space, no tangents.

Cubic Bézier

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$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

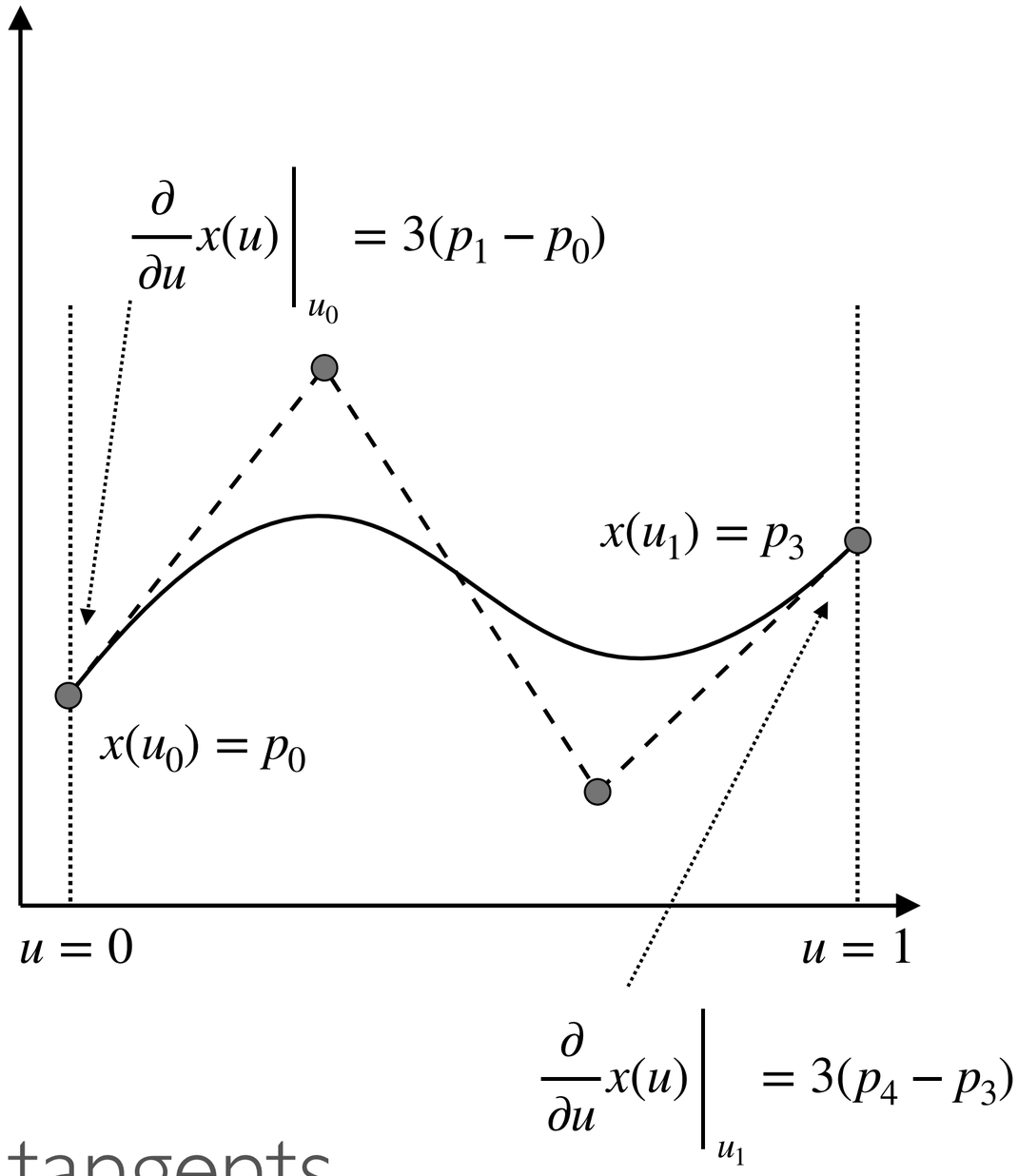
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$$x'(u_0) = h_2 = 3(p_1 - p_2)$$

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$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{M}_{Z \to H} \cdot \mathbf{p}$$



Note: all the control points are actually points in space, no tangents.

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$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x(u_0) = h_0 = p_0$$

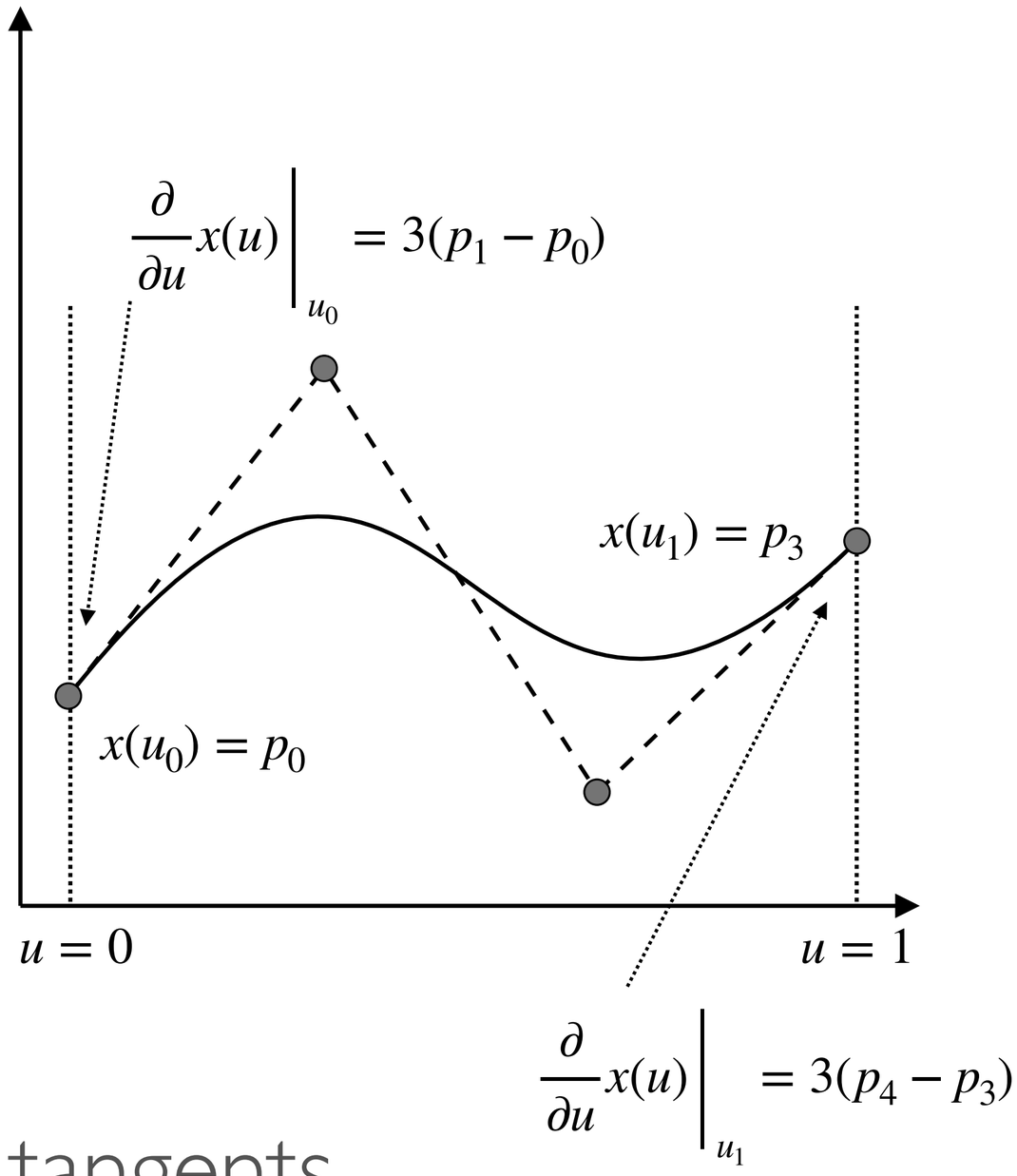
$$x(u_1) = h_1 = p_3$$

$$x'(u_0) = h_2 = 3(p_1 - p_2)$$

$$x'(u_1) = h_3 = 3(p_3 - p_2)$$

Bézier

$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{M}_{Z \to H} \cdot \mathbf{p}$$



Note: all the control points are actually points in space, no tangents.

Cubic Bézier

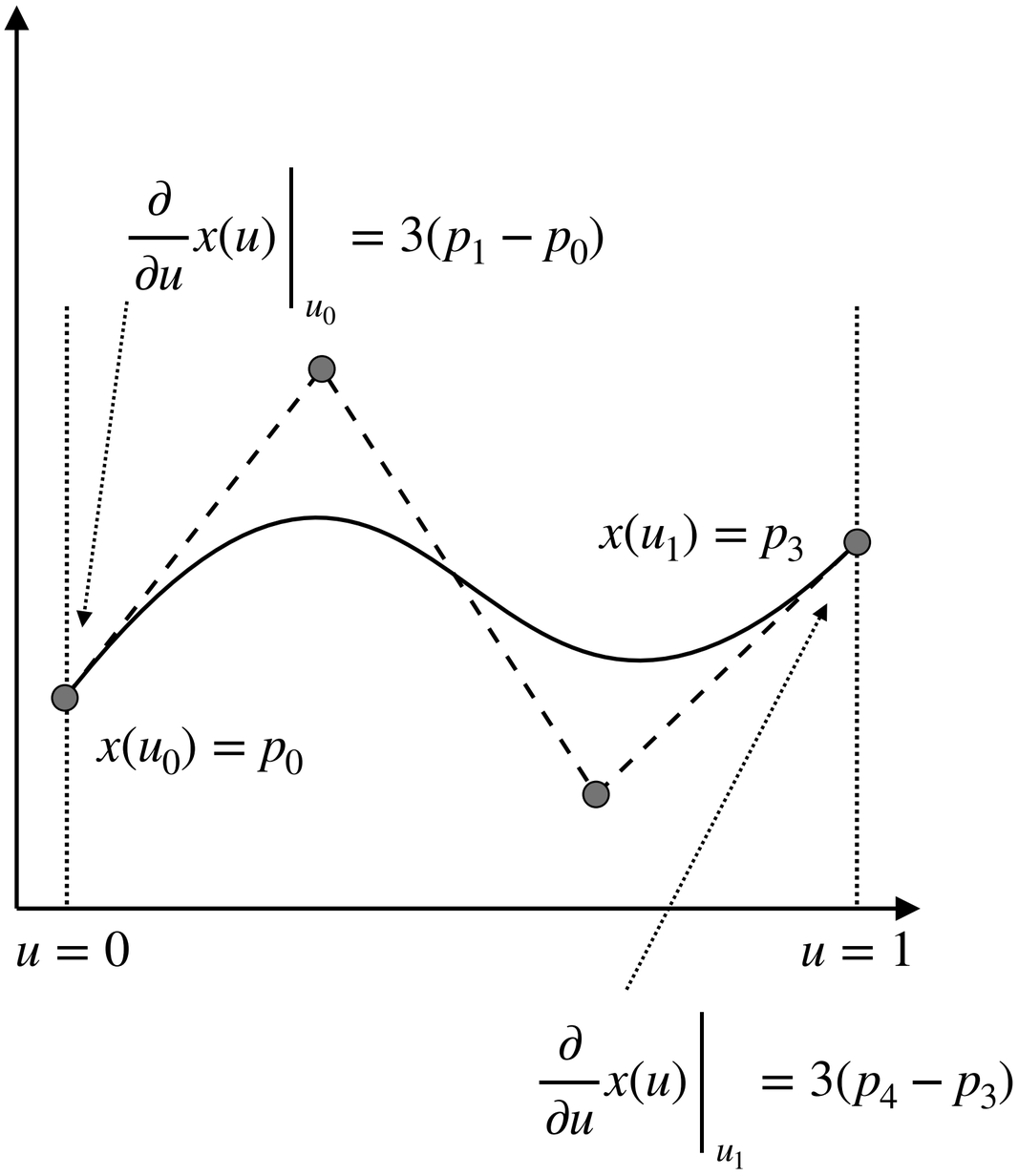
Similar to Hermite, but specify tangents indirectly

$$x(u) = c_0 + c_1 u + c_2 u^2 + c_3 u^3$$

$$x(u_0) = h_0 = p_0$$
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$$\mathbf{M}_{Z \to H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -3 & 3 & 0 & 0 \\ 0 & 0 & -3 & 3 \end{bmatrix}$$

$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_H \cdot \mathbf{M}_{Z \to H} \cdot \mathbf{p}$$



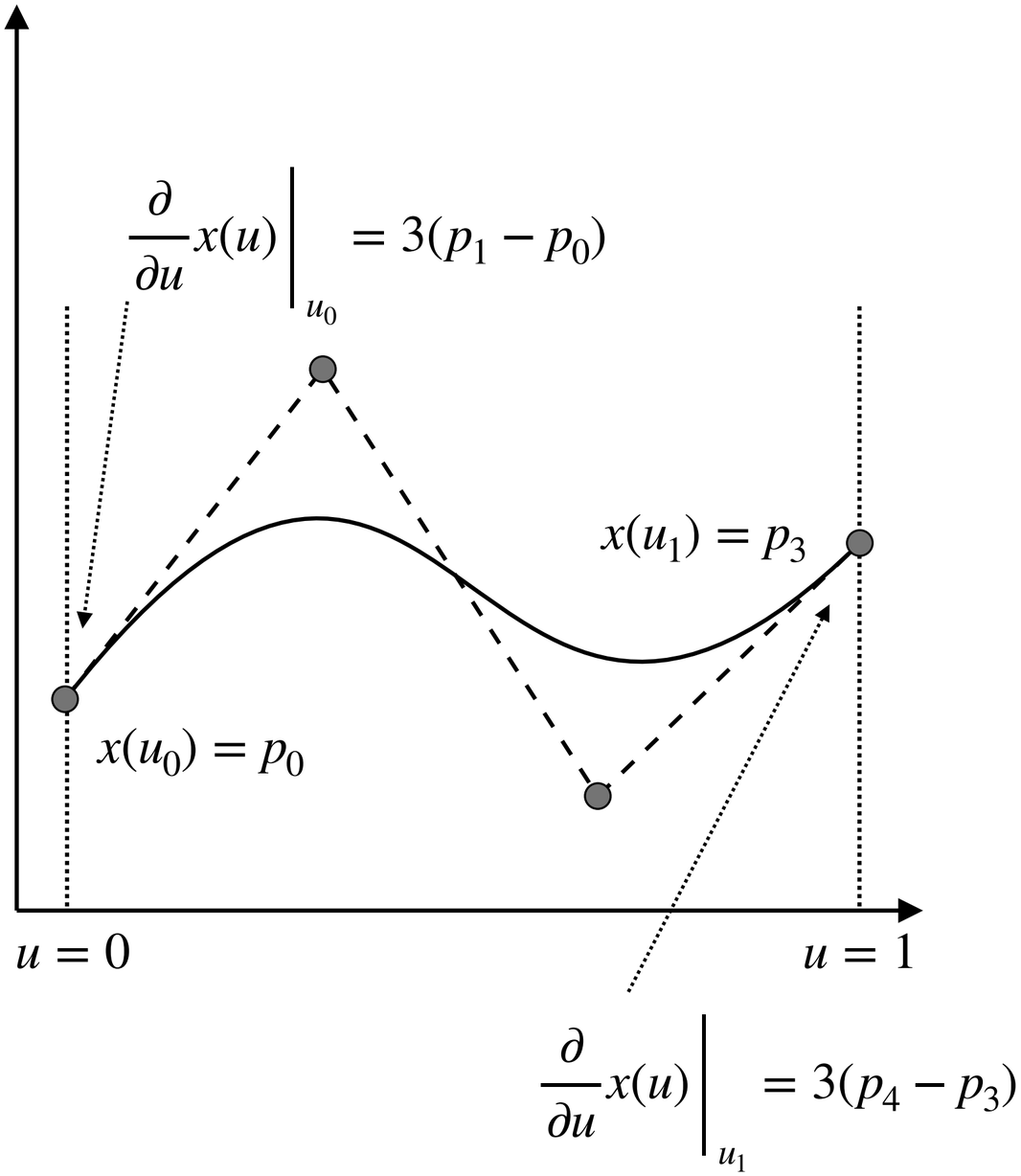
Cubic Bézier

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$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \mathbf{p}$$

$$\boldsymbol{\beta}_Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

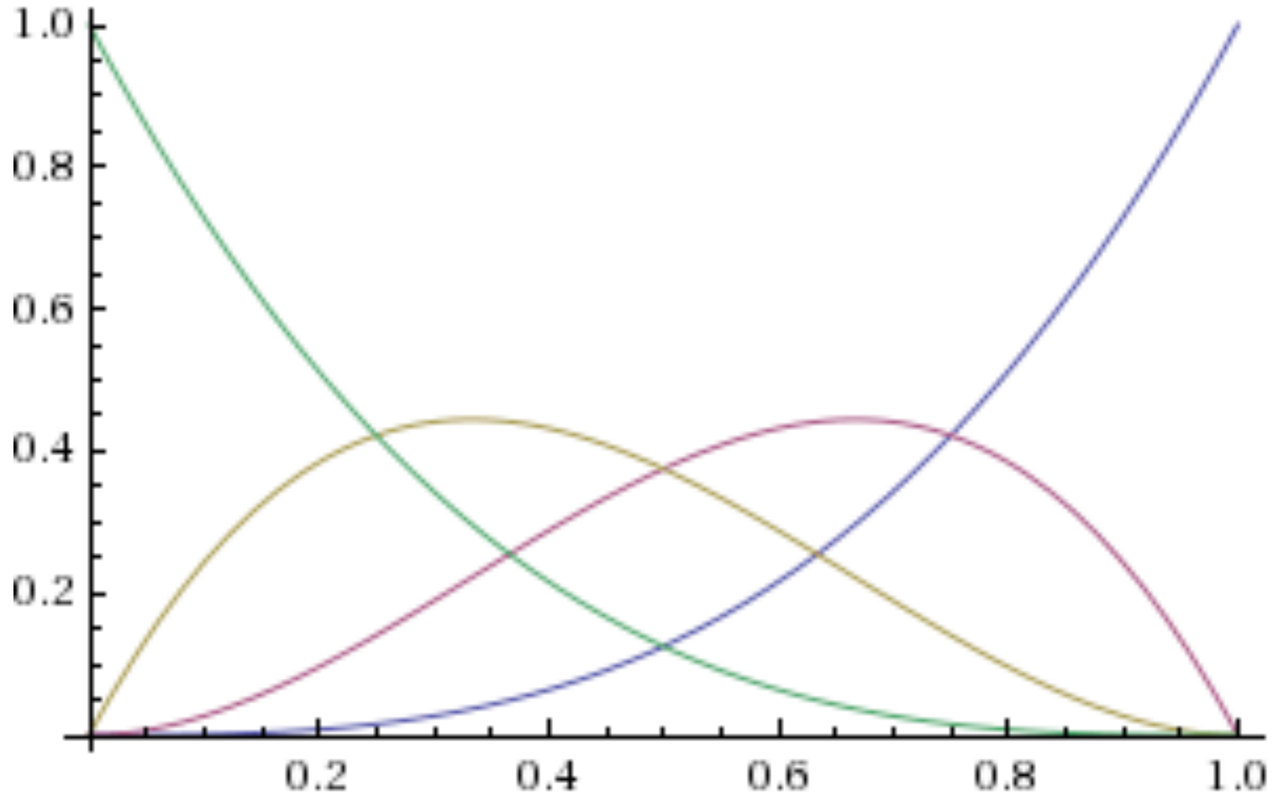


Cubic Bézier

Bézier basis functions

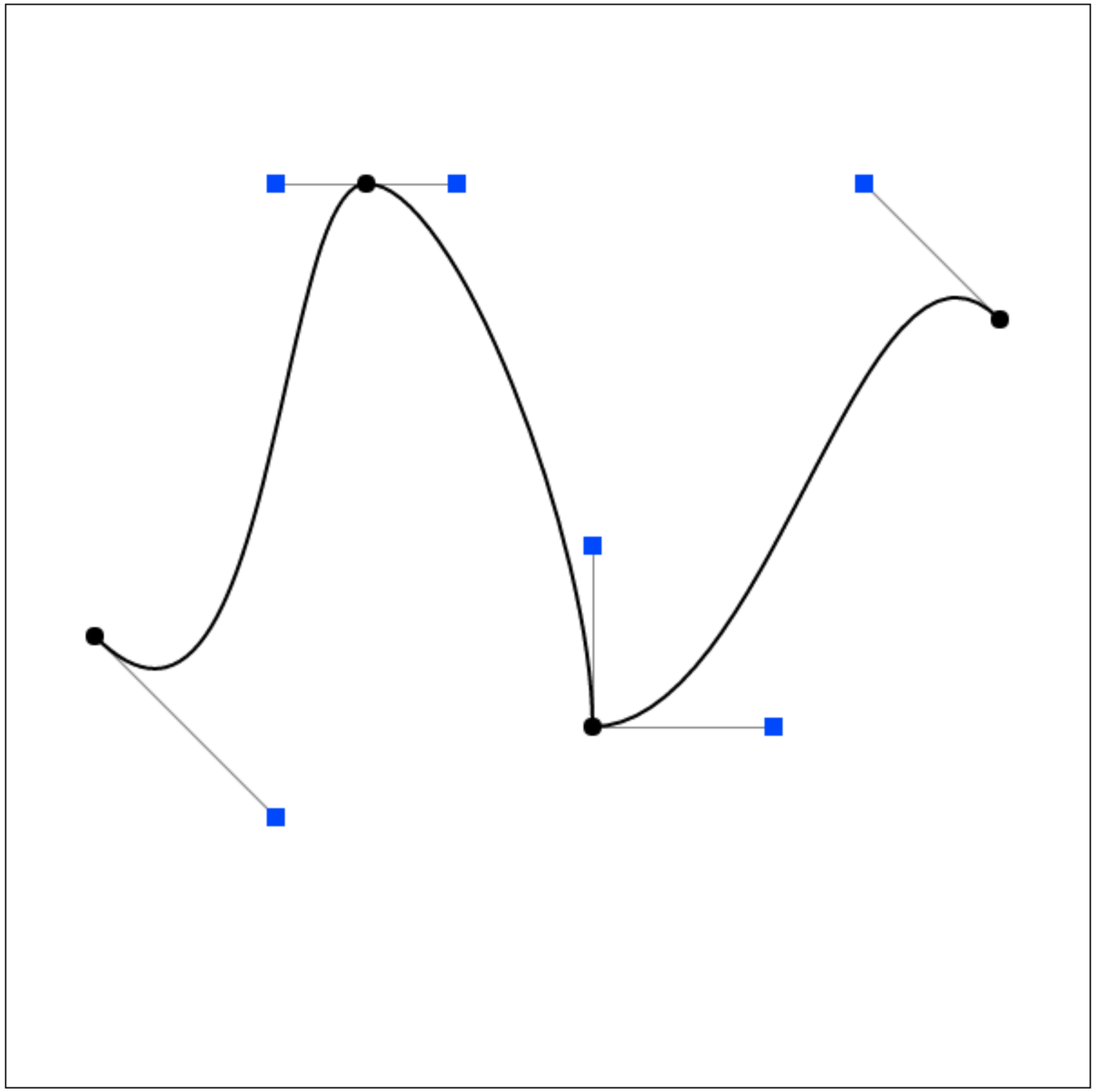
$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \mathbf{p}$$

$$\boldsymbol{\beta}_Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$



$$\mathbf{X}(u) = \begin{bmatrix} 1 - 3u + 3u^2 - 1u^3 \\ 0 + 3u - 6u^2 + 3u^3 \\ 0 + 0u + 3u^2 - 3u^3 \\ 0 + 0u + 0u^2 + 1u^3 \end{bmatrix}^{\mathsf{T}} \mathbf{p}$$

Demo – Piecewise Cubic Bézier Curve



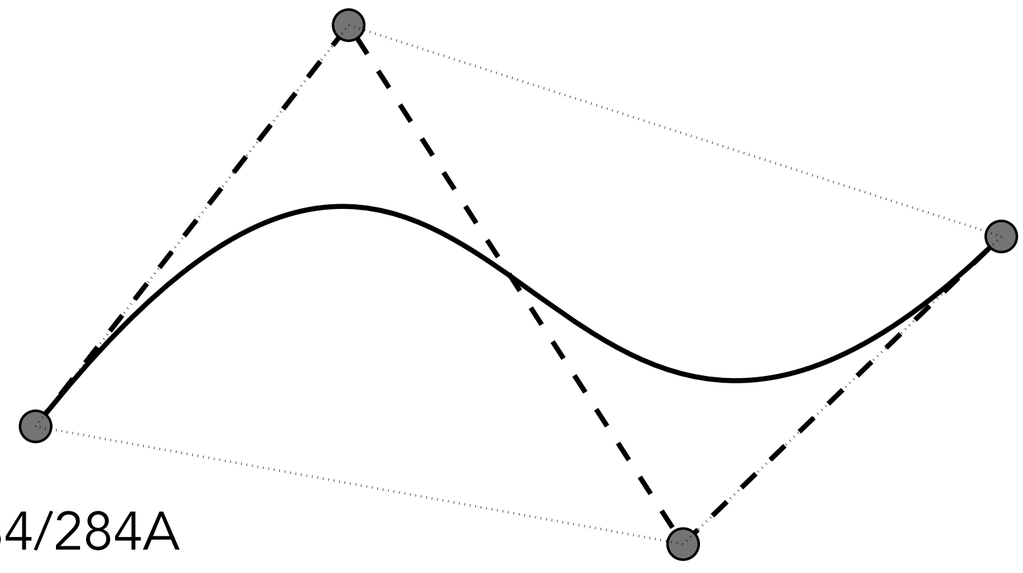
David Eck, http://math.hws.edu/eck/cs424/notes2013/canvas/bezier.html

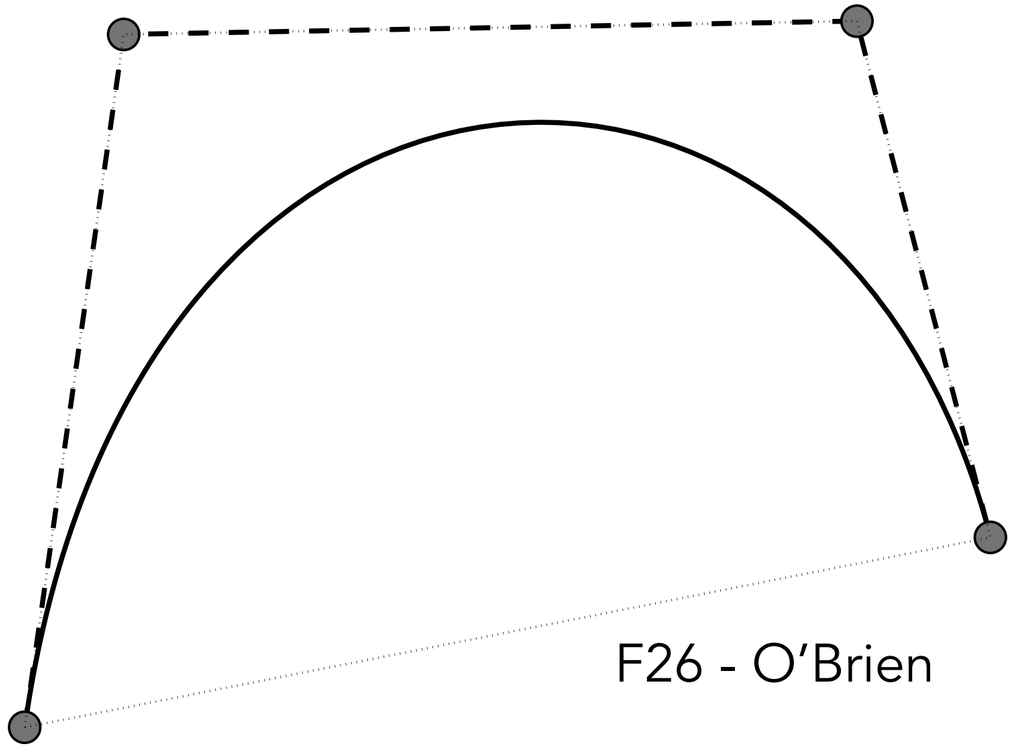
Demo – Piecewise Cubic Bézier Curve

Useful Properties of a Basis

Convex Hull Property

- All points on curve inside convex hull of control points
- Bézier basis has convex hull property!
- Convex hull if the following are true:
  - Basis functions sum to one $\sum_i b_i(u) = 1$
  - Basis functions are positive over domain $b_i(u) \geq 0 \qquad \forall u \in \Omega$



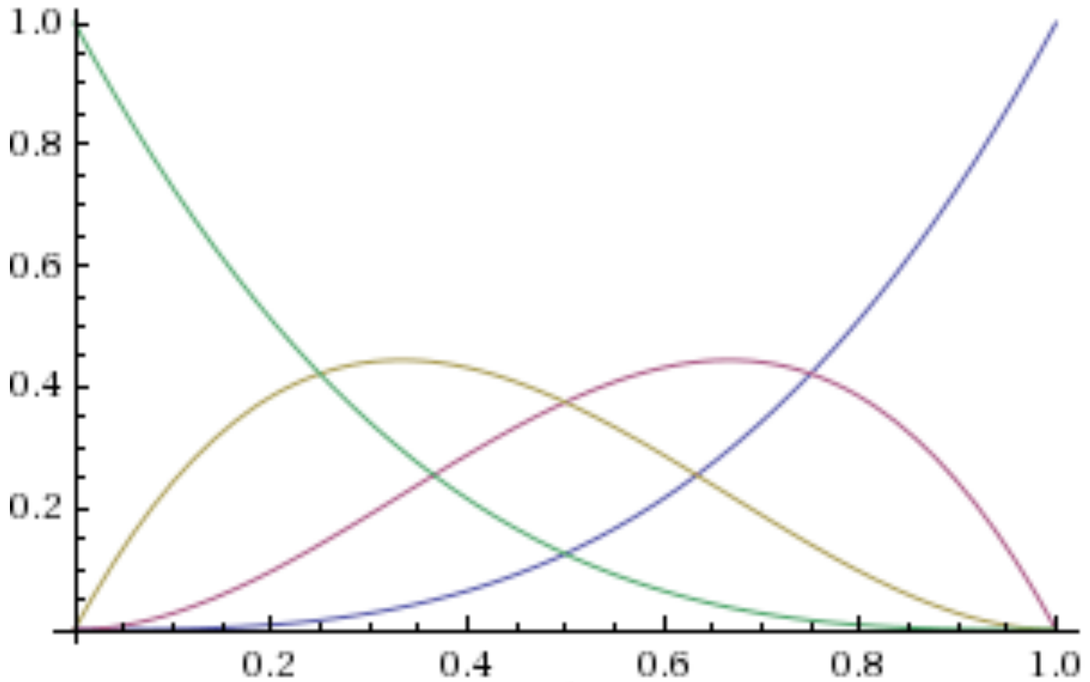


Recall Cubic Bézier

Bézier basis functions

$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \mathbf{p}$$

$$\boldsymbol{\beta}_Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$



All positive in [0..1]

$$x(u) = \begin{bmatrix} 1 - 3u + 3u^2 - 1u^3 \\ 0 + 3u - 6u^2 + 3u^3 \\ 0 + 0u + 3u^2 - 3u^3 \\ 0 + 0u + 0u^2 + 1u^3 \end{bmatrix}^{\mathsf{T}} \mathbf{p}$$

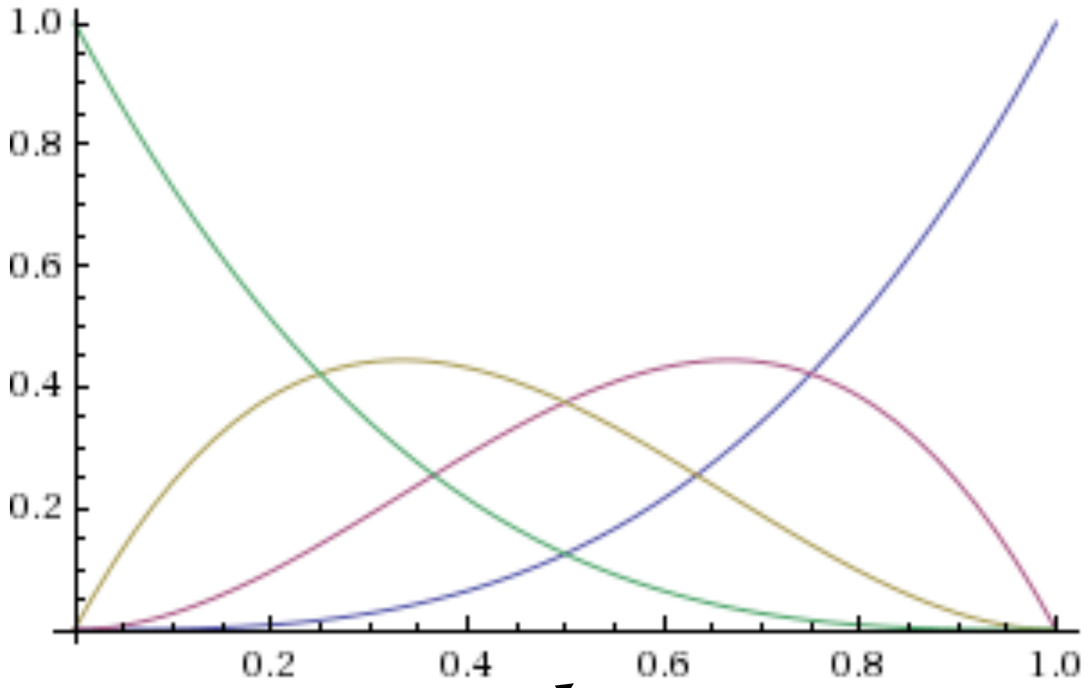
These sum to 1

Recall Cubic Bézier

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These sum to 1

What if that 3 had been something else?

Useful Properties of a Basis

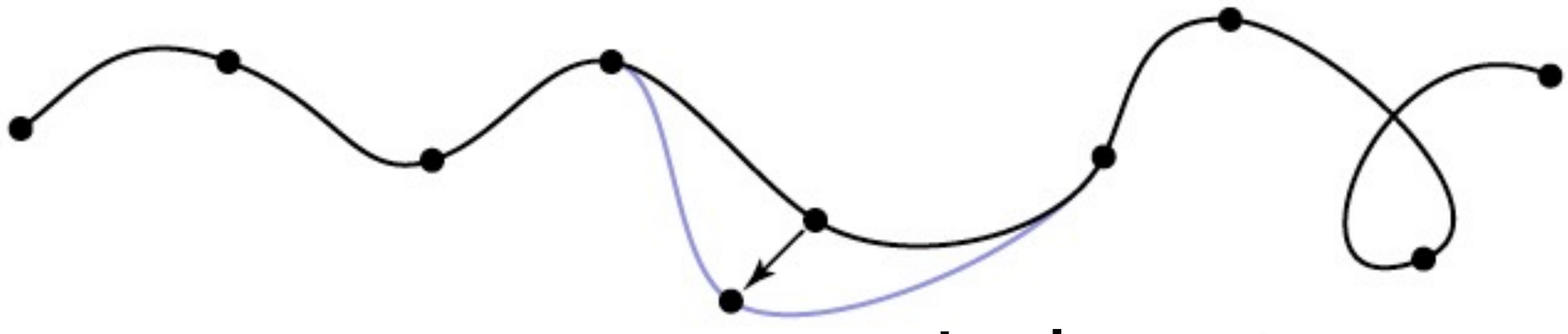
Invariance under class of transforms

- Transforming curve is same as transforming control points
- Bézier basis invariant w.r.t. *affine* transforms
- Bézier NOT invariant for perspective transforms
- NURBS are invariant w.r.t. perspective transforms

$$\mathbf{x}(u) = \boldsymbol{\mathcal{P}}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix} \Leftrightarrow \mathbf{T} \cdot \mathbf{x}(u) = \boldsymbol{\mathcal{P}}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{T} \cdot \mathbf{p}_0 \\ \mathbf{T} \cdot \mathbf{p}_1 \\ \mathbf{T} \cdot \mathbf{p}_2 \\ \mathbf{T} \cdot \mathbf{p}_3 \end{bmatrix}$$

More Useful Properties of a Basis

- **Local support**
  - Changing one control point does not change the entire curve
- **Nice subdivision rules**
- **Orthogonality (** $\int_\Omega b_i(u)b_j(u)\mathrm{d}u = \delta_{ij}$ **)**
- **Fast evaluation scheme**
- **Interpolation -vs- approximation**



Local support

Changing Basis

Switching between different basis functions is done by matrix multiplication:

- Example: $\mathbf{p}_Z = \boldsymbol{\beta}_Z^{-1} \cdot \mathbf{c}$
- Example: $\mathbf{p}_Z = \boldsymbol{\beta}_Z^{-1} \cdot \boldsymbol{\beta}_H \cdot \mathbf{p}_H$
- Only works if two sets of basis functions spans the same space (*i.e.* cubic polynomials)
- Different sets of basis functions have different useful properties

Properties of Bézier Curves

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Interpolates endpoints

- For cubic Bézier: $\mathbf{b}(0) = \mathbf{b}_0; \quad \mathbf{b}(1) = \mathbf{b}_3$

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Tangent to end segments

- Cubic case: $\mathbf{b}'(0) = 3(\mathbf{b}_1 - \mathbf{b}_0); \quad \mathbf{b}'(1) = 3(\mathbf{b}_3 - \mathbf{b}_2)$

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Affine transformation property

- **Transform curve by transforming control points**

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Convex hull property

- Curve is within convex hull of control points

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Well behaved

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Affine transformation property

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Well behaved

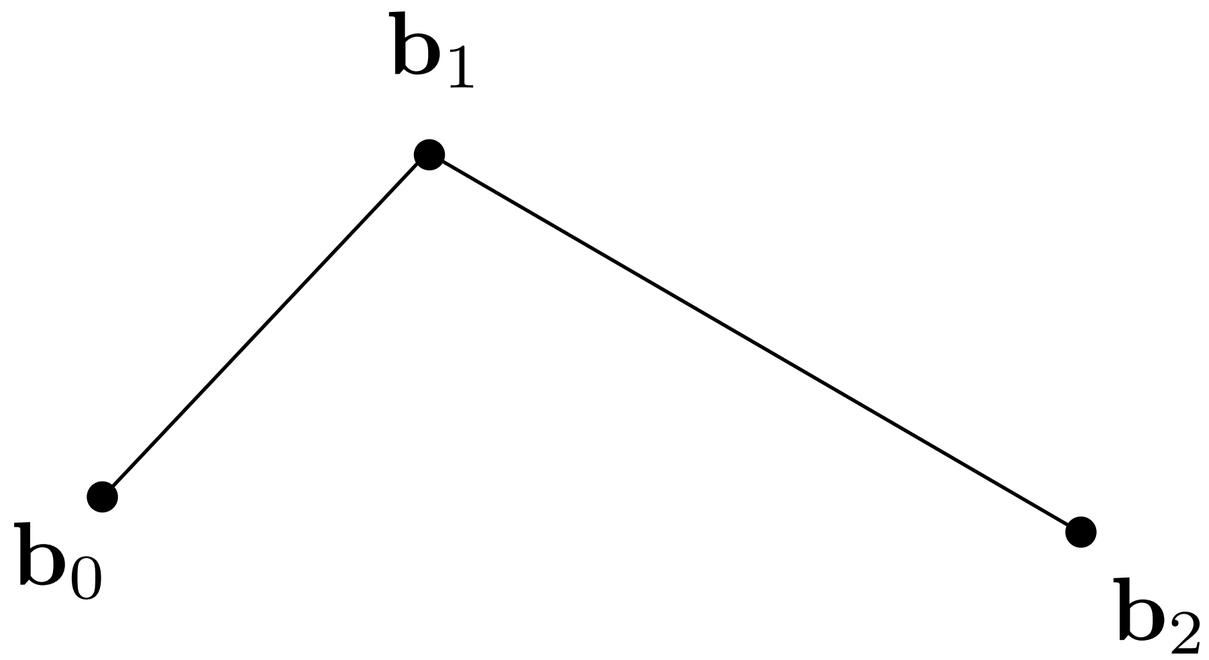
Local control (when joining)

Evaluating Bézier Curves

De Casteljau Algorithm

Bézier Curves – de Casteljau Algorithm

Consider three points (quadratic Bezier)





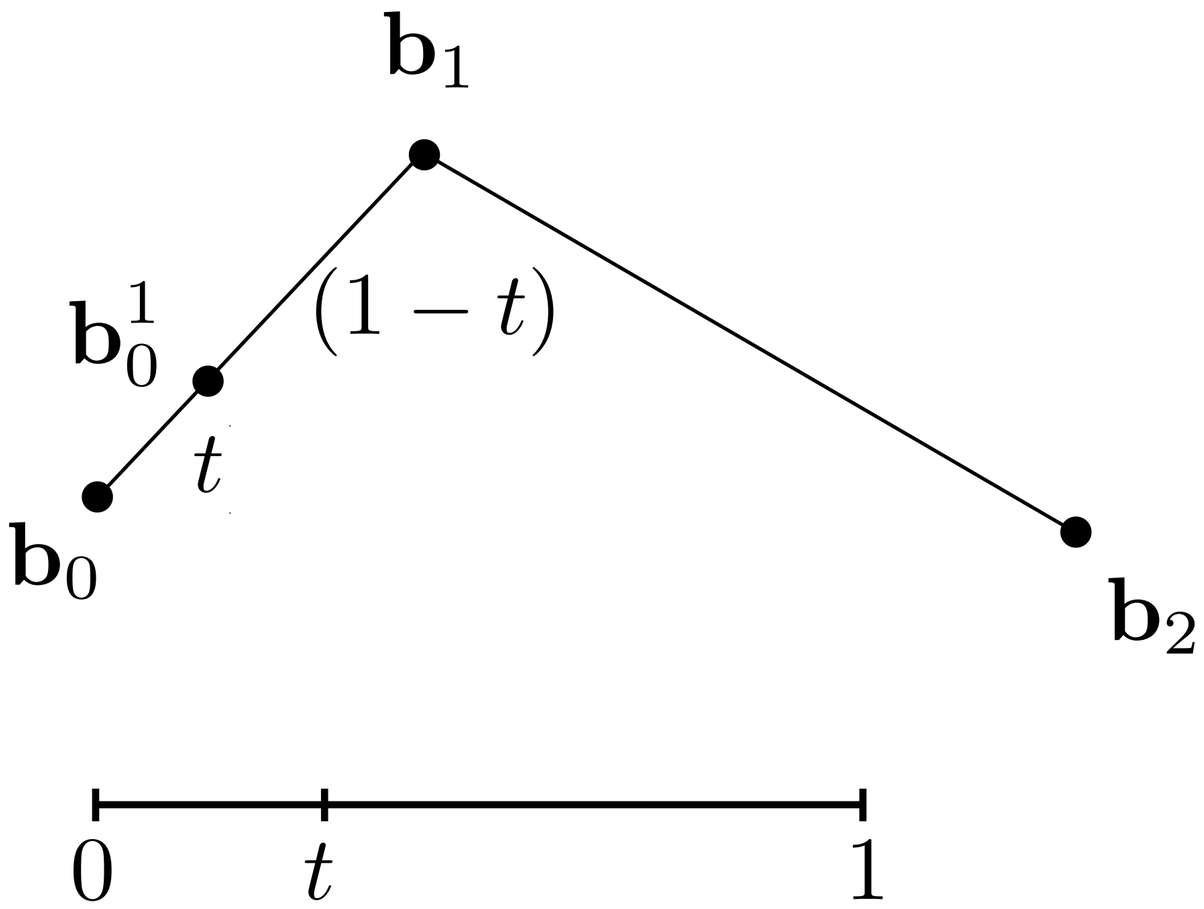
Pierre Bézier
1910 – 1999



Paul de Casteljau
b. 1930

Bézier Curves – de Casteljau Algorithm

Insert a point using linear interpolation



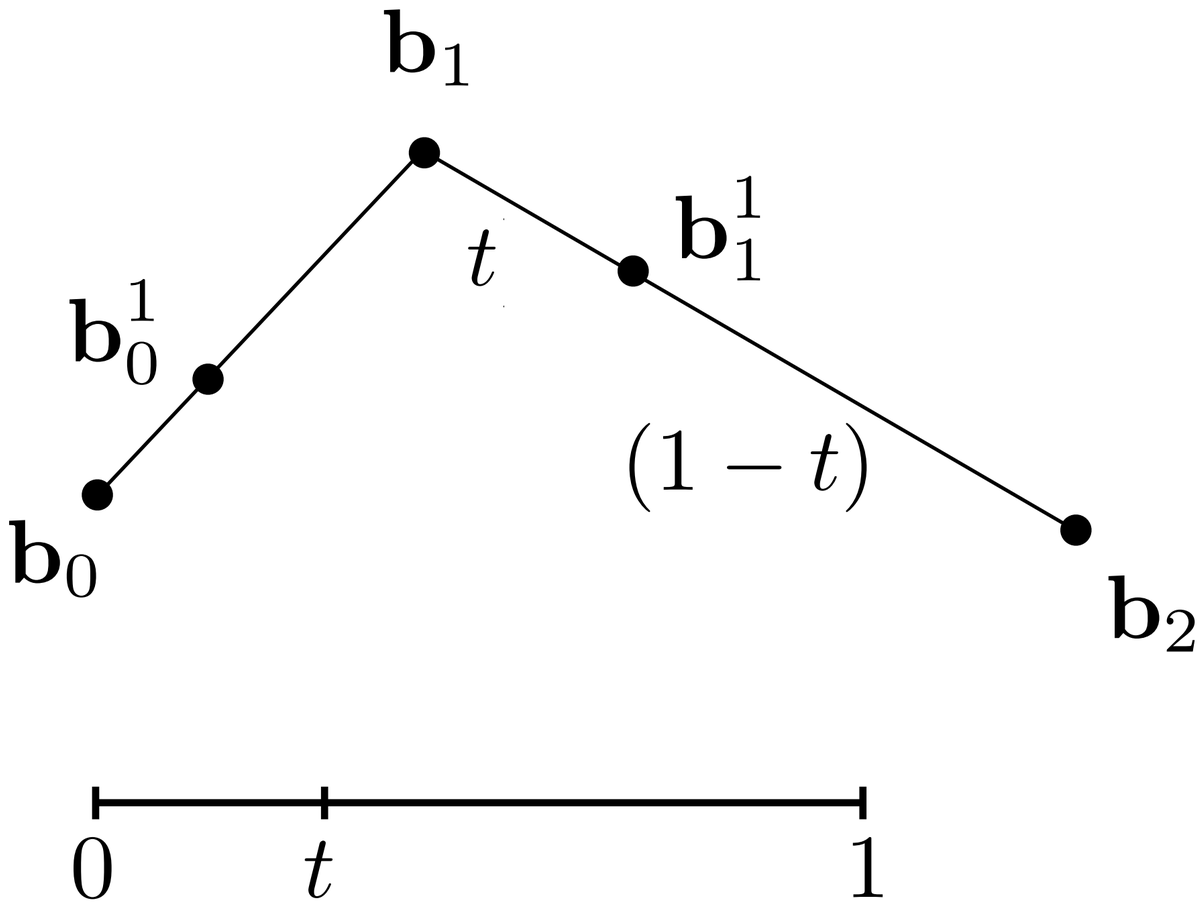


Pierre Bézier
1910 – 1999

Paul de Casteljau
b. 1930

Bézier Curves – de Casteljau Algorithm

Insert on both edges





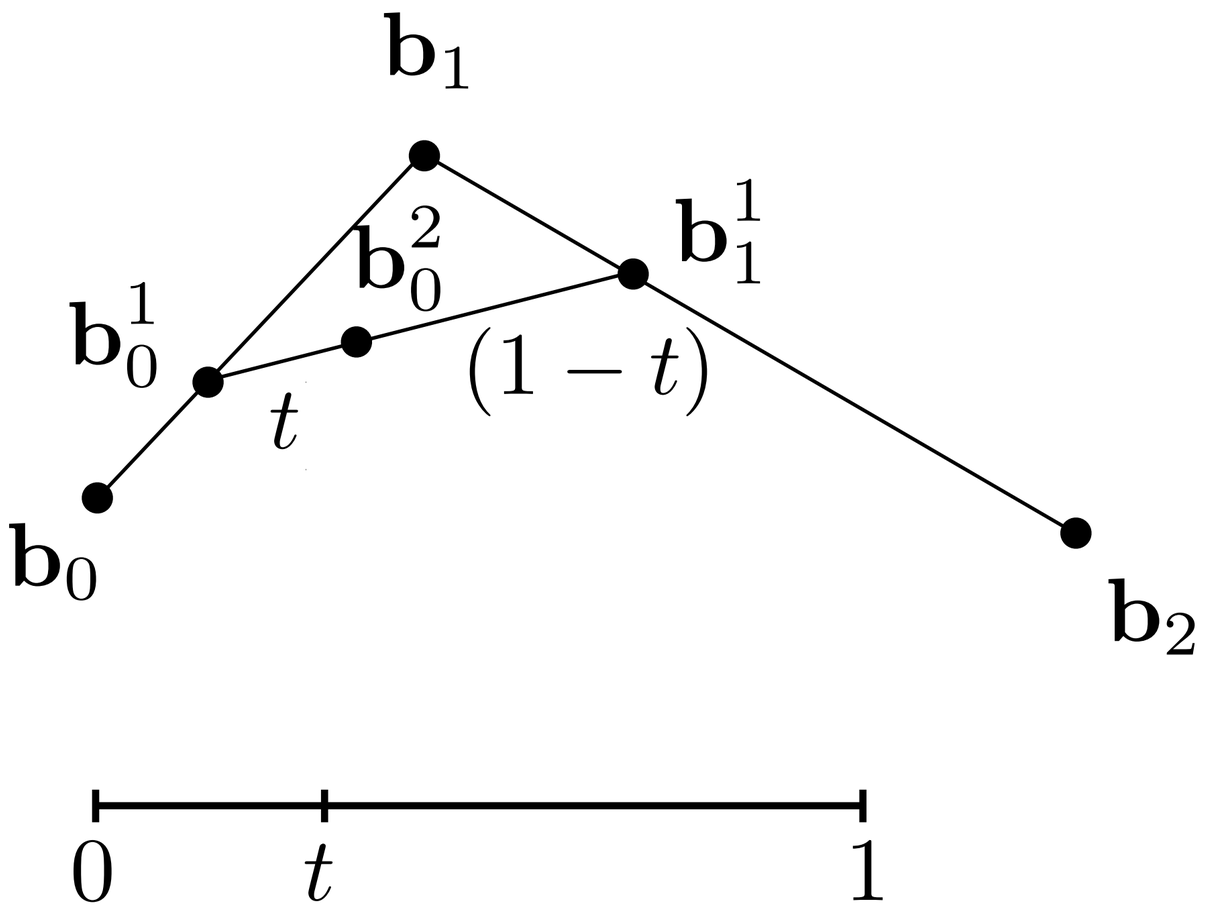
Pierre Bézier
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Bézier Curves – de Casteljau Algorithm

Repeat recursively





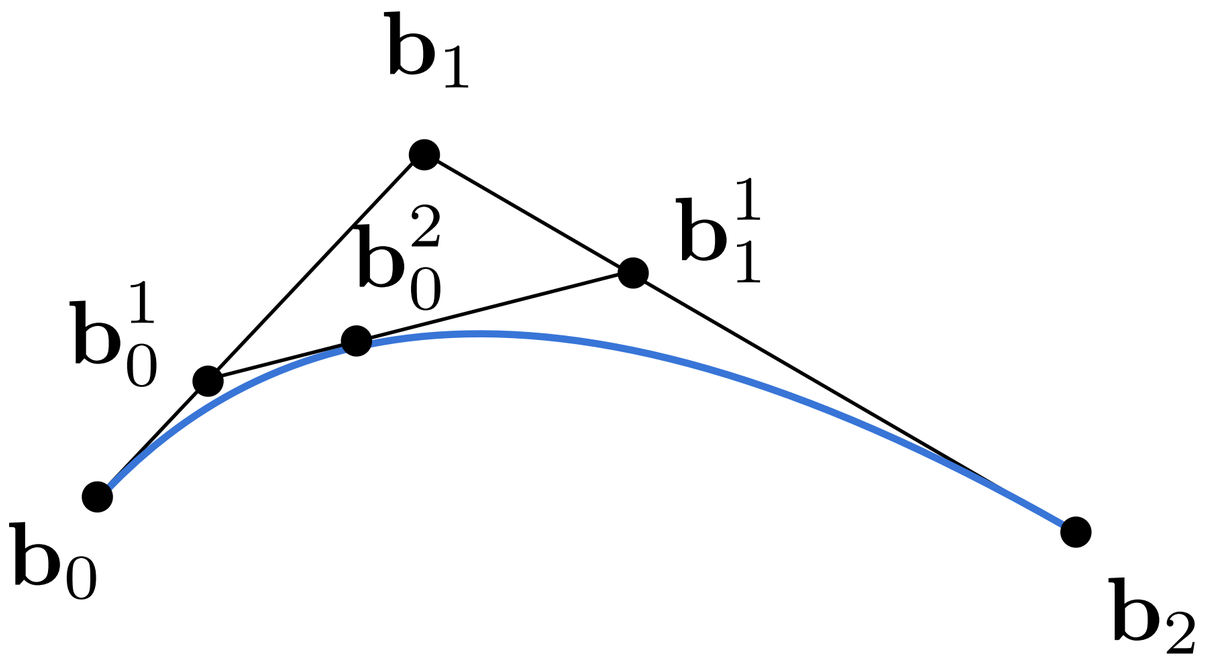
Pierre Bézier
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Bézier Curves – de Casteljau Algorithm

Algorithm defines the curve





Pierre Bézier
1910 – 1999



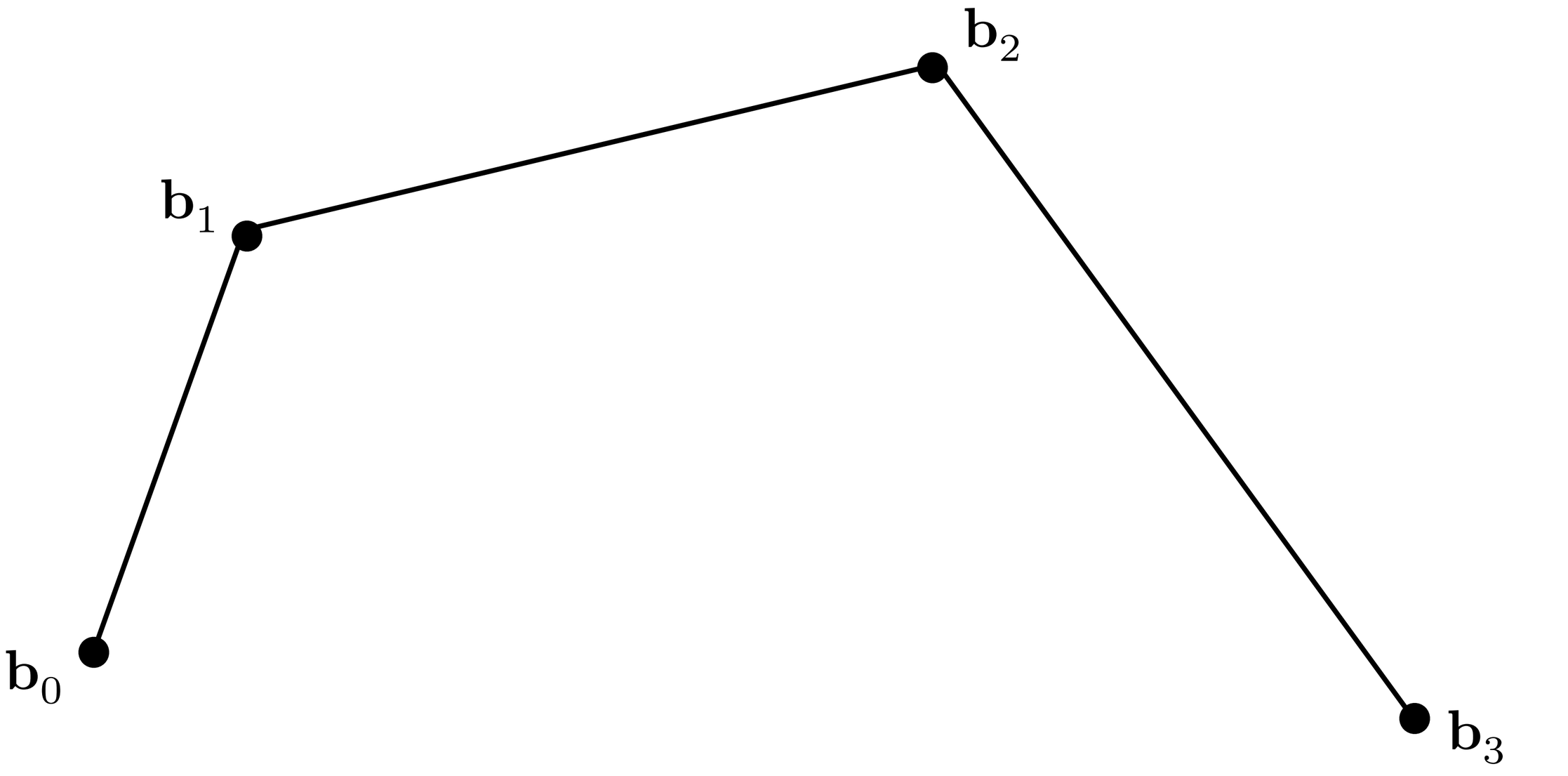
Paul de Casteljau
b. 1930

"Corner cutting" recursive subdivision

Successive linear interpolation

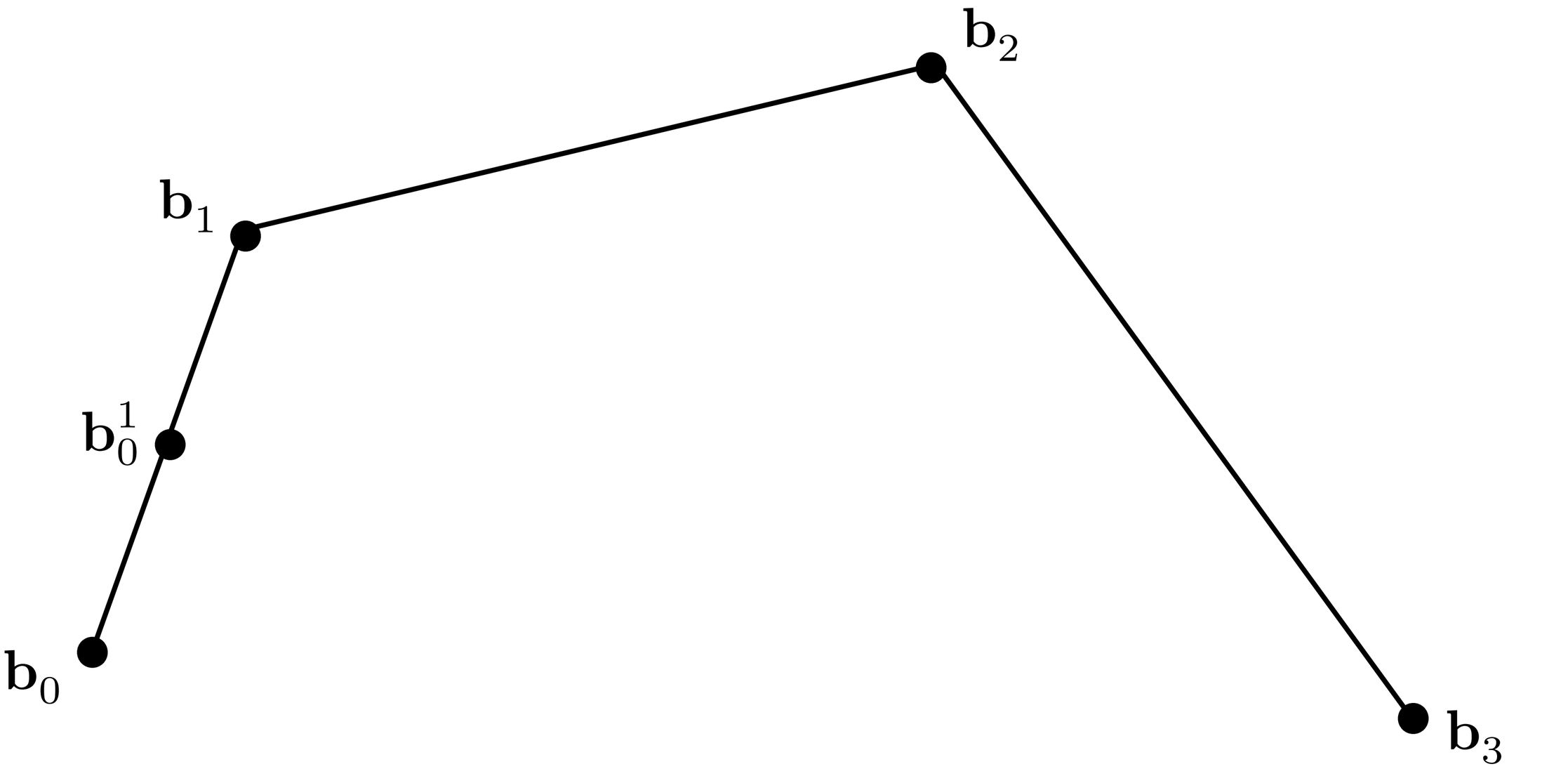
Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations



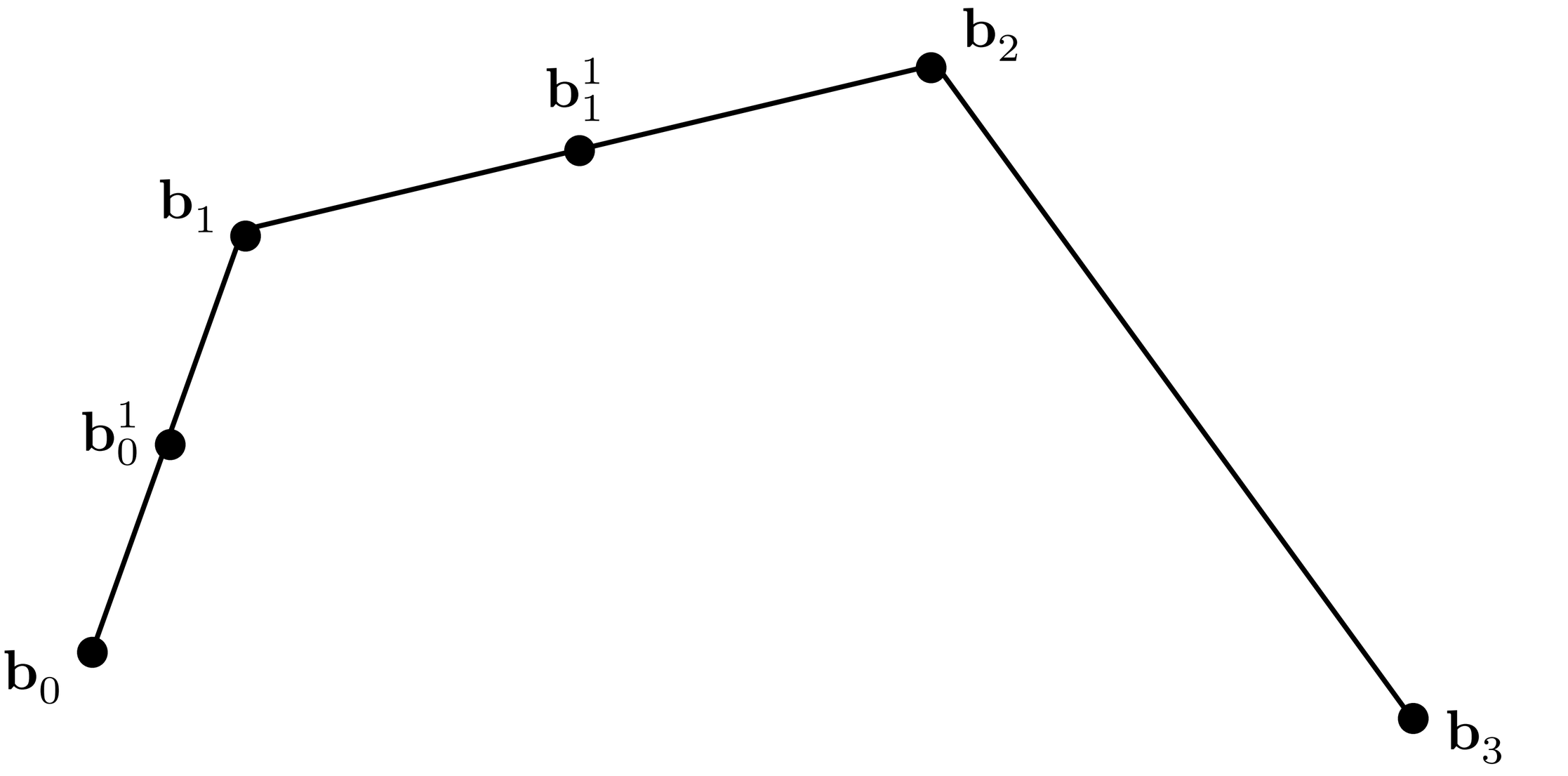
Cubic Bézier Curve – de Casteljau

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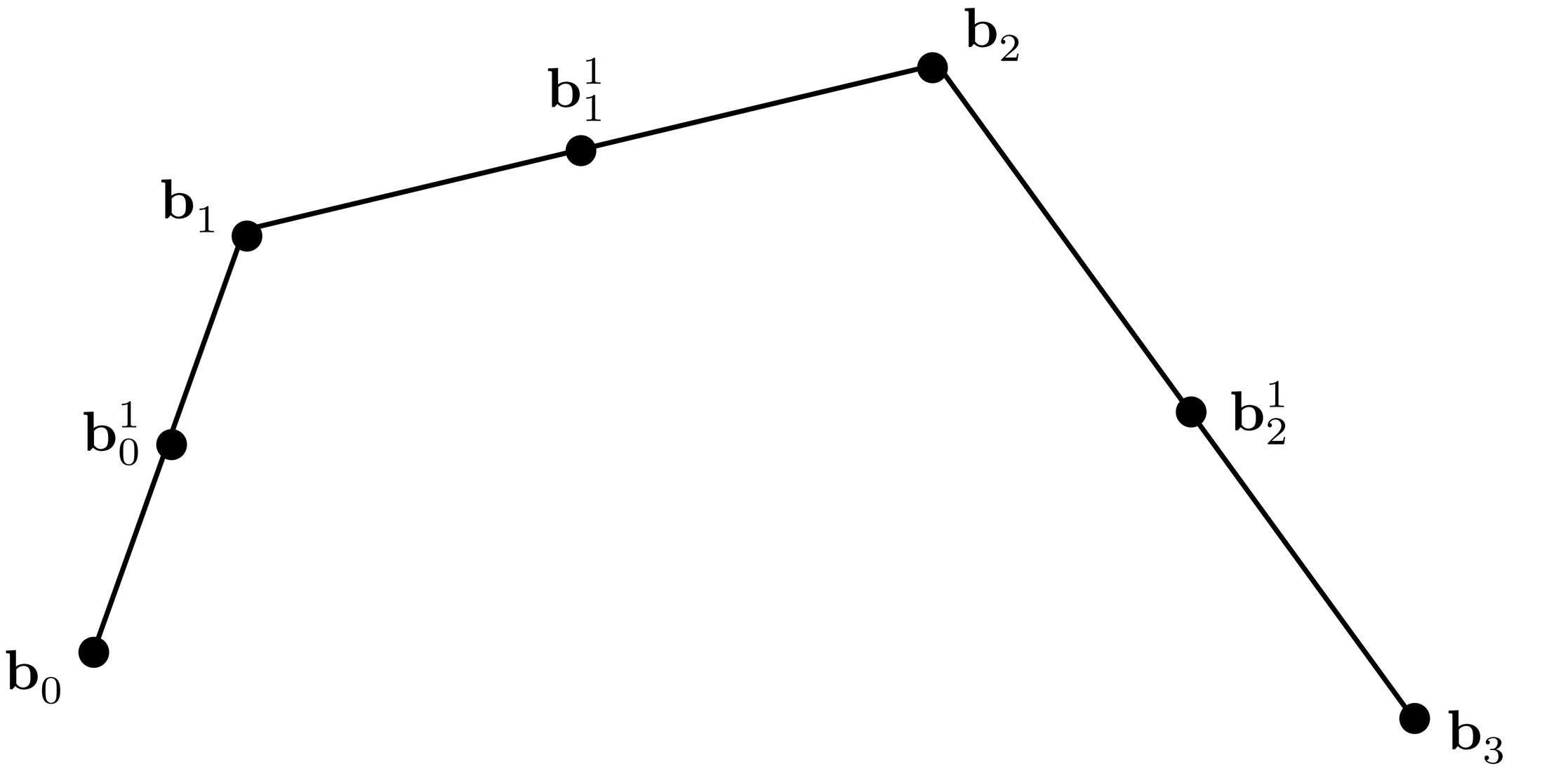
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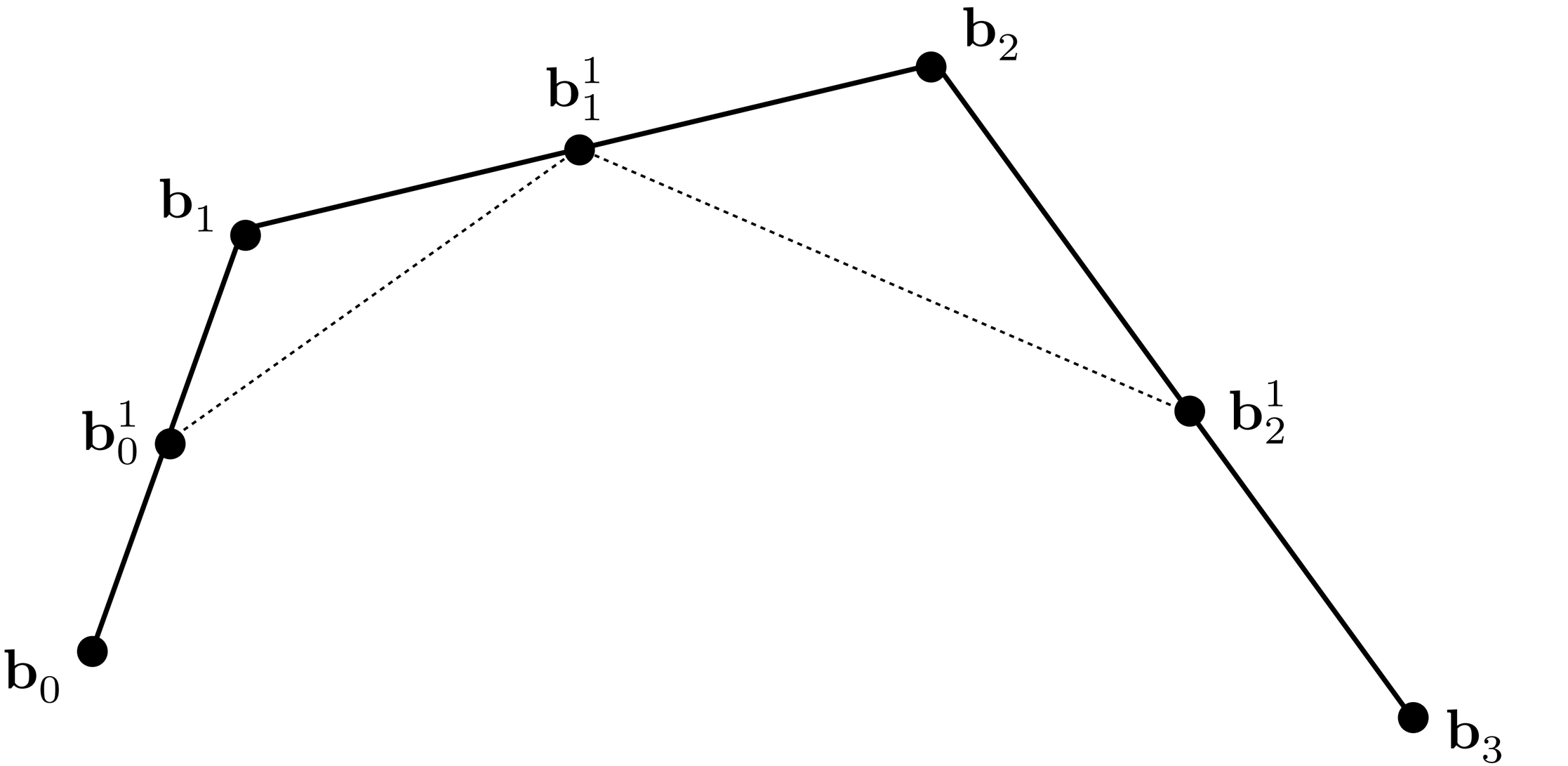
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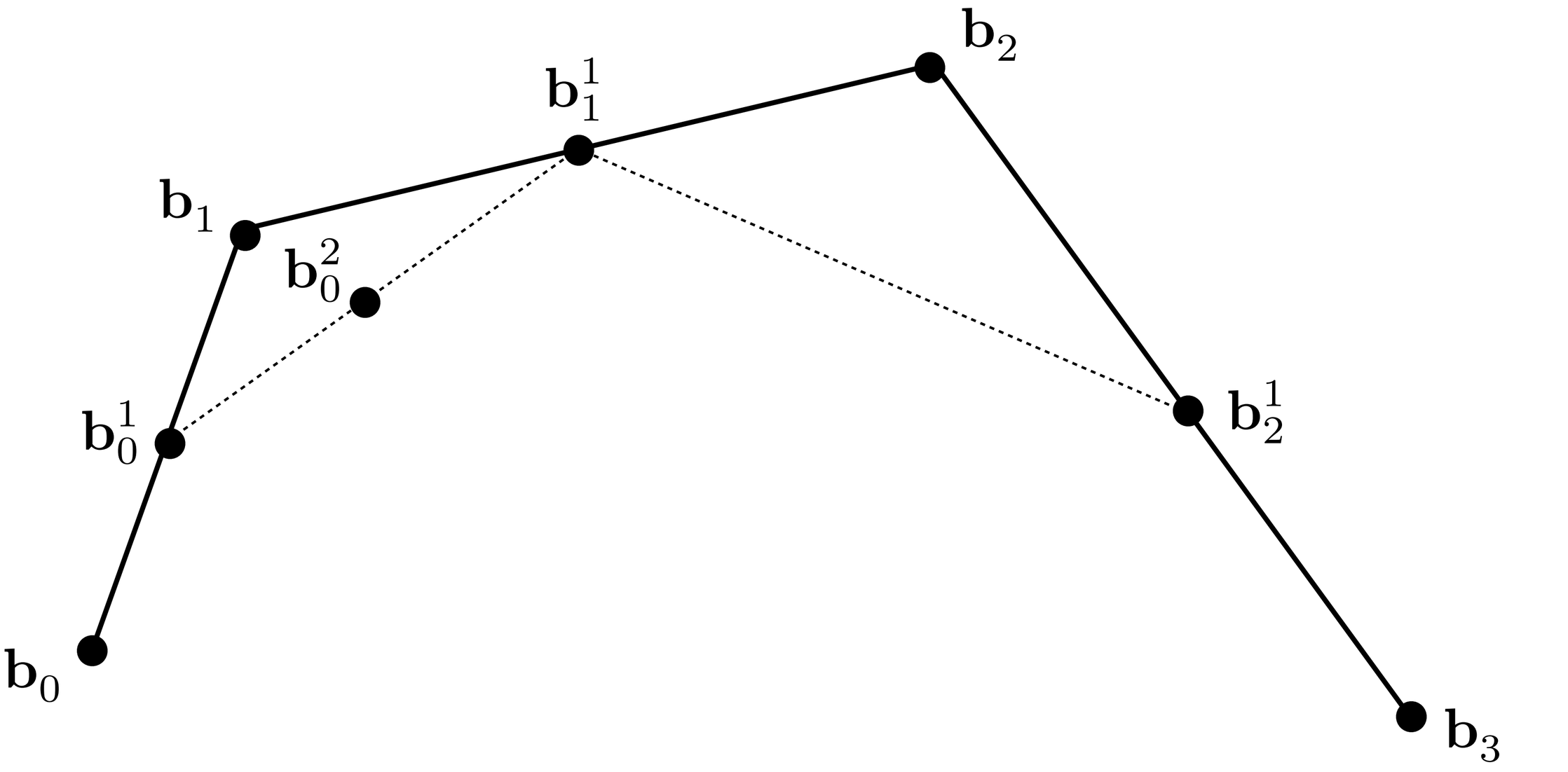
Cubic Bézier Curve – de Casteljau

Consider four points
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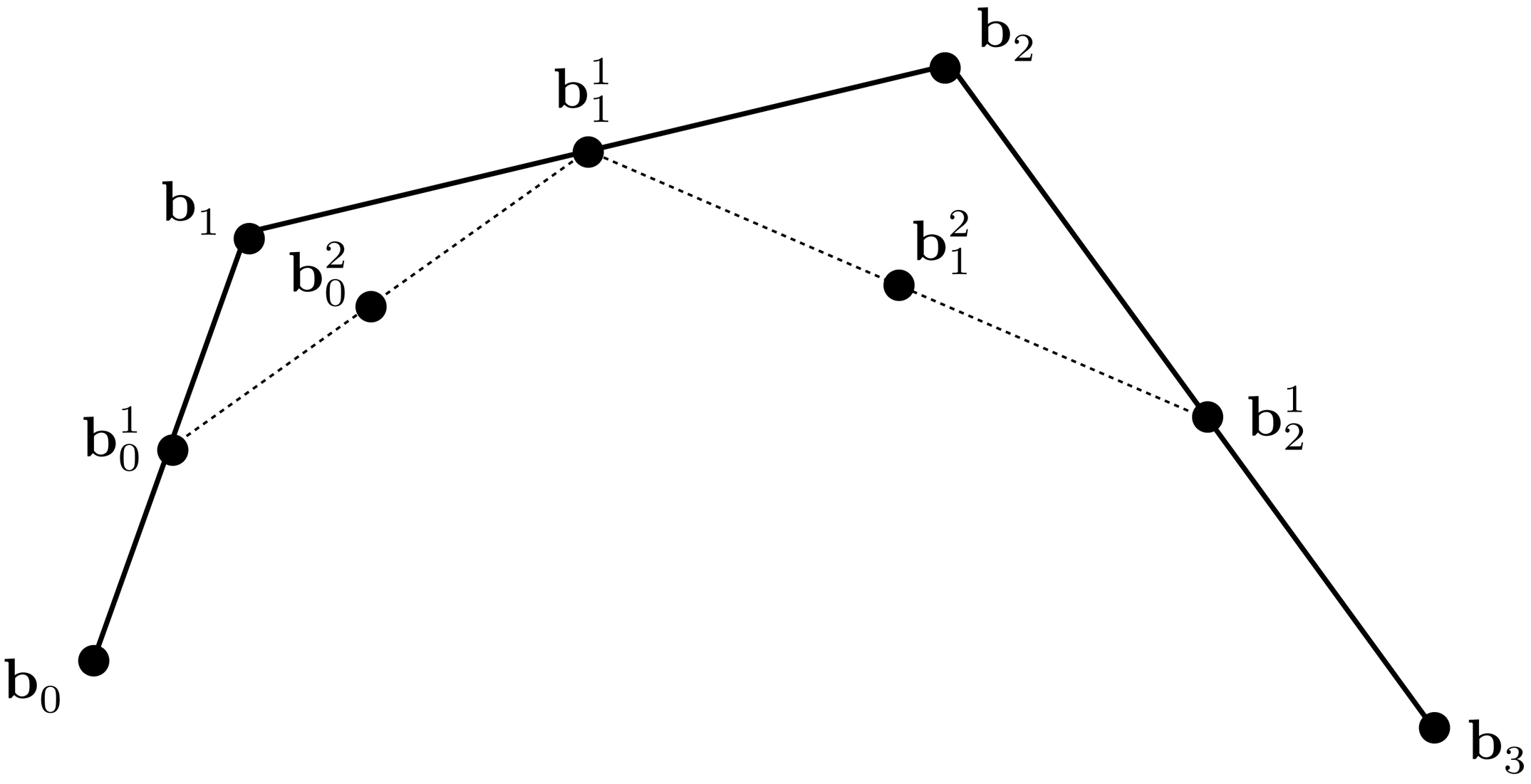
Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations



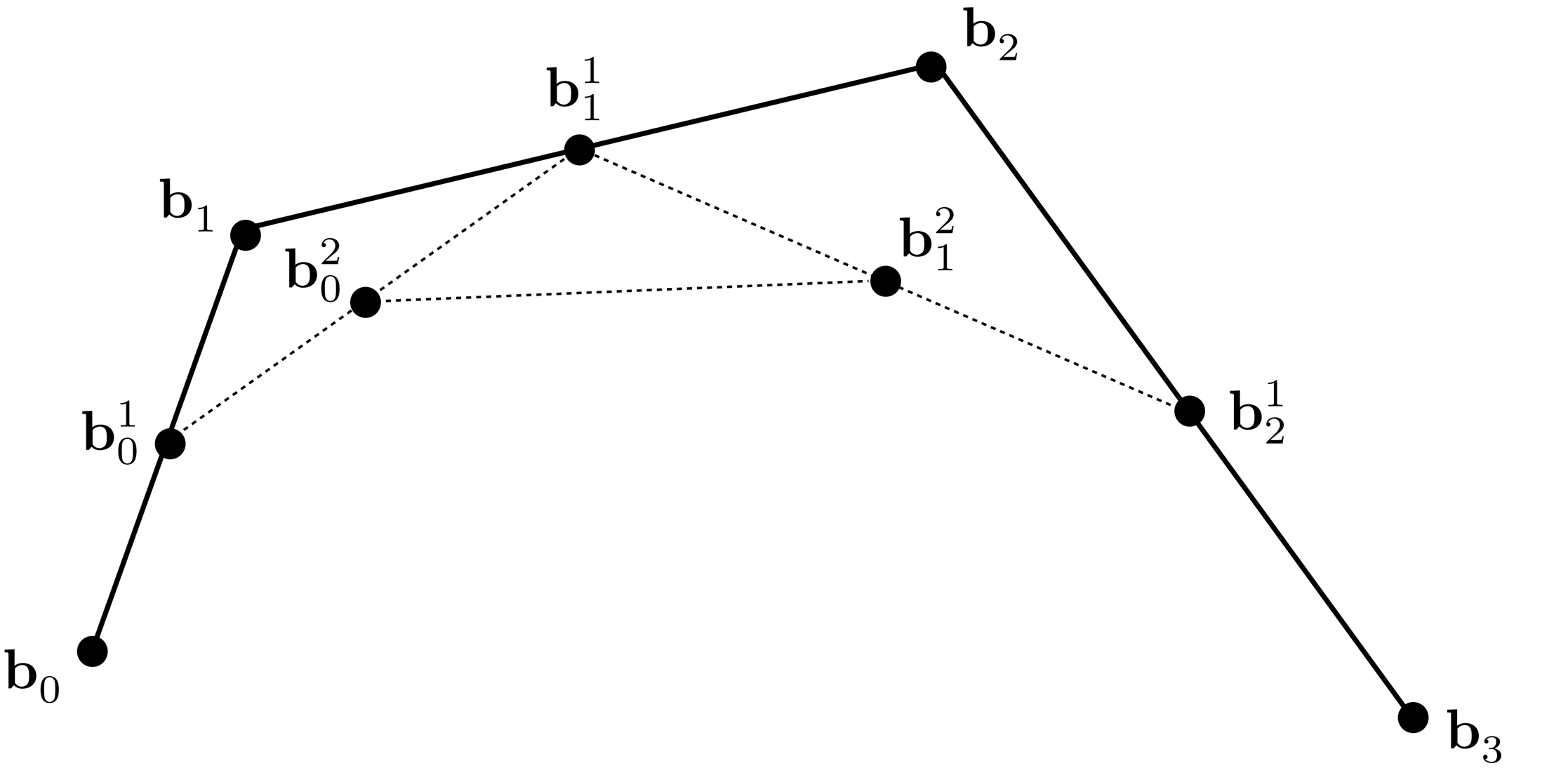
Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations



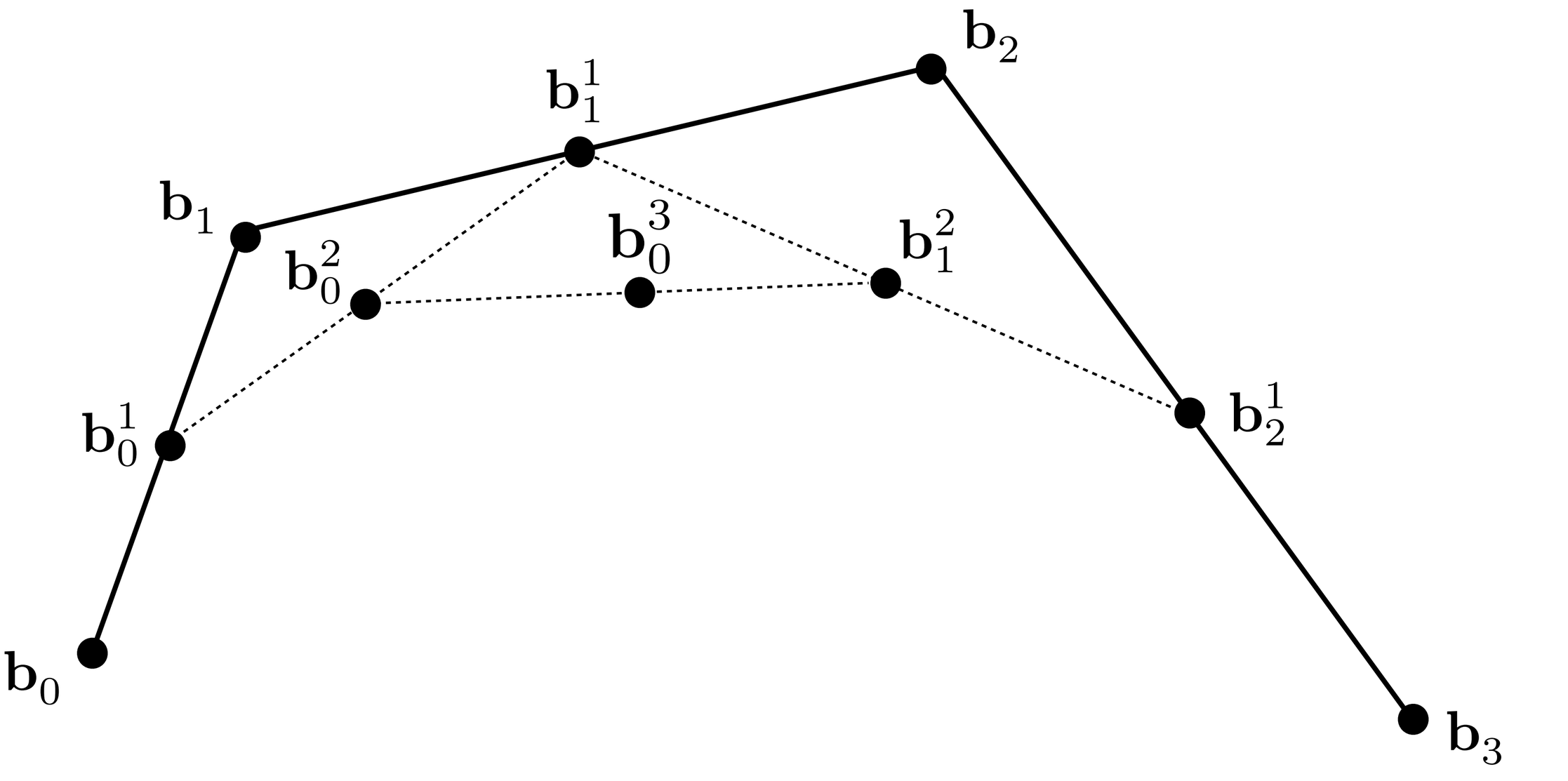
Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations



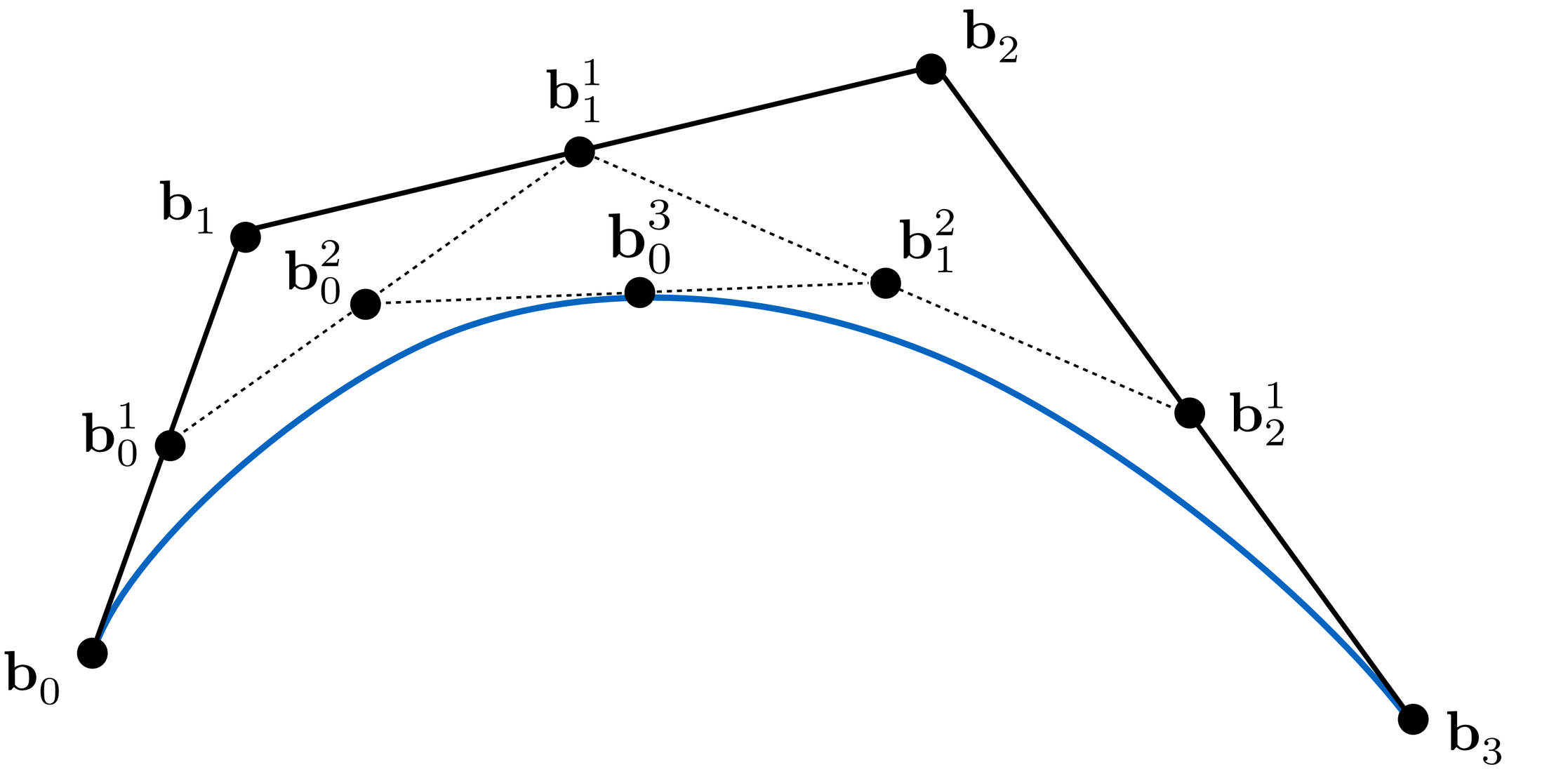
Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations



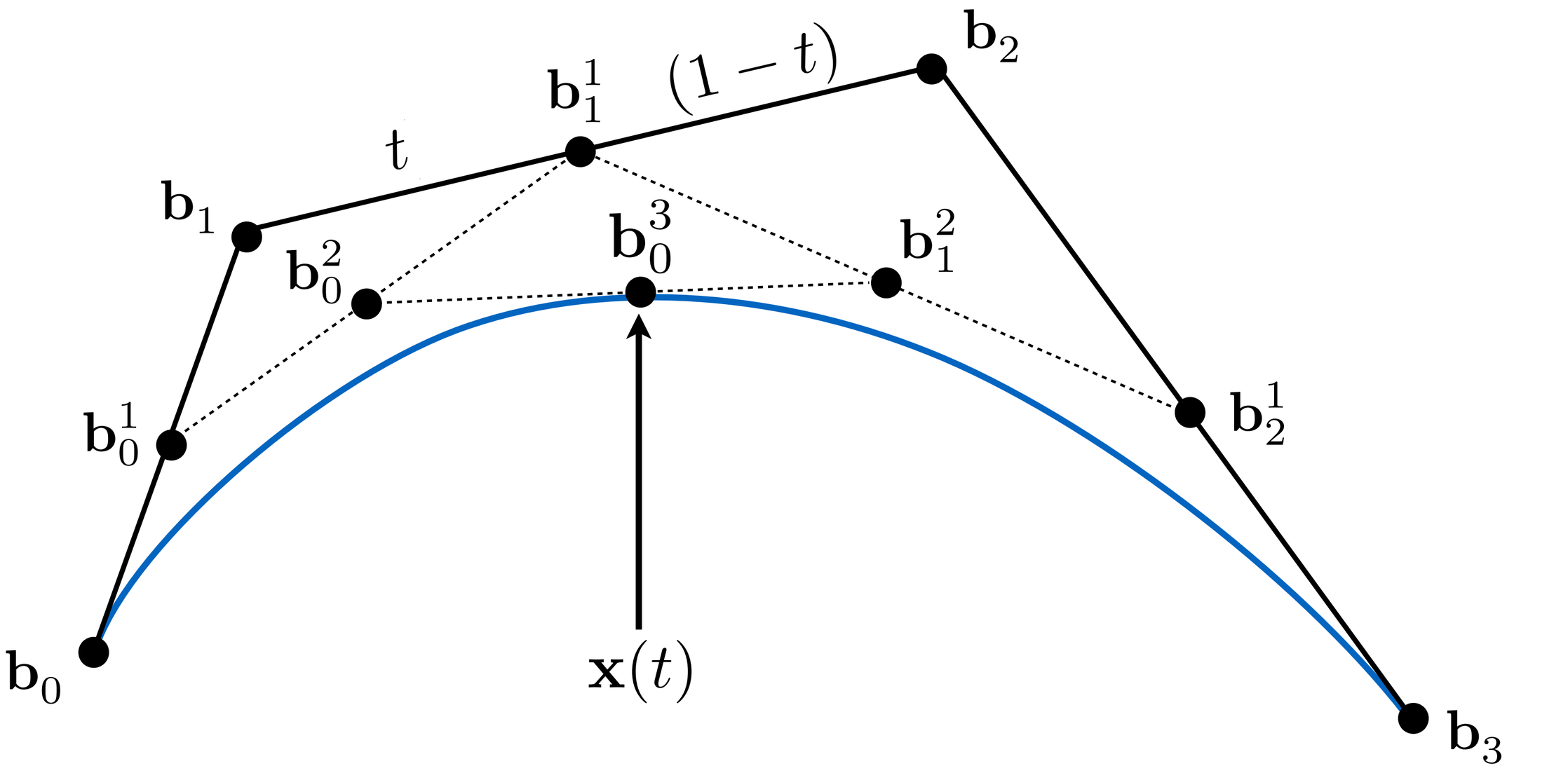
Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations

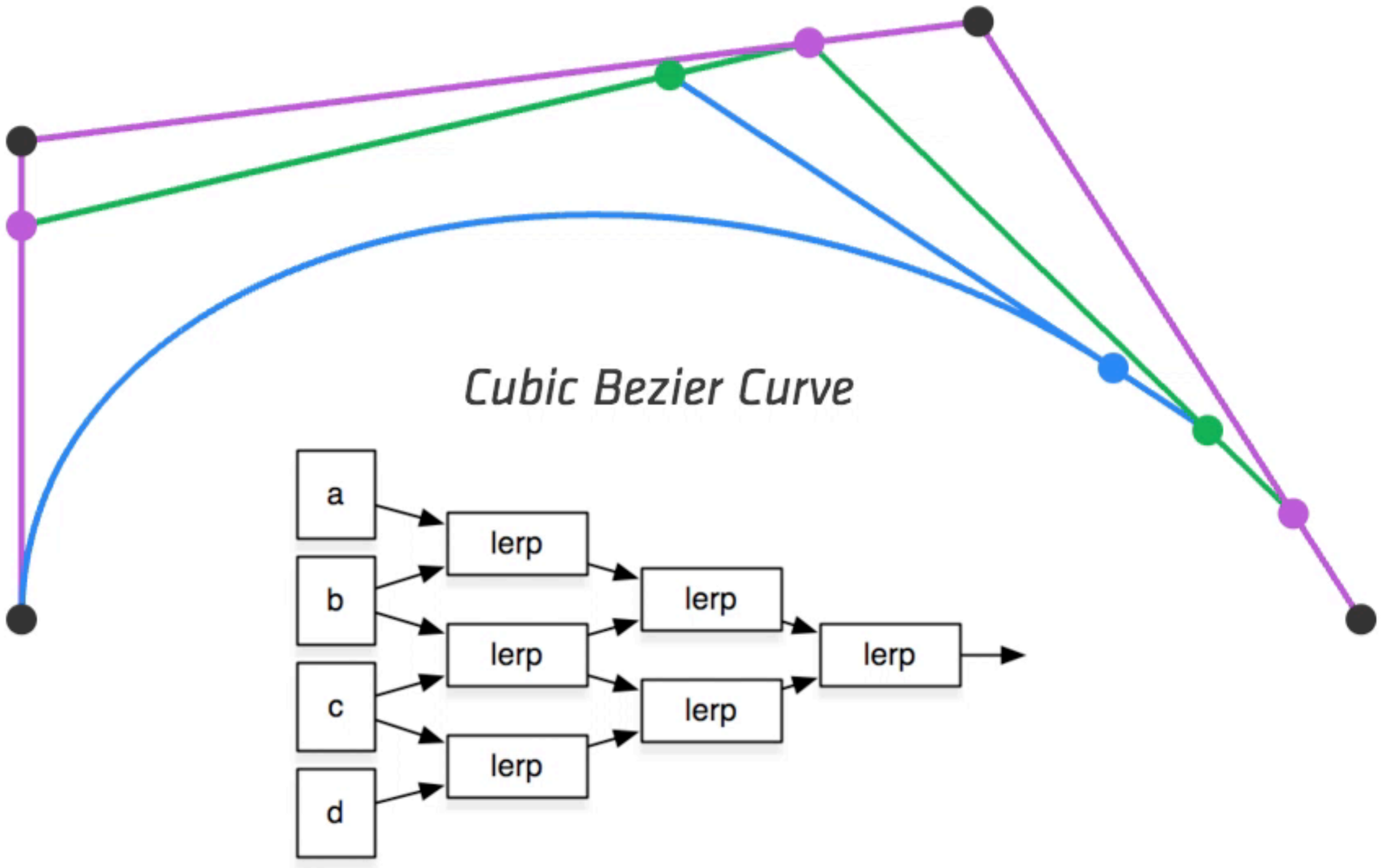


Cubic Bézier Curve – de Casteljau

Consider four points
Same recursive linear interpolations

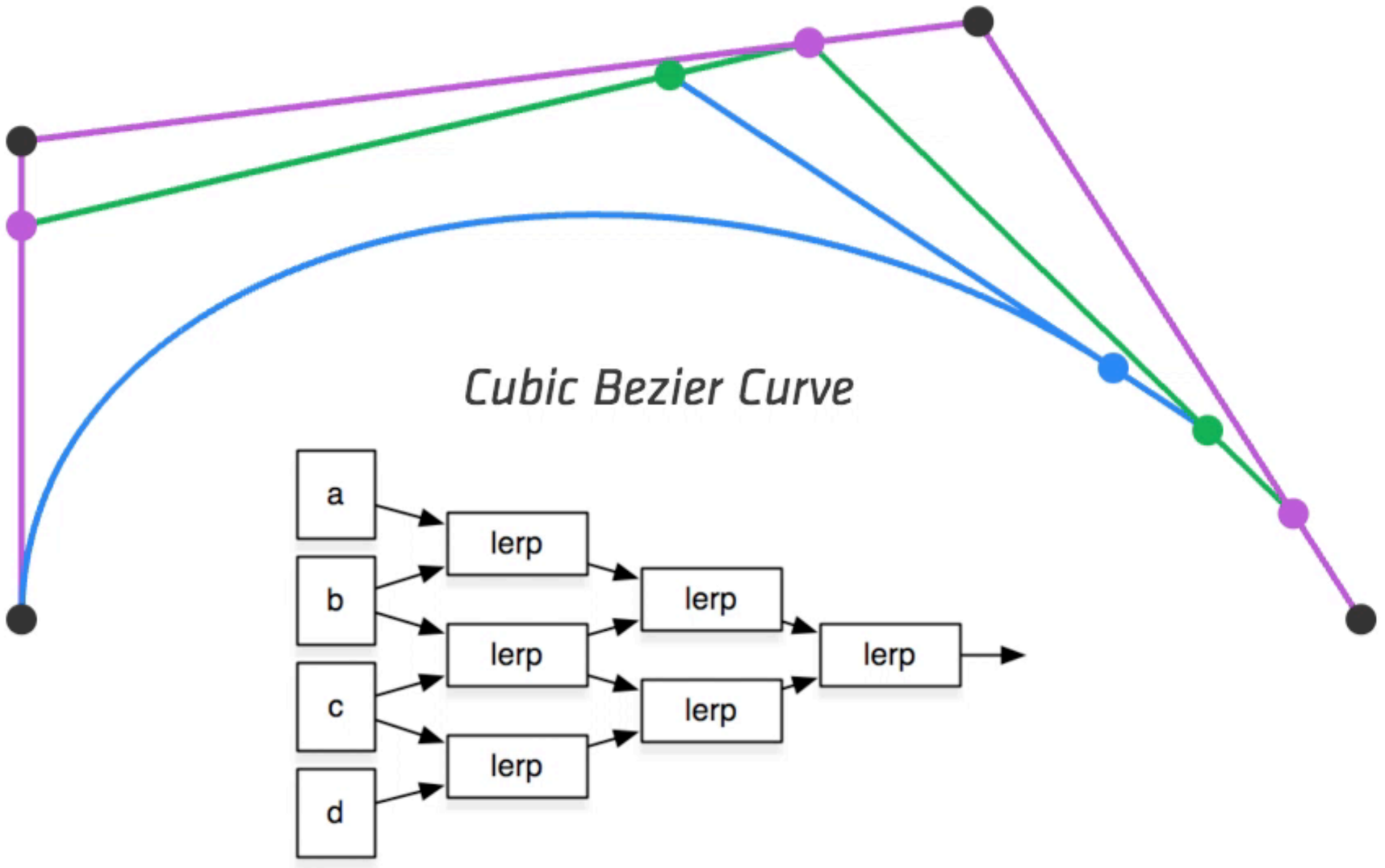


Visualizing de Casteljau Algorithm



Animation: Steven Wittens, Making Things with Maths, http://acko.net

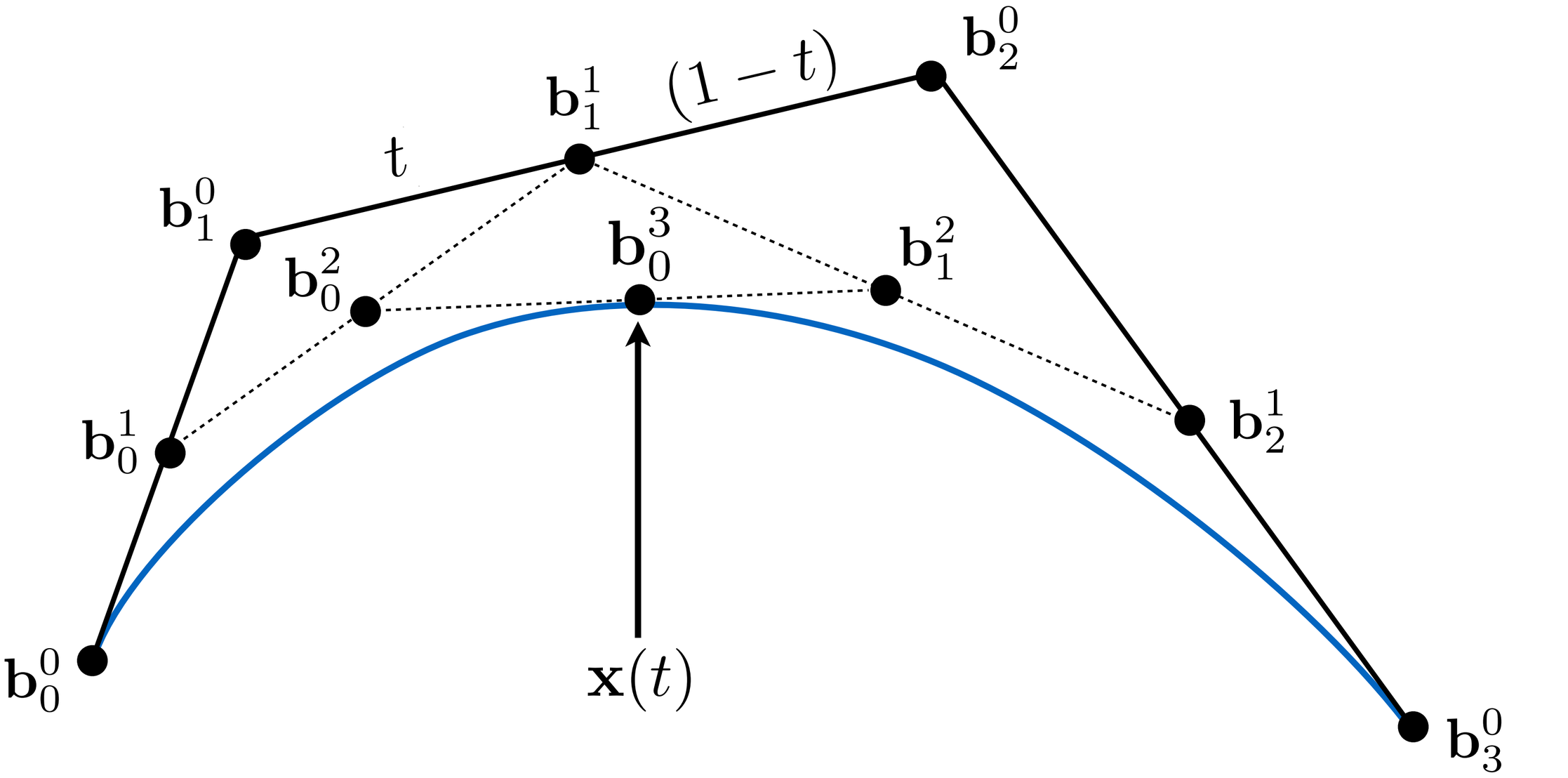
Visualizing de Casteljau Algorithm



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Cubic Bézier Curve – de Casteljau

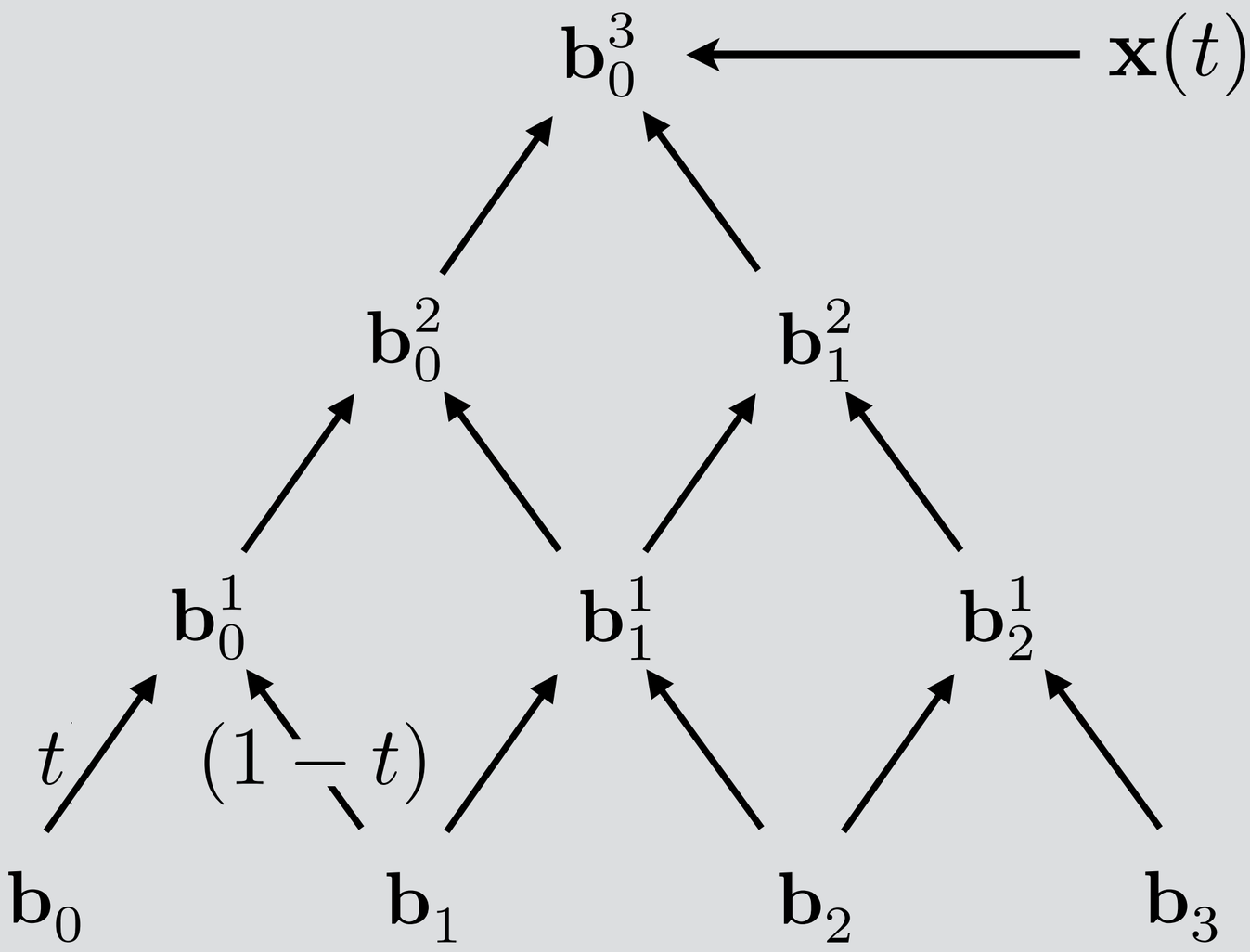
Consider four points
Same recursive linear interpolations

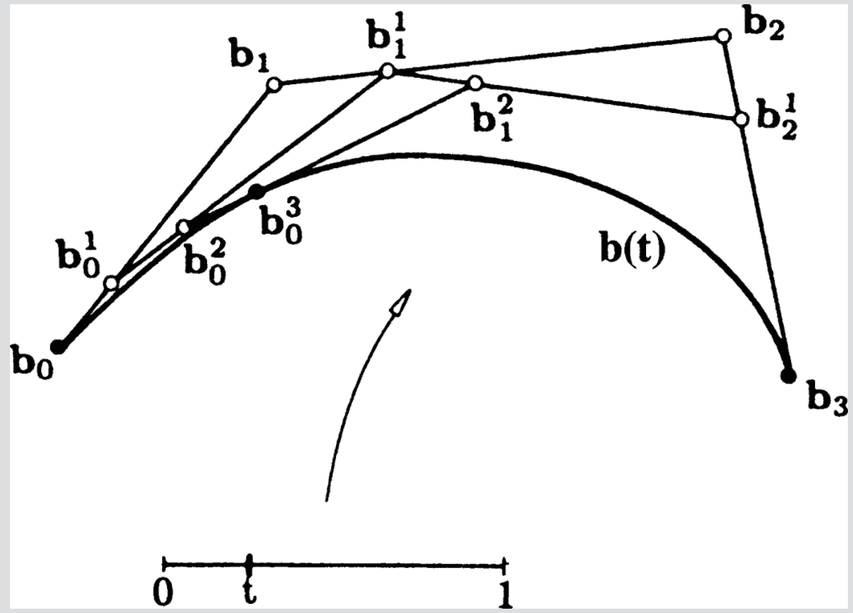


Evaluating Bézier Curves
Algebraic Formula

Bézier Curve – Algebraic Formula

de Casteljau algorithm gives a pyramid of coefficients





Every rightward arrow is multiplication by t,
Every leftward arrow by (1–t)

Bézier Curve – General Algebraic Formula

Bernstein form of a Bézier curve of order n:

$$\mathbf{b}^n(t) = \mathbf{b}_0^n(t) = \sum_{j=0}^{n} \mathbf{b}_j B_j^n(t)$$

Bézier curve order n
(vector polynomial of degree n)

Bernstein polynomial
(scalar polynomial of degree n)

Bézier control points
(vector in R^N)

Bernstein polynomials:

$$B_i^n(t) = \binom{n}{i} t^i (1-t)^{n-i}$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Bézier Curve – Algebraic Formula: Example

Bernstein form of a Bézier curve of order n:

$$\mathbf{b}^n(t) = \mathbf{b}_0^n(t) = \sum_{j=0}^{n} \mathbf{b}_j B_j^n(t)$$

Example: assume n = 3 and we are in R^3

i.e. we could have control points in 3D such as:

$$\mathbf{b}_0 = (0,2,3),\ \mathbf{b}_1 = (2,3,5),\ \mathbf{b}_2 = (6,7,9),\ \mathbf{b}_3 = (3,4,5)$$

These points define a Bezier curve in 3D that is a cubic polynomial in t:

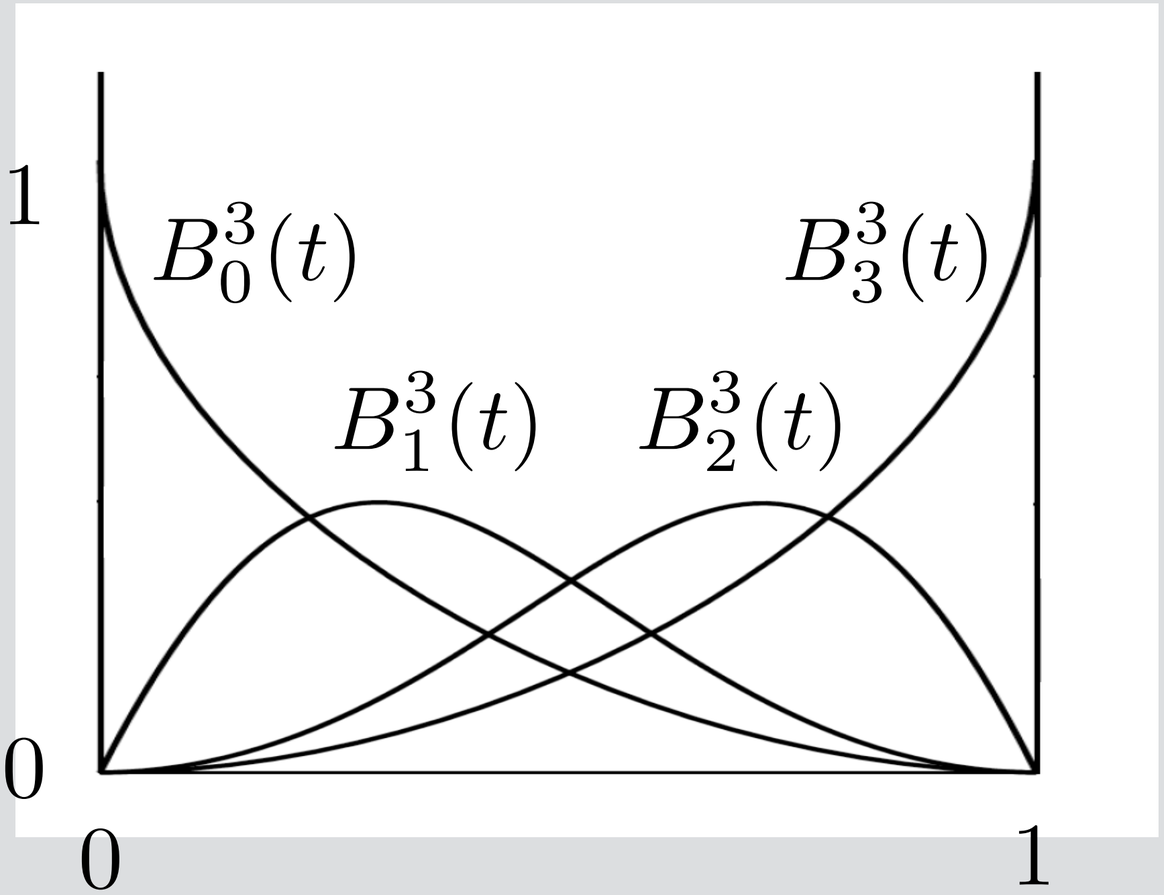
$$\mathbf{b}^n(t) = \mathbf{b}_0\,(1-t)^3 + \mathbf{b}_1\,3t(1-t)^2 + \mathbf{b}_2\,3t^2(1-t) + \mathbf{b}_3\,t^3$$

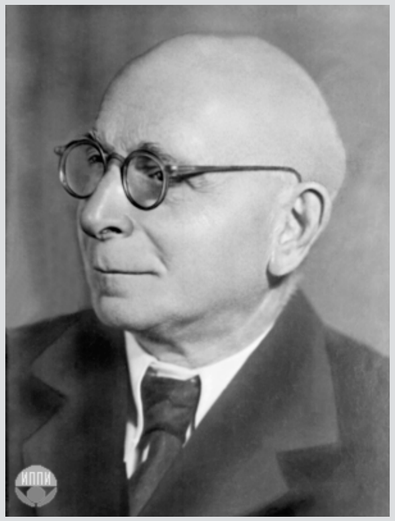
Compare to basis functions in derivation.

Cubic Bézier Basis Functions

Bernstein polynomials:

$$B_i^n(t) = \binom{n}{i} t^i (1-t)^{n-i}$$





Sergei N. Bernstein
1880 – 1968

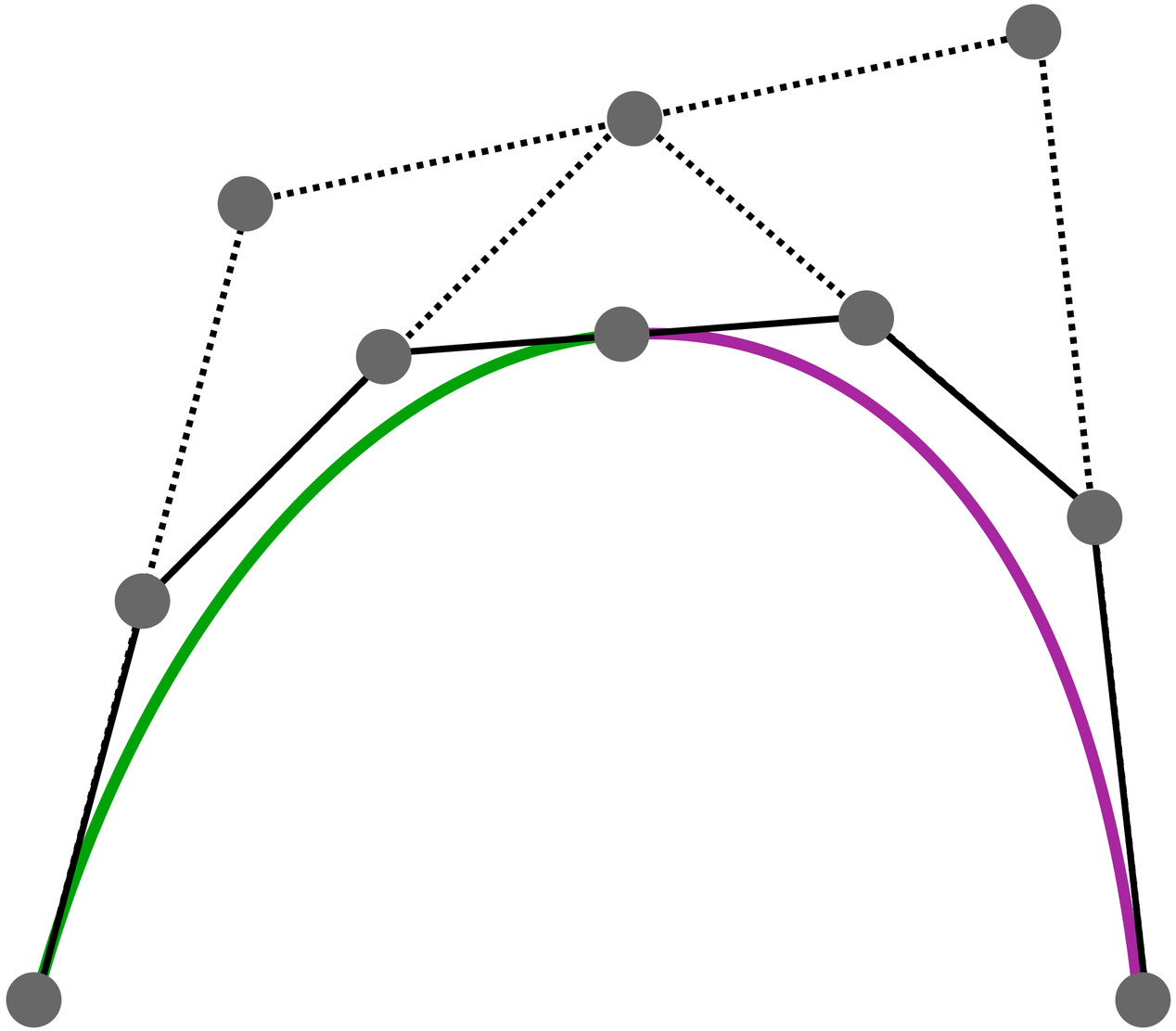
Evaluating Bézier Curves Matrix Formula

Bézier Curve – Matrix Formula

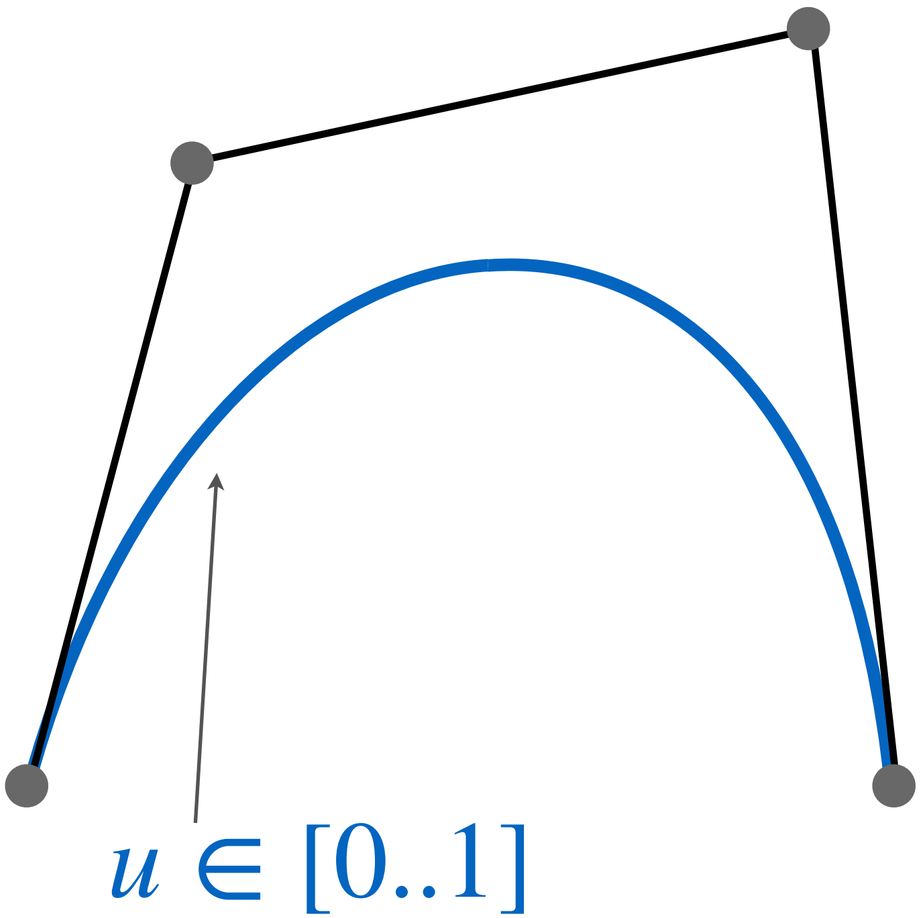
$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix} \qquad \boldsymbol{\beta}_Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

Evaluating Bézier Curves
Subdivision

Method 3: Bézier Subdivision



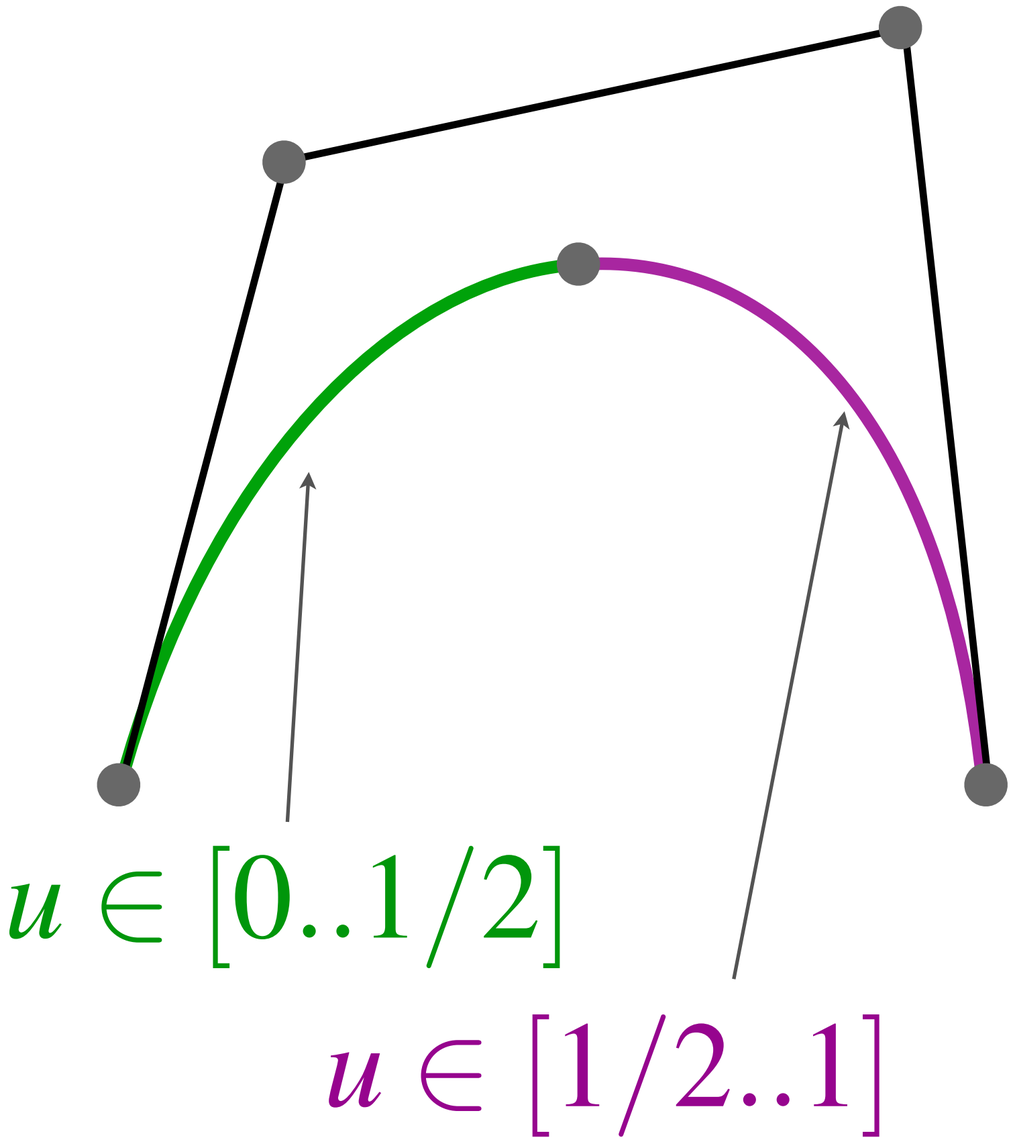
Bézier Subdivision



$$\mathbf{x}(u) = [1, u, u^2, u^3]\,\beta_Z\,\mathbf{P}$$

$$\beta_Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

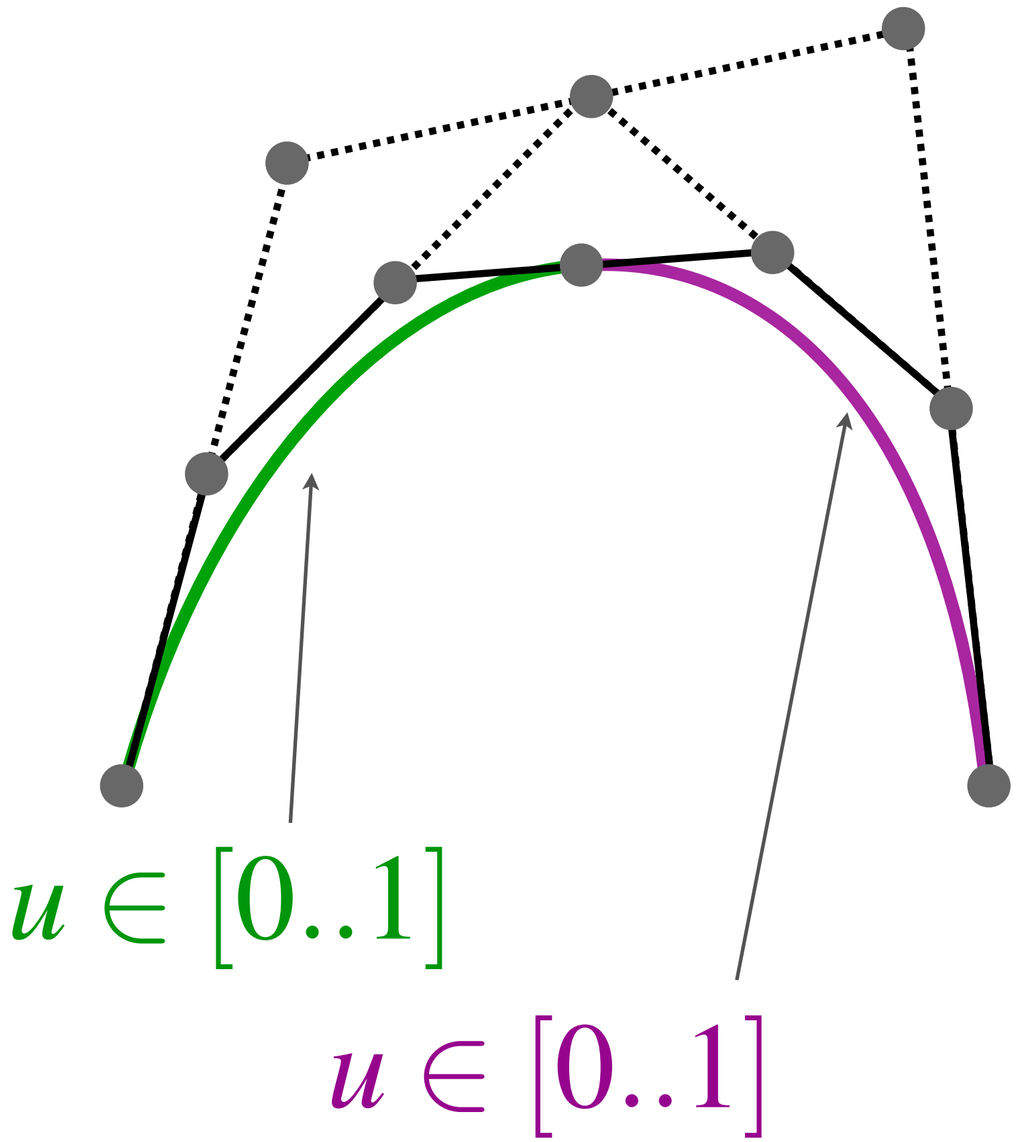
Bézier Subdivision



$$\mathbf{x}(u) = [1, u, u^2, u^3]\,\beta_Z\,\mathbf{P}$$

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Bézier Subdivision



$u \in [0..1]$

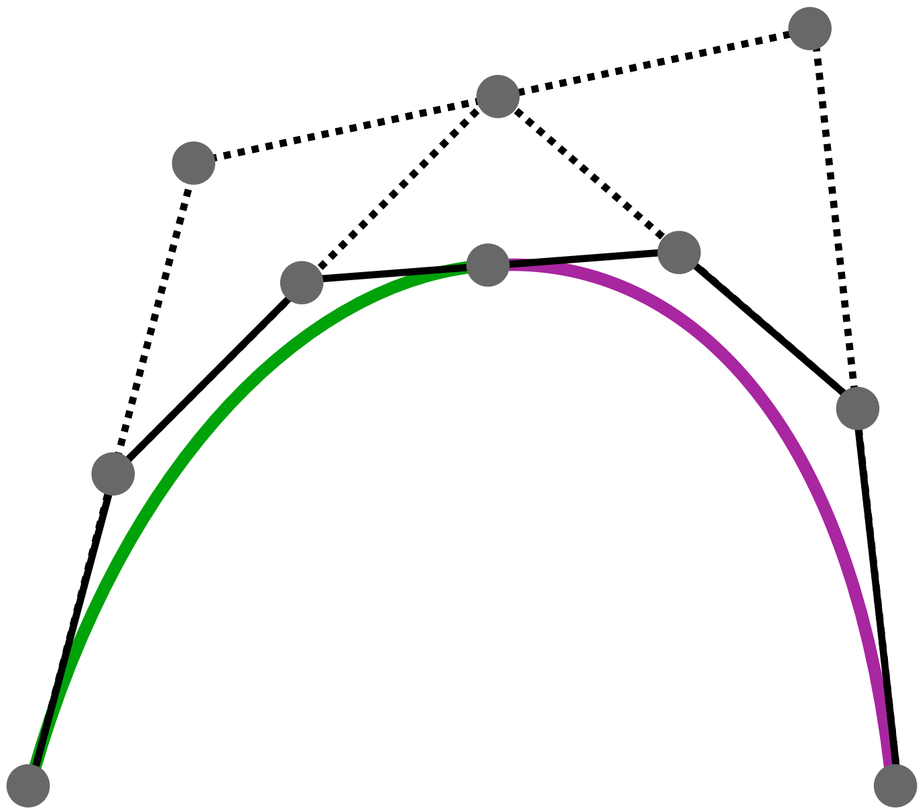
$u \in [0..1]$

$$\mathbf{x}(u) = [1, u, u^2, u^3]\,\beta_Z\,\mathbf{P}$$

Can't change these....

$$\beta_Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 3 & 0 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix}$$

Bézier Subdivision



$$\mathbf{x}(u) = [1, u, u^2, u^3]\,\beta_z\,\mathbf{P} \qquad u \in [0..\frac{1}{2}]$$

$$\mathbf{x}(u) = [1, \frac{u}{2}, \frac{u^2}{4}, \frac{u^3}{8}]\beta_z\mathbf{P} \qquad u \in [0..1]$$

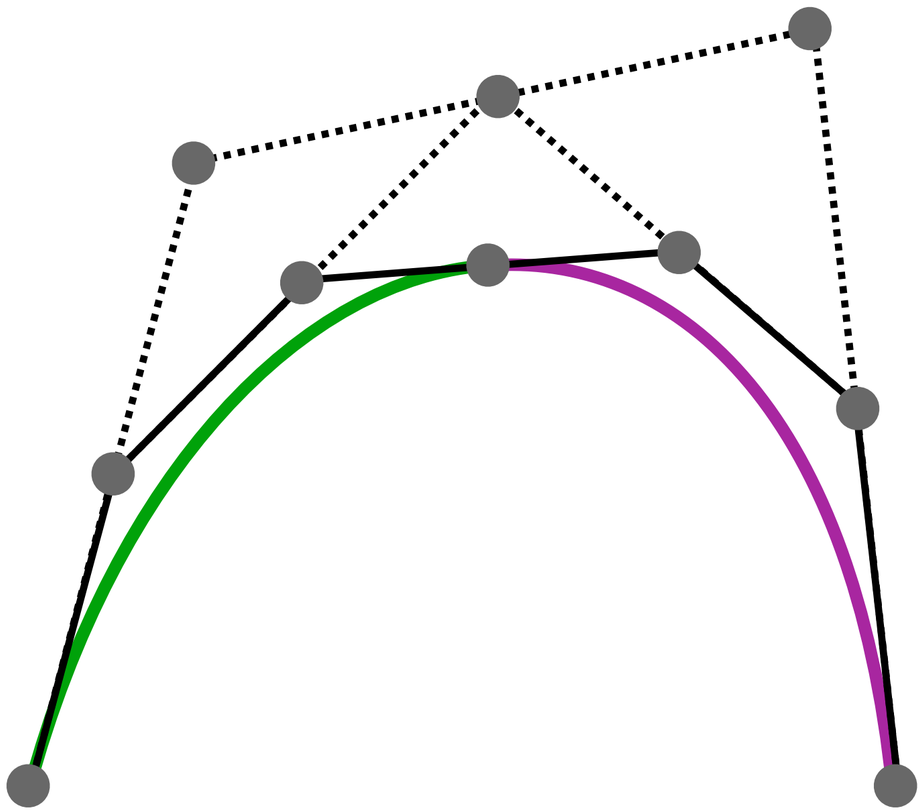
$$\mathbf{x}(u) = [1, u, u^2, u^3]\mathbf{S}_1\beta_z\mathbf{P}$$

$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_z\beta_z^{-1}\mathbf{S}_1\beta_z\mathbf{P}$$

$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_z\mathbf{H}_{z1}\mathbf{P}$$

$$\mathbf{S}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 1/8 \end{bmatrix}$$

Bézier Subdivision



$$\mathbf{x}(u) = [1, u, u^2, u^3]\,\beta_z\,\mathbf{P} \qquad u \in [0..\frac{1}{2}]$$

$$\mathbf{x}(u) = [1, \frac{u}{2}, \frac{u^2}{4}, \frac{u^3}{8}]\beta_z\mathbf{P} \qquad u \in [0..1]$$

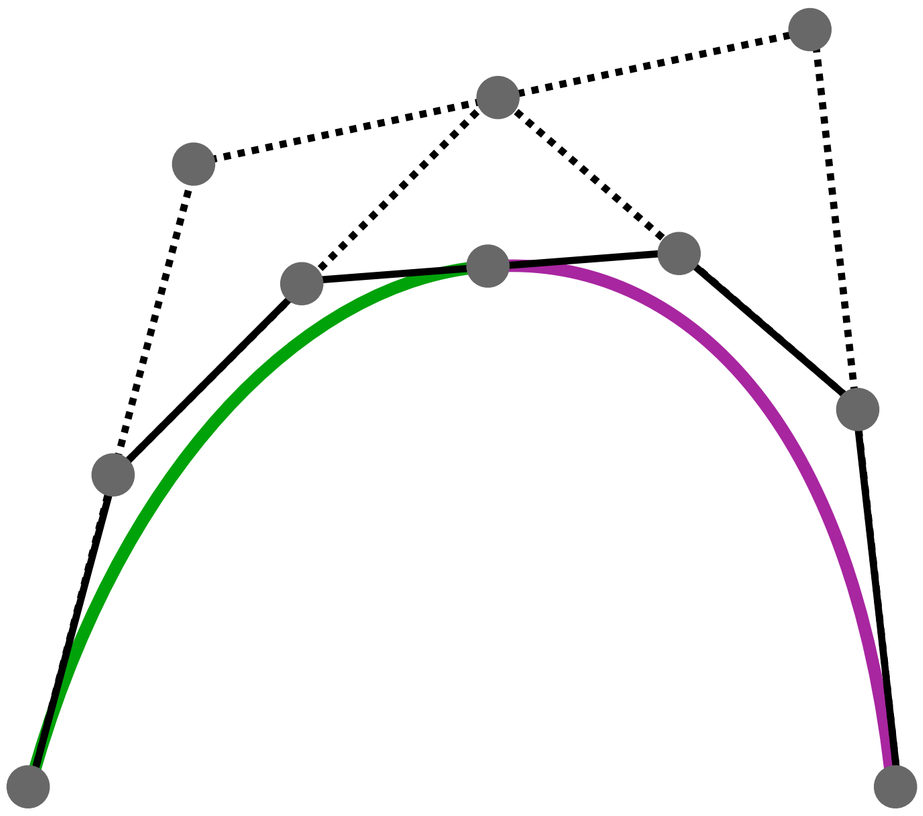
$$\mathbf{x}(u) = [1, u, u^2, u^3]\mathbf{S}_1\beta_z\mathbf{P}$$

$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_z\beta_z^{-1}\mathbf{S}_1\beta_z\mathbf{P}$$

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Bézier Subdivision

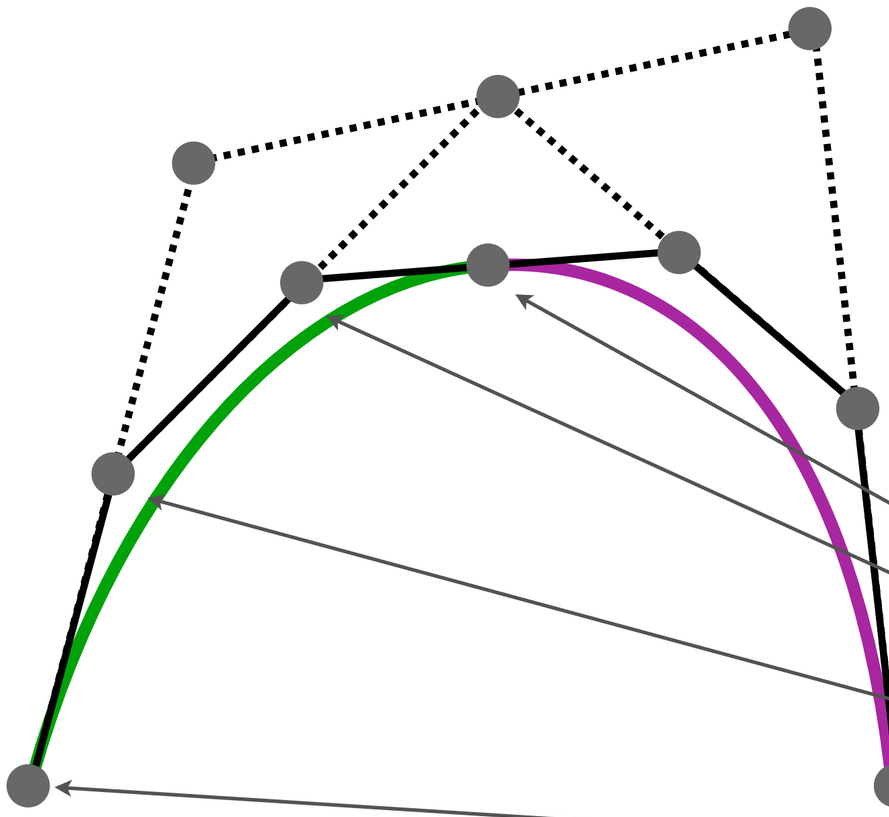


$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_Z \beta_Z^{-1} \mathbf{S}_1 \beta_Z \mathbf{P}$$

$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_Z \mathbf{H}_{Z1} \mathbf{P}$$

$$\mathbf{S}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 1/8 \end{bmatrix}$$

Bézier Subdivision



$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_z\beta_z^{-1}\mathbf{S}_1\beta_z\mathbf{P}$$

$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_z\mathbf{H}_{z1}\mathbf{P}$$

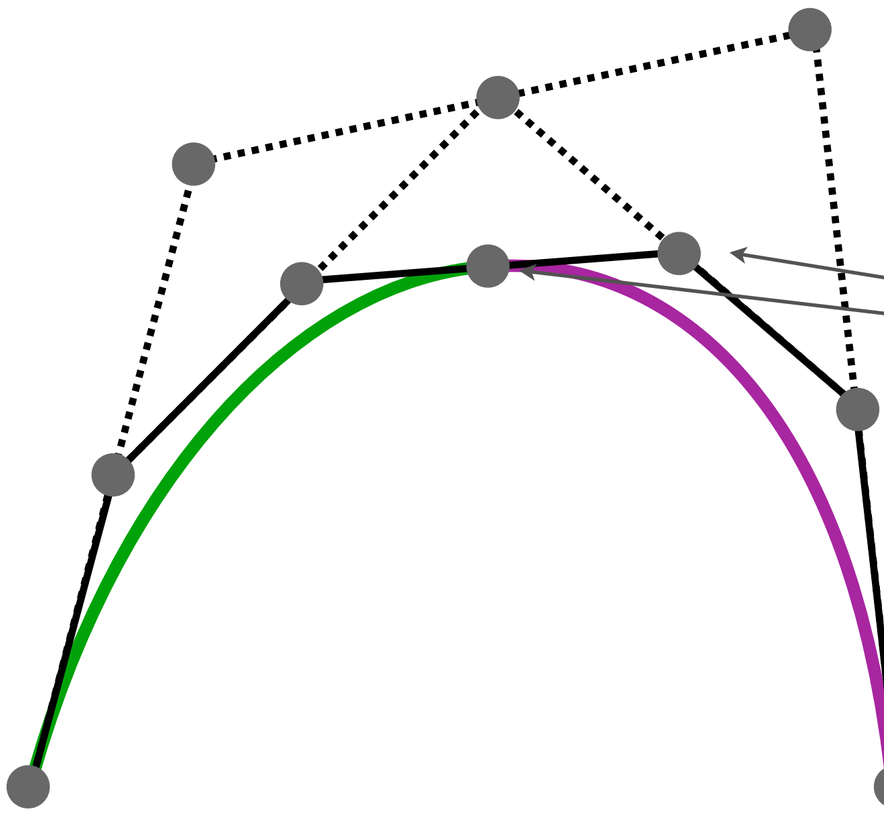
$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_z\mathbf{P}_1$$

$$\mathbf{H}_{z1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \end{bmatrix}$$

$$\mathbf{P}_1 = \mathbf{H}_{z1}\mathbf{P}$$

$$\mathbf{S}_1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1/4 & 0 \\ 0 & 0 & 0 & 1/8 \end{bmatrix}$$

Bézier Subdivision



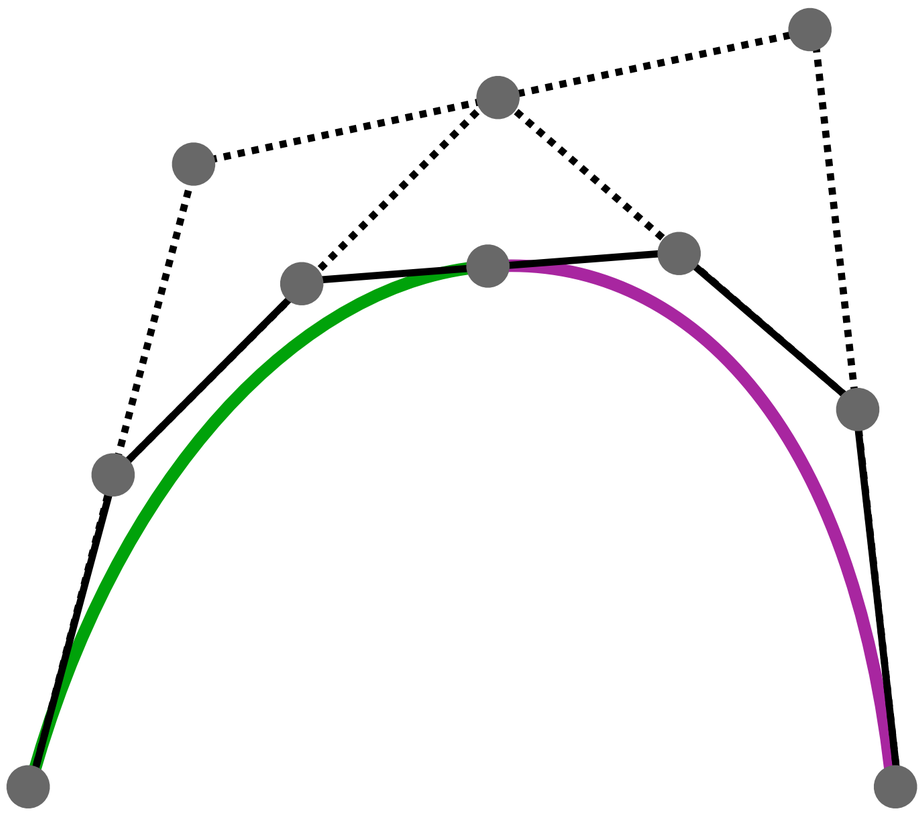
$$\mathbf{x}(u) = [1, u, u^2, u^3]\beta_Z \mathbf{P}_2$$

$$\mathbf{P}_2 = \mathbf{H}_{Z2}\mathbf{P}$$

$$\mathbf{S}_2 = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{4} & \frac{1}{8} \\ 0 & \frac{1}{2} & \frac{1}{2} & \frac{3}{8} \\ 0 & 0 & \frac{1}{4} & \frac{3}{8} \\ 0 & 0 & 0 & \frac{1}{8} \end{bmatrix}$$

$$\mathbf{H}_{Z2} = \begin{bmatrix} \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Bézier Subdivision



$$\mathbf{P}_1 = \mathbf{H}_{Z1}\mathbf{P}$$

$$\mathbf{P}_2 = \mathbf{H}_{Z2}\mathbf{P}$$

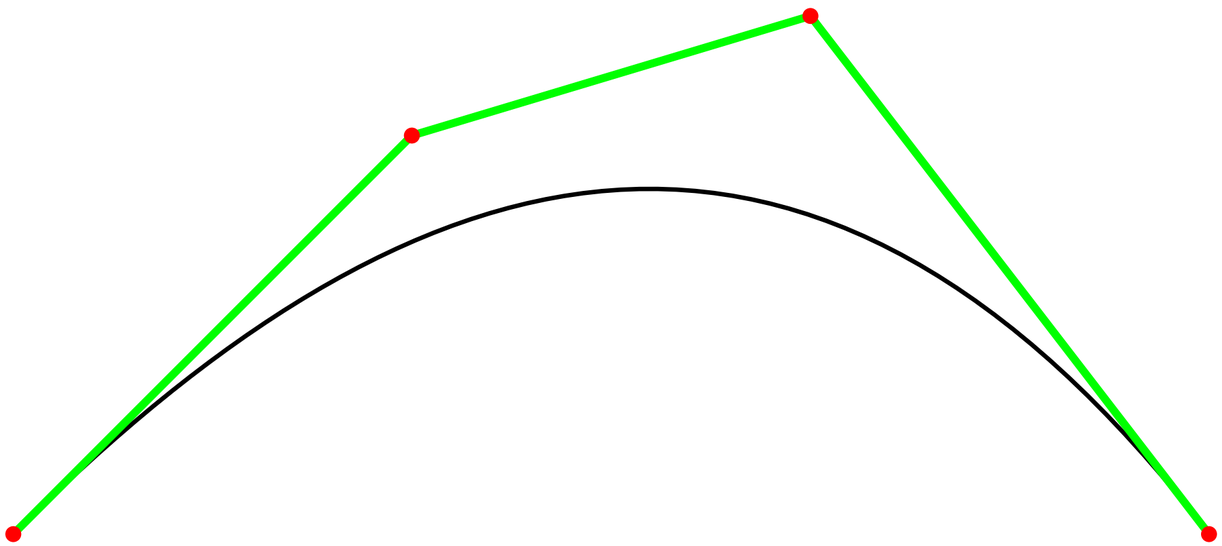
$$\mathbf{H}_{Z1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \end{bmatrix}$$

$$\mathbf{H}_{Z2} = \begin{bmatrix} \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8} \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rows of $\mathbf{H}_{Z\cdot}$ sum to 1 and ≥ 0 $\Rightarrow$ Convex Hull $\Rightarrow$ $\mathbf{P}_1$ and $\mathbf{P}_2 \in$ ConvexHull $(\mathbf{P})$

Curve remains constant $\Rightarrow$ Control Mesh converges to surface!!

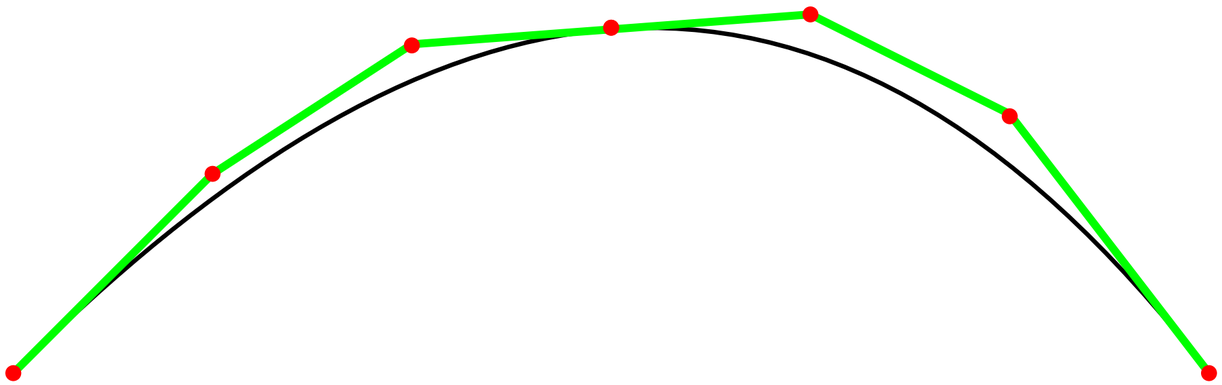
Bézier Subdivision



Level 0

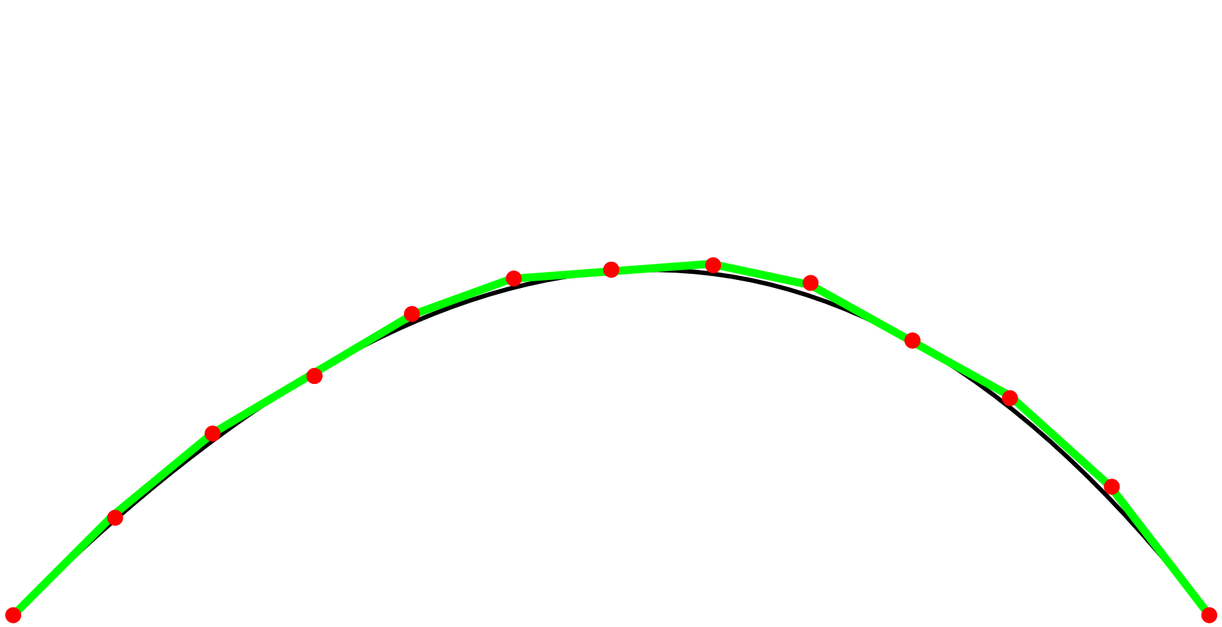
Curve remains constant ⇒ Control Mesh converges to surface!!

Bézier Subdivision



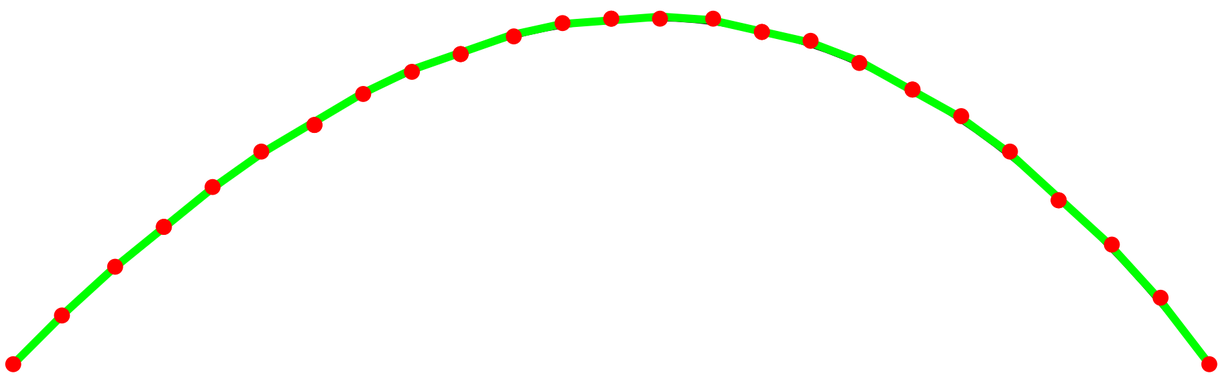
Curve remains constant ⇒ Control Mesh converges to surface!!

Bézier Subdivision

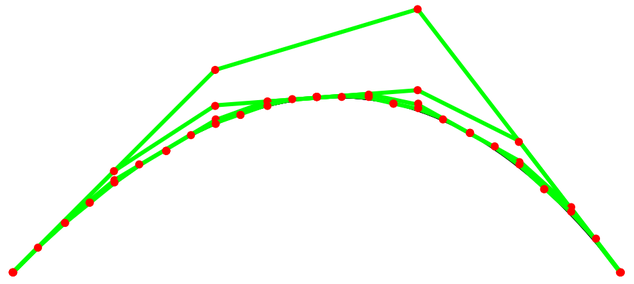


Curve remains constant ⇒ Control Mesh converges to surface!!

Bézier Subdivision



Level 3



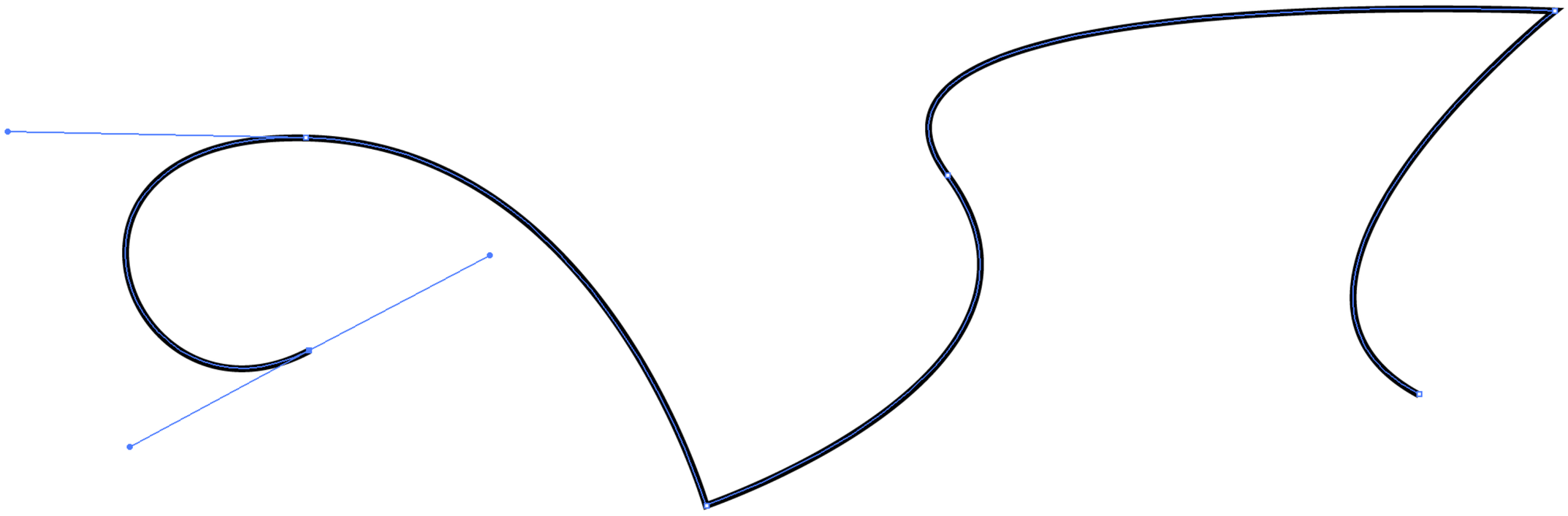
Curve remains constant ⇒ Control Mesh converges to surface!!

Piecewise Bézier Curves (Bézier Spline)

Piecewise Bézier Curves

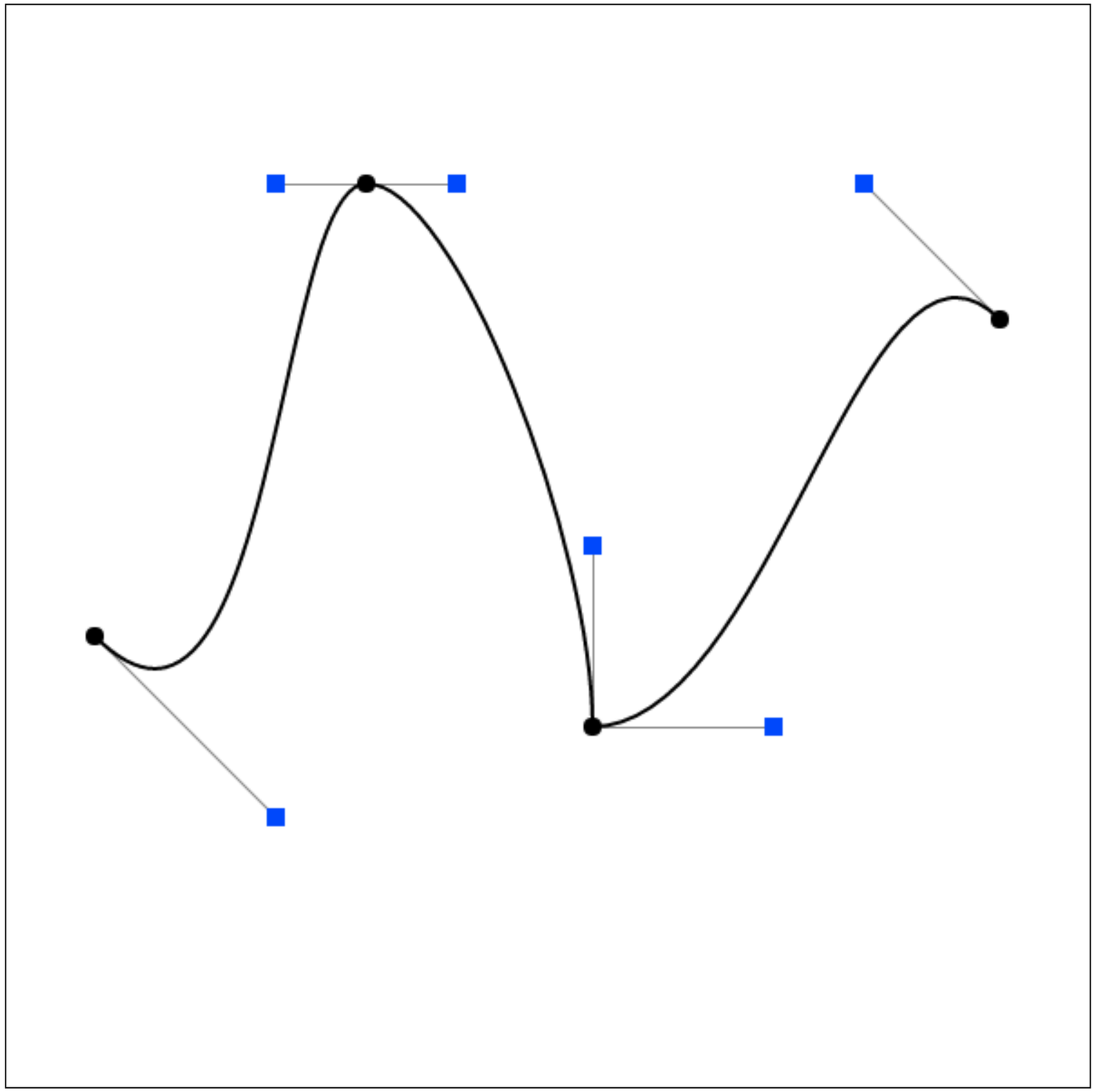
Instead, chain many low-order Bézier curve

Piecewise cubic Bézier the most common technique



Widely used (fonts, paths, Illustrator, Keynote, …)

Demo – Piecewise Cubic Bézier Curve



David Eck, http://math.hws.edu/eck/cs424/notes2013/canvas/bezier.html

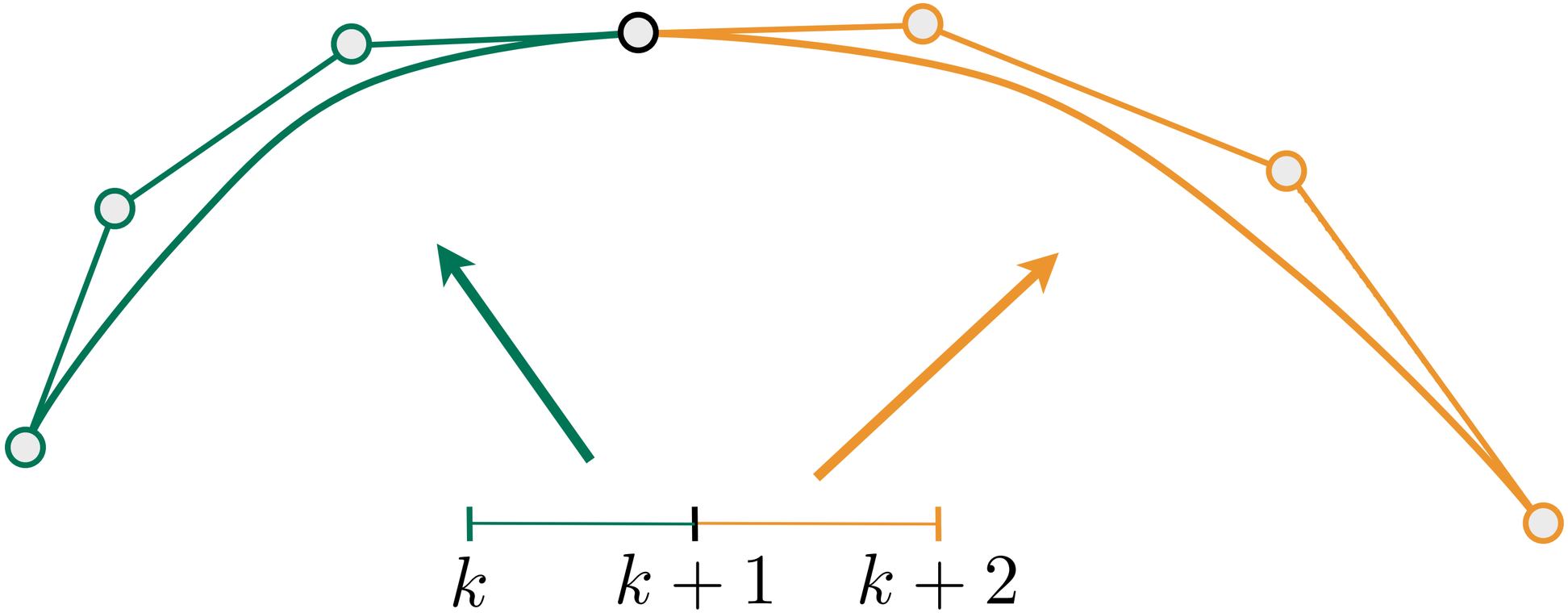
Piecewise Bézier Curve – Continuity

Two Bézier curves

$$\mathbf{a} : [k, k+1] \rightarrow \mathbb{R}^N$$

$$\mathbf{b} : [k+1, k+2] \rightarrow \mathbb{R}^N$$

Assuming integer partitions here, can generalize

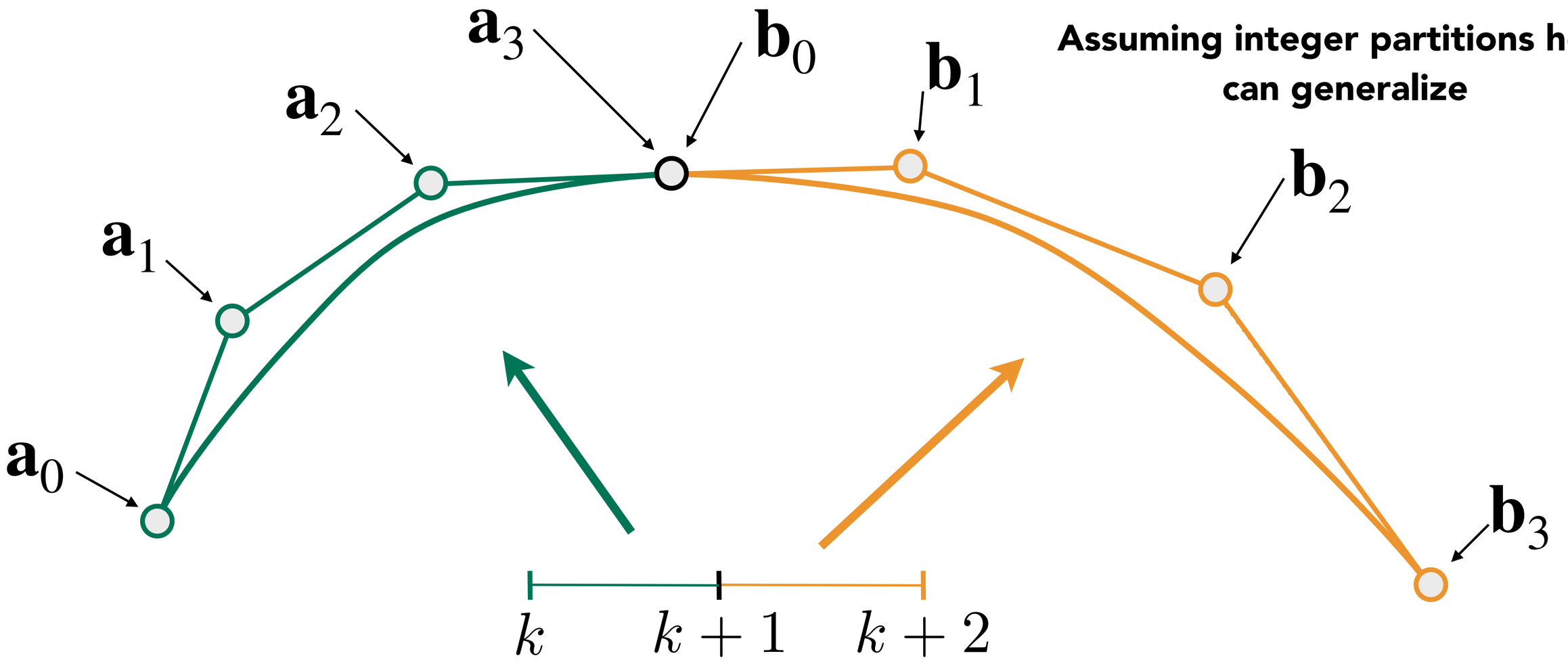


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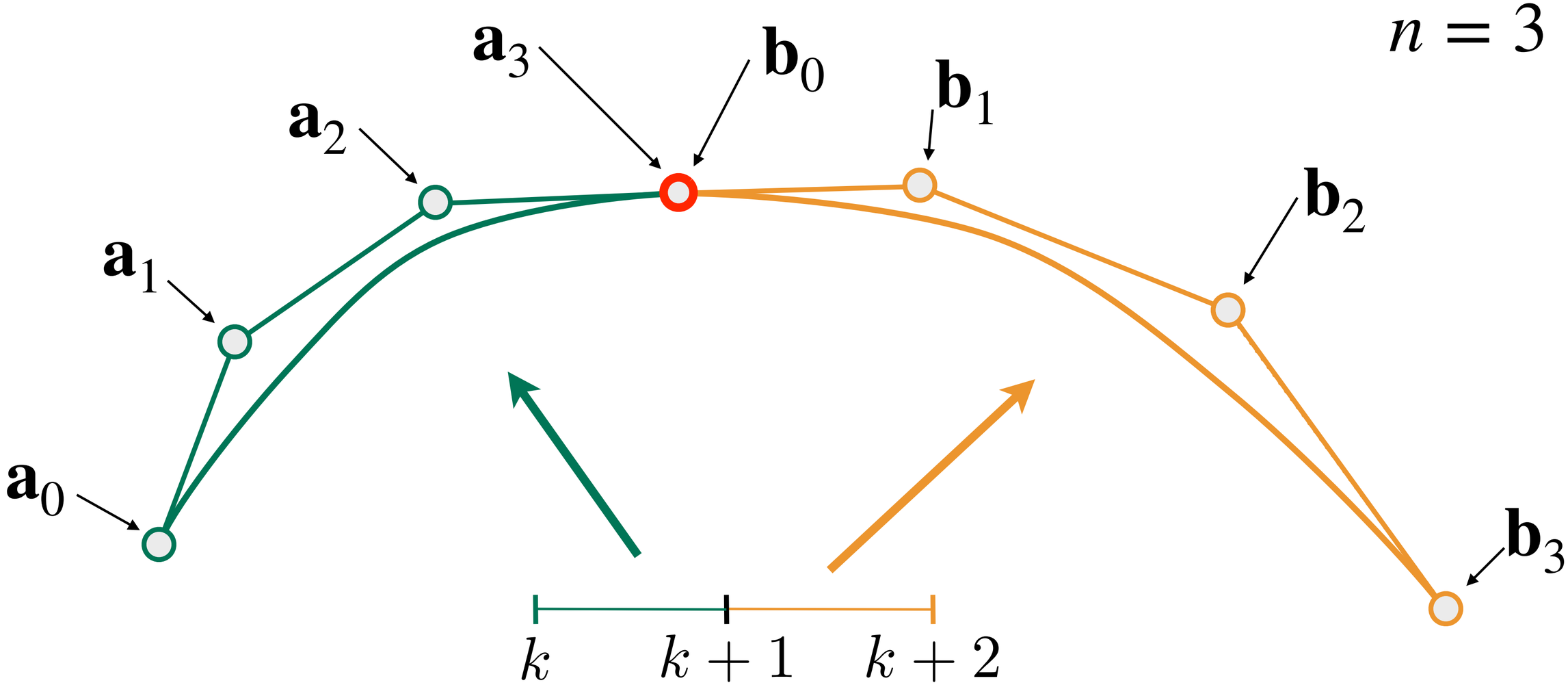
$$\mathbf{b} : [k+1, k+2] \rightarrow \mathbb{R}^N$$



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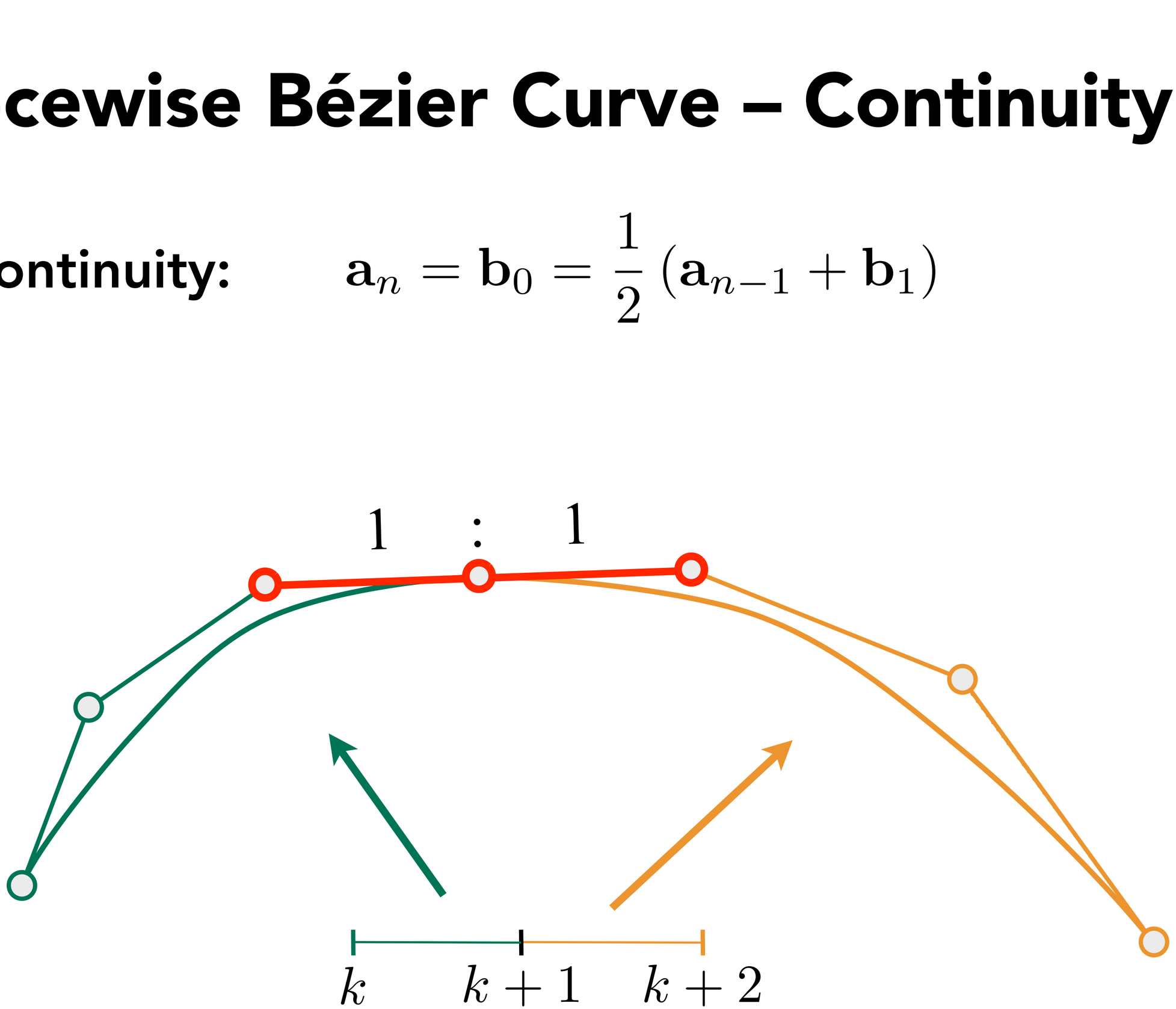
Piecewise Bézier Curve – Continuity

C⁰ continuity: $\mathbf{a}_n = \mathbf{b}_0$



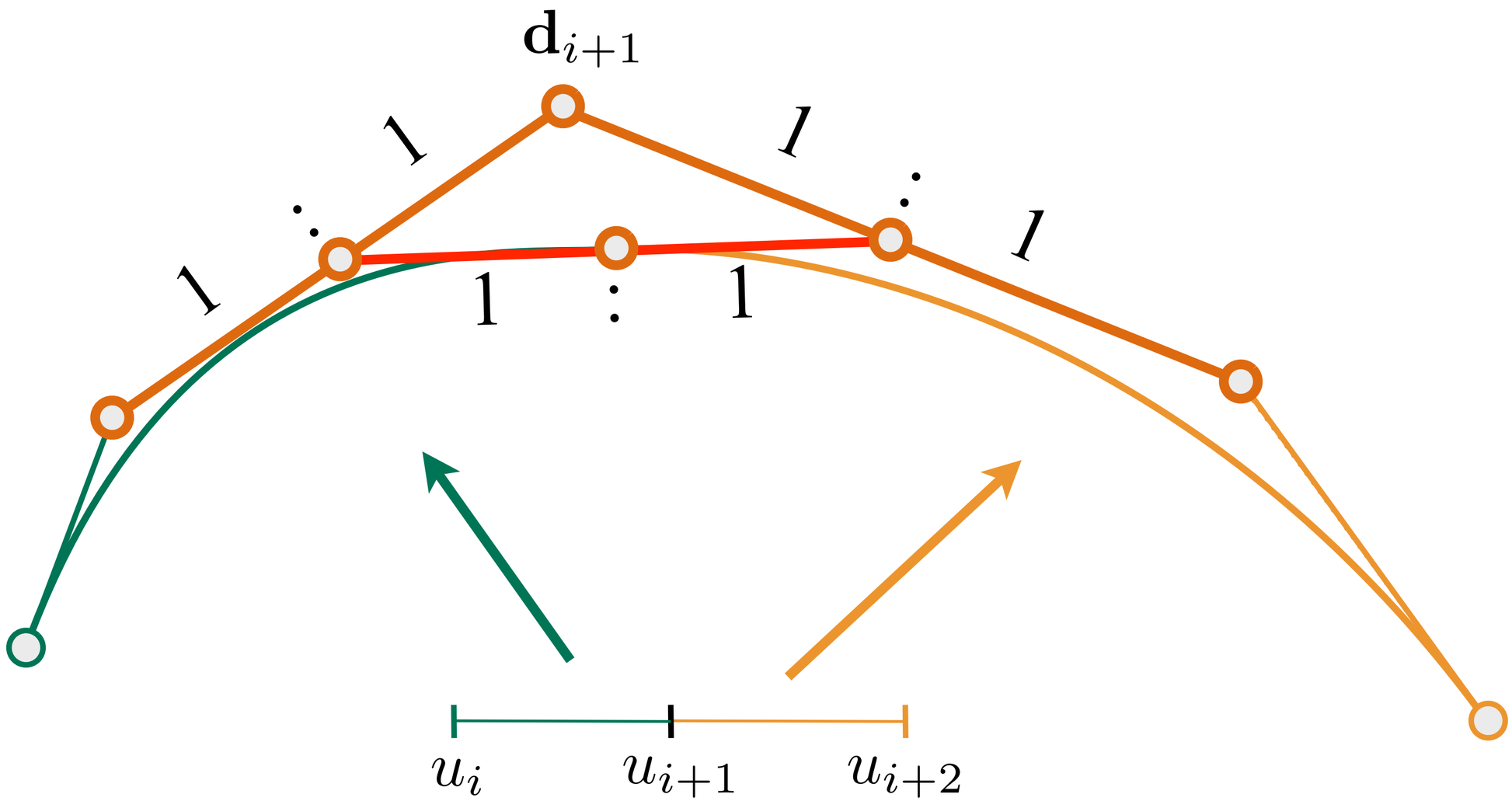
Piecewise Bézier Curve – Continuity

C¹ continuity: $\mathbf{a}_n = \mathbf{b}_0 = \frac{1}{2}\left(\mathbf{a}_{n-1} + \mathbf{b}_1\right)$



New Slide With Improvements [Ng]

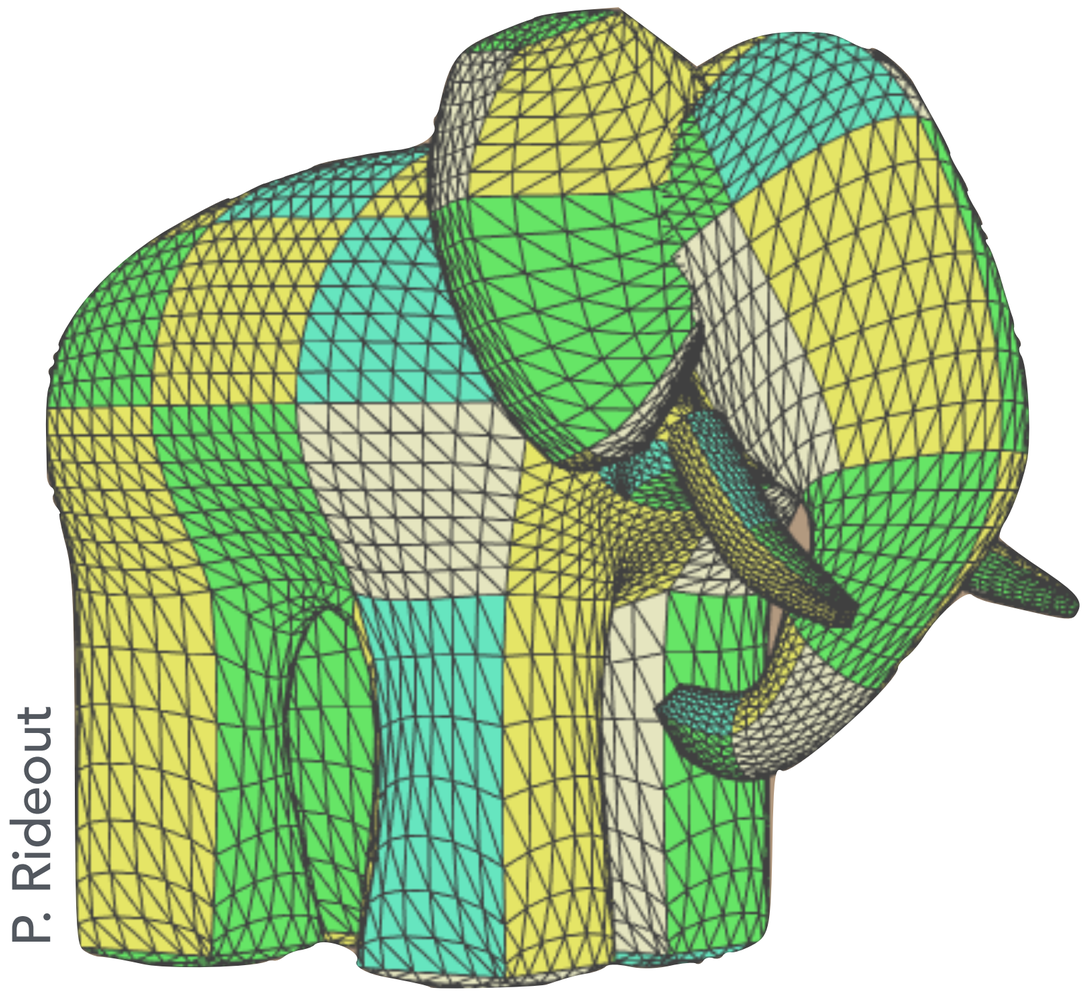
C² continuity: "A-frame" construction



Bézier Surfaces

Bézier Surfaces

Extend Bézier curves to surfaces

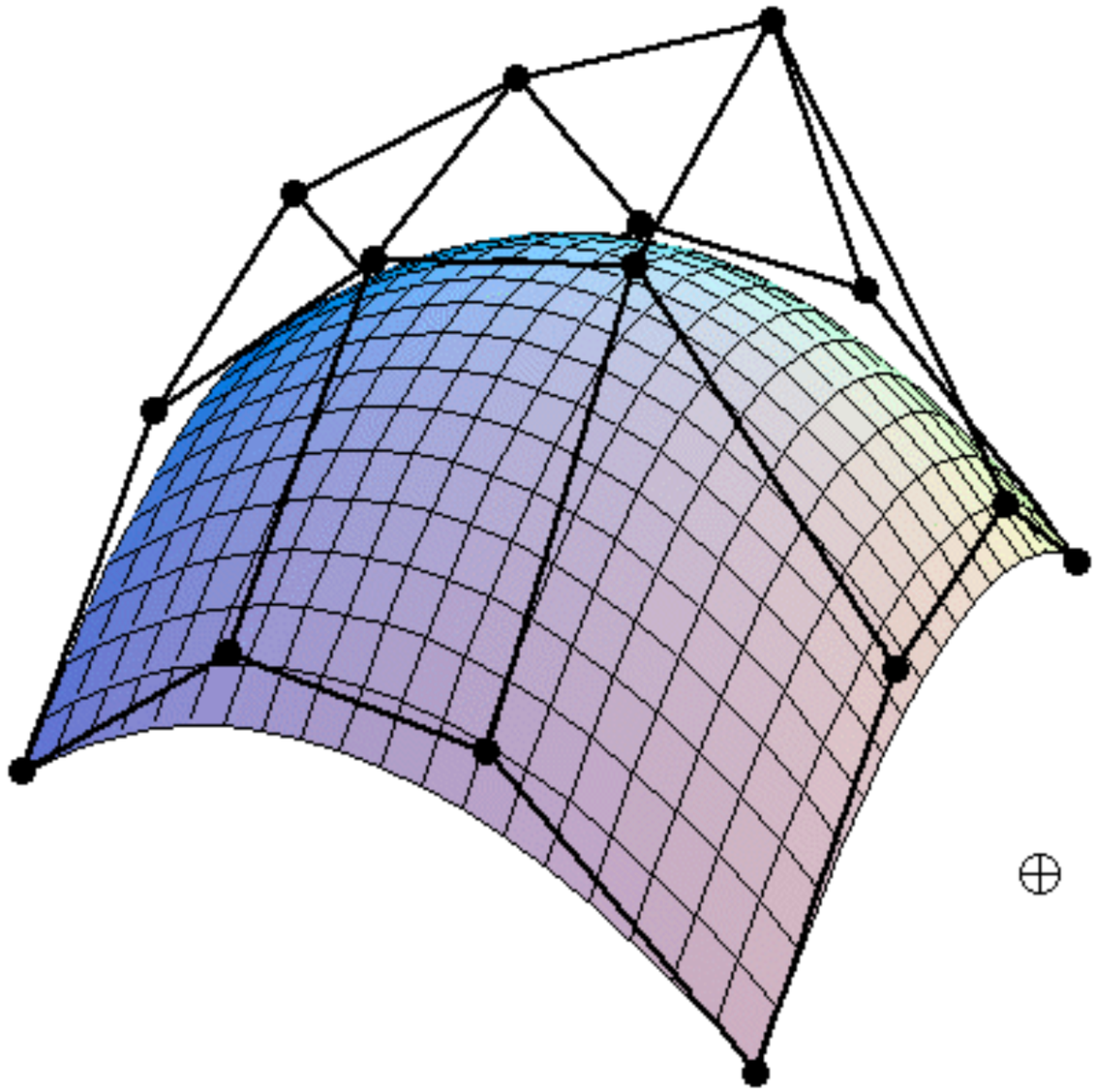


Ed Catmull's "Gumbo" model



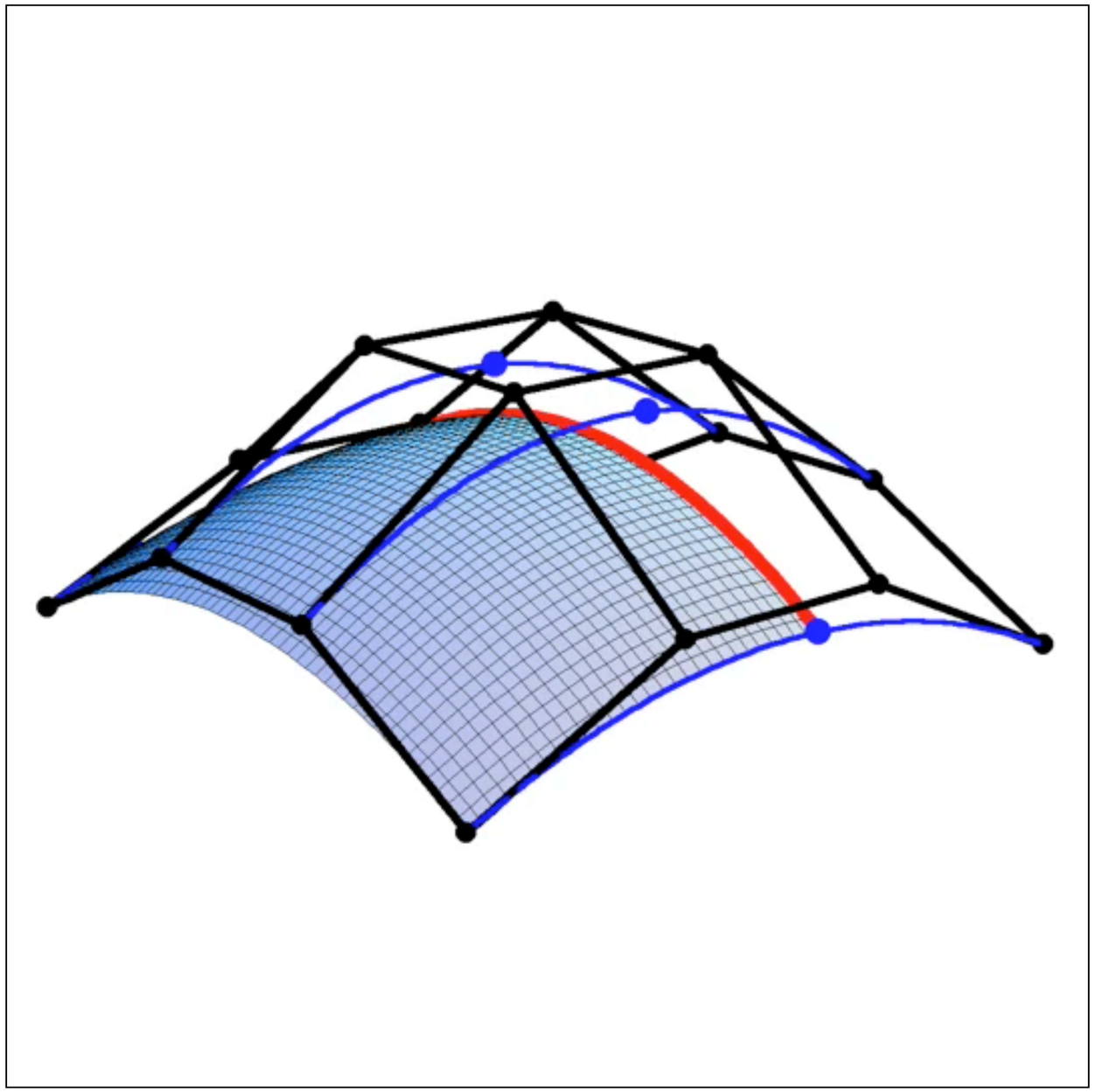
Utah Teapot

Bicubic Bézier Surface Patch



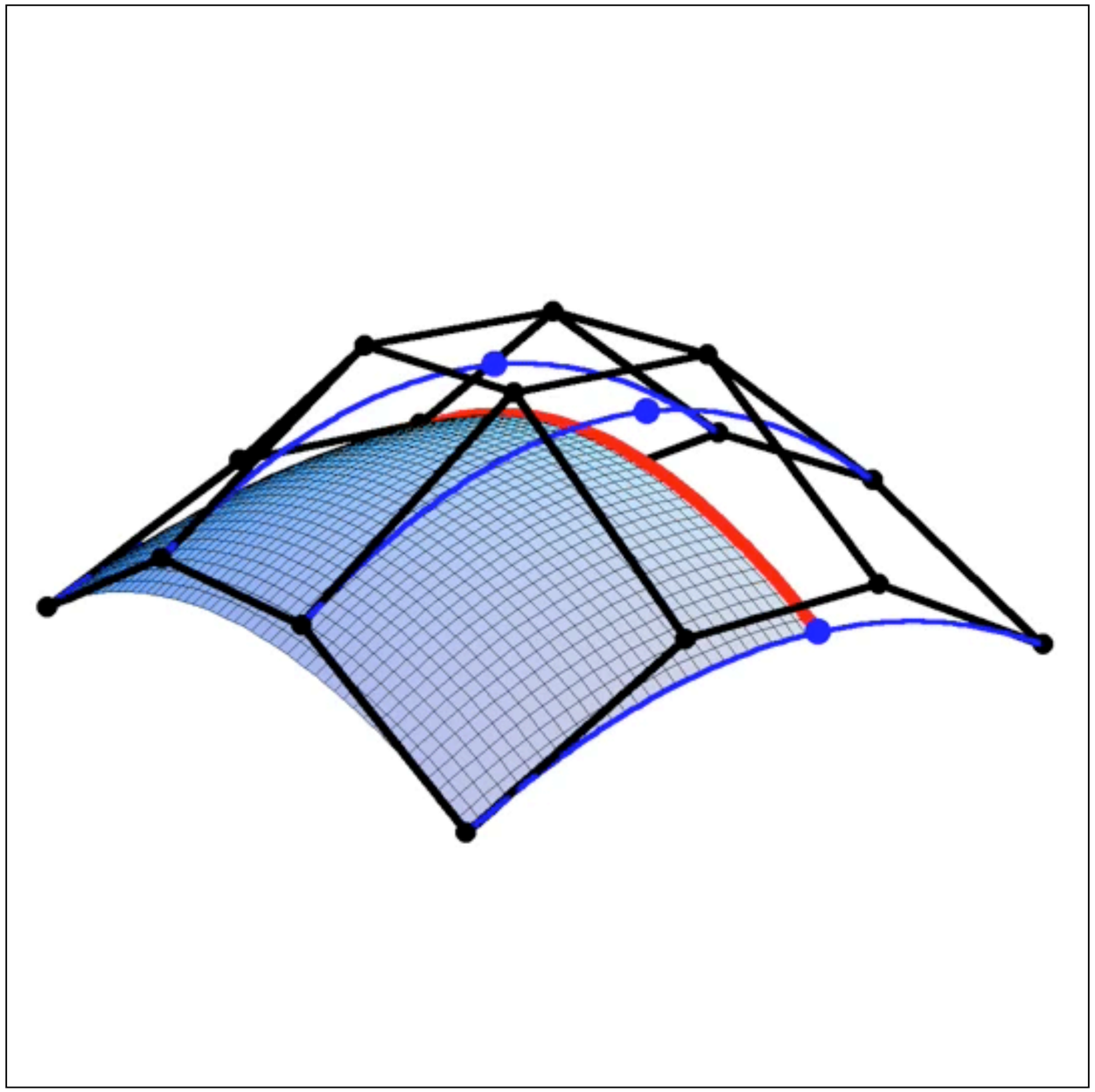
Bezier surface and 4 x 4 array of control points

Bicubic Bézier Surface Patch



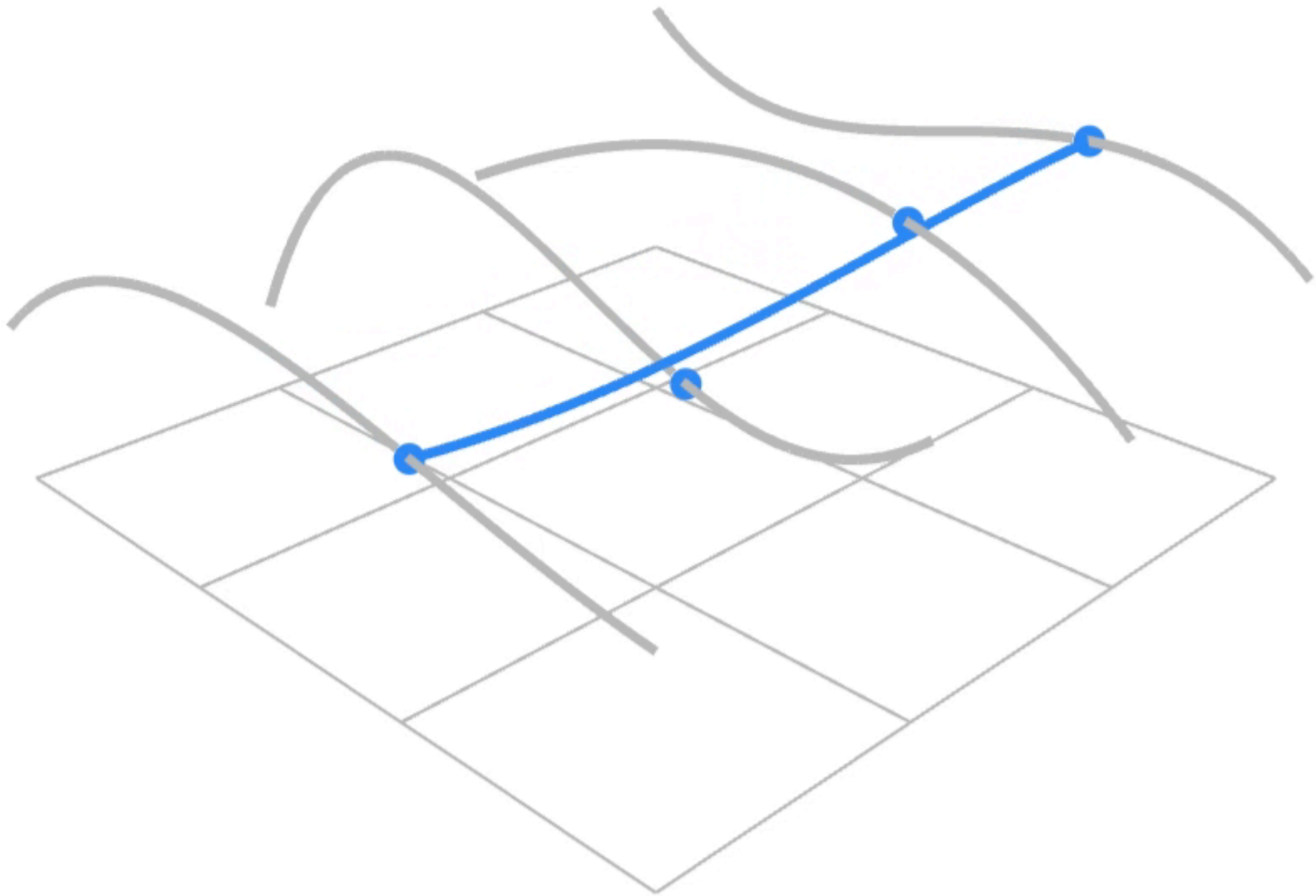
Bezier surface and 4 x 4 array of control points

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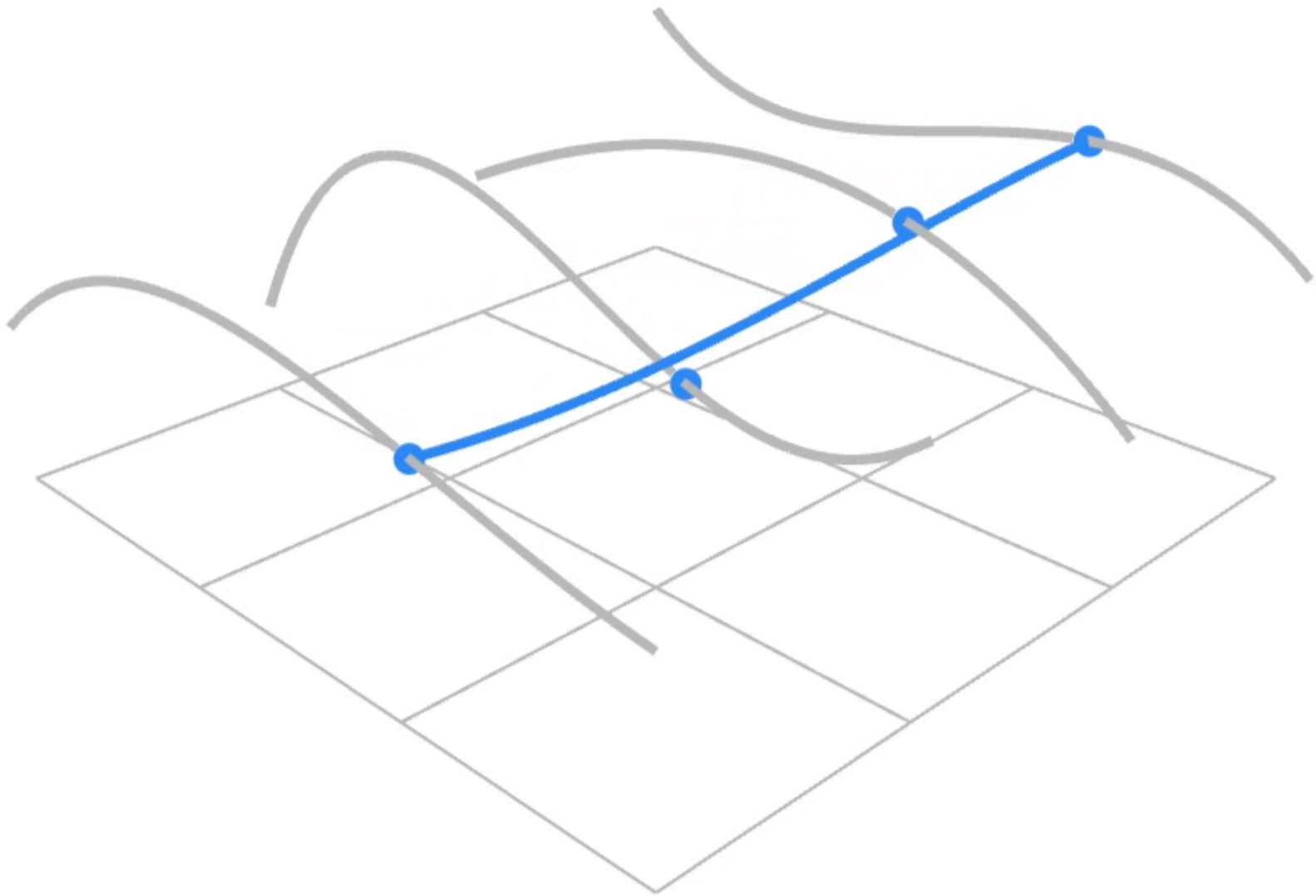
Bezier surface and 4 x 4 array of control points

Visualizing Bicubic Bézier Surface Patch



Animation: Steven Wittens, Making Things with Maths, http://acko.net

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Visualizing Bicubic Bézier Surface Patch

4x4 control points

- Each 4x1 control points in u define a Bezier curve
  - (4 Bezier curves in u)
- Corresponding points on these 4 Bezier curves define 4 control points for a “moving curve” in v
  - This “moving” curve sweeps out the 2D surface

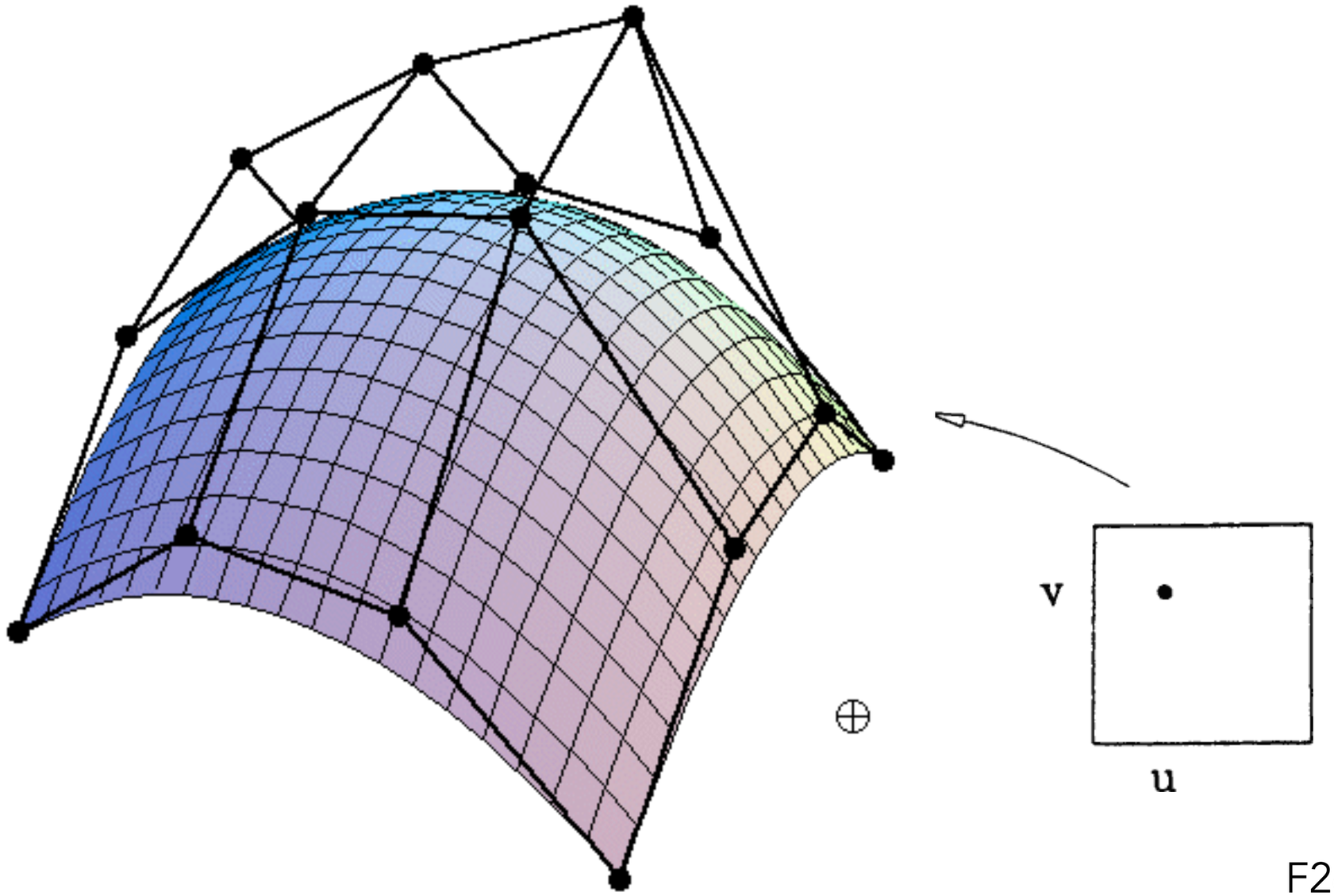
Evaluating Bézier Surfaces

Evaluating Surface Position For Parameters (u,v)

For bi-cubic Bezier surface patch,

Input: 4x4 control points

Output is 2D surface parameterized by (u,v) in $[0,1]^2$

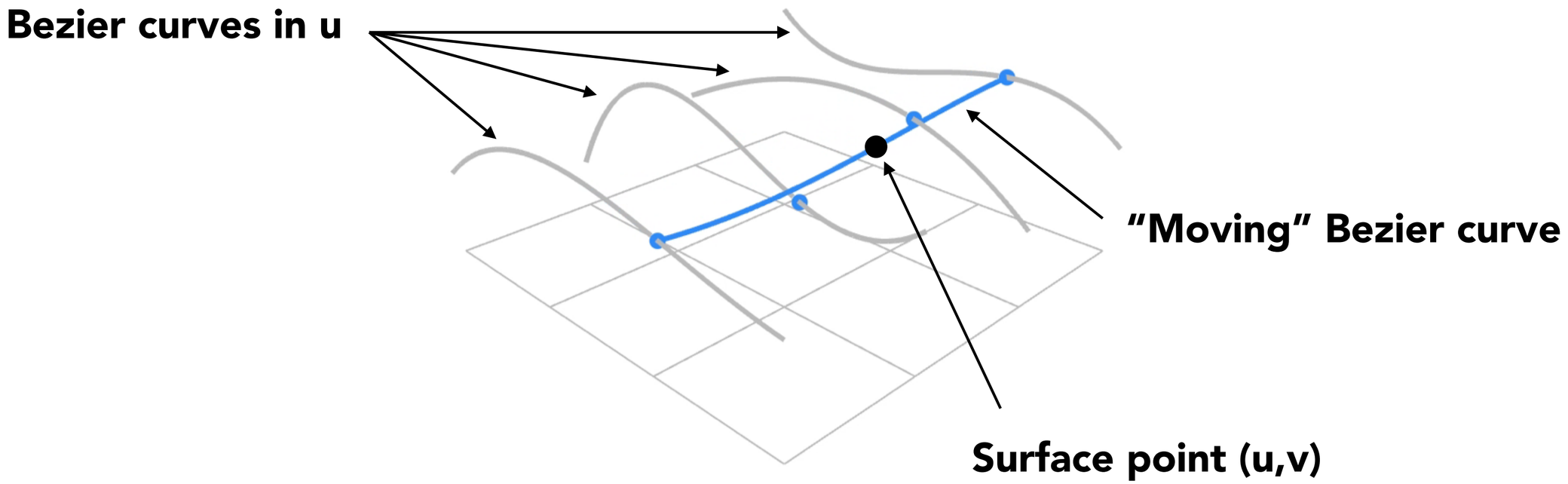


Method 1: Separable 1D de Casteljau Algorithm

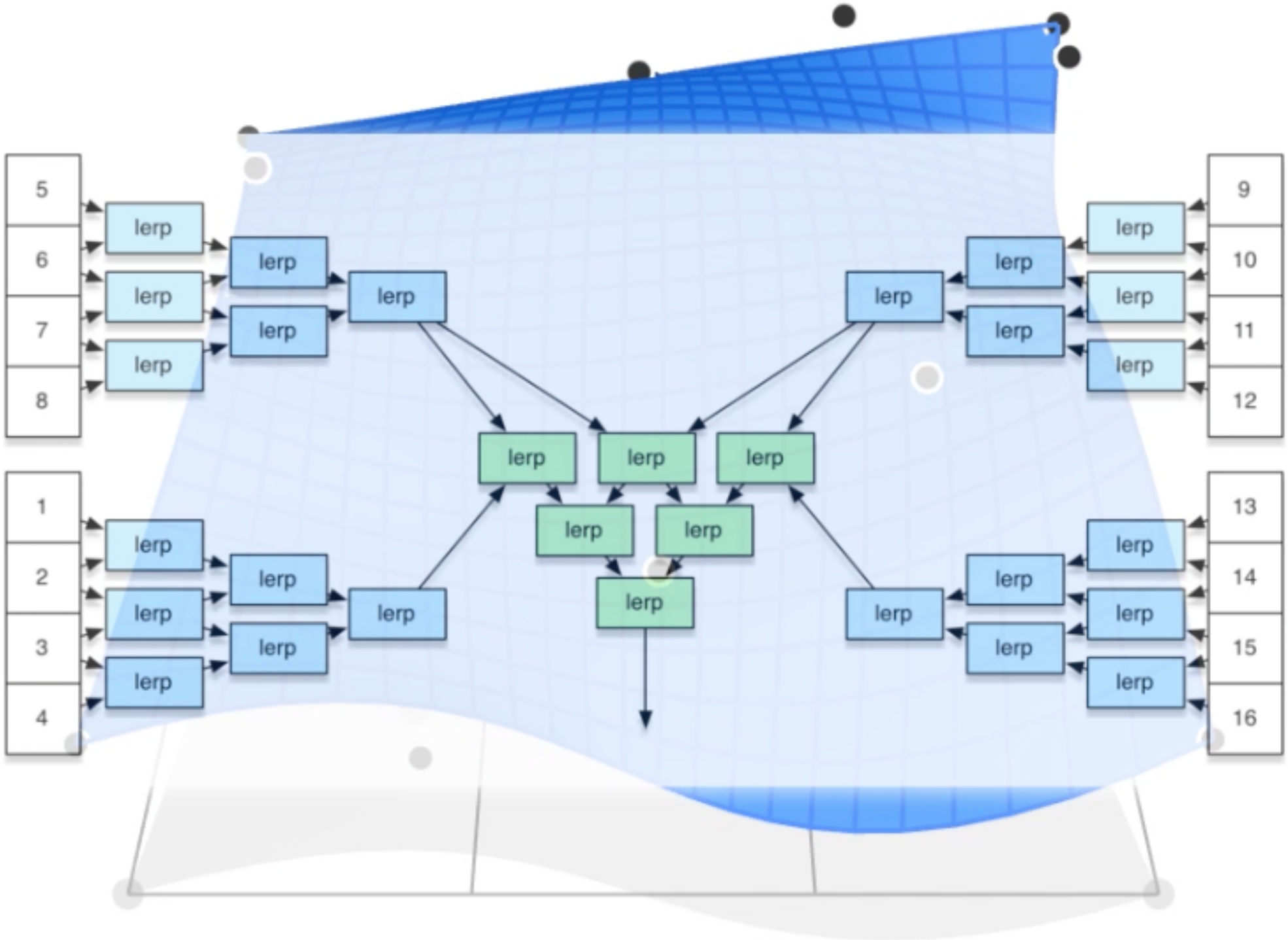
Goal: Evaluate surface position corresponding to (u,v)

(u,v)-separable application of de Casteljau algorithm

- Use de Casteljau to evaluate point u on each of the 4 Bezier curves in u. This gives 4 control points for the “moving” Bezier curve
- Use 1D de Casteljau to evaluate point v on the “moving” curve



Method 1: Separable 1D de Casteljau Algorithm



Method 2: Linear Algebra

$$\mathbf{x}(u) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_0 \\ \mathbf{p}_1 \\ \mathbf{p}_2 \\ \mathbf{p}_3 \end{bmatrix}$$

$$\mathbf{p}_0(v) = \mathscr{P}^3(v) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_{0,0} \\ \mathbf{p}_{0,1} \\ \mathbf{p}_{0,2} \\ \mathbf{p}_{0,3} \end{bmatrix}$$

$$\mathbf{p}_1(v) = \mathscr{P}^3(v) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_{1,0} \\ \mathbf{p}_{1,1} \\ \mathbf{p}_{1,2} \\ \mathbf{p}_{1,3} \end{bmatrix}$$

$$\mathbf{p}_2(v) = \mathscr{P}^3(v) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_{2,0} \\ \mathbf{p}_{2,1} \\ \mathbf{p}_{2,2} \\ \mathbf{p}_{2,3} \end{bmatrix}$$

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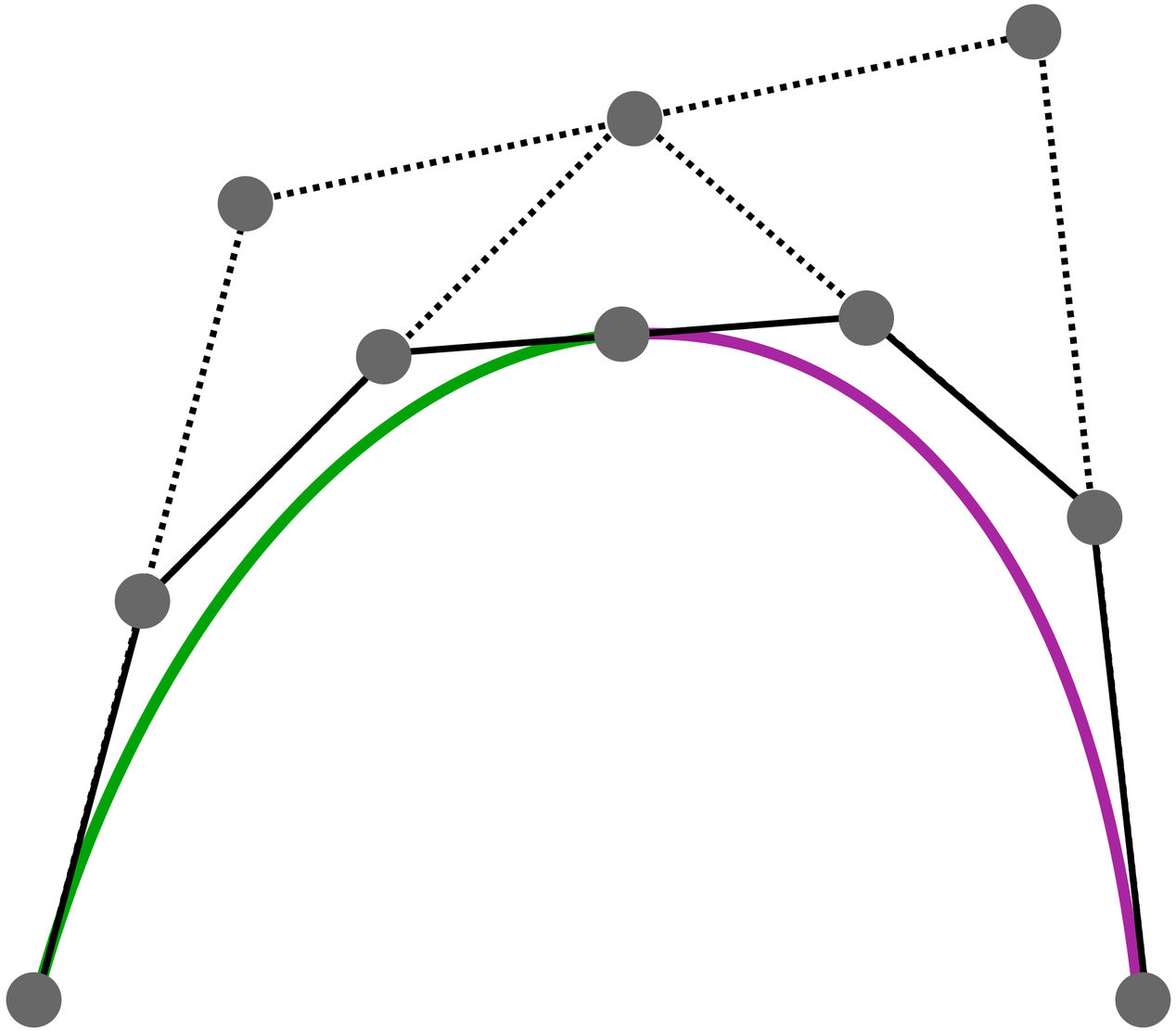
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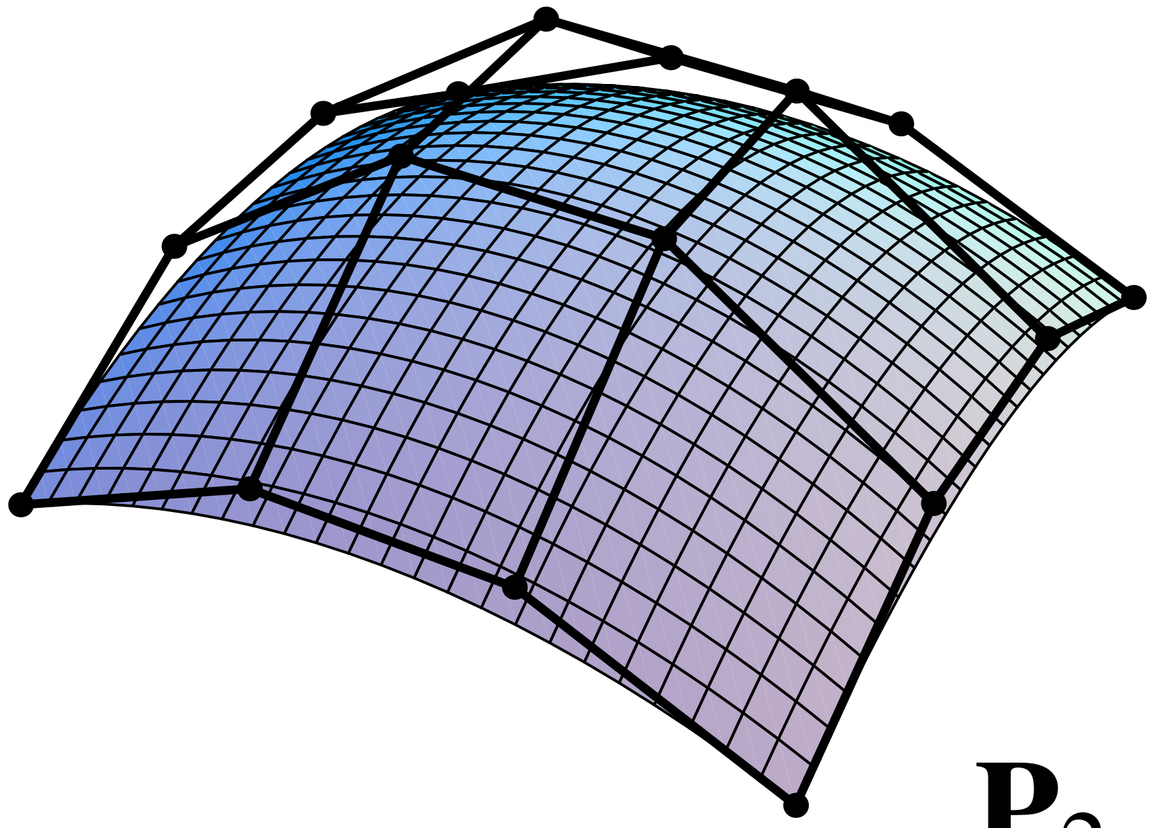
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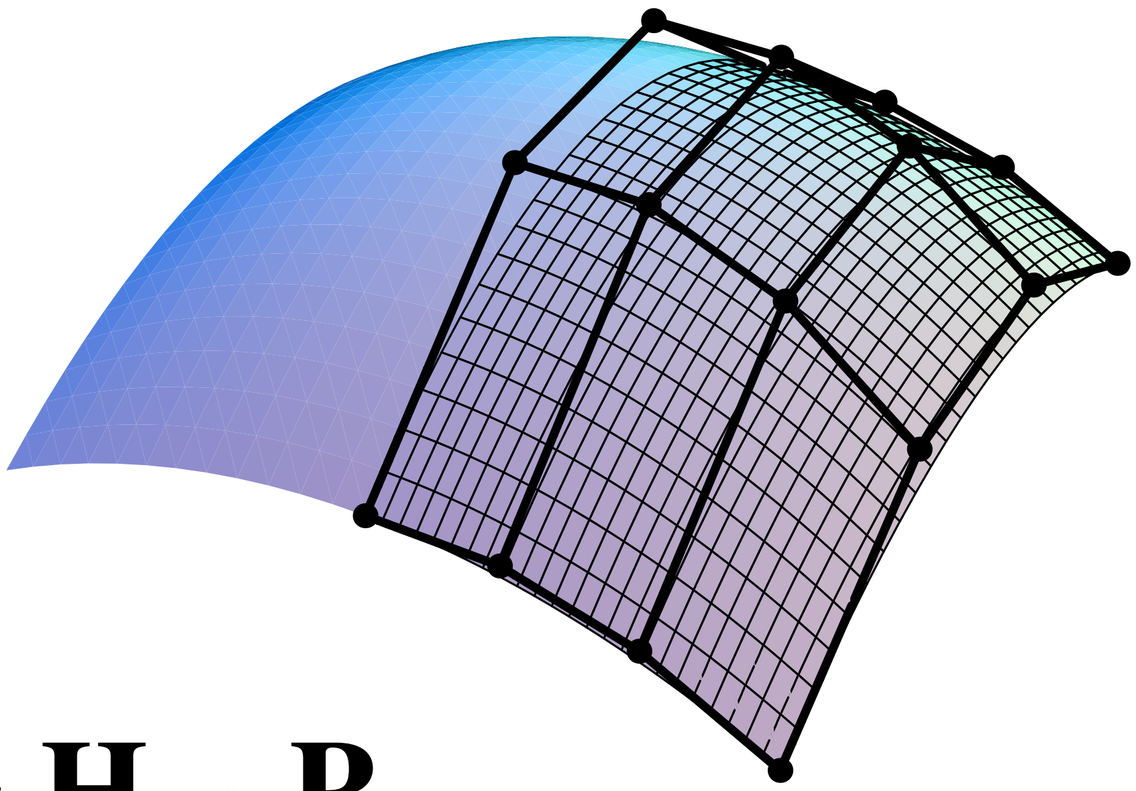
$$\mathbf{x}(u, v) = \mathscr{P}^3(u) \cdot \boldsymbol{\beta}_Z \cdot \begin{bmatrix} \mathbf{p}_{0,0} & \mathbf{p}_{0,1} & \mathbf{p}_{0,2} & \mathbf{p}_{0,3} \\ \mathbf{p}_{1,0} & \mathbf{p}_{1,1} & \mathbf{p}_{1,2} & \mathbf{p}_{1,3} \\ \mathbf{p}_{2,0} & \mathbf{p}_{2,1} & \mathbf{p}_{2,2} & \mathbf{p}_{2,3} \\ \mathbf{p}_{3,0} & \mathbf{p}_{3,1} & \mathbf{p}_{3,2} & \mathbf{p}_{3,3} \end{bmatrix} \cdot \boldsymbol{\beta}_Z^{\mathsf{T}} \cdot \mathscr{P}^3(v)^{\mathsf{T}}$$

Method 3: Bézier Subdivision



Bézier Subdivision



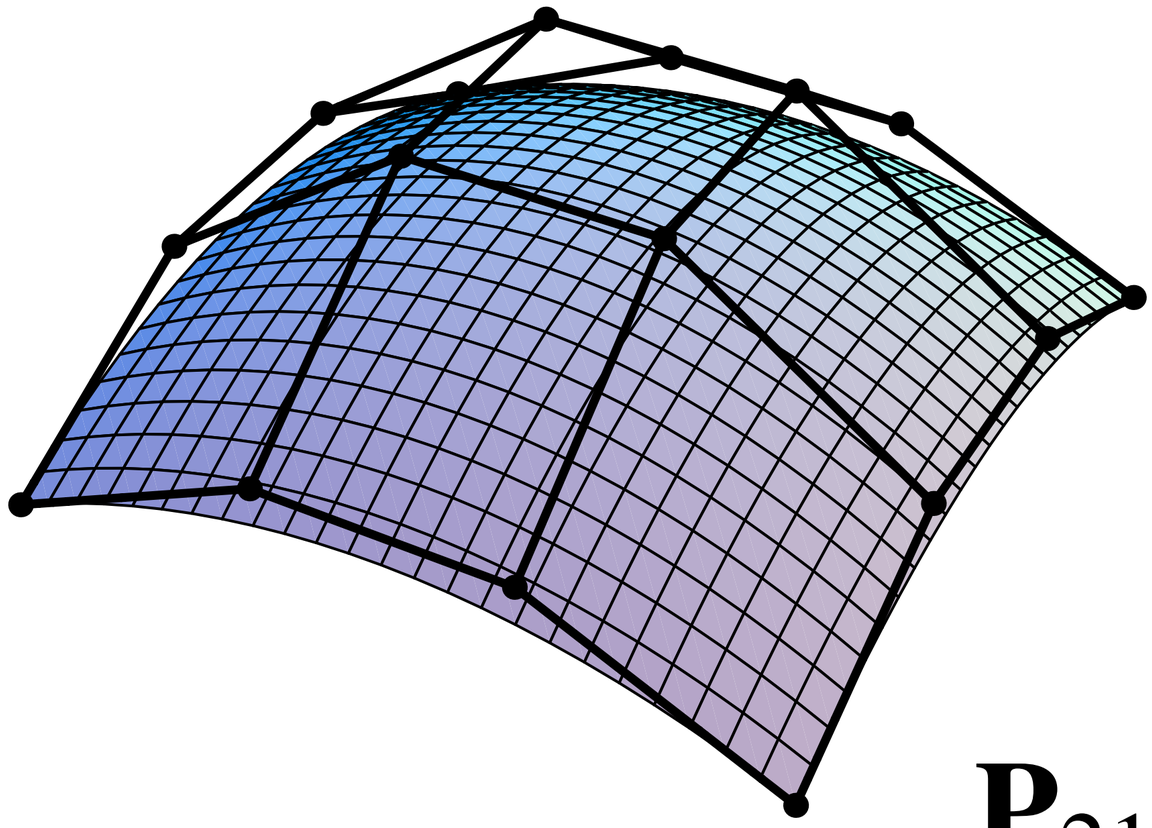


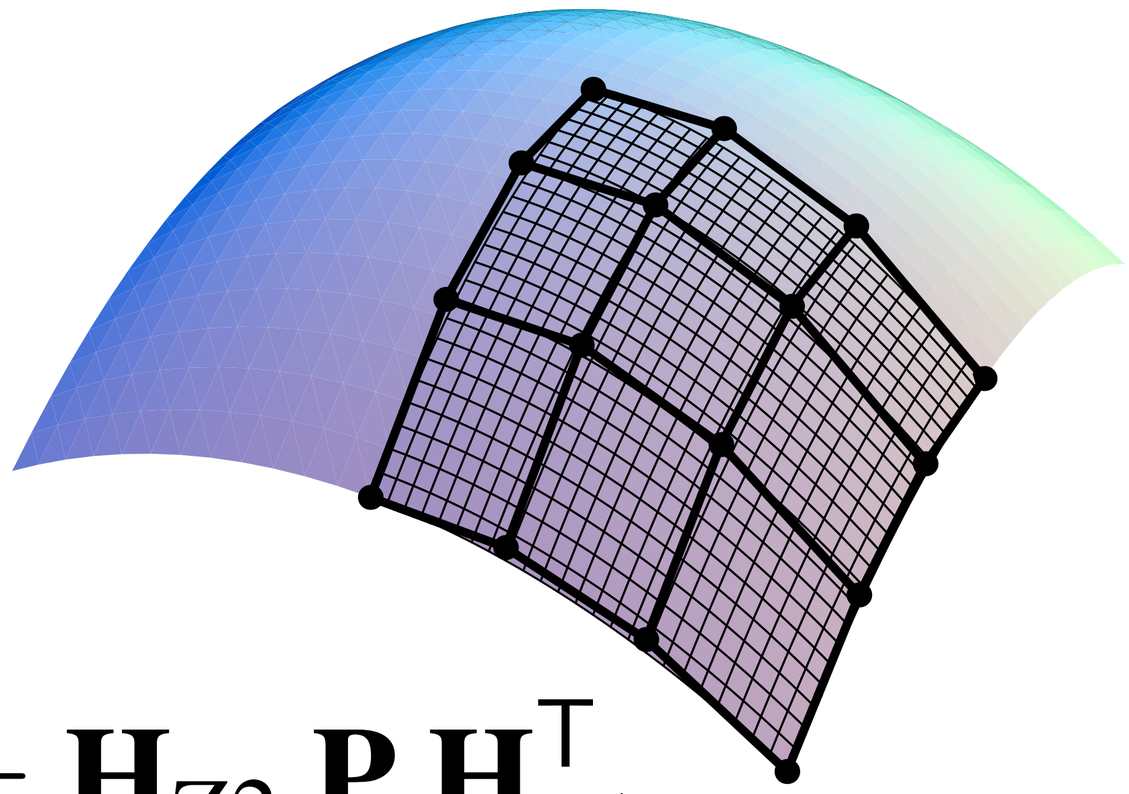
$$\mathbf{P}_{2\cdot} = \mathbf{H}_{Z2}\,\mathbf{P}$$

$$\mathbf{x}(u,v) = [1, u, u^2, u^3]\beta_Z \mathbf{P} \beta_Z^T [1, v, v^2, v^3]^T$$

4 x 4 matrix of control points

Bézier Subdivision





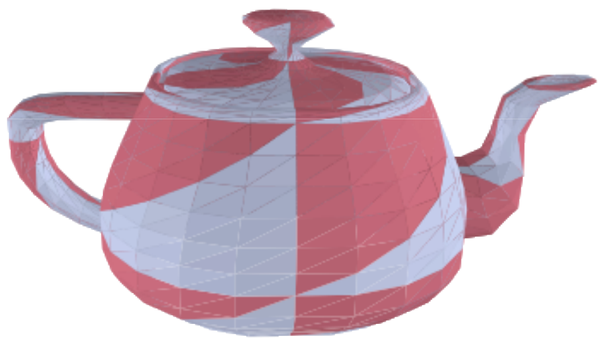
$$\mathbf{P}_{21} = \mathbf{H}_{Z2}\,\mathbf{P}\,\mathbf{H}_{Z1}^{\mathsf{T}}$$

$$\mathbf{x}(u,v) = [1, u, u^2, u^3]\beta_Z \mathbf{P} \beta_Z^{\mathsf{T}} [1, v, v^2, v^3]^{\mathsf{T}}$$

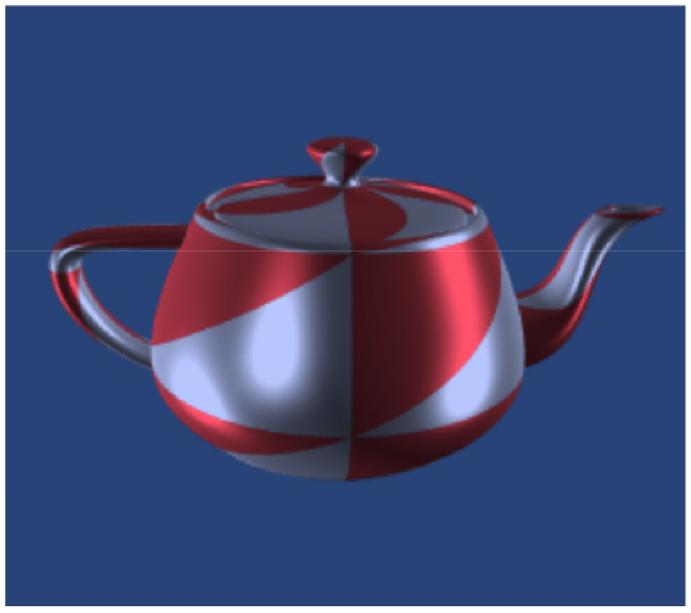
4 x 4 matrix of control points

Raytracing Bézier Surfaces

- Bézier clipping
  - reduces to root finding
  - exploits convex hull property for acceleration
  - many special cases, slow compared to sphere/polygon



control nets



Roth et al., EG 2001

Bézier surface

Method 4: Algebraic Evaluation

Let the moving curve be a degree m Bézier curve

$$\mathbf{b}^m(u) = \sum_{i=0}^{m} \mathbf{b}_i B_i^m(u)$$

(remember, Bernstein polynomials)

$$B_i^n(t) = \binom{n}{i} t^i (1-t)^{n-i}$$

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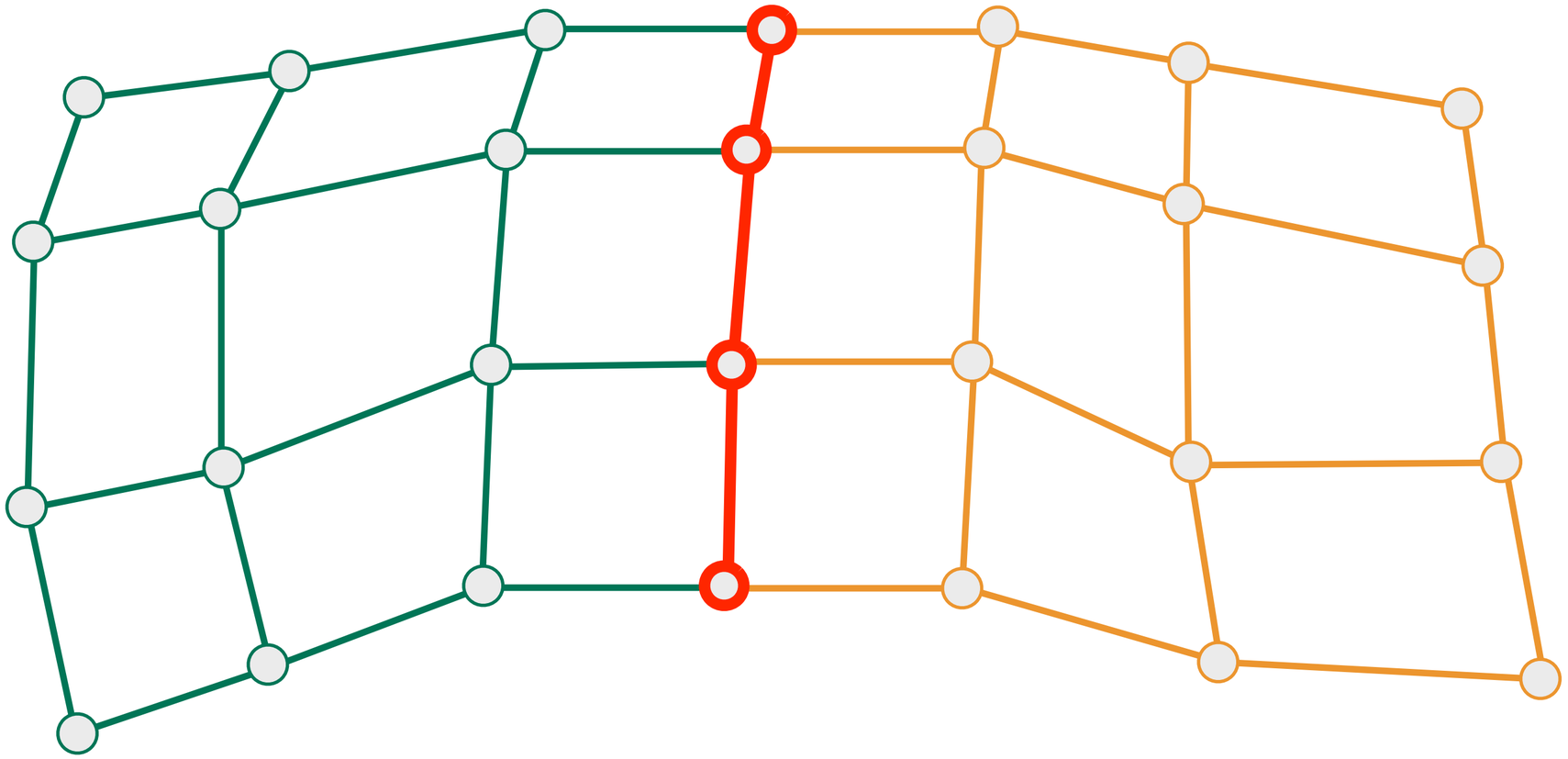
Tensor product Bézier patch

$$\mathbf{b}^{m,n}(u,v) = \sum_{i=0}^{m} \sum_{j=0}^{n} \mathbf{b}_{i,j} B_i^m(u) B_j^n(v)$$

Bézier Surface Continuity

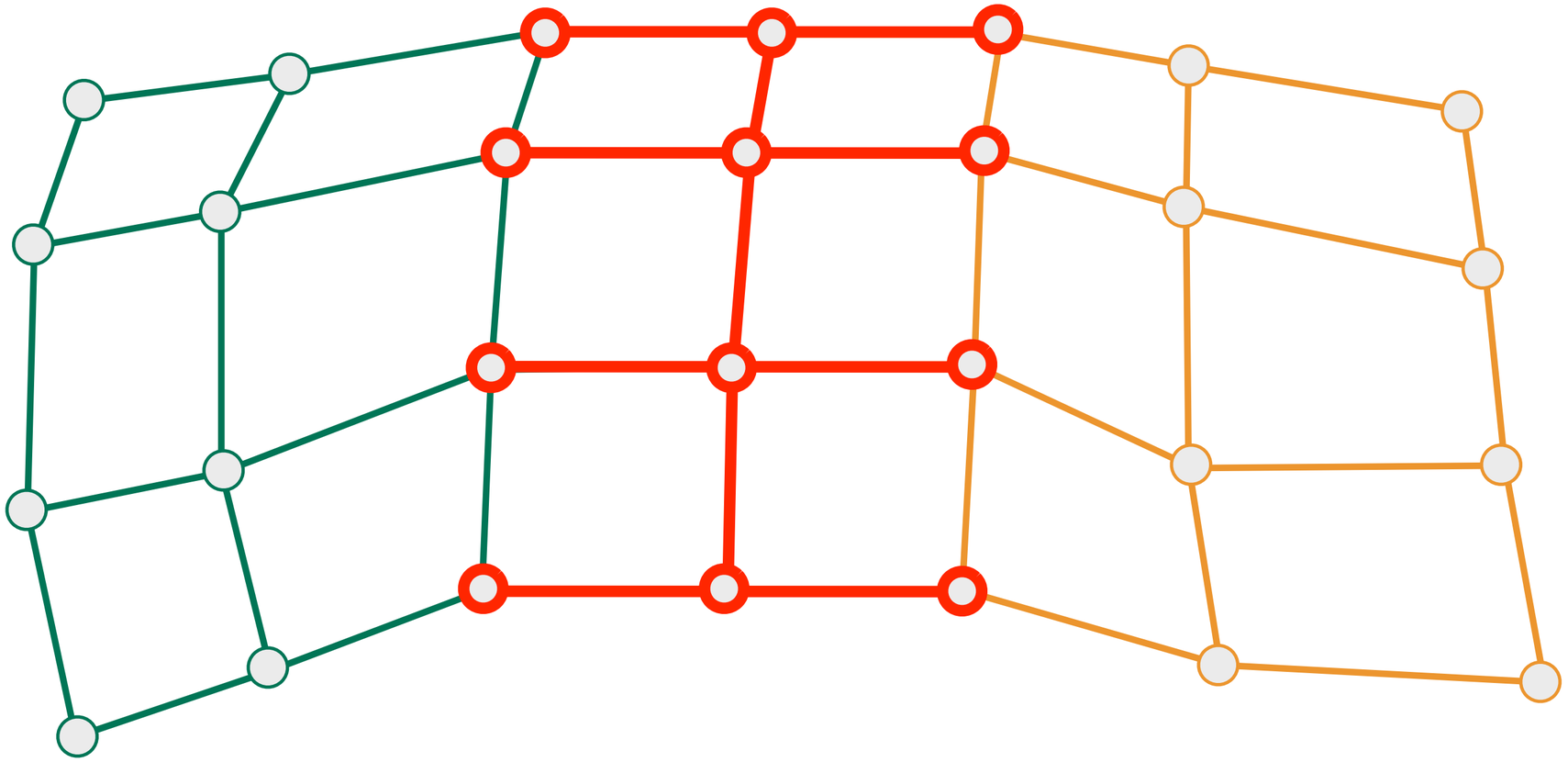
Piecewise Bézier Surfaces

C^0 continuity: Boundary curves



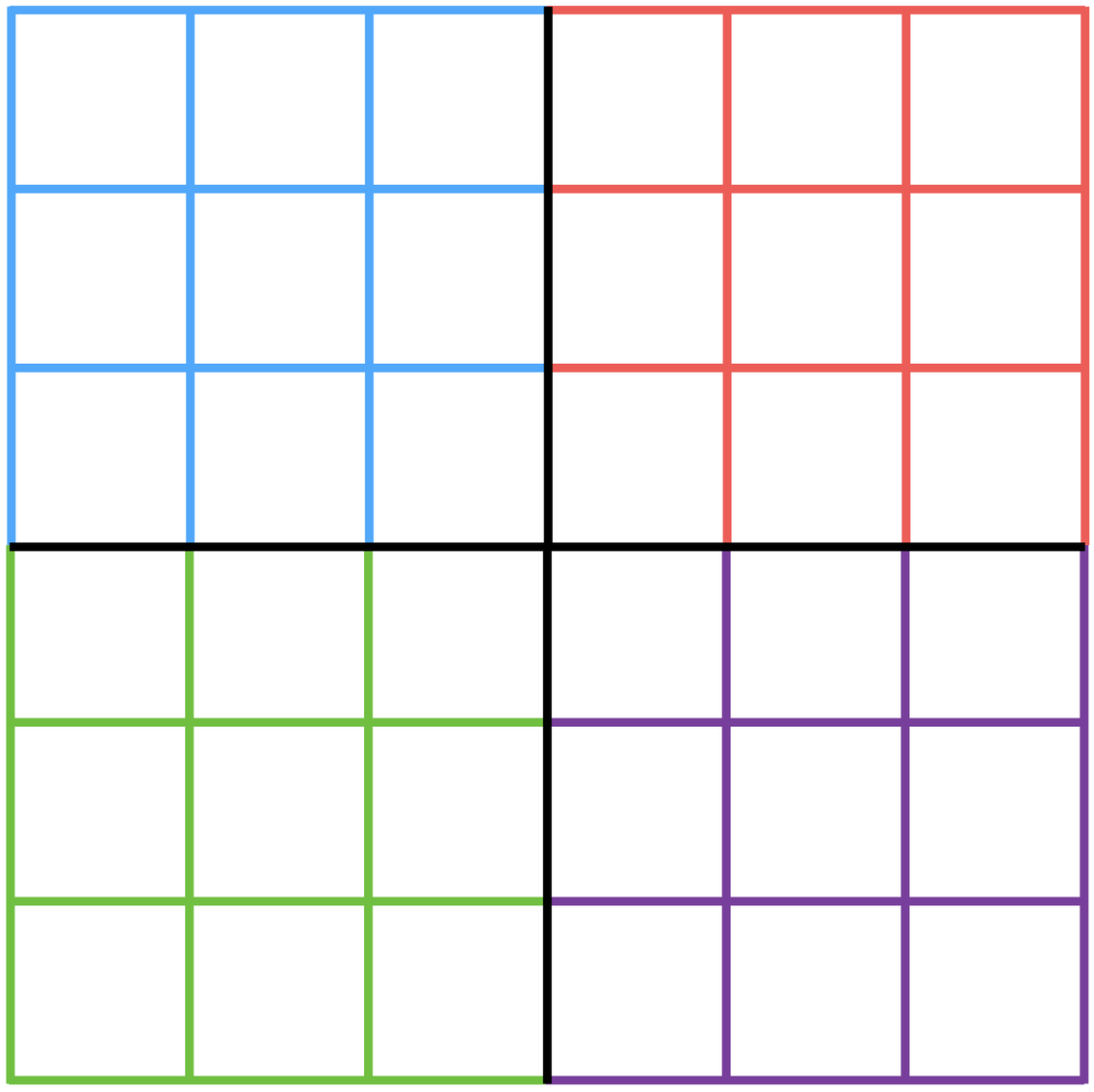
Piecewise Bézier Surfaces

C^1 continuity: Collinearity



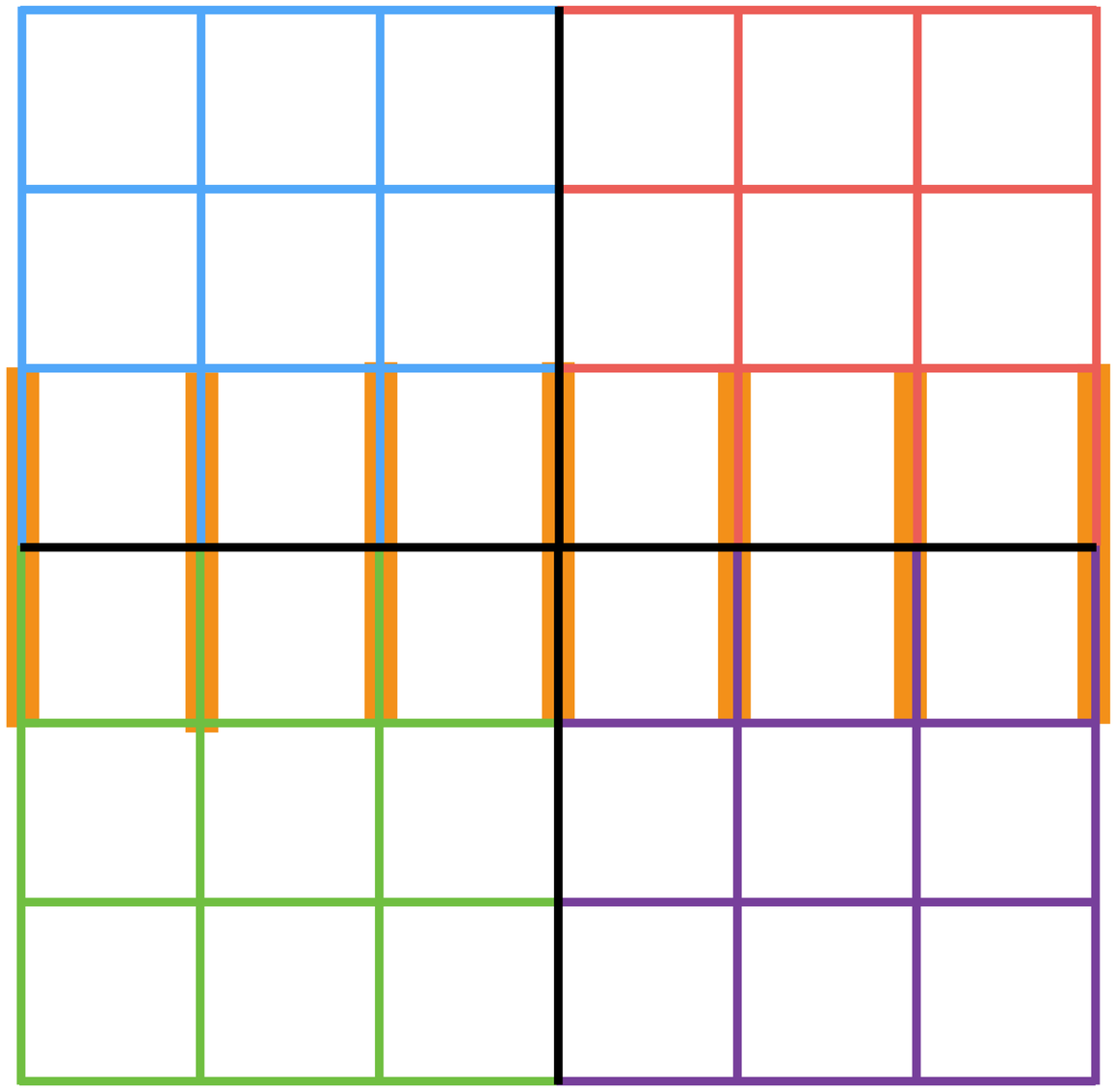
Piecewise Bézier Surfaces

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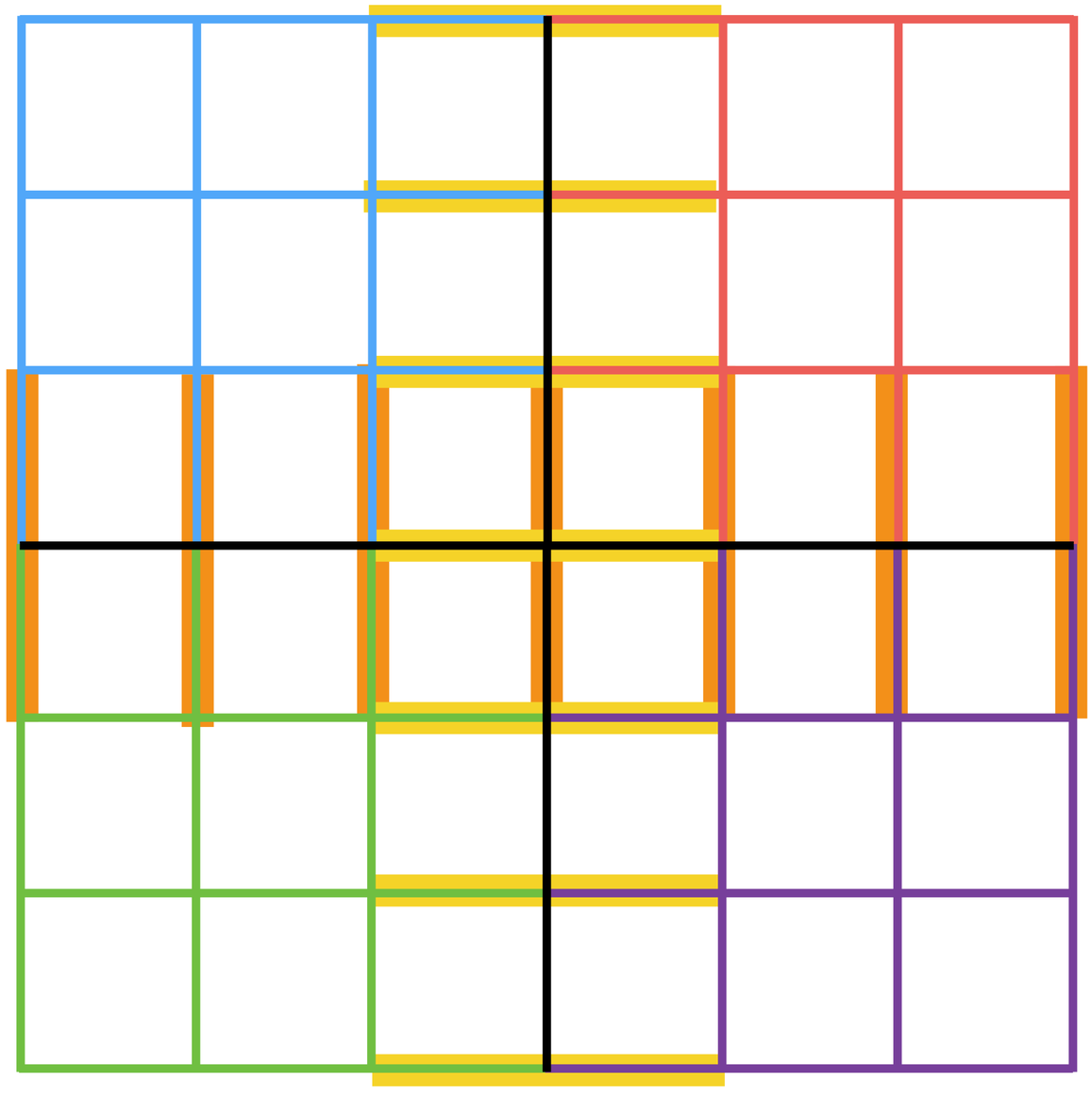
Piecewise Bézier Surfaces

C^1 continuity: Collinearity



Piecewise Bézier Surfaces

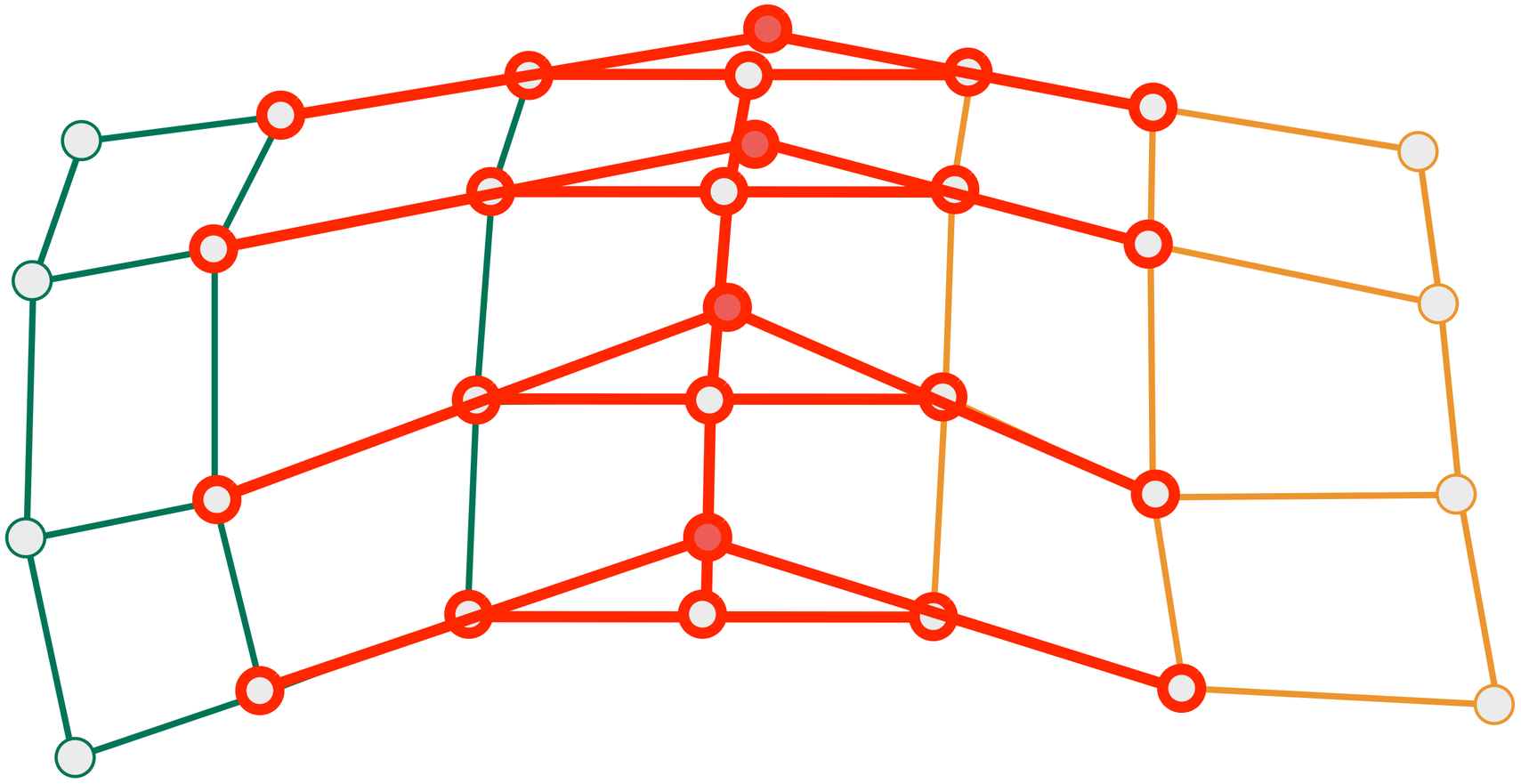
C^1 continuity: Collinearity



Piecewise Bézier Surfaces

C^2 continuity: A-frames

Will this be easy to work with?



Things to Remember

Splines

- Cubic Hermite and Catmull-Rom interpolation
- Matrix representation of cubic polynomials

Bézier curves

- Easy-to-control spline
- Recursive linear interpolation – de Casteljau algorithm
- Properties of Bézier curves
- Piecewise Bézier curve – continuity types and how to achieve

Bézier surfaces

- Bicubic Bézier patches – tensor product surface
- 2D de Casteljau algorithm

Acknowledgments

The CS 184 slide set was developed by:

- Ren Ng
- James O'Brien

With additional contributions from:

- Kayvon Fatahalian
- David Forsyth
- Pat Hanrahan
- Angjoo Kanazawa
- Steve Marschner
- Mark Pauly